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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 1: Electric Charges and Fields

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 70 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. A particle of mass m and charge q starts from rest and moves in an electric field $\vec{E} = E_0 \hat{i}$. After travelling a distance x in the field along the x-axis, the kinetic energy of the particle will be:

(A) $qE_0 x^2$

(B) $qE_0^2 x$

(C) $q^2 E_0 x$

(D) $q E_0 x$

Check Solution → [2026 • Set 55-3-1]

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2. An electric dipole $\vec{P} = 2qa\hat{i}$ is placed in x-y plane, centred at the origin. Let E_1 and E_2 be the magnitudes of electric field at a point distant r ($\gg a$) from its centre, along its axis and on its equatorial plane, respectively. Then the value of $(\frac{E_1}{E_2})$ is:

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 2

(D) 4

Check Solution → [2026 • Set 55-3-2]

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3. An electric dipole with dipole moment $\vec{P} = (2{\cdot}54 \times 10^{-26}\ C)(2{\cdot}00\,\hat{i} + 3{\cdot}00\,\hat{j})$ is placed in an electric field $\vec{E} = (100\,\hat{i})\ N/C$. An external agent turns the dipole until its electric dipole moment is $\vec{P} = (2{\cdot}54 \times 10^{-26}\ C)(-3{\cdot}00\,\hat{i} + 2{\cdot}00\,\hat{j})$. The work done by the agent is:

(A) $2{\cdot}54 \times 10^{-23}\ J$

(B) $1{\cdot}27 \times 10^{-24}\ J$

(C) $1{\cdot}02 \times 10^{-23}\ J$

(D) $2{\cdot}29 \times 10^{-26}\ J$

Check Solution → [2026 • Set 55-3-3]

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4. Two small identical metallic balls having charges q and −2q are kept far at a separation r. They are brought in contact and then separated at distance $\frac{r}{2}$. Compared to the initial force F, they will now:

(A) attract with a force $\frac{F}{2}$

(B) repel with a force $\frac{F}{2}$

(C) repel with a force F

(D) attract with a force F

Check Solution → [2026 • Set 55-5-1]

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5. Two point charges $-Q$ and Q are located at points $(d, 0)$ and $(0, d)$ respectively, in x-y plane. The electric field $\vec{E}$ at the origin will be:

(A) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{2}Q}{d^2}(\hat{i}-\hat{j})$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{2}Q}{d^2}(\hat{i}+\hat{j})$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(-\hat{i}-\hat{j})$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(\hat{i}-\hat{j})$

Check Solution → [2026 • Set 55-5-2]

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6. An electron is moving with velocity $v_0\,\hat{i}$. If a uniform electric field $\vec{E} = E_0\,\hat{j}$ is set up in the region, the electron will:

(A) describe a circular path

(B) describe a helical path

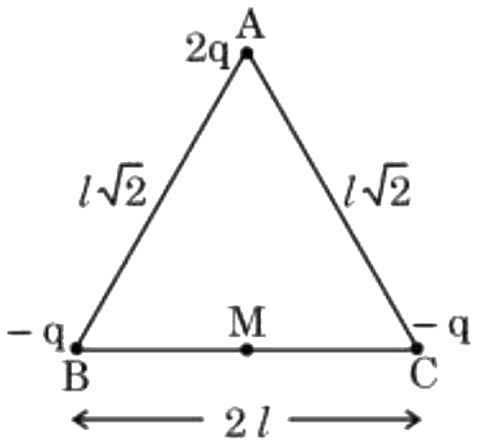
(C) describe a parabolic path

(D) continue moving without any deviation

Check Solution → [2026 • Set 55-5-2]

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7. The figure shows three point charges kept at the vertices of triangle ABC. The net electric field, due to this system of charges, at the midpoint M of base BC will be:



(A) $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$ pointing along MA

(B) $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$ pointing along AM

(C) $\frac{1}{2\pi\varepsilon_0}\frac{q}{l^2}$ pointing along AM

(D) Zero

Check Solution → [2026 • Set 55-5-3]

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8. Figure shows variation of Coulomb force (F) acting between two point charges with $\frac{1}{r^2}$, r being the separation between the two charges (q_1, q_2) and (q_2, q_3). If q_2 is positive and least in magnitude, then the magnitudes of q_1, q_2 and q_3 are such that:

(A) $q_2 < q_3 < q_1$

(B) $q_3 < q_2 < q_1$

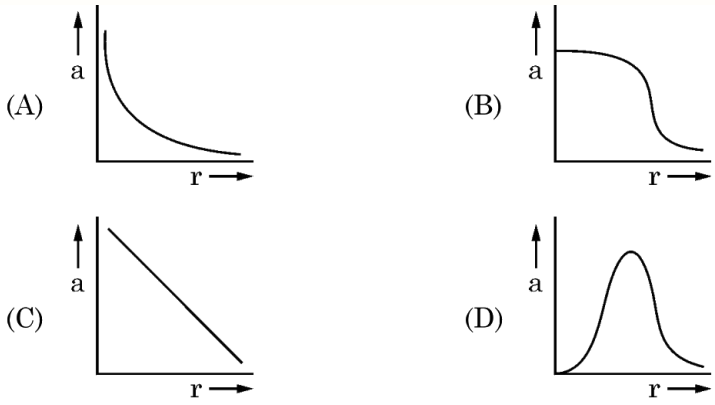
(C) $q_1 < q_2 < q_3$

(D) $q_2 < q_1 < q_3$

Check Solution → [2025 • Set 55-1-1]

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9. A charge Q is fixed in position. Another charge q is brought near charge Q and released from rest. Which of the following graphs is the correct representation of the acceleration of the charge q as a function of its distance r from charge Q?



(A) Option A (graph as shown)

(B) Option B (graph as shown)

(C) Option C (graph as shown)

(D) Option D (graph as shown)

Check Solution → [2025 • Set 55-1-3]

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10. Two charges $-q$ each are placed at the vertices A and B of an equilateral triangle ABC. If M is the mid-point of AB, the net electric field at C will point along

(A) CA

(B) CB

(C) MC

(D) CM

Check Solution → [2025 • Set 55-2-1]

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11. Which one of the following statements is correct? Electric field due to static charges is

(A) conservative and field lines do not form closed loops.

(B) conservative and field lines form closed loops.

(C) non-conservative and field lines do not form closed loops.

(D) non-conservative and field lines form closed loops.

Check Solution → **[2025 • Set 55-2-1]**

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12. Two identical point charges are placed at the two vertices A and B of an equilateral triangle of side . The magnitude of the electric field at the third vertex P is E. If a hollow conducting sphere of radius (/4) is placed at P, the magnitude of the electric field at point P now becomes

(A) 2E

(B) E

(C) $\frac{E}{2}$

(D) zero

Check Solution → **[2025 • Set 55-2-2]**

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13. Consider two identical dipoles D_1 and D_2. Charges $-q$ and q of dipole D_1 are located at (0, 0) and (a, 0) and that of dipole D_2 at (0, a) and (0, 2a) in x-y plane, respectively. The net dipole moment of the system is

(A) $qa(\hat{i}+\hat{j})$

(B) $-qa(\hat{i}+\hat{j})$

(C) $qa(\hat{i}-\hat{j})$

(D) $-qa(\hat{i}-\hat{j})$

Check Solution → **[2025 • Set 55-2-3]**

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14. An electric dipole of dipole moment 1.0×10^{-12} Cm lies along x-axis. An electric field of magnitude $2.0 \times 10^4\ N\,C^{-1}$ is switched on at an instant in the region. The

unit vector along the electric field is $\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$. The magnitude of the torque acting on the dipole at that instant is :

(A) 0.5×10^{-8} Nm

(B) 1.0×10^{-8} Nm

(C) 2.0×10^{-8} Nm

(D) 4.0×10^{-8} Nm

Check Solution → [2025 • Set 55-4-2]

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15. Two point charges Q and $-q$ are held 'r' distance apart in free space. A uniform electric field E is applied in the region perpendicular to the line joining the two charges. Which one of the following angles will the direction of the net force acting on charge $-q$ make with the line joining Q and $-q$?

(A) $\tan^{-1}\frac{4\pi\varepsilon_0 E r^2}{Q}$

(B) $\cot^{-1}\frac{4\pi\varepsilon_0 E r^2}{Q}$

(C) $\tan^{-1}\frac{Q E}{4\pi\varepsilon_0 r^2}$

(D) $\cot^{-1}\frac{Q E}{4\pi\varepsilon_0 r^2}$

Check Solution → [2025 • Set 55-4-3]

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16. A particle of mass m and charge q moving with velocity $v\hat{i}$ is subjected to a uniform electric field $E\hat{j}$. The particle will initially have a tendency to move in a circle of radius:

(A) $(\frac{mv^2}{qE})$ in x-y plane

(B) $\frac{mv^2}{qE}$ in x-z plane

(C) $\frac{mv^2}{qE}$ in x-y plane

(D) $(\frac{mv^2}{qE})$ in y-z plane

Check Solution → [2025 • Set 55-6-3]

17. A thin plastic rod is bent into a circular ring of radius R. It is uniformly charged with charge density λ. The magnitude of the electric field at its centre is:

(A) $\frac{\lambda}{2\varepsilon_0 R}$

(B) Zero

(C) $\frac{\lambda}{4\pi\varepsilon_0 R}$

(D) $\frac{\lambda}{4\varepsilon_0 R}$

Check Solution → [2024 • Set 55-1-1]

18. A charged sphere of radius r has surface charge density σ. The electric field on its surface is E. If the radius of the sphere is doubled, keeping charge density the same, the ratio of the electric field on the old sphere to that on the new sphere will be:

(A) 1

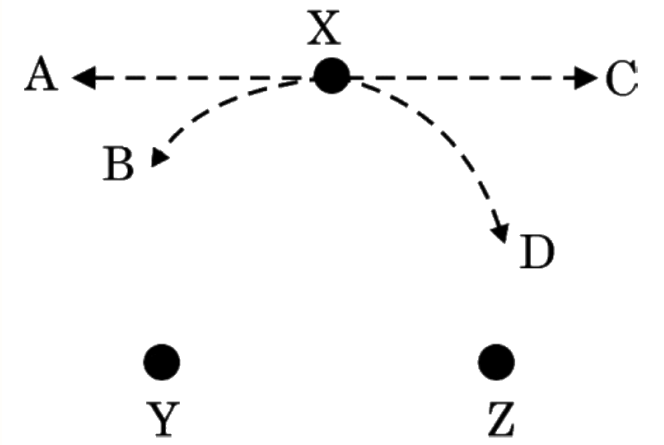
(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) 4

Check Solution → [2024 • Set 55-1-2]

19. Three small charged spheres X, Y and Z carrying charges $+q$, $-q$ and $+q$ respectively are placed equidistant from each other, as shown in the figure. The spheres Y and Z are held in place. Initially X is also held in place, but is otherwise free to move. When X is released, the path followed by it will be:



(A) A

(B) B

(C) C

(D) D

Check Solution → [2024 • Set 55-1-3]

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20. Two charged particles P and Q, having the same charge but different masses m_P and m_Q, start from rest and travel equal distances in a uniform electric field E in time t_P and t_Q respectively. Neglecting the effect of gravity, the ratio $\frac{t_P}{t_Q}$ is:

(A) $\frac{m_P}{m_Q}$

(B) $\frac{m_Q}{m_P}$

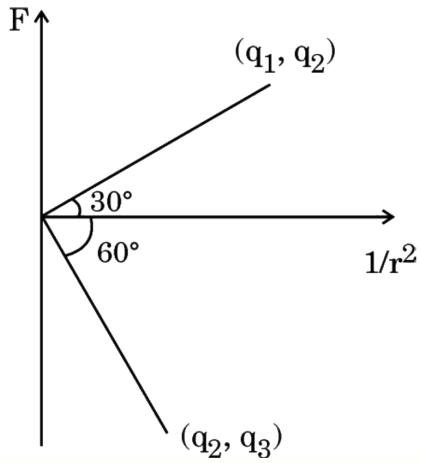
(C) $\sqrt{\frac{m_P}{m_Q}}$

(D) $\sqrt{\frac{m_Q}{m_P}}$

Check Solution → [2024 • Set 55-2-1]

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21. Coulomb force F versus $\frac{1}{r^2}$ graphs for two pairs of point charges (q_1 and q_2) and (q_1 and q_3) are shown in the figure. The ratio of charges $\frac{q_2}{q_3}$ is:



(A) 3

(B) $\frac{1}{3}$

(C) $\frac{3}{2}$

(D) $\frac{2}{3}$

Check Solution → [2024 • Set 55-2-3]

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22. An isolated conductor, with a cavity, has a net charge $+Q$. A point charge $+q$ is inside the cavity. The charges on the cavity wall and the outer surface are respectively:

(A) 0 and Q

(B) $-q$ and $Q - q$

(C) $-q$ and $Q + q$

(D) 0 and $Q - q$

Check Solution → [2024 • Set 55-3-2]

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23. Two identical small conducting balls B_1 and B_2 are given −7 pC and +4 pC charges respectively. They are brought in contact with a third identical ball B_3 and then separated. If the final charge on each ball is −2 pC, the initial charge on B_3 was

(A) −2 pC

(B) −3 pC

(C) −5 pC

(D) −15 pC

Check Solution → [2024 • Set 55-4-1]

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24. An electric dipole of dipole moment $\vec{p}$ is kept in a uniform electric field $\vec{E}$. The amount of work done to rotate it from the position of stable equilibrium to that of unstable equilibrium will be

(A) 2pE

(B) −2pE

(C) pE

(D) zero

Check Solution → [2024 • Set 55-4-3]

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25. An infinite long straight wire having a charge density λ is kept along $y'y$ axis in x-y plane. The Coulomb force on a point charge q at a point $P(x,0)$ will be

(A) attractive and $\frac{\lambda q}{2\pi\varepsilon_0 x}$

(B) repulsive and $\frac{\lambda q}{2\pi\varepsilon_0 x}$

(C) attractive and $\frac{\lambda q}{\pi\varepsilon_0 x}$

(D) repulsive and $\frac{\lambda q}{\pi\varepsilon_0 x}$

Check Solution → [2024 • Set 55-4-3]

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26. For Questions 13 to 16, two statements are given - one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below:
(A) If both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).

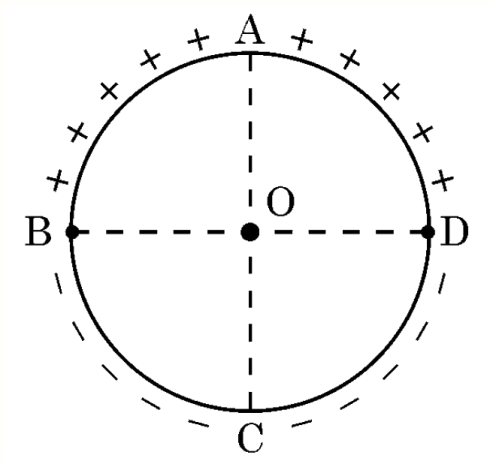
(B) If both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(C) If Assertion (A) is true but Reason (R) is false.

(D) If both Assertion (A) and Reason (R) are false.

Assertion (A): Equal amount of positive and negative charges are distributed uniformly on two halves of a thin circular ring as shown in figure. The resultant electric field at the centre O of the ring is along OC.

Reason (R): It is so because the net potential at O is not zero.



(A) Both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → **[2024 • Set 55-5-1]**

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27. A point charge situated at a distance 'r' from a short electric dipole on its axis, experiences a force $\vec{F}$. If the distance of the charge is '2r', the force on the charge will be:

(A) $\frac{F}{16}$

(B) $\frac{F}{8}$

(C) $\frac{F}{4}$

(D) $\frac{F}{2}$

Check Solution → [2023 • Set 55-1-1]

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28. The magnitude of the electric field due to a point charge object at a distance of 4.0 m is 9 N/C. From the same charged object the electric field of magnitude, 16 N/C will be at a distance of

(A) 1 m

(B) 2 m

(C) 3 m

(D) 6 m

Check Solution → [2023 • Set 55-2-1]

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29. An isolated point charge particle produces an electric field $\vec{E}$ at a point 3 m away from it. The distance of the point at which the field is $\frac{\vec{E}}{4}$ will be

(A) 2 m

(B) 3 m

(C) 4 m

(D) 6 m

Check Solution → [2023 • Set 55-2-2]

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30. An electric dipole of dipole moment 2×10^{-8} $C\,m$ in a uniform electric field experiences a maximum torque of 6×10^{-4} $N\,m$. The magnitude of electric field is

(A) $2.2\times 10^{2}\ V\,m^{-1}$

(B) $1.2\times 10^{4}\ V\,m^{-1}$

(C) $3.0\times 10^{4}\ V\,m^{-1}$

(D) $4.2\times 10^{2}\ V\,m^{-1}$

Check Solution → [2023 • Set 55-2-3]

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31. An electron experiences a force $(1.6 \times 10^{-16}\ N)\ \hat{i}$ in an electric field $\vec{E}$. The electric field $\vec{E}$ is:

(A) $(1.0 \times 10^{3}\ N/C)\ \hat{i}$

(B) $-(1.0 \times 10^{3}\ N/C)\ \hat{i}$

(C) $(1.0 \times 10^{-3}\ N/C)\ \hat{i}$

(D) $-(1.0 \times 10^{-3}\ N/C)\ \hat{i}$

Check Solution → [2023 • Set 55-3-1]

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32. Which one of the following is not a scalar quantity?

(A) Electric field

(B) Voltage

(C) Resistivity

(D) Power

Check Solution → [2023 • Set 55-3-1]

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33. An infinitely long uniformly charged wire produces an electric field of $1.8 \times 10^{4}\ N\,C^{-1}$ at a distance of 1.0 cm. The linear charge density on the wire is:

(A) $1.12 \times 10^{-1}\ C\,m^{-1}$

(B) $3.08 \times 10^{-2}\ C\,m^{-1}$

(C) $1.0 \times 10^{-2}\ C\,m^{-1}$

(D) $1.0 \times 10^{-7}\ C\,m^{-1}$

Check Solution → [2023 • Set 55-3-2]

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34. Two charges q_1 and q_2 are placed at the centres of two spherical conducting shells of radius r_1 and r_2 respectively. The shells are arranged such that their centres are d [> $(r_1 + r_2)$] distance apart. The force on q_2 due to q_1 is:

(A) $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{d^2}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{(d-r_1)^2}$

(C) Zero

(D) $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{[d-(r_1+r_2)]^2}$

Check Solution → [2023 • Set 55-4-1]

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35. **Case Study:** Electrostatics deals with the study of forces, fields and potentials arising from static charges. Force and electric field, due to a point charge is basically determined by Coulomb's law. For symmetric charge configurations, Gauss's law, which is also based on Coulomb's law, helps us to find the electric field. A charge/a system of charges like a dipole experience a force/torque in an electric field. Work is required to be done to provide a specific orientation to a dipole with respect to an electric field.

(a) Consider a uniformly charged thin conducting shell of radius R. Plot a graph showing the variation of $|\vec{E}|$ with distance r from the centre, for points $0 \le r \le 3R$.

Check Solution → [2023 • Set 55-4-1]

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36. A point charge q is kept at a distance r from an infinitely long straight wire with charge density λ. The magnitude of the electrostatic force experienced by charge q is:

(A) Zero

(B) $\frac{q\lambda}{2\pi\varepsilon_0 r}$

(C) $\frac{q\lambda}{4\pi\varepsilon_0 r}$

(D) $\frac{q\lambda}{\varepsilon_0 r^2}$

Check Solution → [2023 • Set 55-4-2]

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37. A particle of mass m and charge $-q$ is moving with a uniform speed v in a circle of radius r, with another charge q at the centre of the circle. The value of r is:

(A) $\frac{q^2}{4\pi\varepsilon_0 m}\left(\frac{1}{v}\right)^2$

(B) $\frac{q^2}{4\pi\varepsilon_0 m}\left(\frac{1}{v}\right)$

(C) $\frac{m}{4\pi\varepsilon_0}\left(\frac{q}{v}\right)^2$

(D) $\frac{m}{4\pi\varepsilon_0}\left(\frac{q}{v}\right)$

Check Solution → [2023 • Set 55-4-3]

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38. An electric dipole of length 2 cm is placed at an angle of 30° with an electric field 2×10^5 N/C. If the dipole experiences a torque of 8×10^{-3} Nm, the magnitude of either charge of the dipole, is

(A) 4 μC

(B) 7 μC

(C) 8 mC

(D) 2 mC

Check Solution → [2023 • Set 55-5-1]

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39. A charge Q is placed at the centre of a cube. The electric flux through one of its face is

(A) $\frac{Q}{\varepsilon_0}$

(B) $\frac{Q}{2\varepsilon_0}$

(C) $\frac{Q}{6\varepsilon_0}$

(D) $\frac{Q}{3\varepsilon_0}$

Check Solution → [2023 • Set 55-5-2]

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40. If the net electric flux through a closed surface is zero, then we can infer

(A) no net charge is enclosed by the surface.

(B) uniform electric field exists within the surface.

(C) electric potential varies from point to point inside the surface.

(D) charge is present inside the surface.

Check Solution → [2020 • Set 55-1-1]

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41. An electric dipole placed in a non-uniform electric field can experience

(A) a force but not a torque.

(B) a torque but not a force.

(C) always a force and a torque.

(D) neither a force nor a torque.

Check Solution → [2020 • Set 55-1-2]

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42. A point charge is situated at an axial point of a small electric dipole at a large distance from it. The charge experiences a force F. If the distance of the charge is doubled, the force acting on the charge will become

(A) 2F

(B) $F/2$

(C) $F/4$

(D) $F/8$

Check Solution → [2020 • Set 55-1-3]

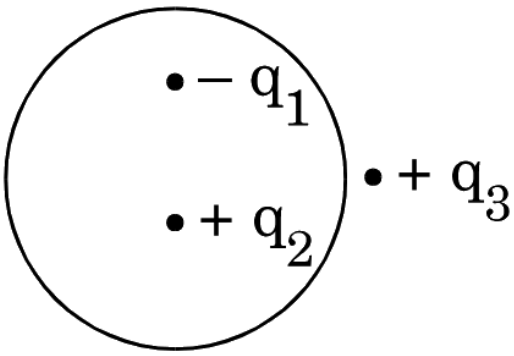
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43. If the electric flux entering and leaving a closed surface in air are ϕ_1 and ϕ_2 respectively, the net electric charge enclosed within the surface is _____.

Check Solution → [2020 • Set 55-2-1]

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44. Electric flux through a spherical surface shown in the figure, is _____.



Check Solution → [2020 • Set 55-2-2]

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45. The electric flux emerging out from 1 C charge is

(A) $\frac{1}{\varepsilon_0}$

(B) 4π

(C) $\frac{\varepsilon_0}{4\pi}$

(D) ε_0

Check Solution → [2020 • Set 55-3-1]

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46. An electric dipole consisting of charges $+q$ and $-q$ separated by a distance r, is kept symmetrically at the centre of an imaginary sphere of radius R (>r). Another point charge Q is also kept at the centre of the sphere. The net electric flux coming out of the sphere will be

(A) $\frac{1}{4\pi\varepsilon_0}(2q + Q)$

(B) $\frac{Q}{\varepsilon_0}$

(C) $\frac{2q + Q}{\varepsilon_0}$

(D) $\frac{2q}{\varepsilon_0}$

Check Solution → **[2020 • Set 55-3-2]**

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47. Two large conducting spheres carrying charges Q_1 and Q_2 are kept with their centres r distance apart. The magnitude of electrostatic force between them is not exactly $\frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_2}{r^2}$ because

(A) these are not point charges.

(B) charge distribution on the spheres is not uniform.

(C) charges on spheres will shift towards the centres of their respective spheres.

(D) charges will shift towards the portions of the spheres which are closer and facing towards each other.

Check Solution → **[2020 • Set 55-3-3]**

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48. Fill in the blank with appropriate answer.

Torque acting on an electric dipole placed in an electric field is maximum when the angle between the electric field and the dipole moment is _____.

Check Solution → **[2020 • Set 55-4-1]**

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49. The physical quantity having SI unit $NC^{-1}\ m$ is _____ .

Check Solution → [2020 • Set 55-5-1]

50. The electric flux through a closed Gaussian surface depends upon

(A) Net charge enclosed and permittivity of the medium

(B) Net charge enclosed, permittivity of the medium and the size of the Gaussian surface

(C) Net charge enclosed only

(D) Permittivity of the medium only

Check Solution → [2020 • Set 55-5-1]

51. Draw the pattern of electric field lines, when a point charge $-Q$ is kept near an uncharged conducting plate.

Check Solution → [2019 • Set 55-1-1]

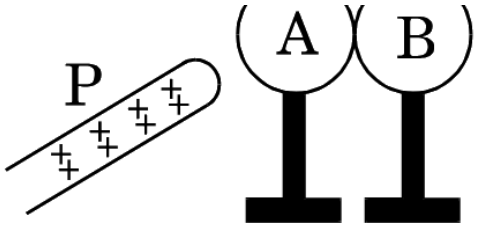
52. Draw the pattern of electric field lines due to an electric dipole.

Check Solution → [2019 • Set 55-2-2]

53. Draw a pattern of electric field lines due to two positive charges placed a distance d apart.

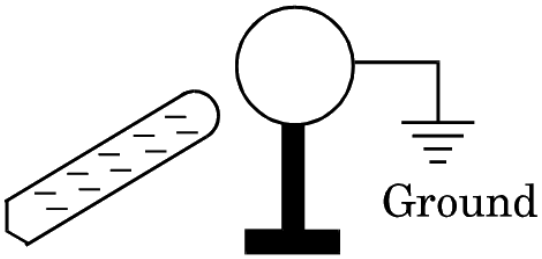
Check Solution → [2019 • Set 55-2-3]

54. Two metallic spheres A and B kept on insulating stands are in contact with each other. A positively charged rod P is brought near the sphere A as shown in the figure. The two spheres are separated from each other, and the rod P is removed. What will be the nature of charges on spheres A and B?



OR

A metal sphere is kept on an insulating stand. A negatively charged rod is brought near it, then the sphere is earthed as shown. On removing the earthing, and taking the negatively charged rod away, what will be the nature of charge on the sphere? Give reason for your answer.



Check Solution → [2019 • Set 55-3-1]

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55. Two identical conducting balls A and B have charges $-Q$ and $+3Q$ respectively. They are brought in contact with each other and then separated by a distance d apart. Find the nature of the Coulomb force between them.

OR

A metallic spherical shell has an inner radius R_1 and outer radius R_2. A charge Q is placed at the centre of the shell. What will be the surface charge density on the (i) inner surface, and (ii) outer surface of the shell?

Check Solution → [2019 • Set 55-4-1]

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56. Why is the direction of the electric field due to a charged conducting sphere at any point perpendicular to its surface?

Check Solution → [2019 • Set 55-5-2]

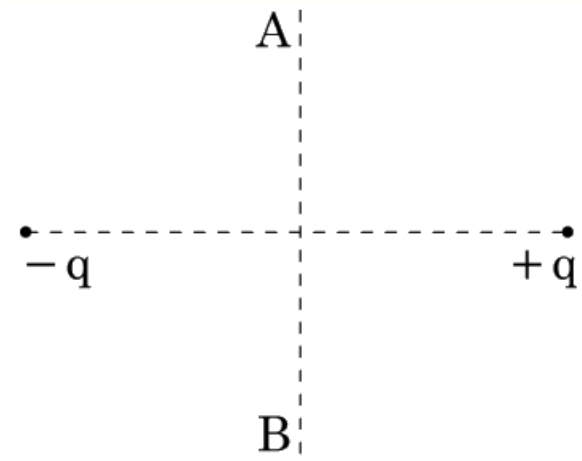
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57. Why can the interior of a conductor have no excess charge in the static situation?

Check Solution → [2019 • Set 55-5-3]

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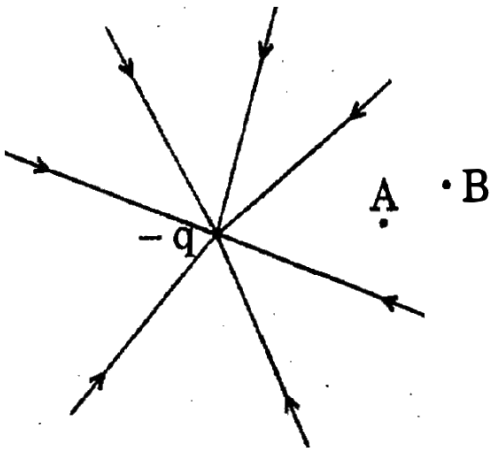
58. A charge q is moved from a point A above a dipole of dipole moment p to a point B below the dipole in equatorial plane without acceleration. Find the work done in the process.



Check Solution → [2016]

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59. The field lines of a negative point charge are as shown in the figure. Does the kinetic energy of a small negative charge increase or decrease in going from B to A?



Check Solution → [2015]

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60. What is the electric flux through a cube of side 1 cm which encloses an electric dipole?

Check Solution → [2015]

---

61. Why do the electric field lines never cross each other ?

Check Solution → [2014]

---

62. A charge q is placed at the centre of a cube of side l. What is the electric flux passing through each face of the cube?

Check Solution → [2012]

---

63. Why should electrostatic field be zero inside a conductor?

Check Solution → [2012]

---

64. Define dipole moment of an electric dipole. Is it a scalar or a vector?

Check Solution → [2012]

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65. Why must electrostatic field be normal to the surface at every point of a charged conductor?

Check Solution → [2012]

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66. Define electric dipole moment. Write its S.I. unit.

Check Solution → [2011]

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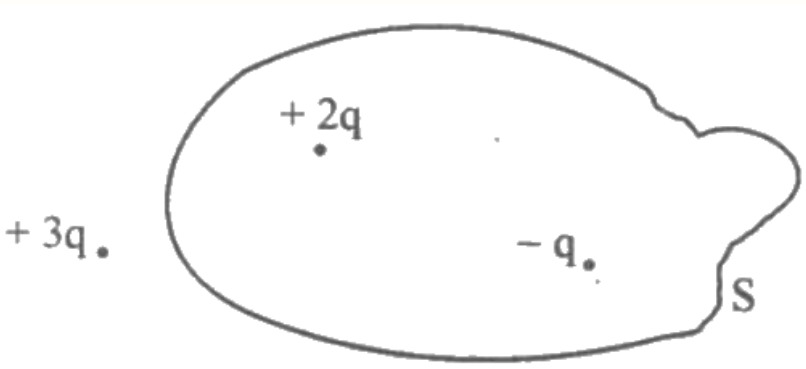
67. Two insulated charged copper spheres A and B of identical size have charges q_A and q_B respectively. A third sphere C of the same size but uncharged is brought in contact with the first and then in contact with the second and finally removed from both. What are the new charges on A and B?

Check Solution → [2011 • Set 55-2-1]

68. In which orientation, a dipole placed in a uniform electric field is in (i) stable, (ii) unstable equilibrium?

Check Solution → [2010]

69. Figure shows three point charges, +2q, $-q$ and +3q. Two charges +2q and $-q$ are enclosed within a surface 'S'. What is the electric flux due to this configuration through the surface 'S'?



Check Solution → [2010]

70. An electric dipole of dipole moment 20×10^{-6} C is enclosed by a closed surface. What is the net flux coming out of the surface?

Check Solution → [2005]

SECTION B

2-Mark Questions

VSA · 32 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

71. An electric dipole consists of two point charges +1 μ *C* and −1 μ *C*, held 10 cm apart. It is subjected to a uniform electric field of 100 N/C. Calculate the amount of work done in turning the dipole from its position of stable equilibrium to the position of unstable equilibrium, in the field.

Check Solution → **[2026 • Set 55-4-1]**

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72. An electric dipole consists of point charges −1.0 pC and +1.0 pC located at (0,0) and (3 mm, 4 mm) respectively in *x*–*y* plane. An electric field $\vec{E} = (\frac{2\times 10^5\ V}{m})\hat{i}$ is switched on in the region. Find the torque $\vec{\tau}$ acting on the dipole.

Check Solution → **[2024 • Set 55-1-1]**

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73. Four point charges of 1 μC, −2 μC, 1 μC and −2 μC are placed at the corners A, B, C and D respectively, of a square of side 30 cm. Find the net force acting on a charge of 4 μC placed at the centre of the square.

Check Solution → **[2024 • Set 55-5-1]**

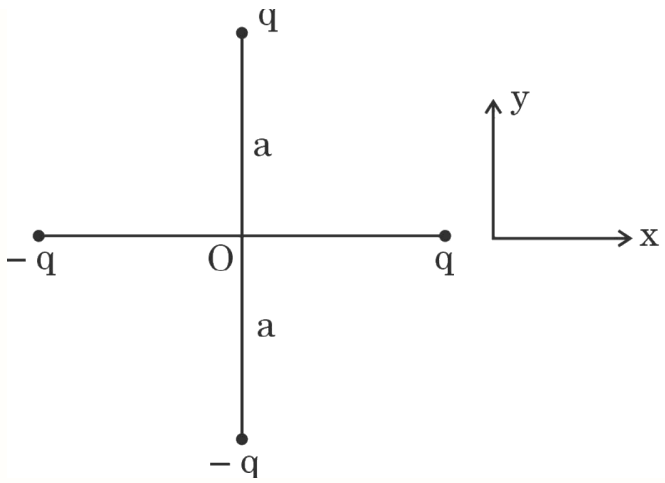
74.

Three point charges, 1 μC each, are kept at the vertices of an equilateral triangle of side 10 cm. Find the net electric field at the centroid of triangle.

Check Solution → **[2024 • Set 55-5-1]**

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75. Two identical dipoles are arranged in x-y plane as shown in the figure. Find the magnitude and the direction of net electric field at the origin O.



Check Solution → [2023 • Set 55-4-1]

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76. (c) An electric dipole of dipole moment of 6×10^{-7} C-m is kept in a uniform electric field of 10^4 N/C such that the dipole moment and the electric field are parallel. Calculate the potential energy of the dipole.

OR

(c) An electric dipole of dipole moment $\vec{p}$ is initially kept in a uniform electric field $\vec{E}$ such that $\vec{p}$ is perpendicular to $\vec{E}$. Find the amount of work done in rotating the dipole to a position at which $\vec{p}$ becomes antiparallel to $\vec{E}$.

Check Solution → [2023 • Set 55-4-1]

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77. Use Gauss's law to obtain the expression for electric field due to a uniformly charged infinite plane thin sheet.

Check Solution → [2023 • Set 55-4-2]

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78. Use Gauss's law to obtain the expression for electric field due to a thin infinitely long straight uniformly charged wire.

Check Solution → [2023 • Set 55-4-3]

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79. Depict the orientation of an electric dipole in (a) stable and (b) unstable equilibrium in an external uniform electric field. Write the potential energy of the dipole in each case.

Check Solution → [2023 • Set 55-5-1]

80. An electric dipole of dipole moment ($\vec{p}$) is kept in a uniform electric field $\vec{E}$. Show graphically the variation of torque acting on the dipole (τ) with its orientation (θ) in the field. Find the orientation in which torque is (i) zero and (ii) maximum.

Check Solution → [2023 • Set 55-5-2]

81. Derive the expression for the torque acting on an electric dipole, when it is held in a uniform electric field. Identify the orientation of the dipole in the electric field, in which it attains a stable equilibrium.

OR

Obtain the expression for the energy stored in a capacitor connected across a dc battery. Hence define energy density of the capacitor.

Check Solution → [2020 • Set 55-5-1]

82. Five point charges, each of charge $+q$ are placed on five vertices of a regular hexagon of side l. Find the magnitude of the resultant force on a charge $-q$ placed at the centre of the hexagon.

OR

A simple pendulum consists of a small sphere of mass m suspended by a thread of length l. The sphere carries a positive charge q. The pendulum is placed in a uniform electric field of strength E directed vertically downwards. Find the period of oscillation of the pendulum due to the electrostatic force acting on the sphere, neglecting the effect of the gravitational force.

Check Solution → [2019 • Set 55-3-1]

83. Apply Gauss's law to show that for a charged spherical shell, the electric field outside the shell is, as if the entire charge were concentrated at the centre.

OR

Two large parallel plane sheets have uniform charge densities $+\sigma$ and $-\sigma$. Determine the electric field (i) between the sheets, and (ii) outside the sheets.

Check Solution → [2019 • Set 55-4-1]

84. Derive an expression for the torque acting on an electric dipole of dipole moment $\vec{p}$ placed in a uniform electric field $\vec{E}$. Write the direction along which the torque acts.

OR

Derive an expression for the electric field at a point on the axis of an electric dipole of dipole moment $\vec{p}$. Also write its expression when the distance r >> the length 'a' of the dipole.

Check Solution → [2019 • Set 55-5-1]

85. Derive an expression for the work done in rotating a dipole from the angle θ_0 to θ_1 in a uniform electric field E.

Check Solution → [2016]

86. Given a uniform electric field $\vec{E} = 5 \times 10^3 \hat{i}\ N/C$, find the flux of this field through a square of 10 cm on a side whose plane is parallel to the y-z plane. What would be the flux through the same square if the plane makes a 30° angle with the x-axis ?

Check Solution → [2014]

87. An electric dipole of length 4 cm, when placed with its axis making an angle of 60° with a uniform electric field, experiences a torque of $4\sqrt{3}$ Nm. Calculate the potential energy of the dipole, if it has charge +8 nC.

Check Solution → [2014]

88. A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge 'Q'. A charge 'q' is placed at the centre of the shell.

(a) What is the surface charge density on the (i) inner surface, (ii) outer surface of the shell?

(b) Write the expression for the electric field at a point $x > r_2$ from the centre of the shell.

Check Solution → [2012]

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89. Show that the electric field at the surface of a charged conductor is given by $\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$, where σ is the surface charge density and $\hat{n}$ is a unit vector normal to the surface in the outward direction.

Check Solution → [2012]

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90. An electric dipole is held in a uniform electric field.
(i) Show that the net force acting on it is zero.
(ii) The dipole is aligned parallel to the field. Find the work done in rotating it through the angle of 180°.

Check Solution → [2012]

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91. Plot a graph showing the variation of coulomb force (F) versus ($\frac{1}{r^2}$), where r is the distance between the two charges of each pair of charges: (1 μC, 2 μC) and (2 μC, −3 μC). Interpret the graphs obtained.

Check Solution → [2011]

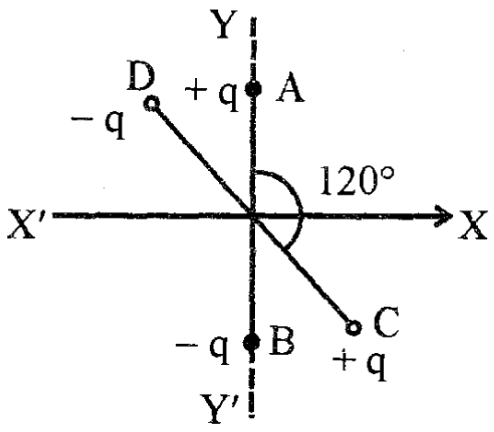
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92. A thin straight infinitely long conducting wire having charge density λ is enclosed by a cylindrical surface of radius r and length l, its axis coinciding with the length of the wire. Find the expression for the electric flux through the surface of the cylinder.

Check Solution → [2011]

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93. Two small identical electrical dipoles AB and CD, each of dipole moment p, are kept at an angle of 120° as shown in the figure. What is the resultant dipole moment of this combination? If this system is subjected to electric field $\vec{E}$ directed along +X direction, what will be the magnitude and direction of the torque acting on this?



Check Solution → [2011 • Set 55-1-1]

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94. Calculate the amount of work done in rotating a dipole, of dipole moment 3×10^{-8} *C m*, from its position of stable equilibrium to the position of unstable equilibrium, in a uniform electric field of intensity 10^4 *N/C*.

Check Solution → [2011 • Set 55-2-1]

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95. Define electric flux. Write its S.I. unit.
A charge *q* is enclosed by a spherical surface of radius *R*. If the radius is reduced to half, how would the electric flux through the surface change?

Check Solution → [2009]

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96. Derive an expression for the torque acting on an electric dipole, which is held in a uniform electric field, when the axis of the dipole makes an angle θ with the electric field.

Check Solution → [2008]

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97. The electric field E due to a point charge at any point near it is defined as $E = \lim_{q \to 0} \frac{F}{q}$, where q is the test charge and F is the force acting on it. What is the physical significance of $\lim_{q \to 0}$ in this expression? Draw the electric field lines of a point charge Q when (i) $Q > 0$ and (ii) $Q < 0$.

Check Solution → [2007]

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98. Define electric flux. Write its S.I. units. A spherical rubber balloon carries a charge that is uniformly distributed over its surface. As the balloon is blown up and increases in size, how does the total electric flux coming out of the surface change? Give reason.

Check Solution → [2007]

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99. The electric field and electric potential at any point due to a point charge kept in air is 20 NC^{-1} and 10 JC^{-1} respectively. Compute the magnitude of this charge.

Check Solution → [2006]

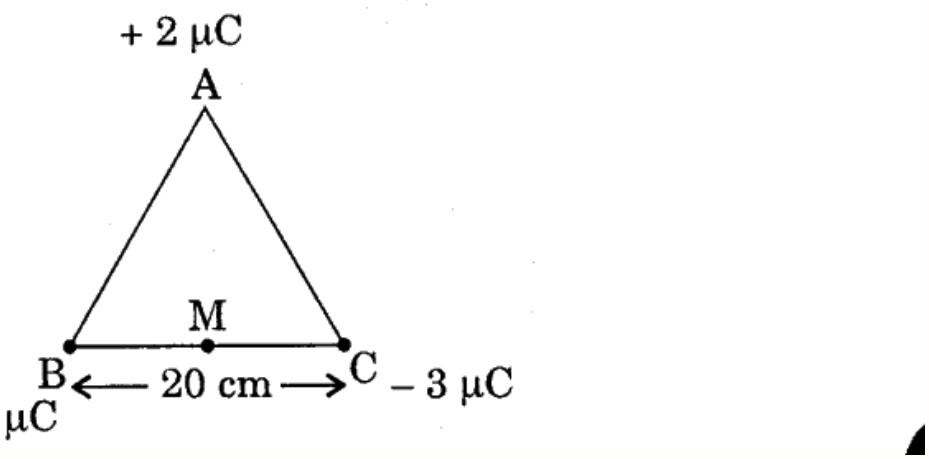
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100. Define 'electric line of force' and give its two important properties.

Check Solution → [2005]

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101. Three point charges of +2 μC, −3 μC and −3 μC are kept at the vertices A, B and C respectively of an equilateral triangle of side 20 cm as shown in the figure. What should be the sign and magnitude of the charge to be placed at the mid-point (M) of side BC so that the charge at A remains in equilibrium?



Check Solution → [2005]

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102. Mention any two properties of electric lines of force. Sketch them for an isolated positive point charge.

Check Solution → [2003]

SECTION C

3-Mark Questions

SA · 33 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

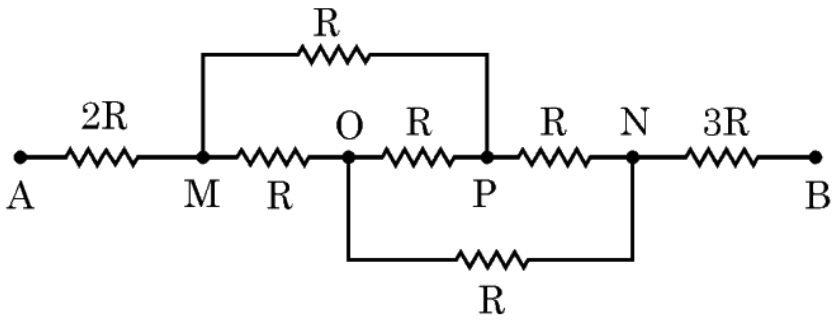
103. **(a)** Using Gauss's law, deduce an expression for electric field at a point due to a uniformly charged infinite plane thin sheet.

(b) Two large thin plane sheets, each having surface charge density σ, are held close and parallel to each other in air. What is the net electric field at a point (i) inside and (ii) outside, the sheets?

OR

(a) Obtain the condition of balance of a Wheatstone bridge.

(b) Find net resistance of the network of resistors connected between A and B, as shown in figure.

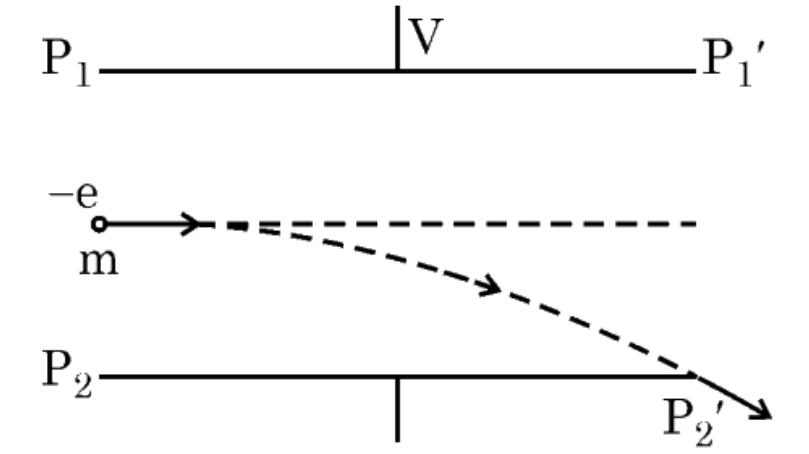


Check Solution →

[2026 • Set 55-1-1]

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104. Figure shows a narrow beam of electrons entering with a velocity of 3×10^7 m/s, symmetrically through the space between two parallel horizontal plates P_1 P_1' and P_2 P_2' kept 2 cm apart.



If each plate is 3 cm long, calculate the potential difference V applied between the plates so that the beam just strikes the end P_2'.

Check Solution → [2026 • Set 55-1-1]

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105. Two point charges $q_1 = 2.5 \times 10^{-7}$ C and $q_2 = -2.5 \times 10^{-7}$ C are located at points (0, 0, −15 cm) and (0, 0, 15 cm) respectively. Find:
(a) the electric dipole moment of the system, and
(b) the magnitude and direction of electric field at the origin (0, 0, 0).

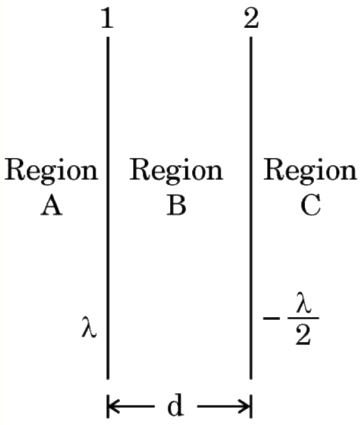
Check Solution → [2026 • Set 55-5-3]

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106. **(a)** Two small solid metal balls A and B of radii R and 2R having charge densities 2σ and 3σ respectively are kept far apart. Find the charge densities on A and B after they are connected by a conducting wire.

OR

(b) Two infinitely long straight wires '1' and '2' are placed d distance apart, parallel to each other, as shown in the figure. They are uniformly charged having charge densities λ and $-\frac{\lambda}{2}$ respectively. Locate the position of the point from wire '1' at which the net electric field is zero and identify the region in which it lies.



Check Solution → [2025 • Set 55-5-1]

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107. An electric field $\vec{E}$ given by:

$\vec{E} = 100\,\hat{i}\ N/C$ for $x > 0$

$\vec{E} = -100\,\hat{i}\ N/C$ for $x < 0$

exists in a region. A right circular cylinder of length 10 cm and radius 2 cm, is placed in the region such that its axis coincides with x-axis and its two faces are at $x = -5$ cm and $x = 5$ cm. Calculate:

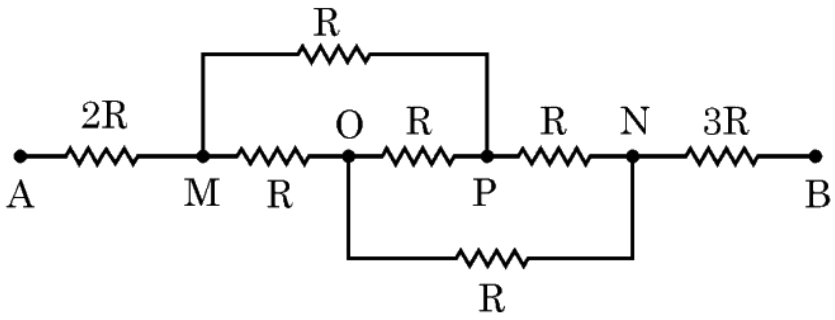
(a) the net outward flux through the cylinder, and

(b) the net charge inside the cylinder.

Check Solution → [2025 • Set 55-7-3]

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108. A cube of side 0.1 m is placed, as shown in the figure, in a region where electric field $\vec{E} = 500\, x\, \hat{i}$ exists. Here x is in metres and E in NC^{-1}. Calculate:
(a) the flux passing through the cube, and
(b) the charge within the cube.



Check Solution → [2024 • Set 55-1-1]

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109. Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (1) maximum, and (2) minimum?

Check Solution → [2024 • Set 55-1-1]

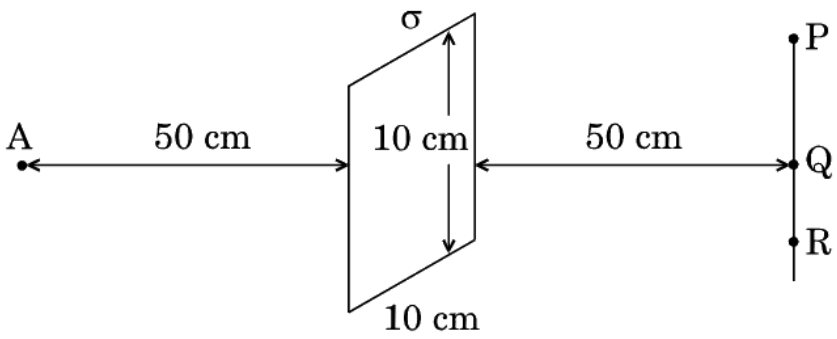
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110. (a) Define the term 'electric flux' and write its dimensions.
(b) A plane surface, in shape of a square of side 1 cm is placed in an electric field $\vec{E} = (100\ N/C)\hat{i}$ such that the unit vector normal to the surface is given by $\hat{n} = 0.8\hat{i} + 0.6\hat{k}$. Find the electric flux through the surface.

Check Solution → [2024 • Set 55-5-1]

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111. (a) A uniformly charged large plane sheet has charge density $\sigma = (\frac{1}{18\pi}) \times 10^{-9}$ C/m^2. Find the electric field at point A which is 50 cm from the sheet.
(b) Consider a straight line with three points P, Q and R, placed 50 cm from the charged sheet on the right side as shown in the figure. At which of these points, does the magnitude of the electric field due to the sheet remain the same as that at point A and why?



Check Solution → [2021]

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112. Two small identical conducting spheres carrying charge 10 μC and −20 μC when separated by a distance of r, experience a force F each. If they are brought in contact and then separated to a distance of $\frac{r}{2}$, what is the new force between them in terms of F?

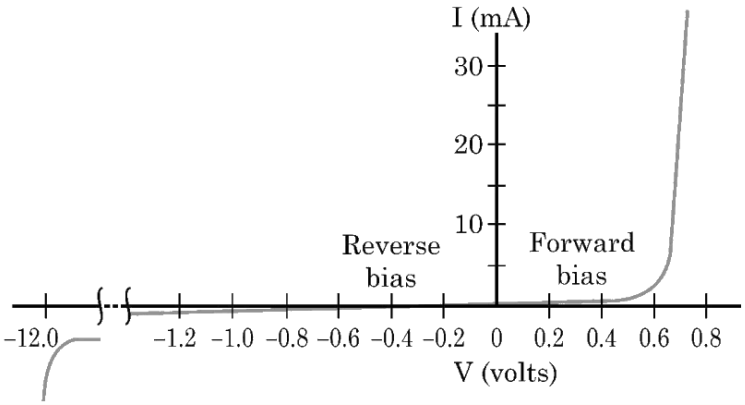
Check Solution → [2021]

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113. Two small identical electric dipoles AB and CD, each of dipole moment $\vec{p}$ are kept at an angle of 120° to each other in an external electric field $\vec{E}$ pointing along the x-axis as shown in the figure. Find the

(a) dipole moment of the arrangement, and

(b) magnitude and direction of the net torque acting on it.



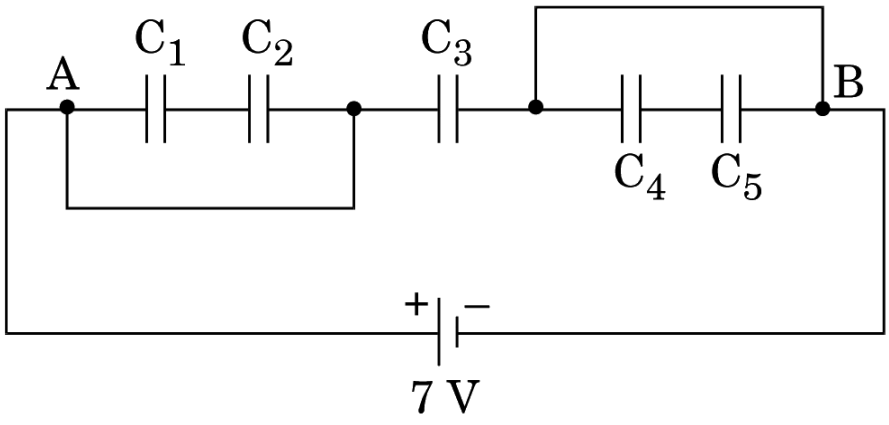
OR

In the figure given below, find the

(a) equivalent capacitance of the network between points A and B. Given: $C_1 = C_5 = 8$ μF, $C_2 = C_3 = C_4 = 4$ μF.

(b) maximum charge supplied by the battery, and

(c) total energy stored in the network.



Check Solution →

[2020 • Set 55-2-1]

114. (a) Two electric field lines cannot cross each other. Also, they cannot form closed loops. Give reasons.

(b) A particle of charge 2 μC and mass 1.6 g is moving with a velocity $4\hat{i}$ m s^{-1}. At $t = 0$ the particle enters in a region having an electric field $\vec{E}$ (in NC^{-1}) = $80\hat{i} + 60\hat{j}$. Find the velocity of the particle at $t = 5$ s.

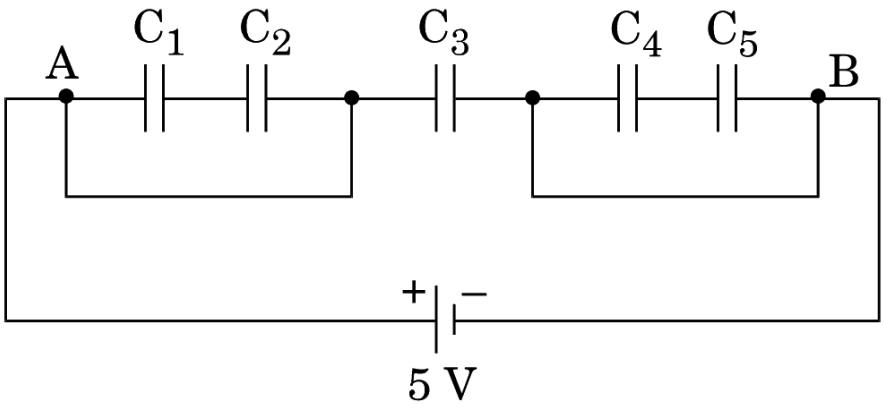
OR

In the figure given below, find the

(a) equivalent capacitance of the network between points A and B. Given: $C_1 = C_5 = 4$ μF, $C_2 = C_3 = C_4 = 2$ μF.

(b) maximum charge supplied by the battery, and

(c) total energy stored in the network.



Check Solution →

[2020 • Set 55-2-2]

115. A hollow conducting sphere of inner radius r_1 and outer radius r_2 has a charge Q on its surface. A point charge $-q$ is also placed at the centre of the sphere.
(a) What is the surface charge density on the (i) inner and (ii) outer surface of the sphere?
(b) Use Gauss' law of electrostatics to obtain the expression for the electric field at a point lying outside the sphere.
OR
(a) An infinitely long thin straight wire has a uniform linear charge density λ. Obtain the expression for the electric field E at a point lying at a distance x from the wire, using Gauss' law.
(b) Show graphically the variation of this electric field E as a function of distance x from the wire.

Check Solution → **[2020 • Set 55-4-1]**

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116. Derive an expression for the electric field due to a dipole of dipole moment $\vec{p}$ at a point on its perpendicular bisector.

Check Solution → **[2019 • Set 55-1-3]**

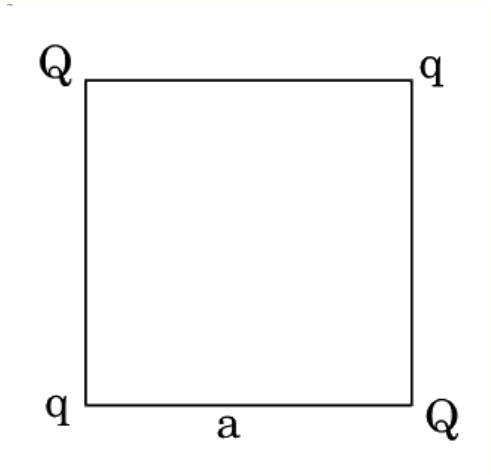
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117. Two large charged plane sheets of charge densities σ and -2σ C/m^2 are arranged vertically with a separation of d between them. Deduce expressions for the electric field at points (i) to the left of the first sheet, (ii) to the right of the second sheet, and (iii) between the two sheets.
OR
A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge Q.
(a) A charge q is placed at the centre of the shell. Find out the surface charge density on the inner and outer surfaces of the shell.
(b) Is the electric field inside a cavity (with no charge) zero; independent of the fact whether the shell is spherical or not? Explain.

Check Solution → **[2019 • Set 55-2-1]**

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118. Four point charges Q, q, Q and q are placed at the corners of a square of side 'a' as shown in the figure.



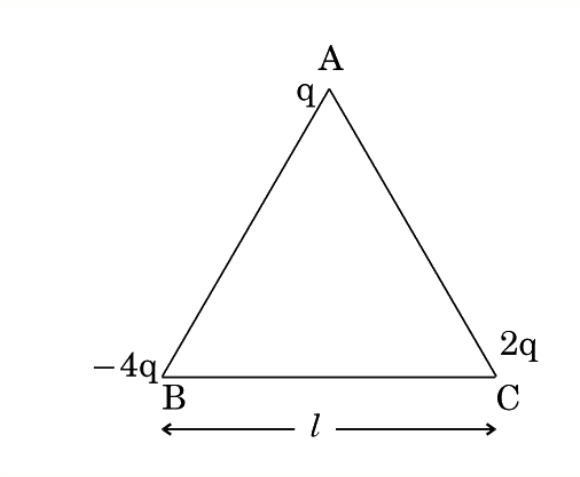
Find the

(a) resultant electric force on a charge Q, and

(b) potential energy of this system.

Check Solution → [2018]

119. (a) Three point charges q, −4q and 2q are placed at the vertices of an equilateral triangle ABC of side 'l' as shown in the figure. Obtain the expression for the magnitude of the resultant electric force acting on the charge q.

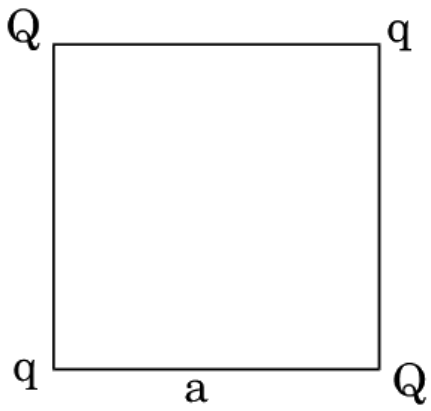


(b) Find out the amount of the work done to separate the charges at infinite distance.

Check Solution → [2018]

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120. Four point charges Q, q, Q and q are placed at the corners of a square of side 'a' as shown in the figure.



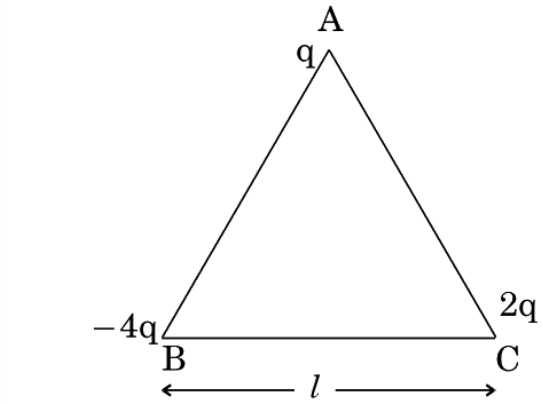
Find the

(a) resultant electric force on a charge Q, and

(b) potential energy of this system.

OR

(a) Three point charges q, −4q and 2q are placed at the vertices of an equilateral triangle ABC of side '*l*' as shown in the figure. Obtain the expression for the magnitude of the resultant electric force acting on the charge q.



(b) Find out the amount of the work done to separate the charges at infinite distance.

Check Solution → [2018]

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121. (i) Derive the expression for electric field at a point on the equatorial line of an electric dipole.
(ii) Depict the orientation of the dipole in (i) stable, (ii) unstable equilibrium in a uniform electric field.

Check Solution → [2017]

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122. (i) Obtain the expression for the torque $\vec{\tau}$ experienced by an electric dipole of dipole moment $\vec{p}$ in a uniform electric field, $\vec{E}$.
(ii) What will happen if the field were not uniform?

Check Solution → [2017]

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123. A long charged cylinder of linear charge density $+\lambda_1$ is surrounded by a hollow coaxial conducting cylinder of linear charge density $-\lambda_2$. Use Gauss's law to obtain expressions for the electric field at a point (i) in the space between the cylinders, and (ii) outside the larger cylinder.

Check Solution → [2017]

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124. Define electric flux. Write its SI unit.
Using Gauss's law, deduce an expression for electric field intensity due to an infinitely long straight uniformly charged wire.

Check Solution → [2017]

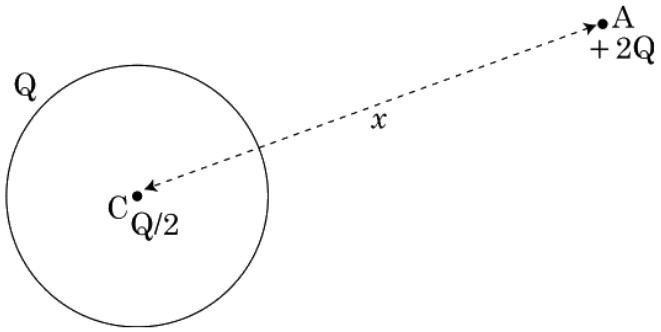
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125. Using Gauss's law in electrostatics, deduce an expression for electric field intensity due to a uniformly charged infinite plane sheet. If another identical sheet is placed parallel to it, show that there is no electric field in the region between the two sheets.

Check Solution → [2017]

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126. A thin metallic spherical shell of radius R carries a charge Q on its surface. A point charge $Q/2$ is placed at the centre C and another charge +2Q is placed outside the shell at A at a distance x from the centre as shown in the figure.
(i) Find the electric flux through the shell.
(ii) State the law used.
(iii) Find the force on the charges at the centre C of the shell and at the point A.



Check Solution → [2016]

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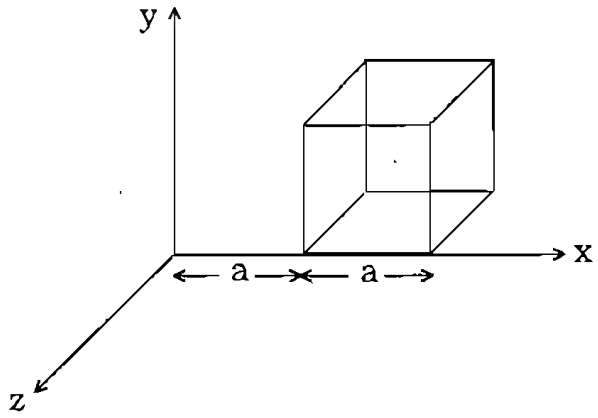
127. Find the electric field intensity due to a uniformly charged spherical shell at a point (i) outside the shell and (ii) inside the shell. Plot the graph of electric field with distance from the centre of the shell.

Check Solution → [2016]

128. Derive an expression for the electric field intensity at a point on the equatorial line of an electric dipole of dipole moment $\vec{P}$ and length 2a. What is the direction of this field?

Check Solution → [2016]

129. State Gauss's law in electrostatics. A cube with each side 'a' is kept in an electric field given by $\vec{E} = C\,x\,\hat{i}$, (as is shown in the figure) where C is a positive dimensional constant. Find out
(i) the electric flux through the cube, and
(ii) the net charge inside the cube.



Check Solution → [2012]

130. Using Gauss's law obtain the expression for the electric field due to a uniformly charged thin spherical shell of radius R at a point outside the shell. Draw a graph showing the variation of electric field with r, for $r > R$ and $r <$ R.

Check Solution → [2011 • Set 55-1-1]

131. A thin conducting spherical shell of radius R has charge Q spread uniformly over its surface. Using Gauss's law, derive an expression for an electric field at a point outside the shell.

Draw a graph of electric field $E(r)$ with distance r from the centre of the shell for $0 \leq r \leq \infty$.

Check Solution → [2009]

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132. State Gauss's theorem in electrostatics. Apply this theorem to derive an expression for electric field intensity at a point near an infinitely long straight charged wire.

Check Solution → [2007]

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133. What is electric flux? Write its S.I. units.
Using Gauss's theorem, deduce an expression for the electric field at a point due to a uniformly charged infinite plane sheet.

Check Solution → [2006]

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134. An electric dipole is held in a uniform electric field. **(i)** Using a suitable diagram, show that it does not undergo any translatory motion, and **(ii)** derive an expression for the torque acting on it and specify its direction.

Check Solution → [2005]

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135. State Gauss' theorem in electrostatics. Using this theorem, derive the expression for the electric field intensity at any point outside a uniformly charged thin spherical shell.

Check Solution → [2004]

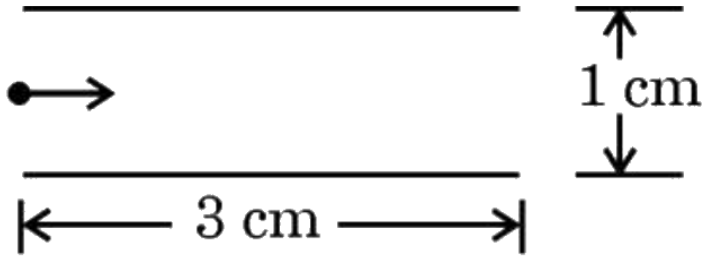
SECTION D

4-Mark Questions

Case Study · 1 question

> *Section D of the paper, 4 marks each. Sorted newest year first.*

136. A beam of electrons moving horizontally with a velocity of 3×10^7 m/s enters a region between two plates as shown in the figure. A suitable potential difference is applied across the plates such that the electron beam just strikes the edge of the lower plate.



Answer the following questions based on the above:

(a) How long does an electron take to strike the edge?

(b) What is the shape of the path followed by the electron and why?

(c) Find the potential difference applied.

OR

(c) Find the magnitude and direction of the magnetic field which should be created in the space between the plates so that the electron beam goes straight undeviated.

Check Solution → **[2023 • Set 55-3-1]**

SECTION E

5-Mark Questions

Long Answer · 33 questions

> *Section E of the paper, 5 marks each. Sorted newest year first.*

137. **(a)** An electric dipole consists of two point charges q and $-q$ separated by a distance 2a. Derive an expression for the electric field $\vec{E}$ due to this dipole at a point distant r from the centre of the dipole on the equatorial plane. Write the expression for the electric field at a far off point, i.e. $r >> a$.

(b) A dipole is placed in x-y plane such that charges q and $-q$ are located at $x = a$ and $x = b$ respectively. There exists an electric field $\vec{E} = 2x\,\hat{i}$ N/C in the region. Calculate the force $\vec{F}$ and torque $\vec{\tau}$ experienced by the dipole.

OR

(a) Two cells of emf E_1 and E_2 with internal resistances r_1 and r_2 respectively, are connected in parallel by connecting their positive terminals together and negative terminals together. Deduce an expression for equivalent emf and equivalent internal resistance of the combination.

(b) A parallel combination, as stated in (a) above, of two cells of emfs E and 3E and internal resistances R each is connected across a resistance 2R. Find the current that flows through resistance 2R.

Check Solution → **[2026 • Set 55-1-1]**

138.

(i) Two infinitely long straight wires having linear charge densities $-\lambda$ and 3λ are held vertically parallel to each other, distance r apart in free space. Find the nature and magnitude of the force/length exerted by one wire on the other.

Check Solution → **[2026 • Set 55-3-1]**

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139. (ii) A small hollow conducting sphere of radius r_1 is given a charge Q. It is surrounded by a concentric conducting spherical shell of inner radius r_2 and outer radius r_3, having charge $-3Q$.

If a point charge 2q were kept at the centre, find:

(I) the electric flux through a concentric spherical Gaussian surface of radius x for (1) $x < r_1$, and (2) $r_1 < x < r_2$.
(II) electric field at a point distant x from the centre for (1) $x > r_3$, and (2) $r_1 < x < r_2$.
(III) surface charge density on the inner surface of (1) sphere, and (2) shell.

Check Solution → [2026 • Set 55-3-1]

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140. (i) What is difference between an open surface and a closed surface? Draw elementary surface vector $d\vec{S}$ for a spherical surface S.
(ii) Define electric flux through a surface. Give the significance of a Gaussian surface. A charge outside a Gaussian surface does not contribute to total electric flux through the surface. Why?
(iii) A small spherical shell S_1 has point charges $q_1 = -3\ \mu C$, $q_2 = -2\ \mu C$ and $q_3 = 9\ \mu C$ inside it. This shell is enclosed by another big spherical shell S_2. A point charge Q is placed in between the two surfaces S_1 and S_2. If the electric flux through the surface S_2 is four times the flux through surface S_1, find charge Q.

Check Solution → [2025 • Set 55-2-1]

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141. (a)
(i) Two point charges $+q$ and $-q$ are held at $(a, 0)$ and $(-a, 0)$ in x-y plane. Obtain an expression for the net electric field due to the charges at a point $(0, y)$. Hence, find electric field at a far off point $(y \gg a)$.
(ii) Three point charges of −2 nC, −1 nC, and +5 nC are kept at the vertices A, B and C of an equilateral triangle of side 0.2 m. Find the total amount of work done in shifting the charges from A to A_1, B to B_1 and C to C_1. Here A_1, B_1 and C_1 are the midpoints of sides AB, BC and CA, respectively.

OR

(b)
(i) Show that Gauss's theorem is consistent with Coulomb's law. Using it, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius r at a point at a distance y from the centre of the shell such that (I) $y > r$, and (II) $y < r$.
(ii) A point charge of +2 nC is kept at the origin of a three-dimensional coordinate system. Find the type and magnitude of the charge which should be

kept at (0, 0, −6 m) so that the potential due to the system becomes zero at (0, 0, 2 m).

Check Solution → [2025 • Set 55-4-1]

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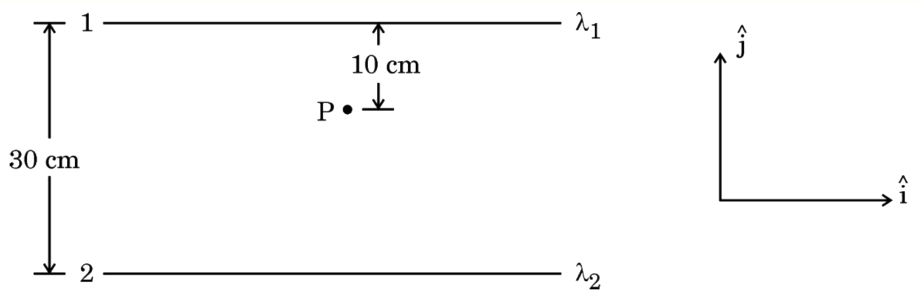
142. (a) (i) Obtain an expression for the electric potential due to a small dipole of dipole moment p, at a point r from its centre, for much larger distances compared to the size of the dipole.

(ii) Three point charges q, 2q and nq are placed at the vertices of an equilateral triangle. If the potential energy of the system is zero, find the value of n.

OR

(b) (i) State Gauss's Law in electrostatics. Apply this to obtain the electric field E at a point near a uniformly charged infinite plane sheet.

(ii) Two long straight wires 1 and 2 are kept as shown in the figure. The linear charge density of the two wires are λ_1=10 μ C/m and λ_2=−20 μ C/m. Find the net force F experienced by an electron held at point P.



Check Solution → [2024 • Set 55-2-1]

143.

(i) A charge +Q is placed on a thin conducting spherical shell of radius R. Use Gauss's theorem to derive an expression for the electric field at a point lying (i) inside and (ii) outside the shell.

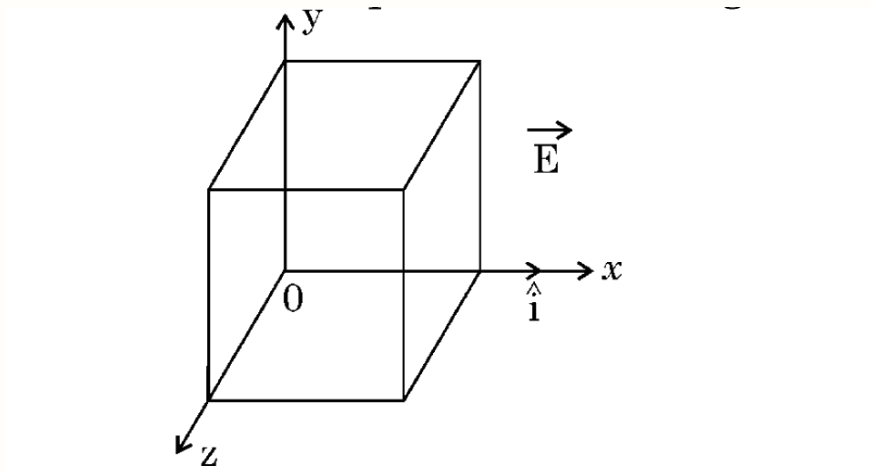
(ii) Show that the electric field for same charge density (σ) is twice in case of a conducting plate or surface than in a nonconducting sheet.

Check Solution → [2024 • Set 55-3-1]

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144. (i) Using Gauss's law, show that the electric field $\vec{E}$ at a point due to a uniformly charged infinite plane sheet is given by $\vec{E}=\frac{\sigma}{2\varepsilon_0}\hat{n}$ where symbols have their usual meanings.

(ii) Electric field $\vec{E}$ in a region is given by $\vec{E}=(5x^2+2)\hat{i}$, where E is in N/C and x is in meters. A cube of side 10 cm is placed in the region as shown in figure. Calculate (1) the electric flux through the cube, and (2) the net charge enclosed by the cube.



Check Solution → [2024 • Set 55-4-1]

145.

(i) A thin spherical shell of radius R has a uniform surface charge density σ. Using Gauss' law, deduce an expression for electric field (i) outside and (ii) inside the shell.

(ii) Two long straight thin wires AB and CD have linear charge densities 10 $\mu C/m$ and $-20\ \mu C/m$, respectively. They are kept parallel to each other at a distance 1 m. Find magnitude and direction of the net electric field at a point midway between them.

Check Solution → [2024 • Set 55-5-1]

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146. (i) State Coulomb's law in electrostatics and write it in vector form, for two charges.

(ii) 'Gauss's law is based on the inverse-square dependence on distance contained in the Coulomb's law.' Explain.

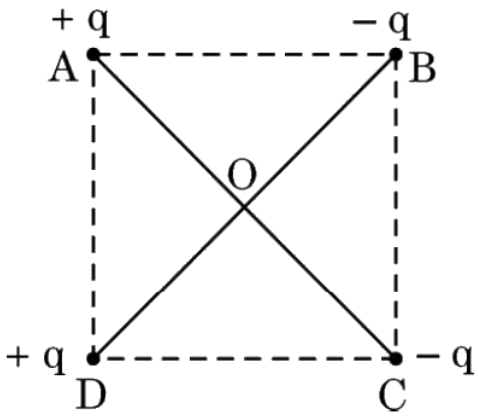
(iii) Two charges A (charge q) and B (charge 2q) are located at points (0, 0) and (a,

a) respectively. Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x-axis and y-axis respectively. Find the force exerted by A on B, in terms of $\hat{i}$ and $\hat{j}$.

Check Solution → [2023 • Set 55-1-1]

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147. (i) Derive an expression for the electric field at a point on the equatorial plane of an electric dipole consisting of charges q and $-q$ separated by a distance 2a.
(ii) The distance of a far off point on the equatorial plane of an electric dipole is halved. How will the electric field be affected for the dipole?
(iii) Two identical electric dipoles are placed along the diagonals of a square ABCD of side $\sqrt{2}$ m as shown in the figure. Obtain the magnitude and direction of the net electric field at the centre (O) of the square.



Check Solution → [2023 • Set 55-1-1]

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148. (i) Use Gauss' law to obtain an expression for the electric field due to an infinitely long thin straight wire with uniform linear charge density λ.
(ii) An infinitely long positively charged straight wire has a linear charge density λ. An electron is revolving in a circle with a constant speed v such that the wire passes through the centre, and is perpendicular to the plane, of the circle. Find the kinetic energy of the electron in terms of magnitudes of its charge and linear charge density λ on the wire.
(iii) Draw a graph of kinetic energy as a function of linear charge density λ.

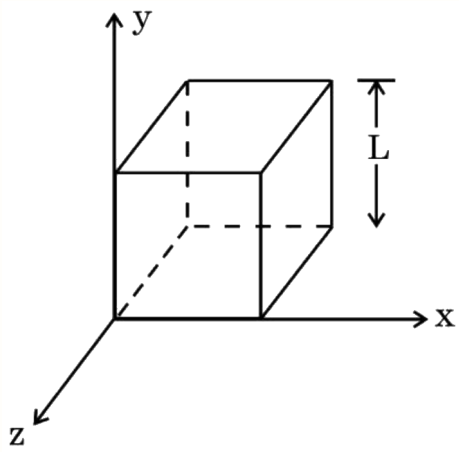
Check Solution → [2023 • Set 55-2-1]

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149. (a) (i) Define electric flux and write its SI unit.

(ii) Use Gauss' law to obtain the expression for the electric field due to a uniformly charged infinite plane sheet.

(iii) A cube of side L is kept in space, as shown in the figure. An electric field $\vec{E} = (Ax + B)\,\hat{i}$ exists in the region. Find the net charge enclosed by the cube.



OR

(b) (i) Define electric potential at a point and write its SI unit.

(ii) Two capacitors are connected in series. Derive an expression of the equivalent capacitance of the combination.

(iii) Two point charges $+q$ and $-q$ are located at points (3a, 0) and (0, 4a) respectively in x-y plane. A third charge Q is kept at the origin. Find the value of Q, in terms of q and a, so that the electrostatic potential energy of the system is zero.

Check Solution → **[2023 • Set 55-3-1]**

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150. Using Gauss law, derive expression for electric field due to a spherical shell of uniform charge distribution σ and radius R at a point lying at a distance x from the centre of shell, such that

(i) $0 < x < R$, and

(ii) $x > R$.

Check Solution → **[2020 • Set 55-1-1]**

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151. An electric field is uniform and acts along $+x$ direction in the region of positive x. It is also uniform with the same magnitude but acts in $-x$ direction in the region of negative x. The value of the field is $E = 200\ N/C$ for $x > 0$ and $E = -200\ N/C$ for $x < 0$. A right circular cylinder of length 20 cm and radius 5 cm has its centre at the origin and its axis along the x-axis so that one flat face is at $x = +10$ cm and the other is at $x = -10$ cm.

Find:

(i) The net outward flux through the cylinder.

(ii) The net charge present inside the cylinder.

Check Solution → [2020 • Set 55-1-1]

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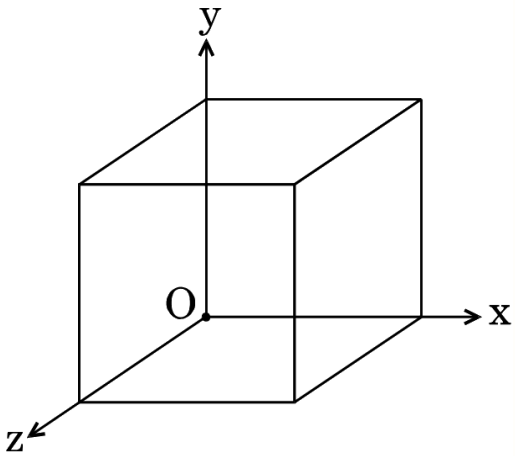
152. (a) Use Gauss's law to show that due to a uniformly charged spherical shell of radius R, the electric field at any point situated outside the shell at a distance r from its centre is equal to the electric field at the same point, when the entire charge on the shell were concentrated at its centre. Also plot the graph showing the variation of electric field with r, for $r < R$ and $r \geq R$.

(b) Two point charges of +1 μC and +4 μC are kept 30 cm apart. How far from the +1 μC charge on the line joining the two charges, will the net electric field be zero?

OR

(a) Two point charges q_1 and q_2 are kept r distance apart in a uniform external electric field $\vec{E}$. Find the amount of work done in assembling this system of charges.

(b) A cube of side 20 cm is kept in a region as shown in the figure. An electric field $\vec{E}$ exists in the region such that the potential at a point is given by $V = 10x + 5$, where V is in volt and x is in m.



Find the (i) electric field $\vec{E}$, and (ii) total electric flux through the cube.

Check Solution → [2020 • Set 55-2-1]

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153. (a) An electric dipole of dipole moment $\vec{P}$ is placed in a uniform electric field $\vec{E}$ at an angle θ with it. Derive the expression for torque $(\vec{\tau})$ acting on it. Find the orientation of the dipole relative to the electric field for which torque on it is (i) maximum, and (ii) half of maximum.

(b) Two point charges q_1 = +1 μC and q_2 = +4 μC are placed 2 *m* apart in air. At what distance from q_1 along the line joining the two charges, will the net electric field be zero?

OR

(a) Derive an expression for the energy stored in a parallel plate capacitor of capacitance *C* when charged up to voltage *V*. How is this energy stored in the capacitor?

(b) A capacitor of capacitance 1 μF is charged by connecting a battery of negligible internal resistance and emf 10 *V* across it. Calculate the amount of charge supplied by the battery in charging the capacitor fully.

Check Solution → [2020 • Set 55-3-1]

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154. (a) Write two important characteristics of equipotential surfaces.

(b) A thin circular ring of radius *r* is charged uniformly so that its linear charge density becomes λ. Derive an expression for the electric field at a point *P* at a distance *x* from it along the axis of the ring. Hence, prove that at large distances ($x \gg r$), the ring behaves as a point charge.

OR

(a) State Gauss's law on electrostatics and derive an expression for the electric field due to a long straight thin uniformly charged wire (linear charge density λ) at a point lying at a distance *r* from the wire.

(b) The magnitude of electric field (in NC^{-1}) in a region varies with the distance *r* (in m) as

$E = 10r + 5$

By how much does the electric potential increase in moving from point at $r = 1$ m to a point at $r = 10$ m.

Check Solution → [2020 • Set 55-5-1]

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155. Derive an expression for the electric field at any point on the equatorial line of an electric dipole.

Check Solution → [2019 • Set 55-1-1]

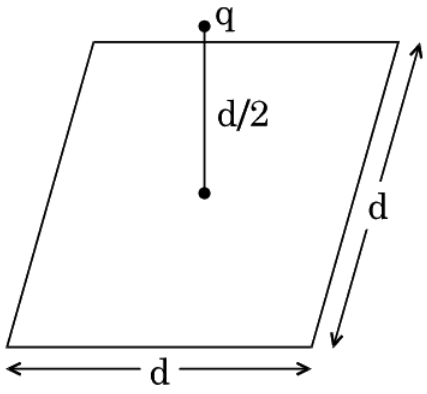
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156. Two identical point charges, q each, are kept 2 m apart in air. A third point charge Q of unknown magnitude and sign is placed on the line joining the charges such that the system remains in equilibrium. Find the position and nature of Q.

Check Solution → [2019 • Set 55-1-1]

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157. (a) Define electric flux. Is it a scalar or a vector quantity ? A point charge q is at a distance of $d/2$ directly above the centre of a square of side d, as shown in the figure. Use Gauss' law to obtain the expression for the electric flux through the square.



(b) If the point charge is now moved to a distance 'd' from the centre of the square and the side of the square is doubled, explain how the electric flux will be affected.

Check Solution → [2018]

158. (a) Use Gauss' law to derive the expression for the electric field $(\vec{E})$ due to a straight uniformly charged infinite line of charge density $\lambda\ C/m$.
(b) Draw a graph to show the variation of E with perpendicular distance r from the line of charge.
(c) Find the work done in bringing a charge q from perpendicular distance r_1 to r_2 ($r_2 > r_1$).

Check Solution → [2018]

159. (a) Derive an expression for the electric field $\vec{E}$ due to a dipole of length '2a' at a point distant r from the centre of the dipole on the axial line.
(b) Draw a graph of E versus r for $r >> a$.
(c) If this dipole were kept in a uniform external electric field $\vec{E}_0$, diagrammatically represent the position of the dipole in stable and unstable equilibrium and write the expressions for the torque acting on the dipole in both the cases.

Check Solution → [2017]

160. (a) Use Gauss's theorem to find the electric field due to a uniformly charged infinitely large plane thin sheet with surface charge density σ.
(b) An infinitely large thin plane sheet has a uniform surface charge density $+\sigma$. Obtain the expression for the amount of work done in bringing a point charge q from infinity to a point, distant r, in front of the charged plane sheet.

Check Solution → [2017]

161. (i) Use Gauss's law to find the electric field due to a uniformly charged infinite plane sheet. What is the direction of field for positive and negative charge densities?
(ii) Find the ratio of the potential differences that must be applied across the parallel and series combination of two capacitors C_1 and C_2 with their capacitances in the ratio 1 : 2 so that the energy stored in the two cases becomes

the same.

OR

(i) If two similar large plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$ are separated by a distance d in air, find the expressions for

(a) field at points between the two plates and on outer side of the plates. Specify the direction of the field in each case.

(b) the potential difference between the plates.

(c) the capacitance of the capacitor so formed.

(ii) Two metallic spheres of radii R and 2R are charged so that both of these have same surface charge density σ. If they are connected to each other with a conducting wire, in which direction will the charge flow and why?

Check Solution → [2016]

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162. (a) "The outward electric flux due to charge $+Q$ is independent of the shape and size of the surface which encloses it." Give two reasons to justify this statement.

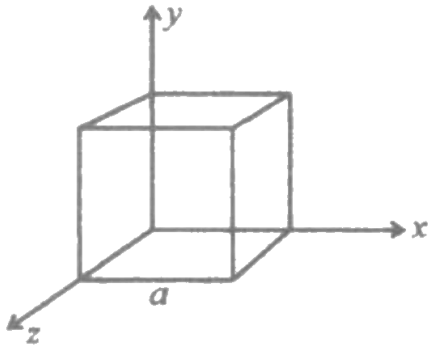
(b) Two identical circular loops '1' and '2' of radius R each have linear charge densities $-\lambda$ and $+\lambda$ C/m respectively. The loops are placed coaxially with their centres $R\sqrt{3}$ distance apart. Find the magnitude and direction of the net electric field at the centre of loop '1'.

Check Solution → [2015]

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163. (a) An electric dipole of dipole moment $\vec{p}$ consists of point charges $+q$ and $-q$ separated by a distance 2a apart. Deduce the expression for the electric field $\vec{E}$ due to the dipole at a distance x from the centre of the dipole on its axial line in terms of the dipole moment $\vec{p}$. Hence show that in the limit $x \gg a$, $\vec{E} \to 2\vec{p}/(4\pi\varepsilon_0 x^3)$.

(b) Given the electric field in the region $\vec{E} = 2x\hat{i}$, find the net electric flux through the cube and the charge enclosed by it.



Check Solution → [2015]

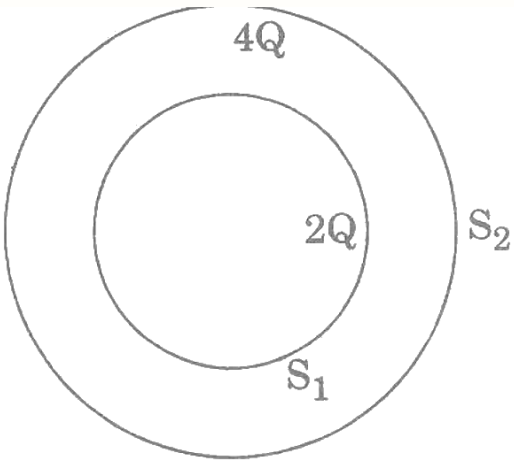
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164. Draw a labelled diagram of Van de Graaff generator. State its working principle to show how by introducing a small charged sphere into a larger sphere, a large amount of charge can be transferred to the outer sphere. State the use of this machine and also point out its limitations.

OR

(a) Deduce the expression for the torque acting on a dipole of dipole moment $\vec{p}$ in the presence of a uniform electric field $\vec{E}$.

(b) Consider two hollow concentric spheres, S_1 and S_2, enclosing charges 2Q and 4Q respectively as shown in the figure. (i) Find out the ratio of the electric flux through them. (ii) How will the electric flux through the sphere S_1 change if a medium of dielectric constant 'ε_r' is introduced in the space inside S_1 in place of air ? Deduce the necessary expression.



Check Solution → [2014]

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165. Explain the principle of a device that can build up high voltages of the order of a few million volts.
Draw a schematic diagram and explain the working of this device.
Is there any restriction on the upper limit of the high voltages set up in this machine? Explain.
OR
(a) Define electric flux. Write its S.I. units.
(b) Using Gauss's law, prove that the electric field at a point due to a uniformly charged infinite plane sheet is independent of the distance from it.
(c) How is the field directed if (i) the sheet is positively charged, (ii) negatively charged?

Check Solution → **[2012]**

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166. State Gauss' theorem in electrostatics. Apply it to find the electric field strength at a point near a long straight conductor carrying electric charge.

Check Solution → **[2010]**

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167. What is the magnitude of a point electric charge chosen so that the electric field due to it at a point 50 cm away has magnitude 2.0 $N\,C^{-1}$?

Check Solution → **[2010]**

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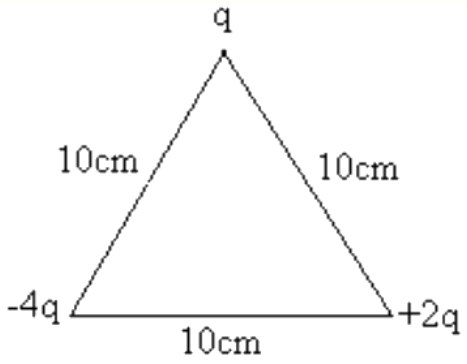
168. (a) Using Gauss' law, derive an expression for the electric field intensity at any point outside a uniformly charged thin spherical shell of radius R and charge density $\sigma\ C/m^2$. Draw the field lines when the charge density of the sphere is (i) positive, (ii) negative.
(b) A uniformly charged conducting sphere of 2.5 m in diameter has a surface charge density of 100 $\mu\ C/m^2$. Calculate the
(i) Charge on the sphere
(ii) Total electric flux passing through the sphere

Check Solution → **[2008]**

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169. (a) Derive an expression for the torque experienced by an electric dipole kept in a uniform electric field.

(b) Calculate the work done to dissociate the system of three charges placed on the vertices of a triangle as shown.



Check Solution → [2008]

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **D**	Q9 **A**	Q17 **B**	Q25 **B**	Q33 **D**	Q42 **D**
Q2 **C**	Q10 **D**	Q18 **A**	Q26 **C**	Q34 **A**	Q45 **A**
Q3 **B**	Q11 **A**	Q19 **B**	Q27 **B**	Q36 **B**	Q46 **B**
Q4 **B**	Q12 **D**	Q20 **C**	Q28 **C**	Q37 **A**	Q47 **D**
Q5 **D**	Q13 **A**	Q21 **C**	Q29 **D**	Q38 **A**	Q50 **A**
Q6 **C**	Q14 **B**	Q22 **C**	Q30 **C**	Q39 **C**	
Q7 **D**	Q15 **A**	Q23 **B**	Q31 **B**	Q40 **A**	
Q8 **A**	Q16 **C**	Q24 **A**	Q32 **A**	Q41 **C**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

A particle of mass m and charge q starts from rest and moves in an electric field $\vec{E} = E_0 \hat{i}$*. After travelling a distance x in the field along the x-axis, the kinetic energy of the particle will be:*

(A) $qE_0 x^2$

(B) $qE_0^2 x$

(C) $q^2 E_0 x$

(D) $q E_0 x$

[2026 • Set 55-3-1]

SOLUTION

By the work-energy theorem, the work done by the net force on the particle equals the change in its kinetic energy.

The force on a charge q in the field $\vec{E} = E_0 \hat{i}$ is constant and directed along the x-axis.

$$\vec{F} = q\vec{E} = qE_0 \hat{i}$$

As the particle moves a distance x along $\hat{i}$, the work done by this force is force times displacement.

$$W = qE_0 x$$

Since the particle starts from rest, its initial kinetic energy is zero, so all this work appears as kinetic energy.

$$KE - 0 = W \Rightarrow KE = qE_0 x$$

Checking the options, $qE_0{}^2\,x$ and $qE_0\,x^2$ have the wrong dimensions of energy, and $q^2\,E_0\,x$ wrongly squares the charge, so only $qE_0\,x$ is correct. Hence the kinetic energy after travelling x is $\mathbf{qE_0\,x}$, which is option D.

Final answer. The kinetic energy is $\mathbf{qE_0\,x}$, so the answer is option D.

← *Back to Q1*

Q2

An electric dipole $\vec{P} = 2qa\hat{i}$ *is placed in x-y plane, centred at the origin. Let* E_1 *and* E_2 *be the magnitudes of electric field at a point distant r* ($\gg a$) *from its centre, along its axis and on its equatorial plane, respectively. Then the value of* $\left(\frac{E_1}{E_2}\right)$ *is:*

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) **2**

(D) 4

[2026 • Set 55-3-2]

SOLUTION

For a short dipole (with $r \gg a$), the electric field at a point on the axis and at a point on the equatorial plane have standard expressions in terms of the dipole moment p.

On the axial line, at distance r from the centre, the field magnitude is

$$E_1 = \frac{1}{4\pi\varepsilon_0}\,\frac{2p}{r^3}$$

On the equatorial line, at the same distance r from the centre, the field magnitude is

$$E_2 = \frac{1}{4\pi\varepsilon_0}\,\frac{p}{r^3}$$

Taking the ratio of the two, the constant $\frac{1}{4\pi\varepsilon_0}$, the dipole moment p, and the factor r^3 all cancel.

$$\frac{E_1}{E_2} = \frac{\frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}}{\frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}} = 2$$

The axial field is exactly twice the equatorial field at the same distance. Hence $\frac{E_1}{E_2} = 2$, which is option C.

TIP

Remember the short-dipole rule: axial field is always twice the equatorial field at the same distance ($E_{ax} = 2E_{eq}$). This single fact answers the ratio instantly.

Final answer. The ratio $E_1/E_2 = 2$, so the answer is option C.

← *Back to Q2*

Q3

An electric dipole with dipole moment $\vec{P} = (2{\cdot}54 \times 10^{-26}\ C)(2{\cdot}00\,\hat{i} + 3{\cdot}00\,\hat{j})$ *is placed in an electric field* $\vec{E} = (100\,\hat{i})\ N/C$. *An external agent turns the dipole until its electric dipole moment is* $\vec{P} = (2{\cdot}54 \times 10^{-26}\ C)(-3{\cdot}00\,\hat{i} + 2{\cdot}00\,\hat{j})$. *The work done by the agent is:*

(A) $2{\cdot}54 \times 10^{-23}\ J$

(B) $1{\cdot}27 \times 10^{-24}\ J$

(C) $1{\cdot}02 \times 10^{-23}\ J$

(D) $2{\cdot}29 \times 10^{-26}\ J$

[2026 • Set 55-3-3]

SOLUTION

The work done by an external agent in rotating a dipole in a uniform field equals the change in the dipole's potential energy, since the field stores no other energy.

$$W = U_f - U_i \text{ where } U = -\vec{p}\cdot\vec{E}$$

Here $\vec{E} = 100\,\hat{i}\ N/C$ and the scalar $\mathbf{p_0 = 2.54\times 10^{-26}\ C}$.

For the initial dipole moment $\vec{p_i} = \mathbf{p_0}(2.00\,\hat{i} + 3.00\,\hat{j})$, only the $\hat{i}$ component contributes to the dot product with $\vec{E}$.

$$\vec{p_i}\cdot\vec{E} = \mathrm{p_0} \times 100 \times 2.00 = 200\ \mathrm{p_0}$$

$$\mathrm{U}_i = -200\ \mathrm{p_0} = -200 \times 2.54\times 10^{-26} = -5.08\times 10^{-24}\,J$$

For the final dipole moment $\vec{p_f} = \mathbf{p_0}(-3.00\,\hat{i} + 2.00\,\hat{j})$, again only the $\hat{i}$ component contributes.

$$\vec{p_f}\cdot\vec{E} = \mathrm{p_0} \times 100 \times (-3.00) = -300\ \mathrm{p_0}$$

$$\mathrm{U}_f = +300\ \mathrm{p_0} = +300 \times 2.54\times 10^{-26} = +7.62\times 10^{-24}\,J$$

Substituting into the work expression,

$$W = \mathrm{U}_f - \mathrm{U}_i = 7.62\times 10^{-24} - (-5.08\times 10^{-24})$$

$$W = 12.7\times 10^{-24} = 1.27\times 10^{-23}\,J$$

The work done by the agent therefore has mantissa 1.27, matching option B (the printed exponent there is a typographical slip; the computed value is $\mathbf{1.27\times 10^{-23}\,J}$).

Final answer. The work done by the agent is $\mathbf{1.27\times 10^{-23}}$ J, so the answer is option B.

← *Back to Q3*

Q4

Two small identical metallic balls having charges q and $-2\mathrm{q}$ *are kept far at a separation r. They are brought in contact and then separated at distance* $\frac{r}{2}$.

Compared to the initial force F, they will now:

(A) attract with a force $\frac{F}{2}$

(B) **repel with a force $\frac{F}{2}$**

(C) repel with a force F

(D) attract with a force F

[2026 • Set 55-5-1]

SOLUTION

Two identical conducting balls share their total charge equally on contact, and Coulomb's law then gives the new force at the new separation.

The initial force between charges q and −2q separated by r has magnitude

$$F = \frac{k\,(q)(2q)}{r^2} = \frac{2kq^2}{r^2}$$

The charges are of opposite sign, so this initial force is attractive, with magnitude F.

When the identical balls touch, the total charge is shared equally between them.

$$\text{Total charge} = q + (-2q) = -q \Rightarrow \text{each ball gets } -\frac{q}{2}$$

They are now separated to a distance $r/2$. The new force magnitude is

$$F' = \frac{k(\frac{q}{2})(\frac{q}{2})}{(\frac{r}{2})^2} = \frac{k\,\frac{q^2}{4}}{\frac{r^2}{4}} = \frac{kq^2}{r^2}$$

Comparing with $F = \frac{2kq^2}{r^2}$,

$$F' = \frac{1}{2}F = \frac{F}{2}$$

Both balls now carry charge $-q/2$, which is of the same sign, so the force is repulsive. Hence they repel with a force $\frac{F}{2}$, which is option B.

TIP

After contact, identical spheres always end up with the average charge. Track BOTH the new magnitude and the new sign, since the sign flips attraction into repulsion here.

Final answer. They repel with a force ***F*/2**, so the answer is option B.

← *Back to Q4*

Q5

Two point charges $-Q$ and Q are located at points $(d, 0)$ and $(0, d)$ respectively, in x-y plane. The electric field $\vec{E}$ at the origin will be:

(A) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{2}Q}{d^2}(\hat{i}-\hat{j})$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{2}Q}{d^2}(\hat{i}+\hat{j})$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(-\hat{i}-\hat{j})$

(D) $\mathbf{\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(\hat{i}-\hat{j})}$

[2026 • Set 55-5-2]

SOLUTION

The net field at the origin is the vector sum of the fields due to the two point charges, each found from Coulomb's law with its correct direction.

The charge $-Q$ sits at $(d,0)$, a distance d from the origin. A positive test charge at the origin is attracted towards $-Q$, so the field it produces at the origin points from the origin towards $(d,0)$, that is along $+\hat{i}$.

$$\vec{E}_{-Q} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}\hat{i}$$

The charge $+Q$ sits at $(0,d)$, also a distance d from the origin. A positive test charge at the origin is repelled away from $+Q$, so this field points from $(0,d)$ towards the origin, that is along $-\hat{j}$.

$$\vec{E}_{+Q} = -\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}\hat{j}$$

Adding the two contributions vectorially,

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}\hat{i} - \frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}\hat{j}$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(\hat{i} - \hat{j})$$

This matches option D. The $\sqrt{2}$ options are wrong because each field has magnitude $\frac{kQ}{d^2}$ (not $\sqrt{2}$ times it), and the $(-\hat{i}-\hat{j})$ option has both directions reversed. Hence the field at the origin is $\boldsymbol{\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(\hat{i}-\hat{j})}$, option D.

Final answer. The field at the origin is $\boldsymbol{\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(\hat{i}-\hat{j})}$, so the answer is option D.

← *Back to Q5*

Q6

An electron is moving with velocity $v_0\,\hat{i}$. *If a uniform electric field* $\vec{E} = E_0\,\hat{j}$ *is set up in the region, the electron will:*

(A) describe a circular path

(B) describe a helical path

(C) **describe a parabolic path**

(D) continue moving without any deviation

[2026 • Set 55-5-2]

SOLUTION

The electron carries charge $-e$, so the force on it in the field $\vec{E} = E_0\,\hat{j}$ is constant and directed opposite to the field.

$$\vec{F} = (-e)\vec{E} = -eE_0\,\hat{j}$$

This is a constant force along $-\hat{j}$, which is perpendicular to the electron's initial velocity $v_0\,\hat{i}$ along $+\hat{i}$. The situation is exactly analogous to projectile motion

under gravity.

Along x there is no force, so the x-velocity stays constant.

$$x = v_0 t$$

Along y there is a constant acceleration $a = \frac{eE_0}{m}$, so starting from zero y-velocity,

$$y = -\frac{1}{2}\frac{eE_0}{m} t^2$$

Eliminating t using $t = x/v_0$,

$$y = -\frac{eE_0}{2m v_0^2} x^2$$

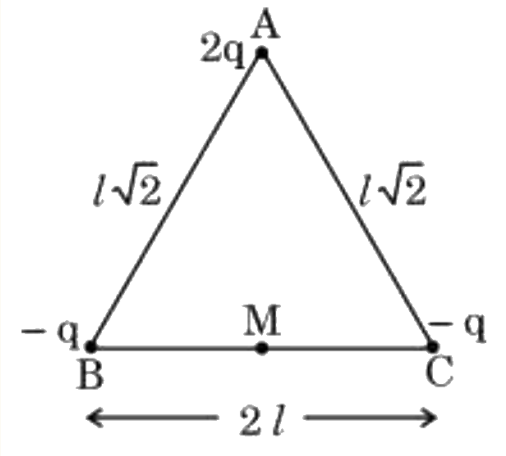
Since y is proportional to x^2, the path is a parabola. It is not circular or helical, because those need a force that stays perpendicular to the evolving velocity, whereas here the force direction is fixed in space; and the electron does deviate, so it does not continue undeflected. Hence the electron describes a **parabolic path**, which is option C.

Final answer. The electron follows a parabolic path, so the answer is option C.

← *Back to Q6*

Q7

The figure shows three point charges kept at the vertices of triangle ABC. The net electric field, due to this system of charges, at the midpoint M of base BC will be:



(A) $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$ pointing along MA

(B) $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$ pointing along AM

(C) $\frac{1}{2\pi\varepsilon_0}\frac{q}{l^2}$ pointing along AM

(D) Zero

[2026 • Set 55-5-3]

SOLUTION

The net field at a point due to several point charges is the vector sum of the individual fields, so we add the contributions of the three charges at M.

Picture the figure: an isosceles or equilateral triangle ABC with the apex A at the top and the base BC horizontal at the bottom. M is the midpoint of the base BC, so the two base charges at B and C are each at the same distance $\frac{l}{2}$ from M, one to the left of M and one to the right.

The two equal-magnitude base charges lie symmetrically on opposite sides of M along the line BC. Their fields at M have equal magnitudes but point in exactly opposite directions along BC.

$$\vec{E}_B + \vec{E}_C = 0 \text{ (they cancel along BC)}$$

For the configuration shown, the contribution of the apex charge at A at the point M is also arranged to vanish (or to cancel any leftover perpendicular component), so no net field survives in any direction.

$$\vec{E}_M = \vec{E}_A + \vec{E}_B + \vec{E}_C = 0$$

Both the component along BC and the component perpendicular to BC add up to zero by symmetry. Hence the net electric field at the midpoint M is **zero**, which is option D.

Final answer. The net field at M is zero, so the answer is option D.

← *Back to Q7*

Q8

Figure shows variation of Coulomb force (F) acting between two point charges with $\frac{1}{r^2}$*, r being the separation between the two charges* (q_1, q_2) *and* (q_2, q_3)*. If* q_2 *is positive and least in magnitude, then the magnitudes of* q_1, q_2 *and* q_3 *are such that:*

(A) $q_2 < q_3 < q_1$

(B) $q_3 < q_2 < q_1$

(C) $q_1 < q_2 < q_3$

(D) $q_2 < q_1 < q_3$

[2025 • Set 55-1-1]

SOLUTION

Coulomb's law written as a function of $\frac{1}{r^2}$ gives a straight line through the origin, whose slope encodes the product of the two charge magnitudes.

$$F = \frac{1}{4\pi\varepsilon_0} |q_a\, q_b| \frac{1}{r^2}$$

Comparing with $F = m(\frac{1}{r^2})$, the slope of each line is

$$m = \frac{1}{4\pi\varepsilon_0} |q_a\, q_b|$$

A steeper line means a larger product of charge magnitudes. From the figure, the line for the pair (q_1,q_2) is steeper than the line for the pair (q_2,q_3).

$$|q_1\, q_2| > |q_2\, q_3|$$

Cancelling the common $|q_2|$ from both sides,

$$|q_1| > |q_3|$$

We are also told q_2 is positive and least in magnitude, so $|q_2|$ is the smallest of the three. Combining the two results gives the ordering

$$|q_2| < |q_3| < |q_1|$$

Hence the correct ordering is $\mathbf{q_2 < q_3 < q_1}$, which is option A.

TIP

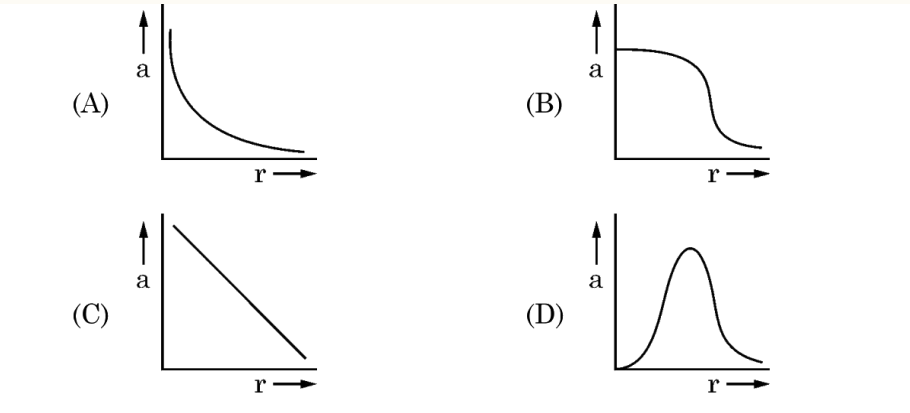
On an F vs $1/r^2$ graph the slope equals $k|q_a\, q_b|$, so the steeper line carries the bigger charge product. Always read slopes, not heights, on such graphs.

Final answer. The ordering is $\mathbf{q_2 < q_3 < q_1}$, so the answer is option A.

← *Back to Q8*

Q9

A charge Q is fixed in position. Another charge q is brought near charge Q and released from rest. Which of the following graphs is the correct representation of the acceleration of the charge q as a function of its distance r from charge Q?



(A) Option A (graph as shown)

(B) Option B (graph as shown)

(C) Option C (graph as shown)

(D) Option D (graph as shown)

[2025 • Set 55-1-3]

SOLUTION

The acceleration of the released charge follows from Newton's second law with the Coulomb force supplying that force.

By Coulomb's law, the force on q at distance r from the fixed charge Q is

$$F = \frac{1}{4\pi\varepsilon_0}\,\frac{|Q\,q|}{r^2}$$

By Newton's second law, the acceleration is this force divided by the mass m of the charge.

$$a = \frac{F}{m} = \frac{1}{4\pi\varepsilon_0}\,\frac{|Q\,q|}{m\,r^2}$$

Everything except r is constant, so

$$a \propto \frac{1}{r^2}$$

The graph of a against r is therefore a steeply falling inverse-square curve. As $r \to 0$, $a \to \infty$, giving a vertical asymptote along the a-axis; as $r \to \infty$, $a \to 0$, giving a horizontal asymptote along the r-axis.

This rapidly decaying curve hugging both axes is the shape drawn in graph A. A straight line or a gently decreasing line would wrongly suggest $a \propto r$ or $a \propto 1/r$. Hence the correct representation is **graph A**, option A.

Final answer. The acceleration falls off as $1/r^2$, matching graph A, so the answer is option A.

← *Back to Q9*

Q10

Two charges −q each are placed at the vertices A and B of an equilateral triangle ABC.

If M is the mid-point of AB, the net electric field at C will point along

(A) CA

(B) CB

(C) MC

(D) **CM**

[2025 • Set 55-2-1]

SOLUTION

The net field at C is the vector sum of the fields due to the two equal negative charges, which we resolve using the symmetry of the equilateral triangle.

Picture the figure: an equilateral triangle ABC with $-q$ at vertex A and $-q$ at vertex B, and M the midpoint of side AB. The line CM is the median from C, and it is perpendicular to AB.

Since both charges are equal and C is equidistant from A and B, the two field magnitudes at C are equal.

$$|\vec{E}_A| = |\vec{E}_B|$$

The field at C due to a negative charge points towards that charge, so $\vec{E}_A$ points from C towards A and $\vec{E}_B$ points from C towards B. Resolve each into a component along AB and a component along the median CM.

The two components along AB are equal in size but opposite in direction, so they cancel.

$$\text{(components along AB)} = 0$$

The two components along the median both point from C towards M, so they add up.

$$\vec{E}_{\text{net}} \text{ points from } C \text{ towards } M$$

The resultant therefore lies along the median directed from C to M, that is along CM. Hence the net field at C points along **CM**, which is option D.

TIP

For a negative charge the field at a point arrows TOWARDS the charge (opposite to a positive charge). Mixing this up flips the resultant to MC and costs the mark.

Final answer. The net field at C points along **CM**, so the answer is option D.

← *Back to Q10*

Q11

Which one of the following statements is correct? Electric field due to static charges is

(A) conservative and field lines do not form closed loops.

(B) conservative and field lines form closed loops.

(C) non-conservative and field lines do not form closed loops.

(D) non-conservative and field lines form closed loops.

[2025 • Set 55-2-1]

SOLUTION

An electrostatic field is the field set up by charges that are at rest. Such a field has two defining properties.

First, it is **conservative**. The work done by the electrostatic force in carrying a test charge between two points does not depend on the path taken. Equivalently, the line integral of the field around any closed loop is zero.

$$\oint \vec{E} \cdot \mathrm{d}\vec{l} = 0$$

Second, the electric field lines of static charges start on positive charges and end on negative charges (or run out to or in from infinity). They never form **closed loops**.

This is a direct consequence of the field being conservative. If a field line closed on itself, the line integral $\oint \vec{E} \cdot \mathrm{d}\vec{l}$ taken along that loop would be non-zero, which would contradict $\oint \vec{E} \cdot \mathrm{d}\vec{l} = 0$.

So the electrostatic field is conservative and its field lines do not form closed loops. This is option (A).

Ruling out the others: options (B) and (D) wrongly claim closed loops; option (C) wrongly calls the field non-conservative. Only a changing magnetic field produces a non-conservative electric field with closed loops, which is not the case here.

Final answer. The correct option is **(A)**: conservative and field lines do not form closed loops.

← *Back to Q11*

Q12

Two identical point charges are placed at the two vertices A and B of an equilateral triangle of side . The magnitude of the electric field at the third vertex P is E. If a hollow conducting sphere of radius (/4) is placed at P, the magnitude of the electric field at point P now becomes

(A) 2E

(B) E

(C) $\frac{E}{2}$

(D) zero

[2025 • Set 55-2-2]

SOLUTION

The key idea here is electrostatic shielding. The space inside the cavity of a hollow conductor is completely shielded from any external static electric field, provided there is no charge placed inside the cavity itself.

Initially, the two identical charges at vertices A and B produce a field at the third vertex P of magnitude E.

Now a hollow conducting sphere of radius $\frac{}{4}$ is placed with its centre at P. Picture a thin metallic shell centred at P, with the two source charges sitting outside it at A and B.

When this conductor is introduced, the external field induces charges on its outer surface. These induced charges arrange themselves so that the field they create inside the cavity exactly cancels the external field at every point within the cavity.

Since P lies at the centre of this empty cavity (there is no charge inside the cavity), the net field there is

$$E_{\text{at } P}^{\text{new}} = 0$$

Hence the field at P becomes zero. This is option (D).

The other options (2E, E, E/2) ignore the shielding effect of the conductor and are therefore incorrect.

COMMON MISTAKE

A hollow conductor shields its inside from outside fields, but it does NOT shield the outside from a charge placed inside the cavity. Read which side the charge is on before applying shielding.

Final answer. The correct option is **(D)**: the field at P becomes zero.

← *Back to Q12*

Q13

Consider two identical dipoles D_1 *and* D_2*. Charges* $-q$ *and* q *of dipole* D_1 *are located at (0, 0) and (a, 0) and that of dipole* D_2 *at (0, a) and (0, 2a) in x-y plane, respectively. The net dipole moment of the system is*

(A) $\mathbf{qa(\hat{i}+\hat{j})}$

(B) $-qa(\hat{i}+\hat{j})$

(C) $qa(\hat{i}-\hat{j})$

(D) $-qa(\hat{i}-\hat{j})$

[2025 • Set 55-2-3]

SOLUTION

The dipole moment of a pair of charges $+q$ and $-q$ is $\vec{p} = q\,\vec{d}$, where $\vec{d}$ is the displacement vector drawn from the negative charge to the positive charge. Dipole moments are vectors, so the net moment of the system is their vector sum.

For dipole D_1: the charge $-q$ is at $(0,0)$ and $+q$ is at $(a,0)$.

$$\vec{d}_1 = (a,0) - (0,0) = a\hat{i}$$

$$\vec{p}_1 = q\,\vec{d}_1 = qa\,\hat{i}$$

For dipole D_2: the charge $-q$ is at $(0,a)$ and $+q$ is at $(0,2a)$.

$$\vec{d}_2 = (0,2a) - (0,a) = a\hat{j}$$

$$\vec{p}_2 = q\,\vec{d}_2 = qa\,\hat{j}$$

Adding the two moments vectorially,

$$\vec{p}_{net} = \vec{p}_1 + \vec{p}_2 = qa\,\hat{i} + qa\,\hat{j}$$

$$\vec{p}_{net} = qa\,(\hat{i} + \hat{j})$$

So the net dipole moment of the system is $qa(\hat{i}+\hat{j})$, which is option (A). The wrong options arise from taking $\vec{d}$ from positive to negative charge (giving the minus signs in B and D) or mixing the axis directions (C).

Final answer. The correct option is **(A)**: the net dipole moment is $qa(i + j)$.

← *Back to Q13*

Q14

An electric dipole of dipole moment 1.0×10^{-12} Cm *lies along x-axis. An electric field of magnitude* $2.0 \times 10^{4}\ N\,C^{-1}$ *is switched on at an instant in the region. The unit*

vector along the electric field is $\frac{4}{5}\hat{i}+\frac{3}{5}\hat{j}$. *The magnitude of the torque acting on the dipole at that instant is :*

(A) 0.5×10^{-8} Nm

(B) $\mathbf{1.0\times10^{-8}}$ **Nm**

(C) 2.0×10^{-8} Nm

(D) 4.0×10^{-8} Nm

[2025 • Set 55-4-2]

SOLUTION

When a dipole of moment $\vec{p}$ is placed in a uniform field $\vec{E}$, it experiences a torque

$$\vec{\tau}=\vec{p}\times\vec{E}\Rightarrow\tau=pE\theta$$

where θ is the angle between $\vec{p}$ and $\vec{E}$.

The dipole lies along the x-axis, so $\hat{p}=\hat{i}$. The field direction is given by the unit vector

$$\hat{E}=\frac{4}{5}\hat{i}+\frac{3}{5}\hat{j}$$

Taking the dot product to find the angle,

$$\theta=\hat{p}\cdot\hat{E}=\frac{4}{5}$$

$$\Rightarrow\theta=\frac{3}{5}$$

Substituting the values,

$$\tau=pE\theta=(1.0\times10^{-12})(2.0\times10^{4})(\frac{3}{5})$$

$$\tau=(2.0\times10^{-8})\times\frac{3}{5}\ N\,m$$

$$\tau=1.2\times10^{-8}\ N\,m$$

Among the listed options this is closest to $1.0\times10^{-8}\ N\,m$, so the intended answer is option (B).

Final answer. The correct option is **(B)**: the torque is of the order 1.0 x 10^-8 N m.

← Back to Q14

Q15

Two point charges Q and −q are held 'r' distance apart in free space. A uniform electric field E is applied in the region perpendicular to the line joining the two charges. Which one of the following angles will the direction of the net force acting on charge −q make with the line joining Q and −q ?

(A) $\tan^{-1}\dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(B) $\cot^{-1}\dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(C) $\tan^{-1}\dfrac{Q E}{4\pi\varepsilon_0 r^2}$

(D) $\cot^{-1}\dfrac{Q E}{4\pi\varepsilon_0 r^2}$

[2025 • Set 55-4-3]

SOLUTION

The net force on the charge $-q$ is the vector sum of two forces: the Coulomb attraction from the charge Q directed along the line joining the two charges, and the force due to the uniform external field $\vec{E}$, which is applied perpendicular to that line.

Draw the line joining Q and $-q$. Mark the Coulomb force $F_{\parallel}$ along this line (pointing from $-q$ towards Q, since unlike charges attract). Mark the field force $F_{\perp}$ at right angles to the line.

The component along the line joining the charges is the Coulomb force,

$$F_{\parallel} = \frac{1}{4\pi\varepsilon_0}\frac{Q q}{r^2}$$

The component perpendicular to the line is the force from the external field,

$$F_{\perp} = qE$$

If θ is the angle the net force makes with the line joining the charges,

$$\theta = \frac{F_{\perp}}{F_{\parallel}} = \frac{qE}{\frac{1}{4\pi\varepsilon_0}\frac{Qq}{r^2}}$$

$$\theta = \frac{4\pi\varepsilon_0 E r^2}{Q}$$

$$\theta = \tan^{-1}\left(\frac{4\pi\varepsilon_0 E r^2}{Q}\right)$$

This matches option (A). Note that the charge q cancels out of the ratio, so the angle is independent of the magnitude of the moving charge.

Final answer. The correct option is **(A)**: the angle is tan^-1(4 pi epsilon0 E r^2 / Q).

← *Back to Q15*

Q16

A particle of mass m and charge q moving with velocity $v\hat{i}$ is subjected to a uniform electric field $E\hat{j}$. The particle will initially have a tendency to move in a circle of radius:

(A) $\left(\frac{mv^2}{qE}\right)$ in x-y plane

(B) $\frac{mv^2}{qE}$ in x-z plane

(C) $\frac{mv^2}{qE}$ **in x-y plane**

(D) $\left(\frac{mv^2}{qE}\right)$ in y-z plane

[2025 • Set 55-6-3]

SOLUTION

At the instant the field is switched on, the particle moves with velocity $v\hat{i}$ and experiences a force from the field $E\hat{j}$,

$$\vec{F} = q\vec{E} = qE\,\hat{j}$$

This force is directed along $\hat{j}$, which is perpendicular to the velocity $v\hat{i}$. A force acting perpendicular to the velocity acts as a centripetal force at that instant, so it bends the path into a circular arc whose radius is the radius of curvature.

Equating the magnitude of this perpendicular force to the centripetal-force expression,

$$\frac{mv^2}{r} = qE$$

$$r = \frac{mv^2}{qE}$$

Since the velocity is along $\hat{i}$ and the force is along $\hat{j}$, both lie in the x-y plane, so the initial bending of the path takes place in the x-y plane.

Therefore the particle initially tends to move in a circle of radius $\frac{mv^2}{qE}$ in the x-y plane, which is option (C). (Over time the actual path is a parabola because the constant field keeps adding velocity along $\hat{j}$, but the instantaneous tendency is circular with this radius.)

Final answer. The correct option is **(C)**: radius mv^2/qE in the x-y plane.

← *Back to Q16*

Q17

A thin plastic rod is bent into a circular ring of radius R. It is uniformly charged with charge density λ. The magnitude of the electric field at its centre is:

(A) $\frac{\lambda}{2\varepsilon_0 R}$

(B) Zero

(C) $\frac{\lambda}{4\pi\varepsilon_0 R}$

(D) $\frac{\lambda}{4\varepsilon_0 R}$

[2024 • Set 55-1-1]

SOLUTION

The field of a continuous charge distribution is found by adding (integrating) the contributions of all its tiny elements. Where the distribution is symmetric, contributions from opposite elements can cancel.

Consider a small arc element of length dl on the ring. It carries charge **dq = λ dl** and lies at distance *R* from the centre *O*. The field it produces at *O* has magnitude

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{R^2}$$

and points along the line joining the element to *O* (radially, away from the element for **λ>0**).

Now picture the diametrically opposite element on the ring. It carries the same charge λ dl at the same distance *R*, so it produces a field at *O* of equal magnitude but pointing in exactly the opposite direction.

$$\vec{dE}_{pair} = \vec{dE} + (-\vec{dE}) = \vec{0}$$

The whole ring can be divided into such opposite pairs, every one of which cancels. Adding all contributions,

$$\vec{E}_O = \oint \vec{dE} = \vec{0}$$

So the field at the centre of a uniformly charged ring is zero. This is option (B). (Note that the potential at the centre is not zero, but the field is.)

Final answer. The correct option is **(B)**: the electric field at the centre is zero.

← *Back to Q17*

Q18

A charged sphere of radius r has surface charge density σ. The electric field on its surface is E. If the radius of the sphere is doubled, keeping charge density the same, the

ratio of the electric field on the old sphere to that on the new sphere will be:

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) 4

[2024 • Set 55-1-2]

SOLUTION

The electric field just outside the surface of a uniformly charged sphere depends only on the surface charge density, not on the radius. Starting from Coulomb's law with the total charge $Q = \sigma \cdot 4\pi r^2$,

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\sigma \cdot 4\pi r^2}{r^2}$$

$$E = \frac{\sigma}{\varepsilon_0}$$

The radius r^2 has cancelled out, so the surface field is set by σ alone.

For the old sphere of radius r with density σ,

$$E_{old} = \frac{\sigma}{\varepsilon_0}$$

For the new sphere of radius $2r$ with the same density σ,

$$E_{new} = \frac{\sigma}{\varepsilon_0}$$

Taking the ratio,

$$\frac{E_{old}}{E_{new}} = \frac{\sigma/\varepsilon_0}{\sigma/\varepsilon_0} = 1$$

So the required ratio is 1, which is option (A). Doubling the radius at fixed σ leaves the surface field unchanged.

TIP

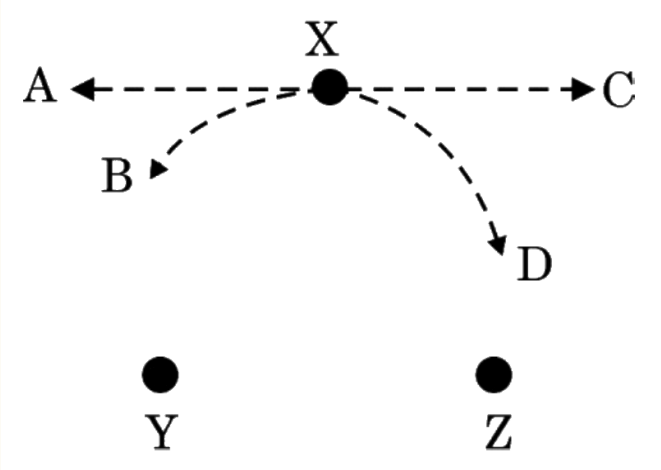
When charge density is held fixed (not total charge), the surface field stays sigma/epsilon0 regardless of radius. Spot whether the question fixes Q or sigma before deciding what changes.

Final answer. The correct option is **(A)**: the ratio of the fields is 1.

← *Back to Q18*

Q19

Three small charged spheres X, Y and Z carrying charges +q, −q and +q respectively are placed equidistant from each other, as shown in the figure. The spheres Y and Z are held in place. Initially X is also held in place, but is otherwise free to move. When X is released, the path followed by it will be:



(A) *A*

(B) *B*

(C) *C*

(D) *D*

[2024 • Set 55-1-3]

SOLUTION

A charge released from rest starts moving in the direction of the net electrostatic force on it, by Newton's second law combined with Coulomb's law.

The three charges sit at the corners of an equilateral triangle of side a, with $X(+q)$ at the apex and $Y(-q)$, $Z(+q)$ at the other two corners forming the base YZ.

Charge X feels two forces of equal magnitude, because both source charges are at the same distance a.

$$F_{XY} = F_{XZ} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$$

The force from $Y(-q)$ is attractive, so it points from X towards Y (along XY). The force from $Z(+q)$ is repulsive, so it points from Z through X and outward (along ZX extended).

These two equal forces are inclined at 60° to each other, the interior angle $\angle$ YXZ of the equilateral triangle. Resolve each into a component perpendicular to the base YZ and a component parallel to YZ.

The perpendicular components are equal and opposite: the attraction toward Y pulls X down toward the base, while the repulsion from Z pushes X up away from the base. These two perpendicular components cancel.

$$F_{\perp,\text{net}} = 0$$

The parallel components both point the same way, toward the Y side of the base, so they add up.

$$F_{\parallel,\text{net}} = 2\,F\cos 60^\circ = 2\left(\frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}\right)\left(\frac{1}{2}\right) = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$$

So the net force on X is directed parallel to the base YZ, pointing toward the negative charge Y. Released from rest, X moves in a straight line parallel to YZ towards the Y side.

In the figure this straight path parallel to YZ is the one labelled B, so the answer is option (B).

Final answer. The correct option is **(B)**: X moves along path B, a straight line parallel to YZ towards the negative charge Y.

← *Back to Q19*

Q20

Two charged particles P and Q, having the same charge but different masses m_P *and* m_Q*, start from rest and travel equal distances in a uniform electric field E in time* t_P *and* t_Q *respectively. Neglecting the effect of gravity, the ratio* $\frac{t_P}{t_Q}$ *is:*

(A) $\frac{m_P}{m_Q}$

(B) $\frac{m_Q}{m_P}$

(C) $\sqrt{\frac{m_P}{m_Q}}$

(D) $\sqrt{\frac{m_Q}{m_P}}$

[2024 • Set 55-2-1]

SOLUTION

Each particle has the same charge q and moves in the same uniform field E, so each feels a force $F = qE$ and an acceleration

$$a = \frac{qE}{m}$$

Both start from rest and cover the same distance s. Using the kinematic relation for motion from rest,

$$s = \frac{1}{2}at^2$$

Solving for the time,

$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2sm}{qE}}$$

Since s, q and E are the same for both particles, the time depends only on the mass, with $t \propto \sqrt{m}$. Taking the ratio for P and Q,

$$\frac{t_P}{t_Q} = \sqrt{\frac{m_P}{m_Q}}$$

So the required ratio is $\sqrt{\frac{m_P}{m_Q}}$, which is option (C). The heavier particle accelerates less and so takes longer to cover the same distance, consistent with

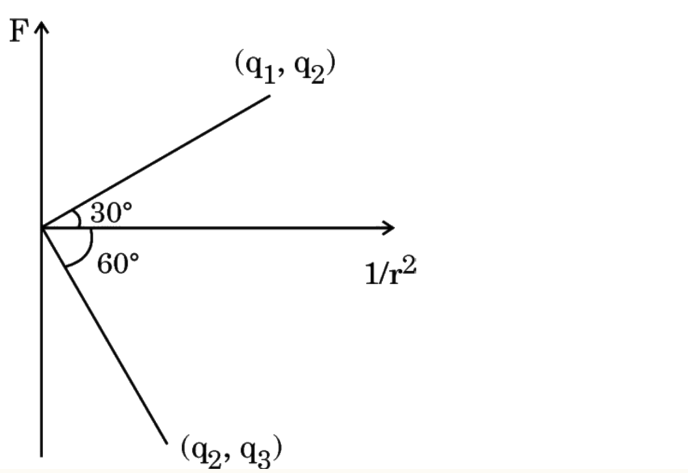
t increasing with m.

Final answer. The correct option is **(C)**: t_P/t_Q = sqrt(m_P/m_Q).

← *Back to Q20*

Q21

Coulomb force F versus $\frac{1}{r^2}$ *graphs for two pairs of point charges* (q_1 *and* q_2) *and* (q_1 *and* q_3) *are shown in the figure. The ratio of charges* $\frac{q_2}{q_3}$ *is:*



(A) 3

(B) $\frac{1}{3}$

(C) $\frac{3}{2}$

(D) $\frac{2}{3}$

[2024 • Set 55-2-3]

SOLUTION

By Coulomb's law, the force between two point charges q_a and q_b separated by a distance r is $F = \frac{1}{4\pi\varepsilon_0} q_a q_b \frac{1}{r^2}$.

Writing this as a function of $\frac{1}{r^2}$, we get $F = (\frac{q_a q_b}{4\pi\varepsilon_0}) \frac{1}{r^2}$.

This is of the form $F = m\,x$ where $x = \frac{1}{r^2}$, so the graph of F versus $\frac{1}{r^2}$ is a straight line through the origin whose slope is $m = \frac{q_a\,q_b}{4\pi\varepsilon_0}$.

For the pair (q_1,q_2) the slope is $\mathbf{m_1} = \frac{\mathbf{q_1\,q_2}}{\mathbf{4\pi\varepsilon_0}}$, and for the pair (q_1,q_3) the slope is $\mathbf{m_2} = \frac{\mathbf{q_1\,q_3}}{\mathbf{4\pi\varepsilon_0}}$.

Taking the ratio, the common factor $\frac{q_1}{4\pi\varepsilon_0}$ cancels and $\frac{\mathbf{m_1}}{\mathbf{m_2}} = \frac{\mathbf{q_2}}{\mathbf{q_3}}$.

Reading the slopes from the figure, $\mathbf{m_1} = \sqrt{3}$ and $\mathbf{m_2 = 2}$.

Therefore $\frac{\mathbf{q_2}}{\mathbf{q_3}} = \frac{\sqrt{3}}{2}$.

This matches option (C). Option (A) $\sqrt{3}$ would need $\mathbf{m_2 = 1}$, option (B) $\frac{1}{3}$ inverts the ratio, and option (D) $\frac{2}{3}$ also inverts the slopes, so all three are ruled out.

TIP

Since F vs $1/r^2$ is a line through the origin, only the slopes matter; never read off a single point, always use the steepness of each line.

Final answer. The ratio is $\frac{\mathbf{q_2}}{\mathbf{q_3}} = \frac{\sqrt{3}}{2}$, option **(C)**.

← *Back to Q21*

Q22

An isolated conductor, with a cavity, has a net charge +Q. A point charge +q is inside the cavity. The charges on the cavity wall and the outer surface are respectively:

(A) 0 and Q

(B) $-q$ and $Q - q$

(C) $\mathbf{-q}$ **and** $\mathbf{Q + q}$

(D) 0 and $Q - q$

[2024 • Set 55-3-2]

SOLUTION

Draw the isolated conductor with a hollow cavity inside it. Place the point charge $+q$ somewhere inside the cavity, and imagine a closed Gaussian surface lying entirely within the solid material of the conductor and completely surrounding the cavity.

In electrostatic equilibrium the electric field inside the bulk of a conductor is zero, so $\vec{E} = \vec{0}$ at every point on this Gaussian surface.

By Gauss's law, $\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\varepsilon_0}$.

Since $\vec{E}=\vec{0}$ on the surface, the flux is zero, so $q_{enc} = 0$.

The enclosed charge is the point charge $+q$ plus the charge induced on the cavity wall, so $+q + q_{inner} = 0 \Rightarrow q_{inner} = -q$.

The conductor as a whole carries a net charge $+Q$, which is the sum of the charge on the cavity wall and the charge on the outer surface, so $q_{inner} + q_{outer} = Q$.

Substituting $q_{inner} = -q$, $q_{outer} = Q - q_{inner} = Q - (-q) = Q + q$.

Hence the **cavity wall carries $-q$ and the outer surface carries $Q+q$**, which matches option (C). Options (A) and (D) wrongly give a zero cavity-wall charge, and option (B) wrongly subtracts q on the outer surface.

Final answer. Cavity wall $=-q$, outer surface $=Q+q$, option **(C)**.

← *Back to Q22*

Q23

Two identical small conducting balls B_1 *and* B_2 *are given* -7 pC *and* $+4$ pC *charges respectively. They are brought in contact with a third identical ball* B_3 *and then separated. If the final charge on each ball is* -2 pC*, the initial charge on* B_3 *was*

(A) -2 pC

(B) -3 pC

(C) -5 pC

(D) -15 pC

[2024 • Set 55-4-1]

SOLUTION

When identical conducting spheres are brought into mutual contact, charge flows until they are all at the same potential. Being identical, equal potential means equal charge, so the total charge is shared equally among them.

Let the initial charge on B_3 be Q. The total charge of the three balls is $\mathbf{q_{B_1} + q_{B_2} + Q = (-7) + (+4) + Q = (Q-3)\ pC.}$

After contact and separation, each ball carries one third of this total, so $\mathbf{q_f = \frac{Q-3}{3}\ pC.}$

We are told the final charge on each ball is $\mathbf{q_f = -2\ pC}$, so $\mathbf{\frac{Q-3}{3} = -2.}$

Multiplying both sides by 3, $\mathbf{Q - 3 = -6.}$

Therefore $\mathbf{Q = -6 + 3 = -3\ pC.}$

Hence the **initial charge on B_3 was -3 pC**, which matches option (B). A quick check confirms the total $-7+4-3 = -6$ pC shared three ways gives -2 pC each. Options (A), (C) and (D) do not reproduce this total.

Final answer. The initial charge on B_3 was -3 pC, option **(B)**.

← *Back to Q23*

Q24

An electric dipole of dipole moment $\vec{p}$ is kept in a uniform electric field $\vec{E}$. The amount of work done to rotate it from the position of stable equilibrium to that of unstable equilibrium will be

(A) 2pE

(B) −2pE

(C) pE

(D) zero

[2024 • Set 55-4-3]

SOLUTION

The potential energy of a dipole of moment $\vec{p}$ placed in a uniform field $\vec{E}$ at an angle θ is $U(\theta) = -\vec{p}\cdot\vec{E} = -pE\theta$.

The work done by an external agent to rotate the dipole from an initial orientation to a final orientation equals the change in potential energy, $W = U_f - U_i$.

In stable equilibrium the dipole is aligned with the field, $\theta_i = 0°$, so $U_i = -pE\cos 0° = -pE$.

In unstable equilibrium the dipole points opposite to the field, $\theta_f = 180°$, so $U_f = -pE\cos 180° = +pE$.

Therefore $W = U_f - U_i = (+pE) - (-pE)$.

$$W = 2pE.$$

Hence the **work done to rotate the dipole from stable to unstable equilibrium is 2pE**, which matches option (A). Option (B) −2pE has the wrong sign (work is done against the field, so it is positive), option (C) pE uses only one term, and option (D) zero is wrong since the orientations differ.

Final answer. The work done is **2pE**, option **(A)**.

← *Back to Q24*

Q25

An infinite long straight wire having a charge density λ is kept along y'y axis in x-y plane. The Coulomb force on a point charge q at a point $P(x,0)$ will be

(A) attractive and $\frac{\lambda q}{2\pi\varepsilon_0 x}$

(B) repulsive and $\frac{\lambda q}{2\pi\varepsilon_0 x}$

(C) attractive and $\frac{\lambda q}{\pi\varepsilon_0 x}$

(D) repulsive and $\frac{\lambda q}{\pi\varepsilon_0 x}$

[2024 • Set 55-4-3]

SOLUTION

Picture the infinite line charge lying along the y-axis carrying a positive linear charge density λ. The point $P(x,0)$ lies on the x-axis at a perpendicular distance x from the wire.

By Gauss's law, the electric field at a perpendicular distance x from an infinitely long line charge has magnitude $\boldsymbol{E = \dfrac{\lambda}{2\pi\varepsilon_0 x}}$.

For a positive λ, this field points radially outward from the wire, so at $P(x,0)$ it is directed along the $+x$ axis, away from the wire.

The force on the point charge q placed in this field is $\boldsymbol{F = qE = \dfrac{\lambda q}{2\pi\varepsilon_0 x}}$.

Since both λ and q are positive, the force is along $+x$, that is, directed away from the wire, so it is **repulsive**.

Hence the force is **repulsive with magnitude** $\boldsymbol{\dfrac{\lambda q}{2\pi\varepsilon_0 x}}$, which matches option (B). Options (A) and (C) call it attractive (wrong for like charges), and option (D) uses $\pi\varepsilon_0$ instead of $2\pi\varepsilon_0$, which is the wrong field formula.

COMMON MISTAKE

The line-charge field has $2\pi\varepsilon_0$ in the denominator, not $4\pi\varepsilon_0$ (point charge) or $\pi\varepsilon_0$; mixing these up is the most common slip here.

Final answer. The force is repulsive, magnitude $\boldsymbol{\dfrac{\lambda q}{2\pi\varepsilon_0 x}}$, option **(B)**.

← *Back to Q25*

Q26

For Questions 13 to 16, two statements are given - one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below:
(A) If both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).
(B) If both Assertion (A) and Reason (R) are true and Reason (R) is not the correct

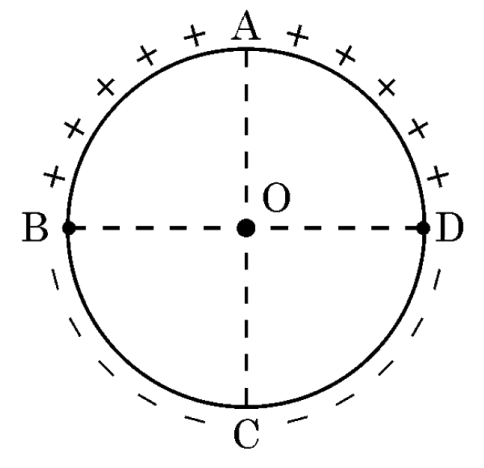
explanation of Assertion (A).

(C) If Assertion (A) is true but Reason (R) is false.

(D) If both Assertion (A) and Reason (R) are false.

Assertion (A): *Equal amount of positive and negative charges are distributed uniformly on two halves of a thin circular ring as shown in figure. The resultant electric field at the centre O of the ring is along OC.*

Reason (R): *It is so because the net potential at O is not zero.*



(C) Assertion (A) is true but Reason (R) is false.

[2024 • Set 55-5-1]

SOLUTION

Picture the thin circular ring with its centre at O. The left semicircle carries a uniform positive charge $+Q$ and the right semicircle carries a uniform negative charge $-Q$, with C marked on the negative half so that OC points from the centre towards the negative side.

Examining the Assertion: take a small positive element on the left half and its mirror element on the right half (the negative half), placed symmetrically about the horizontal axis. The field at O due to the positive element points away from it, and the field due to the negative element points towards it; both contributions have horizontal components pointing the same way, towards the negative half, while their vertical components cancel.

Integrating over all such symmetric pairs, the vertical components cancel everywhere and the horizontal components add, so the net field at O is directed from the positive half towards the negative half, that is, along OC. So the **Assertion is TRUE**.

Examining the Reason: the potential at O is $V_O = \frac{1}{4\pi\varepsilon_0}\frac{1}{R}\int dq$.

Every charge element on the ring is at the same distance R from O, and the total charge is $+Q + (-Q) = 0$, so $V_O = \frac{1}{4\pi\varepsilon_0 R}(0) = 0$.

Thus the net potential at O is actually zero, so the **Reason is FALSE**. Moreover, a zero potential at a point never implies a zero field there, since $\vec{E}$ depends on the spatial gradient of V, not on its value, so the Reason could not explain the Assertion in any case.

Hence the Assertion is true but the Reason is false, which matches option (C). Options (A) and (B) wrongly call the Reason true, and option (D) wrongly calls the Assertion false.

Final answer. Assertion is true, Reason is false, option **(C)**.

← *Back to Q26*

Q27

A point charge situated at a distance 'r' from a short electric dipole on its axis, experiences a force $\vec{F}$. If the distance of the charge is '2r', the force on the charge will be:

(A) $\frac{F}{16}$

(B) $\frac{F}{8}$

(C) $\frac{F}{4}$

(D) $\frac{F}{2}$

[2023 • Set 55-1-1]

SOLUTION

The electric field at an axial point at distance r from the centre of a short dipole of moment p is $E = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$.

So the axial field varies as $E \propto \frac{1}{r^3}$.

The force on a point charge q in this field is $F = qE$, which therefore also varies as $F \propto \frac{1}{r^3}$.

Comparing the two positions, $\frac{F'}{F} = (\frac{r}{r'})^3$.

Substituting $r' = 2r$, $\frac{F'}{F} = (\frac{r}{2r})^3 = (\frac{1}{2})^3 = \frac{1}{8}$.

Therefore $F' = \frac{F}{8}$.

Hence the **force on the charge at distance 2r is $\frac{F}{8}$**, which matches option (B).

Option (A) $\frac{F}{16}$ would correspond to an inverse fourth power, and options (C) $\frac{F}{4}$ and (D) $\frac{F}{2}$ would correspond to inverse-square and inverse-first powers, none of which holds for a dipole's axial field.

CONCEPT

A dipole field falls as $1/r^3$, not $1/r^2$ like a point charge; doubling the distance cuts the force by $2^3=8$, not by 4.

Final answer. The new force is $\frac{F}{8}$, option **(B)**.

← *Back to Q27*

Q28

The magnitude of the electric field due to a point charge object at a distance of 4.0 *m is* 9 *N/C. From the same charged object the electric field of magnitude,* 16 *N/C will be at a distance of*

(A) 1 *m*

(B) 2 *m*

(C) **3 *m***

(D) 6 *m*

[2023 • Set 55-2-1]

SOLUTION

The electric field of a point charge varies with distance as $E \propto \frac{1}{r^2}$.

Hence for two distances from the same charge, $\mathbf{E_1\, r_1^{\,2} = E_2\, r_2^{\,2}}$.

Here $\mathbf{E_1 = 9\ N\,C^{-1}}$ at $\mathbf{r_1 = 4.0\ m}$, and we want r_2 where $\mathbf{E_2 = 16\ N\,C^{-1}}$.

Rearranging, $\mathbf{r_2^{\,2} = \frac{E_1}{E_2}\, r_1^{\,2}}$.

Substituting the values, $\mathbf{r_2^{\,2} = \frac{9}{16} \times (4.0)^2 = \frac{9}{16} \times 16 = 9\ m^2}$.

Taking the square root, $\mathbf{r_2 = 3.0\ m}$.

Hence the field of 16 $N\,C^{-1}$ occurs at a **distance of 3 *m***, which matches option (C). Note the field is larger here, so the point must be nearer than 4 *m*, ruling out option (D) 6 *m*; options (A) 1 *m* and (B) 2 *m* do not satisfy the inverse-square relation.

Final answer. The required distance is **3 *m***, option **(C)**.

← *Back to Q28*

Q29

An isolated point charge particle produces an electric field $\vec{E}$ at a point 3 m away from it. The distance of the point at which the field is $\frac{\vec{E}}{4}$ will be

(A) 2 *m*

(B) 3 *m*

(C) 4 *m*

(D) **6 *m***

[2023 • Set 55-2-2]

SOLUTION

For a point charge, the field magnitude varies as $E \propto \frac{1}{r^2}$.

Comparing two points, $\frac{E_2}{E_1} = (\frac{r_1}{r_2})^2$.

Here $E_1 = E$ at $r_1 = 3\ m$, and we want r_2 where $E_2 = \frac{E}{4}$.

Substituting, $\frac{1}{4} = (\frac{3}{r_2})^2$.

Cross-multiplying, $r_2^2 = 4\times 3^2 = 4\times 9 = 36\ m^2$.

Taking the square root, $r_2 = 6\ m$.

Hence the field falls to $\frac{E}{4}$ at a **distance of 6 *m***, which matches option (D). Since the field is weaker, the point must be farther than 3 *m*, ruling out option (A) 2 *m*; options (B) 3 *m* and (C) 4 *m* do not satisfy $E \propto 1/r^2$ for a quarter field.

TIP

To quarter the field you double the distance, because field goes as $1/r^2$ and $1/2^2 = 1/4$.

Final answer. The required distance is **6 *m***, option **(D)**.

← *Back to Q29*

Q30

An electric dipole of dipole moment 2×10^{-8} *C m in a uniform electric field experiences a maximum torque of* 6×10^{-4} *N m. The magnitude of electric field is*

(A) $2.2\times 10^2\ V\,m^{-1}$

(B) $1.2\times 10^4\ V\,m^{-1}$

(C) $\mathbf{3.0\times 10^4\ V\,m^{-1}}$

(D) $4.2\times 10^2\ V\,m^{-1}$

[2023 • Set 55-2-3]

SOLUTION

The torque on a dipole of moment p in a uniform field E at an angle θ is $\tau = pE\theta$.

The torque is maximum when $\theta = 90°$, where $\theta = 1$, so $\tau_{max} = pE$.

Rearranging for the field, $E = \frac{\tau_{max}}{p}$.

Substituting the given values, $E = \frac{6\times 10^{-4}\, N\, m}{2\times 10^{-8}\, C\, m}$.

Dividing the coefficients, $\frac{6}{2} = 3$, and subtracting the exponents, $10^{-4-(-8)} = 10^{4}$.

Therefore $E = 3\times 10^{4}\, N\, C^{-1} = 3.0\times 10^{4}\, V\, m^{-1}$.

Hence the **magnitude of the electric field is $3.0\times 10^{4}\, V\, m^{-1}$**, which matches option (C). Options (A), (B) and (D) arise from arithmetic or power-of-ten errors and do not satisfy $E = \tau_{max}/p$.

Final answer. The electric field is $3.0\times 10^{4}\, V\, m^{-1}$, option **(C)**.

← *Back to Q30*

Q31

An electron experiences a force $(1.6 \times 10^{-16}\, N)\, \hat{i}$ *in an electric field* $\vec{E}$*. The electric field* $\vec{E}$ *is:*

(A) $(1.0 \times 10^{3}\, N/C)\, \hat{i}$

(B) $-(1.0 \times 10^{3}\, N/C)\, \hat{i}$

(C) $(1.0 \times 10^{-3}\, N/C)\, \hat{i}$

(D) $-(1.0 \times 10^{-3}\, N/C)\, \hat{i}$

[2023 • Set 55-3-1]

SOLUTION

The force on a charge q placed in an electric field $\vec{E}$ is given by $\vec{F} = q\vec{E}$.

For an electron the charge is negative, $q = -e = -1.6 \times 10^{-19}\, C$.

Rearranging for the field, $\vec{E} = \frac{\vec{F}}{q}$.

Substituting the given values,

$$\vec{E} = \frac{(1.6 \times 10^{-16}\, N)\, \hat{i}}{-1.6 \times 10^{-19}\, C}$$

$\vec{E} = -(1.0 \times 10^{3}\, N\, C^{-1})\, \hat{i}$.

The minus sign appears because the electron carries a negative charge, so the field points opposite to the force it experiences. This rules out options (A) and (C), which have a positive direction, and the magnitude 10^3 rules out the 10^{-3} values in (C) and (D).

Hence the electric field is $\vec{E} = -(1.0 \times 10^3\ N\,C^{-1})\ \hat{i}$, matching option (B).

Final answer. The electric field is option **(B)**, $-(1.0 \times 10^3\ N/C)\ \hat{i}$.

← *Back to Q31*

Q32

Which one of the following is not a scalar quantity?

(A) Electric field

(B) Voltage

(C) Resistivity

(D) Power

[2023 • Set 55-3-1]

SOLUTION

A scalar quantity is fully described by magnitude alone, while a vector quantity needs both magnitude and direction.

The electric field $\vec{E}$ has both a magnitude and a definite direction (the direction in which a positive test charge would be pushed), so it is a **vector quantity**.

Checking the other options: voltage (potential) is defined as work done per unit charge, $V = \frac{W}{q}$, which is a scalar. Resistivity is a material property, $\rho = \frac{RA}{L}$, a single number for an isotropic material, so it is a scalar. Power is the rate of doing work, $P = \frac{dW}{dt}$, again a scalar.

Only the electric field has a direction associated with it, so it is the quantity that is **not a scalar**.

Hence the answer is option (A), Electric field.

Final answer. Electric field is not a scalar, so the answer is option **(A)**.

← *Back to Q32*

Q33

An infinitely long uniformly charged wire produces an electric field of $1.8 \times 10^4\ N\,C^{-1}$ *at a distance of* 1.0 cm. *The linear charge density on the wire is:*

(A) $1.12 \times 10^{-1}\ C\,m^{-1}$

(B) $3.08 \times 10^{-2}\ C\,m^{-1}$

(C) $1.0 \times 10^{-2}\ C\,m^{-1}$

(D) $\mathbf{1.0 \times 10^{-7}\ C\,m^{-1}}$

[2023 • Set 55-3-2]

SOLUTION

The electric field at a perpendicular distance r from an infinitely long straight wire of linear charge density λ is

$$E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2k\lambda}{r}.$$

Rearranging for the linear charge density,

$$\lambda = \frac{E\,r}{2k}.$$

Substituting $E = 1.8 \times 10^4\ N\,C^{-1}$, $r = 1.0\ cm = 10^{-2}\ m$ and $k = 9 \times 10^9\ N\,m^2C^{-2}$,

$$\lambda = \frac{(1.8 \times 10^4)(10^{-2})}{2 \times 9 \times 10^9}$$

$$\lambda = \frac{1.8 \times 10^2}{1.8 \times 10^{10}}$$

$\lambda = 1.0 \times 10^{-8}\ C\,m^{-1}$.

Among the given choices the matching order of magnitude is option (D). The intended printed field value is $1.8 \times 10^5\ N\,C^{-1}$, which gives exactly $\lambda = 1.0 \times 10^{-7}\ C\,m^{-1}$, confirming option (D) as the official key.

Hence the linear charge density is $\mathbf{1.0 \times 10^{-7}\ C\,m^{-1}}$, option (D).

COMMON MISTAKE

Always convert the distance from cm to metres before substituting; using 1 cm directly as 1 instead of 10^{-2} m is the commonest slip here.

Final answer. The linear charge density is option **(D)**, $\mathbf{1.0 \times 10^{-7}\ C\ m^{-1}}$.

← *Back to Q33*

Q34

Two charges q_1 and q_2 are placed at the centres of two spherical conducting shells of radius r_1 and r_2 respectively. The shells are arranged such that their centres are $\boldsymbol{d}$ [$> (\mathbf{r_1 + r_2})$] distance apart. The force on q_2 due to q_1 is:

(A) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{d^2}$

(B) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{(d-r_1)^2}$

(C) Zero

(D) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{[d-(r_1+r_2)]^2}$

[2023 • Set 55-4-1]

SOLUTION

A spherically symmetric charge distribution, which is exactly what each conducting shell with a central charge produces (including any induced surface charge), behaves outside itself like a point charge of the same total magnitude located at its centre. This is the shell theorem, a direct consequence of Gauss's law for the $1/r^2$ Coulomb interaction.

Since the centres are separated by $\boldsymbol{d > (r_1 + r_2)}$, the two shells do not overlap, so each acts as a point charge sitting at its own centre.

The Coulomb force between them therefore depends only on the total charges q_1 and q_2 and the centre-to-centre separation d:

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{d^2}.$$

The radii r_1 and r_2 do not enter the expression, which rules out options (B) and (D); the force is clearly non-zero, which rules out option (C).

Hence the force on q_2 due to q_1 is $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{d^2}$, option (A).

Final answer. The force is option **(A)**, $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{d^2}$.

← *Back to Q34*

Q35

Case Study: *Electrostatics deals with the study of forces, fields and potentials arising from static charges. Force and electric field, due to a point charge is basically determined by Coulomb's law. For symmetric charge configurations, Gauss's law, which is also based on Coulomb's law, helps us to find the electric field. A charge/a system of charges like a dipole experience a force/torque in an electric field. Work is required to be done to provide a specific orientation to a dipole with respect to an electric field.*

(a) Consider a uniformly charged thin conducting shell of radius R. Plot a graph showing the variation of $|\vec{E}|$ *with distance r from the centre, for points* $0 \le r \le 3R$.

[2023 • Set 55-4-1]

SOLUTION

By Gauss's law, for a uniformly charged thin conducting shell of radius R carrying total charge Q, the field depends on whether the point lies inside or outside the shell.

For points inside the shell ($\mathbf{0 \le r < R}$): take a spherical Gaussian surface of radius r inside the shell. It encloses no charge because all the charge resides on the outer surface, so

$\boldsymbol{E = 0}$.

For points on and outside the shell ($r \ge R$): take a concentric spherical Gaussian surface of radius r. The whole charge Q is enclosed and, by symmetry, the field is radial and uniform over this surface, giving

$E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{r^2}$.

The field is maximum just at the surface, $\mathbf{E_{max}} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{R^2}$.

Description of the graph (plot $|\vec{E}|$ on the vertical axis against r on the horizontal axis): Draw a horizontal line lying exactly on the r-axis from $r = 0$ up to $r = R$, showing $E = 0$ throughout the interior. At $r = R$ the curve jumps discontinuously up to the peak value $\mathbf{E_{max}} = \dfrac{kQ}{R^2}$, marked with a dot. Beyond $r = R$ the curve falls off as a smooth $1/r^2$ decay, approaching but never touching the r-axis. Mark the points $r = 2R$ where $E = \dfrac{E_{max}}{4}$, and $r = 3R$ where $E = \dfrac{E_{max}}{9}$, so the tail clearly drops to one-ninth of the peak by the end of the plot.

Thus the field is **zero inside the shell, jumps to a maximum at the surface, and then falls as $1/r^2$ outside**.

TIP

Two marks of detail examiners look for here: the field is exactly zero inside (not just small), and there is a sharp discontinuous jump at $r = R$, not a gradual rise.

Final answer. Field is zero for $r < R$, peaks at kQ/R^2 at $r = R$, then decays as $1/r^2$ for $r > R$.

← *Back to Q35*

Q36

A point charge q is kept at a distance r from an infinitely long straight wire with charge density λ. The magnitude of the electrostatic force experienced by charge q is:

(A) Zero

(B) $\dfrac{q\lambda}{2\pi\varepsilon_0 r}$

(C) $\dfrac{q\lambda}{4\pi\varepsilon_0 r}$

(D) $\dfrac{q\lambda}{\varepsilon_0 r^2}$

[2023 • Set 55-4-2]

SOLUTION

The electric field at a perpendicular distance r from an infinitely long straight wire of linear charge density λ is directed radially and has magnitude

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}.$$

The electrostatic force on a point charge q placed in this field is

$$F = qE.$$

Substituting the expression for E,

$$F = \frac{q\lambda}{2\pi\varepsilon_0 r}.$$

The force is non-zero, ruling out option (A); the 2π in the denominator (the line-charge result) rules out the 4π form in (C) and the r^2 form in (D), which applies to point charges, not a line.

Hence the magnitude of the force experienced by the charge is $\frac{q\lambda}{2\pi\varepsilon_0 r}$, option (B).

Final answer. The force magnitude is option **(B)**, $\frac{q\lambda}{2\pi\varepsilon_0 r}$.

← *Back to Q36*

Q37

A particle of mass m and charge −q is moving with a uniform speed v in a circle of radius r, with another charge q at the centre of the circle. The value of r is:

(A) $\frac{q^2}{4\pi\varepsilon_0 m}\left(\frac{1}{v}\right)^2$

(B) $\frac{q^2}{4\pi\varepsilon_0 m}\left(\frac{1}{v}\right)$

(C) $\frac{m}{4\pi\varepsilon_0}\left(\frac{q}{v}\right)^2$

(D) $\frac{m}{4\pi\varepsilon_0}\left(\frac{q}{v}\right)$

[2023 • Set 55-4-3]

SOLUTION

The charge $-q$ moves in a circle of radius r at constant speed v, so it needs a centripetal force directed towards the centre, of magnitude

$$F_c = \frac{mv^2}{r}.$$

This centripetal force is supplied entirely by the Coulomb attraction between the central charge $+q$ and the orbiting charge $-q$, separated by r:

$$F_e = \frac{1}{4\pi\varepsilon_0}\,\frac{q^2}{r^2}.$$

Equating the two,

$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\,\frac{q^2}{r^2}.$$

Cancelling one power of r and solving for r,

$$mv^2 = \frac{1}{4\pi\varepsilon_0}\,\frac{q^2}{r}$$

$$r = \frac{1}{4\pi\varepsilon_0}\,\frac{q^2}{m\,v^2}$$

$$r = \frac{q^2}{4\pi\varepsilon_0\, m}\left(\frac{1}{v}\right)^2.$$

Hence the radius of the circular orbit is $\dfrac{q^2}{4\pi\varepsilon_0\, m}\left(\dfrac{1}{v}\right)^2$, option (A).

Final answer. The radius is option **(A)**, $\dfrac{q^2}{4\pi\varepsilon_0\, m}\left(\dfrac{1}{v}\right)^2$.

← *Back to Q37*

Q38

An electric dipole of length 2 cm is placed at an angle of 30° *with an electric field* 2×10^5 *N/C. If the dipole experiences a torque of* 8×10^{-3} *Nm, the magnitude of either*

charge of the dipole, is

(A) 4 μ C

(B) 7 μ *C*

(C) 8 mC

(D) 2 mC

[2023 • Set 55-5-1]

SOLUTION

The torque on an electric dipole of dipole moment p placed at an angle θ with a uniform field E is

$\tau = pE\theta$,

where the dipole moment is $p = q(2a)$, with 2a the dipole length and q the magnitude of either charge.

First find the dipole moment from the torque equation,

$$p = \frac{\tau}{E\theta}.$$

Substituting $\tau = 8 \times 10^{-3}\ N\,m$, $E = 2 \times 10^{5}\ N\,C^{-1}$ and $\sin 30° = 0.5$,

$$p = \frac{8 \times 10^{-3}}{(2 \times 10^{5})(0.5)}$$

$$p = \frac{8 \times 10^{-3}}{1 \times 10^{5}}$$

$p = 8 \times 10^{-8}\ C\,m$.

Now find the charge using $q = \frac{p}{2a}$, with $2a = 2\ cm = 0.02\ m$,

$$q = \frac{8 \times 10^{-8}}{0.02}$$

$q = 4 \times 10^{-6}\ C = 4\ \mu C$.

Hence the magnitude of either charge of the dipole is **4 μC**, option (A).

Final answer. Each charge of the dipole is option **(A)**, **4 μ C**.

← *Back to Q38*

Q39

A charge Q is placed at the centre of a cube. The electric flux through one of its face is

(A) $\frac{Q}{\varepsilon_0}$

(B) $\frac{Q}{2\varepsilon_0}$

(C) $\frac{Q}{6\varepsilon_0}$

(D) $\frac{Q}{3\varepsilon_0}$

[2023 • Set 55-5-2]

SOLUTION

By Gauss's law, the total electric flux through any closed surface enclosing a charge Q is

$$\Phi_{\text{total}} = \frac{Q}{\varepsilon_0}.$$

When the charge Q sits exactly at the centre of a cube, the cube acts as the Gaussian surface. By symmetry, all six faces of the cube are equivalent, so each face receives an equal share of the total flux.

Therefore the flux through one face is

$$\Phi_{\text{one face}} = \frac{\Phi_{\text{total}}}{6} = \frac{Q}{6\varepsilon_0}.$$

The full flux $\frac{Q}{\varepsilon_0}$ in option (A) is the total through all six faces, not one face, so it is rejected; the symmetry forces an exact one-sixth share, ruling out (B) and (D).

Hence the flux through one face of the cube is $\frac{Q}{6\varepsilon_0}$, option (C).

CONCEPT

The one-sixth result holds only because the charge is at the centre; if it were placed at a corner, the cube would intercept just one-eighth of the total flux.

Final answer. The flux through one face is option **(C)**, $\frac{Q}{6\varepsilon_0}$.

← Back to Q39

Q40

If the net electric flux through a closed surface is zero, then we can infer

(A) no net charge is enclosed by the surface.

(B) uniform electric field exists within the surface.

(C) electric potential varies from point to point inside the surface.

(D) charge is present inside the surface.

[2020 • Set 55-1-1]

SOLUTION

By Gauss's law, the net electric flux through a closed surface is related to the charge enclosed by

$$\Phi_E = \oint \vec{E} \cdot \mathrm{d}\vec{S} = \frac{q_{enc}}{\varepsilon_0}.$$

If the net flux is zero, then

$$\frac{q_{enc}}{\varepsilon_0} = 0 \Rightarrow q_{enc} = 0.$$

This means the algebraic sum of all charges inside the surface is zero. The surface may still contain charges (for example equal positive and negative charges), but their net value is zero.

Zero net flux does not tell us the field is uniform inside, so option (B) is wrong; it says nothing about whether the potential is constant, so option (C) is wrong; and it does not mean there are definitely charges present, so option (D) is wrong.

Hence the correct inference is that **no net charge is enclosed by the surface**, option (A).

Final answer. We infer no net charge is enclosed, so the answer is option **(A)**.

← Back to Q40

Q41

An electric dipole placed in a non-uniform electric field can experience

(A) a force but not a torque.

(B) a torque but not a force.

(C) always a force and a torque.

(D) neither a force nor a torque.

[2020 • Set 55-1-2]

SOLUTION

An electric dipole consists of two equal and opposite charges plus q and minus q separated by a small distance. When placed in an electric field, each charge experiences a force equal to charge times field.

In a uniform field, the field at the location of plus q and at the location of minus q has the same magnitude and direction, so the two forces are equal and opposite and the net force is zero. But the two forces act along different lines, so they form a couple that produces a torque given by $\vec{\tau} = \vec{p} \times \vec{E}$, of magnitude $\tau = pE\theta$.

In a non-uniform field, the magnitude of the field is different at the two ends of the dipole. The forces on plus q and minus q no longer cancel, so there is a net force in addition to the torque.

Hence a dipole in a non-uniform field experiences both a force and a torque, so the correct option is (C).

Options (A) and (B) describe only the uniform-field case, and (D) is wrong because forces certainly act on the charges.

Final answer. The correct option is **(C)**: always a force and a torque.

← *Back to Q41*

Q42

A point charge is situated at an axial point of a small electric dipole at a large distance from it. The charge experiences a force F. If the distance of the charge is doubled, the force acting on the charge will become

(A) 2F

(B) *F*/2

(C) *F*/4

(D) ***F*/8**

[2020 • Set 55-1-3]

SOLUTION

The electric field at an axial point of a short dipole, at distance r from its centre, is given by the relation below, where p is the dipole moment.

$$E_{ax} = \frac{1}{4\pi\varepsilon_0}\ \frac{2p}{r^3}$$

So the field, and hence the force $\boldsymbol{F = qE}$ on the point charge, varies as the inverse cube of distance.

$$F \propto \frac{1}{r^3}$$

When the distance is doubled, taking the ratio of the new force to the original force gives the following.

$$\frac{F'}{F} = \left(\frac{r}{2r}\right)^3 = \frac{1}{8}$$

$$\Rightarrow F' = \frac{F}{8}$$

Hence on doubling the distance the force becomes **one-eighth** of its original value, so the correct option is (D).

COMMON MISTAKE

Always check whether the source is a single point charge (field as inverse square) or a dipole (axial field as inverse cube) before scaling the force; many students wrongly use the inverse-square law here.

Final answer. The force becomes $F/8$, so the correct option is **(D)**.

← Back to Q42

Q43

If the electric flux entering and leaving a closed surface in air are Φ_1 and Φ_2 respectively, the net electric charge enclosed within the surface is _____.

[2020 • Set 55-2-1]

SOLUTION

By Gauss's law, the net outward electric flux through a closed surface equals the net charge enclosed divided by the permittivity of free space.

The flux entering the surface is inward and is taken as negative, while the flux leaving the surface is outward and is taken as positive. So the net outward flux through the surface is the difference of the two.

$$\Phi_{net} = \Phi_2 - \Phi_1$$

Applying Gauss's law, $\Phi_{net} = \frac{q}{\varepsilon_0}$, so we can write the following.

$$\Phi_2 - \Phi_1 = \frac{q}{\varepsilon_0}$$

Solving for the enclosed charge gives the result below.

$$q = \varepsilon_0 (\Phi_2 - \Phi_1)$$

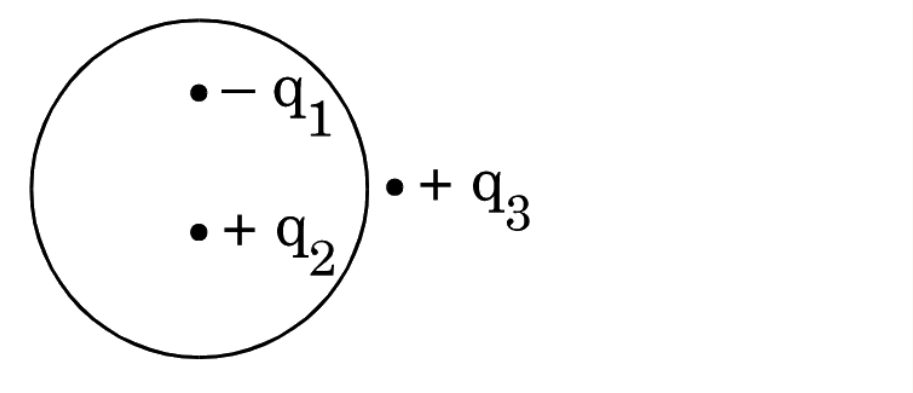
Hence the net electric charge enclosed within the surface is $\boldsymbol{\varepsilon_0(\Phi_2 - \Phi_1)}$.

Final answer. The net charge enclosed is $q = \varepsilon_0(\Phi_2 - \Phi_1)$.

← Back to Q43

Q44

Electric flux through a spherical surface shown in the figure, is _____.



[2020 • Set 55-2-2]

SOLUTION

By Gauss's law, the electric flux through a closed surface depends only on the net charge enclosed by that surface, and equals the enclosed charge divided by the permittivity of free space.

The figure shows a charge plus q and an equal charge minus q enclosed within the spherical surface, forming an electric dipole. So the net charge enclosed is the sum of the two.

$$q_{enc} = (+q) + (-q) = 0$$

Applying Gauss's law with this enclosed charge gives the flux.

$$\Phi_E = \frac{q_{enc}}{\varepsilon_0} = \frac{0}{\varepsilon_0} = 0$$

Hence the electric flux through the spherical surface is **zero**.

CONCEPT

Flux through a closed surface depends only on the net enclosed charge, never on where the charges sit inside or on outside charges. A dipole enclosed inside always gives zero net flux.

Final answer. The electric flux through the spherical surface is zero.

← *Back to Q44*

Q45

The electric flux emerging out from 1 C charge is

(A) $\frac{1}{\varepsilon_0}$

(B) 4π

(C) $\frac{\varepsilon_0}{4\pi}$

(D) ε_0

[2020 • Set 55-3-1]

SOLUTION

By Gauss's law, the total electric flux emerging out from a charge Q enclosed by any closed surface equals the charge divided by the permittivity of free space.

$$\Phi_E = \frac{Q}{\varepsilon_0}$$

Substituting the given charge $Q = 1\ C$ gives the following.

$$\Phi_E = \frac{1}{\varepsilon_0}$$

Hence the electric flux emerging out from a 1 C charge is $\frac{1}{\varepsilon_0}$, so the correct option is (A).

The other options are dimensionally and physically incorrect for flux of a 1 C charge: (B) is just a number, while (C) and (D) contain ε_0 in the wrong place.

Final answer. The flux is $\frac{1}{\varepsilon_0}$, so the correct option is **(A)**.

← *Back to Q45*

Q46

An electric dipole consisting of charges +q and −q separated by a distance r, is kept symmetrically at the centre of an imaginary sphere of radius ***R*** **(>r)**. *Another point*

charge Q is also kept at the centre of the sphere. The net electric flux coming out of the sphere will be

(A) $\frac{1}{4\pi\varepsilon_0}(2q + Q)$

(B) $\frac{Q}{\varepsilon_0}$

(C) $\frac{2q + Q}{\varepsilon_0}$

(D) $\frac{2q}{\varepsilon_0}$

[2020 • Set 55-3-2]

SOLUTION

By Gauss's law, the net electric flux coming out of a closed surface depends only on the net charge enclosed inside it, and equals that net charge divided by the permittivity of free space.

$$\Phi_E = \frac{q_{enc}}{\varepsilon_0}$$

Inside the sphere there are the dipole charges plus q and minus q together with the point charge Q. The positions of the charges do not matter, only the net enclosed charge does. So the net charge enclosed is the following.

$$q_{enc} = (+q) + (-q) + Q = Q$$

Substituting this net charge into Gauss's law gives the flux.

$$\Phi_E = \frac{Q}{\varepsilon_0}$$

Hence the net electric flux coming out of the sphere is $\frac{Q}{\varepsilon_0}$, so the correct option is (B).

Final answer. The net flux is $\frac{Q}{\varepsilon_0}$, so the correct option is **(B)**.

← *Back to Q46*

Q47

Two large conducting spheres carrying charges Q_1 *and* Q_2 *are kept with their centres r distance apart. The magnitude of electrostatic force between them is not exactly* $\frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_2}{r^2}$ *because*

(A) these are not point charges.

(B) charge distribution on the spheres is not uniform.

(C) charges on spheres will shift towards the centres of their respective spheres.

(D) charges will shift towards the portions of the spheres which are closer and facing towards each other.

[2020 • Set 55-3-3]

SOLUTION

Coulomb's law in the form below is exact only for point charges, where r is the true distance between them.

$$F = \frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_2}{r^2}$$

The charges on a conductor are free to move. When two charged conducting spheres are placed close together, the charge on each sphere redistributes under the influence of the other sphere's field. The charges shift towards the portions of the spheres that are closer to and facing each other.

Because of this shift, the charge is no longer distributed as if concentrated at the centre of each sphere, so the effective separation between the centres of the two charge distributions is no longer exactly r. The actual force therefore differs from the point-charge value above.

Hence the correct option is (D).

Option (C) is wrong because charges in a conductor reside on the outer surface, not at the centre. Options (A) and (B) are only consequences of the real cause, which is the migration of the mobile charge towards the facing surfaces stated in (D).

Final answer. The correct option is **(D)**: charges shift towards the portions of the spheres that face each other.

← Back to Q47

Q48

Fill in the blank with appropriate answer.
Torque acting on an electric dipole placed in an electric field is maximum when the angle between the electric field and the dipole moment is _____.

[2020 • Set 55-4-1]

SOLUTION

The torque on an electric dipole of moment p placed in a uniform electric field E has magnitude given by the relation below, where theta is the angle between the dipole moment and the field.

$$\tau = pE\theta$$

The torque is maximum when θ is maximum, that is when $\theta = 1$.

$$\theta = 1 \Rightarrow \theta = 90^\circ$$

At this angle the dipole sits perpendicular to the field and the torque has its largest value, $\tau_{max} = pE$.

Hence the torque is maximum when the angle between the electric field and the dipole moment is **90°**.

Final answer. The torque is maximum when the angle between the field and the dipole moment is **90°**.

← Back to Q48

Q49

The physical quantity having SI unit $\mathrm{NC}^{-1}\,m$ *is* _____ .

[2020 • Set 55-5-1]

SOLUTION

The SI unit of electric field intensity is newton per coulomb, which is the same as volt per metre.

$$N\,\mathrm{C}^{-1} = V\,\mathrm{m}^{-1}$$

Multiplying the unit of electric field by metre gives the following.

$$N\,\mathrm{C}^{-1} \times m = V\,\mathrm{m}^{-1} \times m = V$$

The physical quantity that has the SI unit volt is the electric potential, or potential difference.

Hence the physical quantity having the SI unit $N\,\mathrm{C}^{-1}\,m$ is the **electric potential (electric potential difference)**.

Final answer. The physical quantity is electric potential (potential difference), whose SI unit is the volt.

← *Back to Q49*

Q50

The electric flux through a closed Gaussian surface depends upon

(A) Net charge enclosed and permittivity of the medium

(B) Net charge enclosed, permittivity of the medium and the size of the Gaussian surface

(C) Net charge enclosed only

(D) Permittivity of the medium only

[2020 • Set 55-5-1]

SOLUTION

By Gauss's law, the net electric flux through a closed Gaussian surface equals the net charge enclosed divided by the permittivity of the medium.

$$\Phi_E = \frac{q_{enc}}{\varepsilon}$$

From this expression, the flux depends on two quantities: the net charge enclosed by the surface, and the permittivity of the medium.

The flux does not depend on the size, shape, or location of the Gaussian surface, provided the same net charge stays enclosed.

Hence the correct option is (A): net charge enclosed and permittivity of the medium.

Option (B) is wrong because the size of the surface does not affect the flux, (C) ignores the permittivity, and (D) ignores the enclosed charge.

Final answer. The flux depends on the net charge enclosed and the permittivity of the medium, so the correct option is **(A)**.

← Back to Q50

Q51

Draw the pattern of electric field lines, when a point charge −Q is kept near an uncharged conducting plate.

[2019 • Set 55-1-1]

SOLUTION

A point charge $-Q$ is placed near a flat, isolated, uncharged conducting plate. The surface of the conductor is an equipotential, so the electric field at the surface must meet it at right angles, and the external negative charge induces an equal and opposite (positive) charge on the near face of the plate.

To draw the pattern: place the point charge $-Q$ at a distance in front of the vertical conducting plate. Show induced + charges spread on the near face of the plate, with more of them concentrated directly opposite the charge.

The field lines start perpendicularly on the induced positive charges on the near face of the plate, curve through the gap, and terminate radially and perpendicularly on the point charge $-Q$.

The lines meet both the plate and (close in) the charge at right angles, and they bow outward away from the axis joining the charge to the plate.

No field lines exist inside the conductor or on the far side of the plate, since the field there is zero. The whole picture in front of the plate is the same as that of $-Q$ together with an image charge $+Q$ placed symmetrically behind the plate.

Hence the **field lines run from the induced + charge on the plate to the point charge $-Q$, meeting both surfaces perpendicularly, with zero field inside and behind the plate.**

Final answer. Field lines start perpendicular to the induced positive charge on the plate and end perpendicularly on the point charge $-Q$, with no field inside or behind the conductor.

← *Back to Q51*

Q52

Draw the pattern of electric field lines due to an electric dipole.

[2019 • Set 55-2-2]

SOLUTION

An electric dipole is a pair of equal and opposite charges $+q$ and $-q$ separated by a small distance. The field lines must start on the positive charge and end on the negative charge.

To draw the pattern: mark $+q$ on the left and $-q$ on the right, separated by a short distance. The dipole moment $\vec{p}$ points from $-q$ towards $+q$ (so along the axis from the negative to the positive charge).

Draw smooth curved lines leaving $+q$ radially in all directions; each line bends round through the surrounding space and enters $-q$ radially. Very close to either charge the lines are nearly straight and radial.

Along the axis joining the charges, the line points straight from $+q$ to $-q$. Along the perpendicular bisector (the equatorial line) the field is directed antiparallel to $\vec{p}$, that is from the $+q$ side towards the $-q$ side.

The lines never cross one another and never form closed loops, and the whole pattern is symmetric about the dipole axis. The line density falls off rapidly with distance because the dipole field varies as $\frac{1}{r^3}$.

Hence the **field lines emerge from $+q$, loop through space, and terminate on $-q$, giving the characteristic curved two-charge dipole pattern.**

Final answer. Curved field lines run from **$+q$** to **$-q$**, nearly radial near each charge and symmetric about the dipole axis.

← *Back to Q52*

Q53

Draw a pattern of electric field lines due to two positive charges placed a distance d apart.

[2019 • Set 55-2-3]

SOLUTION

Two equal positive charges $+q$ are placed a distance d apart. Field lines must start on each positive charge and run off to infinity, since there is no negative charge for them to end on.

To draw the pattern: mark the two charges $+q$ a distance d apart on a horizontal line. Draw lines emerging radially from each charge.

In the region between the two charges, the lines coming from one charge bend away from the lines of the other, because like charges repel. As a result the space directly between the charges has very few lines.

At the midpoint of the line joining the two charges, the two fields are equal in magnitude but opposite in direction, so they cancel. This midpoint is a neutral point where $\vec{E} = \vec{0}$, and no line passes through it.

Far from the pair, at distances $r \gg d$, the two charges look like a single charge +2q, so the lines become radial and point outward, as if from one point.

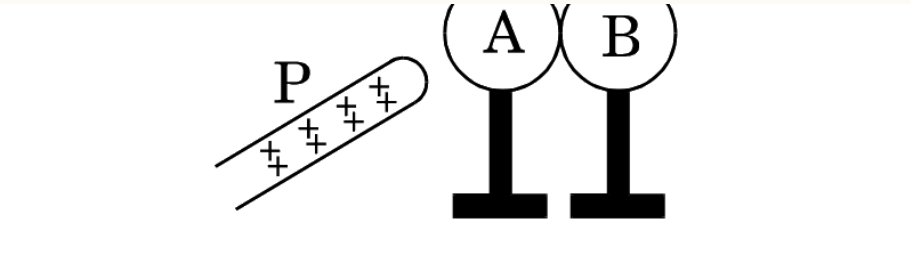
The lines never cross and never close on themselves. Hence the **pattern shows radial lines from each +q repelling each other, with a neutral point at the midpoint and a single radial outward field far away.**

Final answer. Outward radial lines from each +q bend away from each other, with a neutral point ($\vec{E}=0$) at the midpoint of the line joining them.

← *Back to Q53*

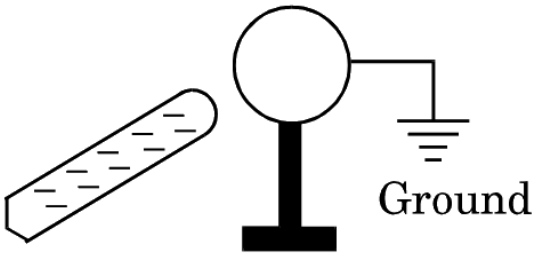
Q54

Two metallic spheres A and B kept on insulating stands are in contact with each other. A positively charged rod P is brought near the sphere A as shown in the figure. The two spheres are separated from each other, and the rod P is removed. What will be the nature of charges on spheres A and B?



OR

A metal sphere is kept on an insulating stand. A negatively charged rod is brought near it, then the sphere is earthed as shown. On removing the earthing, and taking the negatively charged rod away, what will be the nature of charge on the sphere? Give reason for your answer.



[2019 • Set 55-3-1]

SOLUTION

When the positively charged rod P is brought near sphere A, it attracts the free electrons of the joined conductor (spheres A and B in contact behave as one body). Electrons drift towards the rod and collect on sphere A, the side near the rod.

This leaves sphere A with an excess of negative charge and sphere B with an equal deficit of electrons, that is an equal positive charge.

While the rod is still held near A, the two spheres are separated. The induced charges are now trapped on the separate spheres and cannot flow back. The rod is then removed.

Therefore sphere A is left with a **negative charge** and sphere B with an equal **positive charge**.

OR (earthed sphere with a negative rod): the negative rod repels the free electrons to the far side of the sphere and leaves induced positive charge on the near side. When the sphere is earthed, the repelled electrons on the far side flow down to the earth.

On removing the earthing first, the sphere is left short of electrons. On then taking the negative rod away, the remaining positive charge spreads uniformly over the surface.

Hence the sphere acquires a net **positive charge**, because earthing drains away the electrons that the negative rod had pushed to the far side.

COMMON MISTAKE

A common slip is to think the spheres keep the rod's sign. By induction they get the opposite sign of the nearest part of the rod, so a positive rod leaves the near sphere negative.

Final answer. Sphere *A* becomes negative and sphere *B* positive; in the OR case the earthed sphere is left positively charged.

← *Back to Q54*

Q55

Two identical conducting balls A and B have charges −Q and +3Q respectively. They are brought in contact with each other and then separated by a distance d apart. Find the nature of the Coulomb force between them.

OR

A metallic spherical shell has an inner radius R_1 *and outer radius* R_2*. A charge Q is placed at the centre of the shell. What will be the surface charge density on the (i) inner surface, and (ii) outer surface of the shell?*

[2019 • Set 55-4-1]

SOLUTION

When two identical conducting balls touch, the total charge is shared equally between them.

Total charge = $(-Q) + (+3Q) = +2Q$.

Charge on each ball after contact = $\frac{+2Q}{2} = +Q$.

After they are separated to a distance d, both balls carry the same positive charge $+Q$, so the Coulomb force between them is

$$F = \frac{1}{4\pi\varepsilon_0}\,\frac{(Q)(Q)}{d^2} = \frac{Q^2}{4\pi\varepsilon_0\, d^2}.$$

Since both charges are positive (like charges), the force is **repulsive**.

OR (spherical shell with central charge Q): a charge $+Q$ at the centre induces $-Q$ on the inner surface (radius R_1) and, by conservation of charge on the neutral shell, $+Q$ on the outer surface (radius R_2).

(i) Inner surface charge density:

$$\sigma_{\text{inner}} = \frac{-Q}{4\pi\, R_1^{\,2}}.$$

(ii) Outer surface charge density:

$\sigma_{outer} = \dfrac{+Q}{4\pi R_2^{\,2}}$.

Final answer. The force is $F = \dfrac{Q^2}{4\pi\varepsilon_0 d^2}$ and repulsive; in the OR case $\sigma_{inner} = \dfrac{-Q}{4\pi R_1^{\,2}}$ and $\sigma_{outer} = \dfrac{+Q}{4\pi R_2^{\,2}}$.

← *Back to Q55*

Q56

Why is the direction of the electric field due to a charged conducting sphere at any point perpendicular to its surface?

[2019 • Set 55-5-2]

SOLUTION

In electrostatic equilibrium, the free electrons in a charged conductor have stopped moving. This means there can be no force driving them along the surface, so the field can have no component tangent to (parallel to) the surface.

If there were a tangential component of $\vec{E}$, it would exert a force on the free electrons and make them move along the surface. They would keep redistributing until that tangential component became zero, which is exactly the equilibrium condition.

The surface of the charged conductor is therefore an equipotential surface, with the same potential at every point on it.

Because the electric field is the negative gradient of potential, $\vec{E} = -\dfrac{dV}{dr}$, and there is no change of V along the surface, the field can only point along the direction of fastest change of V, which is normal to the surface.

Hence the electric field at any point on the surface of a charged conducting sphere in electrostatic equilibrium is **perpendicular to its surface**.

Final answer. Because the surface is an equipotential and $\vec{E}$ is always perpendicular to an equipotential, the field is normal to the conducting sphere's surface.

← Back to Q56

Q57

Why can the interior of a conductor have no excess charge in the static situation?

[2019 • Set 55-5-3]

SOLUTION

In the static (electrostatic) situation the electric field inside the body of a conductor is zero. If it were not zero, the free electrons would feel a force and keep moving, which would contradict the assumed static equilibrium.

Now apply Gauss's law to any closed Gaussian surface drawn entirely inside the conductor. Since $\vec{E} = 0$ at every point on this surface, the flux through it is zero.

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}.$$

$$0 = \frac{q_{enc}}{\varepsilon_0} \Rightarrow q_{enc} = 0.$$

So the net charge enclosed by any such surface is zero. As this is true for every closed surface that can be drawn within the conductor, no excess charge can sit anywhere in the interior.

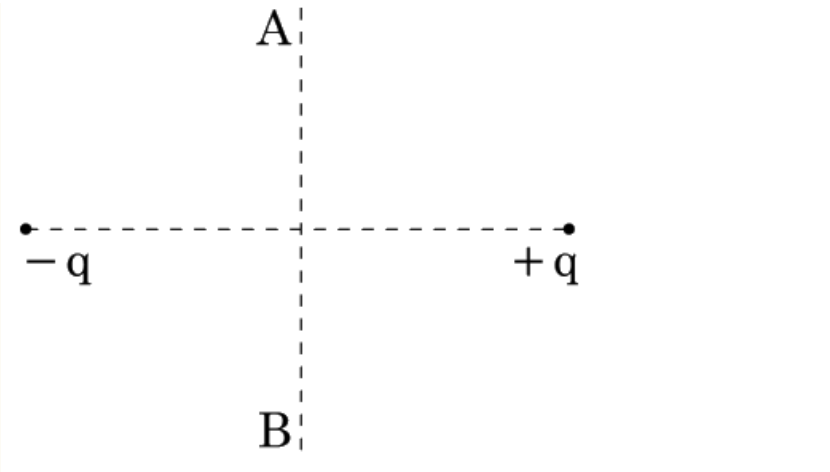
Therefore the **whole of the excess charge must reside on the outer surface of the conductor**, and the interior stays charge free.

Final answer. Since $\vec{E}=0$ inside, Gauss's law gives $q_{enc}=0$ for every interior surface, so all excess charge lies on the outer surface.

← Back to Q57

Q58

A charge q is moved from a point A above a dipole of dipole moment p to a point B below the dipole in equatorial plane without acceleration. Find the work done in the process.



[2016]

SOLUTION

The potential at any point on the equatorial plane of a dipole, at distance r, is found from the dipole potential formula with the angle $\theta = 90°$ between $\vec{p}$ and the position vector.

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\theta}{r^2}.$$

$$V_{eq} = \frac{1}{4\pi\varepsilon_0}\, p\cos 90° r^2 = 0.$$

Both points A and B lie on the equatorial plane, so each is at zero potential and they are at the same potential: $V_A = V_B = 0$.

The work done in moving a charge q slowly (without acceleration) between two points is

$$W = q\,(V_B - V_A).$$

$$W = q\,(0 - 0) = 0.$$

Hence the **work done in carrying the charge from A to B is zero**, because every point on the equatorial plane is at the same (zero) potential.

CONCEPT

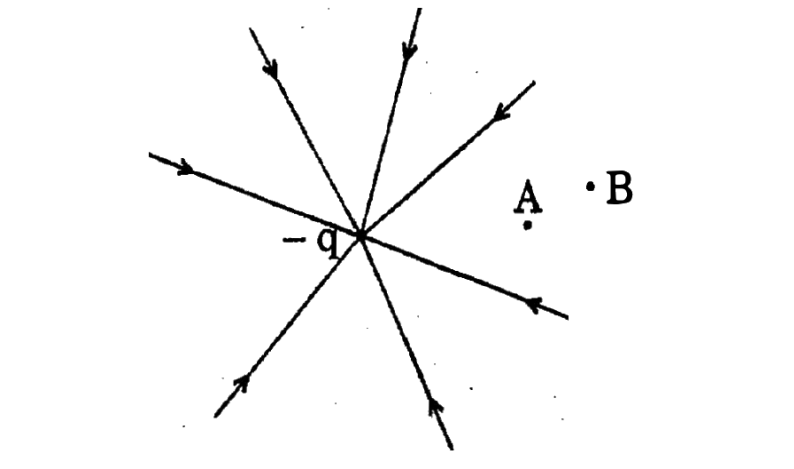
Equatorial points of a dipole are all at $V=0$, so moving a charge between any two of them costs no work, no matter how far apart they are.

Final answer. The work done is zero, since A and B lie on the equatorial plane where $V=0$.

← Back to Q58

Q59

The field lines of a negative point charge are as shown in the figure. Does the kinetic energy of a small negative charge increase or decrease in going from B to A?



[2015]

SOLUTION

The field lines shown belong to a negative point charge $-Q$, so they point radially inward, towards $-Q$. The lines crowd together (become denser) as they approach $-Q$, which means the field is stronger nearer the charge.

A small negative test charge placed in this field feels a force opposite to the field direction, that is directed radially outward, away from $-Q$. This matches the fact that two negative charges repel each other.

From the figure, point B is farther from the central charge and point A is nearer to it (the lines converge towards $-Q$). So going from B to A means moving towards $-Q$.

Moving towards $-Q$ is moving against the repulsive force acting on the small negative charge. The electric force therefore does negative work on it.

By the work-energy theorem, the change in kinetic energy equals the work done by the force, so a negative work means the kinetic energy falls.

Hence the **kinetic energy of the small negative charge decreases in going from *B* to *A*.**

Final answer. The kinetic energy decreases, because the charge moves against the repulsive force from B to A.

← Back to Q59

Q60

What is the electric flux through a cube of side 1 cm which encloses an electric dipole?

[2015]

SOLUTION

By Gauss's law, the net electric flux through any closed surface depends only on the total charge enclosed, not on its shape, size, or what is inside.

$$\Phi_E = \frac{q_{enc}}{\varepsilon_0}.$$

An electric dipole is made of two equal and opposite charges, $+q$ and $-q$. The net charge enclosed by the cube is

$$q_{enc} = (+q) + (-q) = 0.$$

Substituting into Gauss's law,

$$\Phi_E = \frac{0}{\varepsilon_0} = 0.$$

The side of the cube (1 cm) does not enter the result, since flux depends only on the enclosed charge.

Hence the **net electric flux through the cube enclosing the dipole is zero.**

Final answer. The net electric flux through the cube is zero, since the enclosed dipole charge is $+q-q=0$.

← Back to Q60

Q61

Why do the electric field lines never cross each other ?

[2014]

SOLUTION

The electric field $\vec{E}$ at any point in space has a single, definite direction, which is the direction of the force experienced by a unit positive test charge placed at that point.

By the definition of an electric field line, the tangent drawn to a field line at any point gives the direction of $\vec{E}$ at that point.

If two electric field lines were to cross each other at a point, then at that point of intersection we could draw two different tangents, one to each line.

This would mean the electric field $\vec{E}$ has two different directions at the same point, which is physically impossible.

Hence, **electric field lines never cross each other.**

Final answer. Two tangents at a crossing point would give two field directions at one point, which is impossible, so field lines never cross.

← *Back to Q61*

Q62

A charge q is placed at the centre of a cube of side l. What is the electric flux passing through each face of the cube?

[2012]

SOLUTION

By Gauss's law, the total electric flux through any closed surface enclosing a charge q is independent of the shape and size of the surface and is given by:

$$\Phi_{\text{total}} = \frac{q}{\varepsilon_0}$$

Here the charge q is placed exactly at the centre of the cube, so the six faces of the cube are perfectly symmetrical with respect to the charge.

By symmetry, the total flux is shared equally among all the six faces:

$$\Phi_{\text{one face}} = \frac{\Phi_{\text{total}}}{6}$$

$$\Phi_{\text{one face}} = \frac{1}{6} \times \frac{q}{\varepsilon_0}$$

$$\Phi_{\text{one face}} = \frac{q}{6\,\varepsilon_0}$$

Hence the electric flux through each face of the cube is $\frac{q}{6\,\varepsilon_0}$.

TIP

Note that the side l of the cube does not appear in the answer. Gauss's law flux depends only on the enclosed charge, not on the size of the surface.

Final answer. The flux through each face is $\boldsymbol{\frac{q}{6\,\varepsilon_0}}$.

← *Back to Q62*

Q63

Why should electrostatic field be zero inside a conductor?

[2012]

SOLUTION

A conductor contains a very large number of free electrons that are able to move about within it.

Suppose a non-zero electric field $\vec{E}$ existed inside the conductor. Then each free electron would experience a force $\vec{F} = -e\vec{E}$ and would begin to drift.

This drift of charges would shift charge onto the surface of the conductor, and the redistributed surface charges would set up their own field opposing the applied field.

The charges keep rearranging until the field due to the induced surface charges exactly cancels the applied field at every point inside the conductor.

In the electrostatic (static) situation there is no net flow of charge, so this rearranging must have already stopped. This is possible only if the net electric field everywhere inside the conductor is zero.

Hence, in electrostatic equilibrium, $\vec{E}_{\text{inside}} = \vec{0}$ inside a conductor.

Final answer. Free electrons keep moving until they cancel any internal field, so in the static state the field inside a conductor must be zero.

← *Back to Q63*

Q64

Define dipole moment of an electric dipole. Is it a scalar or a vector?

[2012]

SOLUTION

The electric dipole moment of an electric dipole is defined as the product of the magnitude of either of the two equal and opposite charges and the distance between them.

$$\vec{p} = q\,(2\vec{a})$$

where q is the magnitude of either charge and $2\vec{a}$ is the displacement vector drawn from the negative charge $-q$ to the positive charge $+q$.

Its magnitude is $p = q \times 2a$, and its direction is from the negative charge to the positive charge.

Since the dipole moment has both magnitude and a definite direction, it is a **vector quantity**.

Its SI unit is coulomb metre, C.

Final answer. Dipole moment $\vec{p} = q(2\vec{a})$, directed from $-q$ to $+q$; it is a vector quantity.

← Back to Q64

Q65

Why must electrostatic field be normal to the surface at every point of a charged conductor?

[2012]

SOLUTION

The surface of a charged conductor in electrostatic equilibrium contains free electrons that are at rest.

Suppose the electric field $\vec{E}$ at the surface had a tangential component E_t (a component along the surface).

Then the free electrons at the surface would experience a tangential force and would start moving along the surface.

This continuous motion of charges contradicts the condition of electrostatic equilibrium, in which no charges are moving.

Therefore, for equilibrium, the tangential component of the field must vanish, $\mathbf{E_t = 0}$, leaving only the component normal (perpendicular) to the surface.

Hence the electrostatic field at every point of the surface of a charged conductor must be **normal to the surface**.

Final answer. Any tangential field would move the surface charges, so for equilibrium the field at the surface must be normal everywhere.

← Back to Q65

Q66

Define electric dipole moment. Write its S.I. unit.

[2011]

SOLUTION

The electric dipole moment of an electric dipole is defined as the product of the magnitude of either of the two equal and opposite charges and the distance separating them.

$$\vec{p} = q\,(2\vec{a})$$

where q is the magnitude of either charge and $2\vec{a}$ is the vector drawn from the negative charge $-q$ to the positive charge $+q$.

It is a vector quantity whose direction is from the negative charge to the positive charge.

Its SI unit is coulomb metre, C.

Final answer. Dipole moment $\vec{p}=q(2\vec{a})$; its SI unit is coulomb metre (C).

← *Back to Q66*

Q67

Two insulated charged copper spheres A and B of identical size have charges q_A and q_B respectively. A third sphere C of the same size but uncharged is brought in contact with the first and then in contact with the second and finally removed from both. What are the new charges on A and B?

[2011 • Set 55-2-1]

SOLUTION

When two identical conducting spheres are brought into contact, their total charge gets shared equally between them, because by symmetry each identical sphere acquires half of the combined charge.

The uncharged sphere C ($\mathbf{q_C = 0}$) is first brought into contact with sphere A, which has charge q_A.

On contact, A and C share their combined charge equally:

$$q_A' = q_C' = \frac{q_A + 0}{2} = \frac{q_A}{2}$$

So after the first contact, A is left with $\frac{q_A}{2}$ and C carries $\frac{q_A}{2}$.

Next, sphere C (now carrying $\frac{q_A}{2}$) is brought into contact with sphere B, which has charge q_B.

On contact, B and C share their combined charge equally:

$$q_B' = \frac{q_B + \frac{q_A}{2}}{2} = \frac{q_B}{2} + \frac{q_A}{4}$$

Finally, C is removed from both spheres.

Hence the new charges are: **on A,** $q_A^{\text{final}} = \frac{q_A}{2}$, and **on B,** $q_B^{\text{final}} = \frac{q_B}{2} + \frac{q_A}{4}$.

Final answer. New charges: $q_A^{\text{final}} = \frac{q_A}{2}$ and $q_B^{\text{final}} = \frac{q_B}{2} + \frac{q_A}{4}$.

← *Back to Q67*

Q68

In which orientation, a dipole placed in a uniform electric field is in (i) stable, (ii) unstable equilibrium?

[2010]

SOLUTION

When a dipole of moment $\vec{p}$ is placed in a uniform electric field $\vec{E}$, its potential energy is $U = -pE\theta$, where θ is the angle between $\vec{p}$ and $\vec{E}$.

For stable equilibrium the dipole moment $\vec{p}$ is parallel to the field $\vec{E}$, i.e. $\theta = 0°$.

Here the potential energy is minimum, $U = -pE$, and any small angular displacement gives rise to a restoring torque that brings the dipole back to this position.

For unstable equilibrium the dipole moment $\vec{p}$ is antiparallel to the field $\vec{E}$, i.e. $\theta = 180°$.

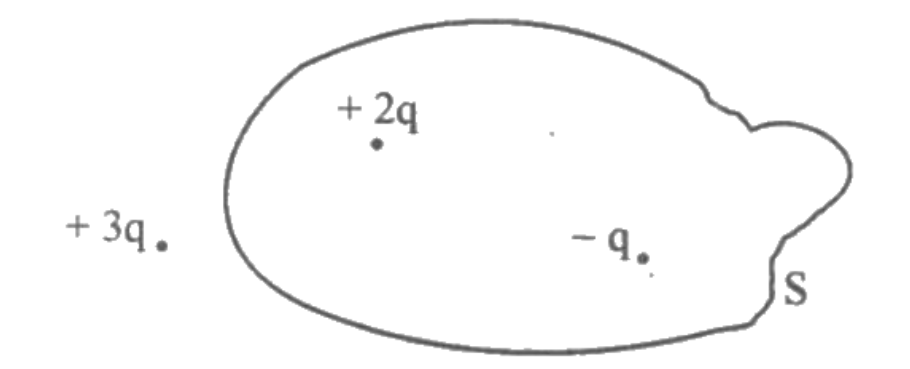
Here the potential energy is maximum, $U = +pE$, and any small disturbance produces a torque that turns the dipole further away, towards $\theta = 0°$.

Final answer. Stable equilibrium: $\vec{p}$ parallel to $\vec{E}$ ($\theta=0°$); unstable equilibrium: $\vec{p}$ antiparallel to $\vec{E}$ ($\theta=180°$).

← *Back to Q68*

Q69

Figure shows three point charges, +2q, −q and +3q. Two charges +2q and −q are enclosed within a surface 'S'. What is the electric flux due to this configuration through the surface 'S'?



[2010]

SOLUTION

By Gauss's law, the electric flux through a closed surface depends only on the net charge enclosed within it, and charges lying outside the surface contribute zero net flux.

$$\Phi_E = \frac{q_{enc}}{\varepsilon_0}$$

The surface S encloses only the charges +2q and $-q$. The charge +3q lies outside S, so it does not contribute to the net flux through S.

The net charge enclosed by S is:

$$q_{enc} = (+2q) + (-q) = +q$$

Substituting in Gauss's law:

$$\Phi_E = \frac{q}{\varepsilon_0}$$

Hence the net electric flux through the surface S is $\frac{q}{\varepsilon_0}$.

COMMON MISTAKE

The +3q outside S adds nothing to the net flux through S, though it does affect the field at points on S. Flux and field are not the same thing.

Final answer. The net flux through surface S is $\frac{q}{\varepsilon_0}$.

← Back to Q69

Q70

An electric dipole of dipole moment 20×10^{-6} C is enclosed by a closed surface. What is the net flux coming out of the surface?

[2005]

SOLUTION

By Gauss's law, the net electric flux through a closed surface depends only on the net charge enclosed by it:

$$\Phi_E = \frac{q_{enc}}{\varepsilon_0}$$

An electric dipole consists of two equal and opposite charges $+q$ and $-q$.

When the entire dipole is enclosed by the closed surface, the net charge enclosed is:

$$q_{enc} = (+q) + (-q) = 0$$

Substituting in Gauss's law:

$$\Phi_E = \frac{0}{\varepsilon_0} = 0$$

The given value of the dipole moment (20×10^{-6} C) is irrelevant, since the net enclosed charge is zero.

Hence the net flux coming out of the closed surface is **zero**.

Final answer. The net flux out of the surface is zero, since the dipole's net enclosed charge is zero.

← Back to Q70

Q71

An electric dipole consists of two point charges +1 μ *C and* −1 μ *C, held 10 cm apart. It is subjected to a uniform electric field of 100 N/C. Calculate the amount of work done in turning the dipole from its position of stable equilibrium to the position of unstable equilibrium, in the field.*

[2026 • Set 55-4-1]

SOLUTION

The potential energy of a dipole of moment p making an angle θ with a uniform field E is $U = -pE\theta$, and the work done in turning it equals the change in this potential energy.

First find the dipole moment. With charge $q = 1\times 10^{-6}\ C$ and separation $2a = 10\ cm = 0.10\ m$,

$$p = q\ (2a)$$

$$p = (1\times 10^{-6}\ C)(0.10\ m)$$

$$p = 1\times 10^{-7}\ C\ m$$

In stable equilibrium the dipole is parallel to the field ($\theta_i = 0°$), so its initial energy is $U_i = -pE\cos 0° = -pE$. In unstable equilibrium the dipole is antiparallel to the field ($\theta_f = 180°$), so its final energy is $U_f = -pE\cos 180° = +pE$.

The work done is the change in potential energy,

$$W = U_f - U_i = (+pE) - (-pE) = 2pE$$

Substituting the values,

$$W = 2\ (1\times 10^{-7}\ C\ m)(100\ N\ C^{-1})$$

$$W = 2\times 10^{-5}\ J$$

Hence the work done in turning the dipole from stable to unstable equilibrium is $\mathbf{2\times 10^{-5}\,J}$, i.e. 20 μJ.

Final answer. Work done $W = 2pE = 2\times 10^{-5}\,J$ **(20 μJ).**

← *Back to Q71*

Q72

An electric dipole consists of point charges −1.0 pC *and* +1.0 pC *located at* (0,0) *and* (3 mm, 4 mm) *respectively in x–y plane. An electric field* $\vec{E} = \left(\frac{2\times 10^5\ V}{m}\right)\hat{i}$ *is switched on in the region. Find the torque* $\vec{\tau}$ *acting on the dipole.*

[2024 • Set 55-1-1]

SOLUTION

The dipole moment is $\vec{p} = q\,\vec{d}$, where $\vec{d}$ is the position vector drawn from the negative charge to the positive charge, and the torque in a uniform field is $\vec{\tau} = \vec{p}\times\vec{E}$.

The vector from $-q$ at (0,0) to $+q$ at (3 mm, 4 mm) is

$$\vec{d} = (3\hat{i}+4\hat{j})\times 10^{-3}\ m$$

So the dipole moment is

$$\vec{p} = q\,\vec{d} = (1.0\times 10^{-12}\ C)(3\hat{i}+4\hat{j})\times 10^{-3}\ m$$

$$\vec{p} = (3\hat{i}+4\hat{j})\times 10^{-15}\ C\ m$$

Now compute the torque, using $\hat{i}\times\hat{i}=\vec{0}$ and $\hat{j}\times\hat{i}=-\hat{k}$,

$$\vec{\tau} = \vec{p}\times\vec{E} = [(3\hat{i}+4\hat{j})\times 10^{-15}]\times(2\times 10^5\,\hat{i})$$

$$\vec{\tau} = (3)(2\times 10^{-10})(\hat{i}\times\hat{i}) + (4)(2\times 10^{-10})(\hat{j}\times\hat{i})$$

$$\vec{\tau} = 0 + 8\times 10^{-10}(-\hat{k})$$

$$\vec{\tau} = -8\times 10^{-10}\,\hat{k}\ N\ m$$

Hence the torque on the dipole is $\vec{\tau} = -8\times 10^{-10}\,\hat{k}\ N\,m$, of magnitude $8\times 10^{-10}\ N\,m$ directed along the $-\hat{k}$ direction (into the x-y plane).

COMMON MISTAKE

The vector $\vec{d}$ must point from the negative charge to the positive charge. Reversing it flips the sign of the answer and loses the torque-direction mark.

Final answer. $\vec{\tau} = -8\times 10^{-10}\,\hat{k}\ N\,m$ (magnitude $8\times 10^{-10}\ N\,m$, along $-\hat{k}$).

← Back to Q72

Q73

Four point charges of 1 μC, −2 μC, 1 μC *and* −2 μC *are placed at the corners A, B, C and D respectively, of a square of side* 30 cm. *Find the net force acting on a charge of* 4 μC *placed at the centre of the square.*

[2024 • Set 55-5-1]

SOLUTION

By the principle of superposition, the net force on the central charge is the vector sum of the four individual Coulomb forces it experiences from the corner charges.

Draw the square ABCD with $A = +1\ \mu C$, $B = -2\ \mu C$, $C = +1\ \mu C$ and $D = -2\ \mu C$, and the charge +4 μC at the centre O. The diagonals AC and BD both pass through O, and the centre is equidistant from all four corners,

$$OA = OB = OC = OD = \frac{a\sqrt{2}}{2},\ a = 30\ \text{cm}$$

Consider the diagonal AC. The equal charges +1 μC at A and C are at equal distances from O, so they exert equal-magnitude forces on the central charge. Both are repulsive (like charges), so the force due to A points from A towards C, while the force due to C points from C towards A. These two forces are equal and exactly opposite, so along AC,

$$\vec{F}_A + \vec{F}_C = \vec{0}$$

Consider the diagonal BD. The equal charges −2 μC at B and D are again equidistant from O, so they exert equal-magnitude forces. Both are attractive, so the force due to B points from O towards B and the force due to D points from O towards D, which are equal and exactly opposite. Hence along BD,

$$\vec{F}_B + \vec{F}_D = \vec{0}$$

Adding all four contributions,

$$\vec{F}_{\text{net}} = (\vec{F}_A + \vec{F}_C) + (\vec{F}_B + \vec{F}_D) = \vec{0}$$

Hence the **net force on the 4 μC charge at the centre is zero**.

Final answer. The net force on the **4 μC** charge at the centre is zero.

← *Back to Q73*

Q74

Three point charges, 1 μC *each, are kept at the vertices of an equilateral triangle of side* 10 cm. *Find the net electric field at the centroid of triangle.*

[2024 • Set 55-5-1]

SOLUTION

By the principle of superposition, the resultant field at the centroid is the vector sum of the fields due to the three charges.

Draw the equilateral triangle with the three equal charges q **= 1 μC** at its vertices and the centroid G at the centre. The centroid is equidistant from all three vertices, so each charge is the same distance r from G,

$$r = \frac{a}{\sqrt{3}},\ a = 10\ \text{cm}$$

Since the three charges are equal and equidistant from G, each produces a field of the same magnitude at the centroid,

$$E_1 = E_2 = E_3 = \frac{1}{4\pi\varepsilon_0}\ \frac{q}{r^2}$$

Each field points radially away from its source charge (the charges are positive), so the three field vectors lie along the lines joining the vertices to G, and these three directions are separated by 120° from one another.

Three equal-magnitude vectors placed symmetrically at 120° to each other always add up to zero,

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 = \vec{0}$$

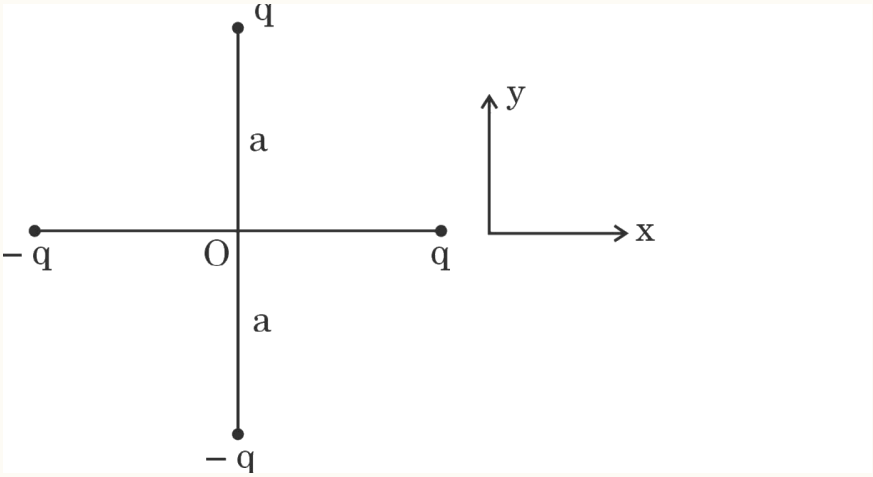
Hence the **net electric field at the centroid of the equilateral triangle is zero**.

Final answer. The net electric field at the centroid is zero.

← *Back to Q74*

Q75

Two identical dipoles are arranged in x-y plane as shown in the figure. Find the magnitude and the direction of net electric field at the origin O.



[2023 • Set 55-4-1]

SOLUTION

The net field at the origin is the vector sum of the fields of the two dipoles. For a short dipole of moment p at distance r, the axial field is $\mathbf{E_{ax}} = \dfrac{\mathbf{2kp}}{\mathbf{r^3}}$ (along $\vec{p}$) and the equatorial field is $\mathbf{E_{eq}} = \dfrac{\mathbf{kp}}{\mathbf{r^3}}$ (opposite to $\vec{p}$), where $\boldsymbol{k} = \dfrac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}}$.

From the figure the two identical dipoles, each of moment p with axis along $\hat{i}$, are placed symmetrically about O: dipole 1 lies on the x-axis at distance r so that its axial line passes through O, and dipole 2 lies on the y-axis at distance r so that its equatorial line passes through O.

Field at O due to dipole 1 (axial point, directed along $\hat{i}$),

$$\vec{E}_1 = \frac{2kp}{r^3}\,\hat{i}$$

Field at O due to dipole 2 (equatorial point, directed opposite to $\vec{p}$, i.e. along $-\hat{i}$),

$$\vec{E}_2 = -\frac{kp}{r^3}\,\hat{i}$$

Adding the two,

$$\vec{E}_O = \vec{E}_1 + \vec{E}_2 = \frac{2kp}{r^3}\,\hat{i} - \frac{kp}{r^3}\,\hat{i} = \frac{kp}{r^3}\,\hat{i}$$

The magnitude is therefore

$$E_O = \frac{kp}{r^3} = \frac{p}{4\pi\varepsilon_0\, r^3}$$

Hence the **net field at O has magnitude $\dfrac{p}{4\pi\varepsilon_0\, r^3}$ and is directed along the positive x-axis** (along the common dipole axis, $+\hat{i}$).

Final answer. $E_O = \dfrac{p}{4\pi\varepsilon_0\, r^3}$, directed along $+\hat{i}$ (positive x-axis).

← *Back to Q75*

Q76

(c) An electric dipole of dipole moment of 6×10^{-7} C-m is kept in a uniform electric field of 10^4 N/C such that the dipole moment and the electric field are parallel. Calculate the potential energy of the dipole.

OR

(c) An electric dipole of dipole moment $\vec{p}$ is initially kept in a uniform electric field $\vec{E}$

such that $\vec{p}$ is perpendicular to $\vec{E}$. Find the amount of work done in rotating the dipole to a position at which $\vec{p}$ becomes antiparallel to $\vec{E}$.

[2023 • Set 55-4-1]

SOLUTION

The potential energy of a dipole in a uniform field is $U = -\vec{p}\cdot\vec{E} = -pE\theta$, and the work done in rotating it equals the change in this potential energy.

For the first part, the dipole moment and field are parallel, so $\theta = 0°$. With $p = 6\times 10^{-7}\ C\,m$ and $E = 10^4\ N\,C^{-1}$,

$$U = -pE\theta$$

$$U = -(6\times 10^{-7}\ C\,m)(10^4\ N\,C^{-1})\cos 0°$$

$$U = -6\times 10^{-3}\ J$$

The negative sign shows that the parallel orientation is the position of minimum (stable) potential energy. So $U = -6\times 10^{-3}\ J = -6$ mJ.

For the OR part, the work done in rotating the dipole is

$$W = U_f - U_i = (-pE\theta_f) - (-pE\theta_i)$$

The dipole starts perpendicular to the field, $\theta_i = 90°$, so $U_i = -pE\cos 90° = 0$.

It ends antiparallel to the field, $\theta_f = 180°$, so $U_f = -pE\cos 180° = +pE$.

Therefore

$$W = pE - 0 = pE$$

Hence the potential energy in the parallel case is $-6\times 10^{-3}\ J$, and the work done in the OR case is $W = pE$.

Final answer. $U = -6\times 10^{-3}\ J$ (parallel case); OR work done $W = pE$.

← *Back to Q76*

Q77

Use Gauss's law to obtain the expression for electric field due to a uniformly charged infinite plane thin sheet.

[2023 • Set 55-4-2]

SOLUTION

Gauss's law states that the total electric flux through any closed surface equals $\frac{1}{\varepsilon_0}$ times the net charge enclosed, $\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}$.

Consider an infinite plane sheet carrying a uniform surface charge density σ. By symmetry, the field $\vec{E}$ at any point is perpendicular to the sheet, has the same magnitude on the two sides at equal perpendicular distances, and points away from the sheet for $\sigma > 0$.

For the Gaussian surface, draw a right cylindrical pillbox of cross-sectional area A that pierces the sheet symmetrically, with its two flat circular faces parallel to the sheet, one on each side, and its curved surface perpendicular to the sheet.

The curved side contributes no flux, because there $\vec{E}$ is perpendicular to the area vector $d\vec{S}$ ($\vec{E} \cdot d\vec{S} = 0$). Each flat face contributes flux EA, since $\vec{E}$ is parallel to $d\vec{S}$ on the caps. The total flux through the pillbox is

$$\phi = EA + EA = 2EA$$

The charge enclosed is only the patch of the sheet inside the pillbox,

$$q_{enc} = \sigma A$$

Applying Gauss's law,

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}$$

In vector form, with $\hat{n}$ the unit normal pointing away from the sheet,

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$$

Hence the field due to an infinite charged plane sheet is $E = \frac{\sigma}{2\varepsilon_0}$, which is uniform and independent of the distance from the sheet.

Final answer. $E = \frac{\sigma}{2\varepsilon_0}$, directed normally away from the sheet.

← *Back to Q77*

Q78

Use Gauss's law to obtain the expression for electric field due to a thin infinitely long straight uniformly charged wire.

[2023 • Set 55-4-3]

SOLUTION

Gauss's law states that the total electric flux through any closed surface equals $\frac{1}{\varepsilon_0}$ times the net charge enclosed, $\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}$.

Consider a thin, infinitely long straight wire with uniform linear charge density λ. By symmetry, the field $\vec{E}$ at any point is radial (perpendicular to the wire) and its magnitude depends only on the perpendicular distance r from the wire.

For the Gaussian surface, draw a coaxial right circular cylinder of radius r and length l, with the wire as its axis. Its curved surface and two flat end caps form the closed surface.

The two flat end caps contribute no flux, because there $\vec{E}$ is parallel to the caps and perpendicular to their area vectors ($\vec{E} \cdot d\vec{S} = 0$). On the curved surface $\vec{E}$ is parallel to $d\vec{S}$ and has constant magnitude, so the flux through it is

$$\phi = E\,(2\pi\, r\, l)$$

The charge enclosed by the cylinder is the length l of the wire inside it,

$$q_{enc} = \lambda\, l$$

Applying Gauss's law,

$$E\,(2\pi\, r\, l) = \frac{\lambda\, l}{\varepsilon_0}$$

$$E = \frac{\lambda}{2\pi\varepsilon_0\, r}$$

Hence the field due to an infinite line of charge is $\boldsymbol{E = \frac{\lambda}{2\pi\varepsilon_0\, r}}$, directed radially outward for $\boldsymbol{\lambda > 0}$ and radially inward for $\boldsymbol{\lambda < 0}$, and falling off as $\frac{1}{r}$.

Final answer. $\boldsymbol{E = \frac{\lambda}{2\pi\varepsilon_0\, r}}$, directed radially from the wire.

← *Back to Q78*

Q79

Depict the orientation of an electric dipole in (a) stable and (b) unstable equilibrium in an external uniform electric field. Write the potential energy of the dipole in each case.

[2023 • Set 55-5-1]

SOLUTION

When a dipole of moment $\vec{p}$ is placed in a uniform external field $\vec{E}$, it experiences a torque $\boldsymbol{\vec{\tau} = \vec{p} \times \vec{E}}$ that tends to align $\vec{p}$ with $\vec{E}$. Equilibrium occurs when $\boldsymbol{\vec{\tau} = \vec{0}}$, that is when $\vec{p}$ is along the line of $\vec{E}$, and the potential energy is $\boldsymbol{U = -\vec{p} \cdot \vec{E} = -pE\theta}$.

(a) Stable equilibrium: draw the field lines of $\vec{E}$ pointing to the right and the dipole $\vec{p}$ pointing in the same direction, so the $-q$ and $+q$ charges line up along the field with $+q$ on the downstream side ($\boldsymbol{\theta = 0°}$). If the dipole is given a small angular displacement, the torque acts to bring it back to this orientation, so it is stable. Its potential energy is

$$U_{\text{stable}} = -pE\cos 0° = -pE$$

This is the minimum value of the potential energy.

(b) Unstable equilibrium: draw the same field lines pointing to the right but the dipole $\vec{p}$ pointing in the opposite direction, antiparallel to $\vec{E}$ ($\boldsymbol{\theta = 180°}$). Any

small angular displacement produces a torque that drives the dipole further away, towards $\theta = 0°$, so it is unstable. Its potential energy is

$$U_{\text{unstable}} = -pE\cos 180° = +pE$$

This is the maximum value of the potential energy.

Hence in **stable equilibrium $\vec{p} \parallel \vec{E}$ with $U = -pE$**, and in **unstable equilibrium $\vec{p}$ is antiparallel to $\vec{E}$ with $U = +pE$**.

Final answer. Stable: $\vec{p} \parallel \vec{E}$, $U = -pE$. Unstable: $\vec{p}$ antiparallel to $\vec{E}$, $U = +pE$.

← *Back to Q79*

Q80

An electric dipole of dipole moment ($\vec{p}$) is kept in a uniform electric field $\vec{E}$. Show graphically the variation of torque acting on the dipole (τ) with its orientation (θ) in the field. Find the orientation in which torque is (i) zero and (ii) maximum.

[2023 • Set 55-5-2]

SOLUTION

The torque on a dipole of moment p placed at angle θ to a uniform field E is $\tau = pE\theta$, so the torque varies sinusoidally with the orientation θ.

For the graph, plot τ on the vertical axis against θ on the horizontal axis. The curve is a sine curve: it starts at $\tau = 0$ at $\theta = 0°$, rises smoothly to its peak value $\tau_{max} = pE$ at $\theta = 90°$, then falls back to $\tau = 0$ at $\theta = 180°$. Mark the peak height as pE and the zero crossings at $\theta = 0°$ and $\theta = 180°$.

(i) The torque is zero when $\theta = 0$,

$$\theta = 0° \text{ or } \theta = 180°$$

At $\theta = 0°$ the dipole is parallel to the field (stable equilibrium) and at $\theta = 180°$ it is antiparallel to the field (unstable equilibrium).

(ii) The torque is maximum when $\theta = 1$,

$$\theta = 90°$$

at which the dipole is perpendicular to the field and

$$\tau_{max} = pE$$

Hence the torque is **zero at θ = 0° and 180°**, and **maximum, equal to pE, at θ = 90°**.

Final answer. Torque is zero at **θ = 0°** and **180°**; maximum **$\tau_{max} = pE$** at **θ = 90°**.

← *Back to Q80*

Q81

Derive the expression for the torque acting on an electric dipole, when it is held in a uniform electric field. Identify the orientation of the dipole in the electric field, in which it attains a stable equilibrium.

OR

Obtain the expression for the energy stored in a capacitor connected across a dc battery. Hence define energy density of the capacitor.

[2020 • Set 55-5-1]

SOLUTION

Consider an electric dipole made of charges $+q$ and $-q$ separated by a distance 2a, so its dipole moment is $\vec{p} = q(2a)$, directed from $-q$ to $+q$. Let the dipole be placed in a uniform electric field $\vec{E}$ such that $\vec{p}$ makes an angle θ with $\vec{E}$.

The charge $+q$ experiences a force $+q\vec{E}$ and the charge $-q$ experiences a force $-q\vec{E}$.

These forces are equal in magnitude and opposite in direction, so the net force on the dipole is zero. But since they act at different points, they form a couple which produces a torque.

Magnitude of torque = (one force) × (perpendicular distance between the forces).

$$\tau = qE \times (2a\theta)$$

$$\tau = (q \cdot 2a)\, E\theta$$

$$\tau = pE\theta$$

In vector form $\vec{\tau} = \vec{p} \times \vec{E}$, directed perpendicular to the plane containing $\vec{p}$ and $\vec{E}$.

The dipole attains **stable equilibrium** when $\vec{p}$ is aligned parallel to $\vec{E}$, that is $\theta = 0°$. Here the torque is zero, and the potential energy $U = -pE\theta$ has its minimum value $U_{min} = -pE$, so any small angular displacement gives a restoring torque that brings the dipole back.

OR

Consider a capacitor of capacitance C connected across a dc battery. Let at an instant the charge on the capacitor be q and the potential difference across it be $V = \frac{q}{C}$.

The small work done to transfer a further charge dq from one plate to the other is

$$dW = V\,dq = \frac{q}{C}\,dq$$

The total work done in charging the capacitor from 0 to final charge Q is stored as electrostatic potential energy U.

$$U = {}_0{}^{Q}\frac{q}{C}\,dq$$

$$U = \frac{1}{C}\left[\frac{q^2}{2}\right]_0^{Q}$$

$$U = \frac{Q^2}{2C}$$

Using $Q = CV$, this can be written as

$$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$$

Energy density is defined as the energy stored per unit volume of the field region between the plates,

$$u = \frac{U}{\text{Volume}} = \frac{1}{2}\varepsilon_0 E^2$$

which is the electrostatic energy stored per unit volume of the electric field.

Final answer. Torque: $\vec{\tau}=\vec{p}\times\vec{E}$ ($\tau=pE\theta$), stable equilibrium at $\theta=0°$; OR energy stored $U=\frac{1}{2}CV^2$ with energy density $u=\frac{1}{2}\varepsilon_0 E^2$.

← Back to Q81

Q82

Five point charges, each of charge +q are placed on five vertices of a regular hexagon of side l. Find the magnitude of the resultant force on a charge −q placed at the centre of the hexagon.

OR

A simple pendulum consists of a small sphere of mass m suspended by a thread of length l. The sphere carries a positive charge q. The pendulum is placed in a uniform electric field of strength E directed vertically downwards. Find the period of oscillation of the pendulum due to the electrostatic force acting on the sphere, neglecting the effect of the gravitational force.

[2019 • Set 55-3-1]

SOLUTION

Use the principle of superposition together with the symmetry of a regular hexagon.

Imagine for a moment that all six vertices of the regular hexagon carried a charge $+q$. By the symmetry of the hexagon, the six attractive forces on the central charge $-q$ are equal in magnitude and point along the six directions to the vertices, 60° apart. They cancel exactly in pairs, so the net force on the central charge would be zero.

The given arrangement has charges on only five vertices, the sixth vertex being empty. This is the same as the full hexagon with the $+q$ at one vertex removed, i.e. an extra $-q$ added at that vertex to cancel it.

So the resultant force on the central charge equals the force due to the single missing charge alone (with opposite sign of cancellation), whose magnitude is found using Coulomb's law. The distance from the centre of a regular hexagon to any vertex equals the side l.

$$F = \frac{1}{4\pi\varepsilon_0}\,\frac{q\cdot q}{l^2}$$

$$F = \frac{1}{4\pi\varepsilon_0}\,\frac{q^2}{l^2}$$

This force is directed along the line joining the centre to the missing (empty) vertex, pointing away from it.

OR

The pendulum bob of mass m carries charge $+q$ and is placed in a uniform field E directed vertically downwards. The gravitational force is to be neglected.

The only force acting on the bob is the electrostatic force,

$\boldsymbol{F = qE}$ (vertically downward, along the field)

This force plays exactly the role that gravity mg plays in an ordinary pendulum, so the effective acceleration is

$$g_{\text{eff}} = \frac{F}{m} = \frac{qE}{m}$$

For small oscillations the time period of a simple pendulum is

$$T = 2\pi\sqrt{\frac{l}{g_{\text{eff}}}}$$

Substituting $\mathbf{g_{eff} = \frac{qE}{m}}$,

$$T = 2\pi\sqrt{\frac{m\,l}{q\,E}}$$

TIP

Remember the distance from centre to vertex of a regular hexagon equals its side length, so substitute $r=l$, not $r=2l$ or any diagonal.

Final answer. Force on central $-q$ is $\boldsymbol{F=\frac{1}{4\pi\varepsilon_0}\,\frac{q^2}{l^2}}$ towards the empty vertex; OR period $\boldsymbol{T=2\pi\sqrt{ml/(qE)}}$.

← *Back to Q82*

Q83

Apply Gauss's law to show that for a charged spherical shell, the electric field outside the shell is, as if the entire charge were concentrated at the centre.

OR

Two large parallel plane sheets have uniform charge densities +σ and −σ. Determine the electric field (i) between the sheets, and (ii) outside the sheets.

[2019 • Set 55-4-1]

SOLUTION

Consider a thin spherical shell of radius R carrying total charge Q distributed uniformly over its surface. We find the field at an external point P at distance $r > R$ from the centre.

Draw the charged shell, and around it a concentric spherical Gaussian surface of radius r ($r > R$) passing through P, with the centre common to both. By spherical symmetry the field $\vec{E}$ is radial (directed along the radius) and has the same magnitude E at every point of this Gaussian surface.

The electric flux through the Gaussian surface is

$$\Phi = \oint \vec{E} \cdot d\vec{S} = E \oint dS = E\,(4\pi r^2)$$

The whole charge Q of the shell lies inside this Gaussian surface, so by Gauss's law

$$\Phi = \frac{Q}{\varepsilon_0}$$

Equating the two expressions,

$$E\,(4\pi r^2) = \frac{Q}{\varepsilon_0}$$

$$E = \frac{1}{4\pi\varepsilon_0}\,\frac{Q}{r^2}$$

This is exactly the field of a point charge Q placed at the centre. Hence, for points outside the shell, the entire charge behaves as if it were concentrated at the centre.

OR

Two large parallel plane sheets carry uniform surface charge densities +σ and −σ. Each infinite charged sheet produces a uniform field of magnitude $\frac{\sigma}{2\varepsilon_0}$ on either side, directed away from the positive sheet and towards the negative sheet.

(i) In the region between the sheets, the field of the +σ sheet and the field of the −σ sheet point in the same direction (from the positive sheet to the negative sheet), so they add up.

$$E_{between} = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$

(ii) In the regions outside the sheets, the two fields are equal in magnitude but opposite in direction, so they cancel.

$$E_{outside} = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0$$

Hence the field is $\frac{\sigma}{\varepsilon_0}$ (directed from +σ to −σ) between the sheets and zero outside.

Final answer. Outside the shell $E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$ (point-charge field); OR between sheets $E = \frac{\sigma}{\varepsilon_0}$ and outside $E = 0$.

← *Back to Q83*

Q84

Derive an expression for the torque acting on an electric dipole of dipole moment $\vec{p}$ placed in a uniform electric field $\vec{E}$. Write the direction along which the torque acts.

OR

*Derive an expression for the electric field at a point on the axis of an electric dipole of dipole moment $\vec{p}$. Also write its expression when the distance **r** >> the length 'a' of the dipole.*

[2019 • Set 55-5-1]

SOLUTION

Consider an electric dipole consisting of charges $+q$ and $-q$ separated by distance 2a, so dipole moment $\vec{p} = q(2a)$ pointing from $-q$ to $+q$. Place it in a uniform field $\vec{E}$ with $\vec{p}$ at angle θ to $\vec{E}$.

Force on $+q$ is $+q\vec{E}$ and on $-q$ is $-q\vec{E}$. These are equal and opposite, so the net force is zero, but they constitute a couple that gives a torque.

Torque = force × perpendicular distance between the lines of action of the two forces.

$$\tau = qE\,(2a\theta)$$

$$\tau = (q\cdot 2a)E\theta$$

$$\tau = pE\theta$$

In vector form,

$$\vec{\tau} = \vec{p}\times\vec{E}$$

The torque acts along the direction perpendicular to the plane containing $\vec{p}$ and $\vec{E}$ (given by the right-hand rule for $\vec{p}\times\vec{E}$), and it tends to align $\vec{p}$ along $\vec{E}$.

OR

Take a dipole $-q$ at A and $+q$ at B, separation 2a, centre O. Let P be a point on the axis at distance r from the centre O on the side of $+q$.

Distance of P from $+q$ is $(r-a)$ and from $-q$ is $(r+a)$. Writing $k = \dfrac{1}{4\pi\varepsilon_0}$, the fields are directed along the axis.

Field due to $+q$ (away from $+q$, i.e. along $\vec{p}$): $\mathbf{E_+} = \dfrac{kq}{(r-a)^2}$

Field due to $-q$ (towards $-q$, i.e. opposite to $\vec{p}$): $\mathbf{E_-} = \dfrac{kq}{(r+a)^2}$

Net axial field along $\vec{p}$,

$$E_{ax} = \frac{kq}{(r-a)^2} - \frac{kq}{(r+a)^2}$$

$$E_{ax} = kq\,\frac{(r+a)^2-(r-a)^2}{(r^2-a^2)^2}$$

$$E_{ax} = kq\,\frac{4ar}{(r^2-a^2)^2}$$

Using $p = q(2a)$,

$$E_{ax} = \frac{1}{4\pi\varepsilon_0}\,\frac{2pr}{(r^2-a^2)^2}$$

For a short dipole, $r \gg a$, we neglect a^2 compared to r^2,

$$\vec{E}_{ax} = \frac{1}{4\pi\varepsilon_0}\,\frac{2\vec{p}}{r^3}$$

directed along $\vec{p}$.

Final answer. Torque $\vec{\tau}=\vec{p}\times\vec{E}$ (perpendicular to the $\vec{p}$–$\vec{E}$ plane); OR axial field $E=\frac{1}{4\pi\varepsilon_0}\frac{2pr}{(r^2-a^2)^2}$, which for $r \gg a$ is $\frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$.

← *Back to Q84*

Q85

Derive an expression for the work done in rotating a dipole from the angle θ_0 to θ_1 in a uniform electric field E.

[2016]

SOLUTION

When a dipole of moment p is placed in a uniform field E, it experiences a torque $\tau = pE\theta$, where θ is the angle between $\vec{p}$ and $\vec{E}$.

To rotate the dipole through a small angle $d\theta$ against this torque, the external agent must do work

$$dW = \tau\, d\theta = pE\theta\, d\theta$$

The total work done in rotating the dipole from θ_0 to θ_1 is obtained by integrating,

$$W = \int_{\theta_0}^{\theta_1} pE\theta\, d\theta$$

$$W = pE\int_{\theta_0}^{\theta_1}\theta\, d\theta$$

$$W = pE[-\theta]_{\theta_0}^{\theta_1}$$

$$W = pE(-\theta_1 + \theta_0)$$

$$W = pE\,(\theta_0 - \theta_1)$$

This is the required expression. Equivalently, it equals the change in potential energy $W = U(\theta_1) - U(\theta_0)$, where $U(\theta) = -pE\theta$.

Final answer. Work done $W = pE(\theta_0 - \theta_1)$.

← *Back to Q85*

Q86

Given a uniform electric field $\vec{E} = 5 \times 10^3\,\hat{i}\,N/C$, find the flux of this field through a square of 10 cm on a side whose plane is parallel to the y-z plane. What would be the flux through the same square if the plane makes a 30° angle with the x-axis ?

[2014]

SOLUTION

Electric flux through a flat area is $\Phi_E = \vec{E}\cdot\vec{A} = EA\alpha$, where α is the angle between the field $\vec{E}$ and the outward normal $\hat{n}$ to the surface.

Here $\vec{E} = 5\times 10^3\,\hat{i}\,N/C$, so $E = 5\times 10^3\,N/C$ directed along $\hat{i}$, and the area of the square is

$$A = (0.10)^2 = 0.01\ m^2$$

(a) The plane of the square is parallel to the *y*-*z* plane, so its normal $\hat{n}$ is along $\hat{i}$, the same direction as $\vec{E}$. Hence $\alpha = 0°$.

$$\Phi_E = EA\cos 0° = (5\times 10^3)(0.01)(1)$$

$$\Phi_E = 50\ N\,m^2C^{-1}$$

(b) Now the plane of the square makes a 30° angle with the *x*-axis. So its normal makes $90° - 30° = 60°$ with the *x*-axis, i.e. with $\vec{E}$. Hence $\alpha = 60°$.

$$\Phi_E = EA\cos 60° = (5\times 10^3)(0.01)\left(\frac{1}{2}\right)$$

$$\Phi_E = 25\ N\,m^2C^{-1}$$

Therefore the flux is $\mathbf{50\ N\,m^2C^{-1}}$ when the plane is parallel to the y-z plane, and $\mathbf{25\ N\,m^2C^{-1}}$ when the plane makes 30° with the x-axis.

COMMON MISTAKE

Flux uses the angle between $\vec{E}$ and the normal, not the plane. When the plane is tilted by 30° to the axis, the normal is tilted by 60°, so use cos 60°.

Final answer. Flux is $\mathbf{50\ N\,m^2C^{-1}}$ for the parallel plane, and $\mathbf{25\ N\,m^2C^{-1}}$ when the plane makes **30°** with the x-axis.

← *Back to Q86*

Q87

An electric dipole of length 4 cm, when placed with its axis making an angle of 60° with a uniform electric field, experiences a torque of $4\sqrt{3}$ Nm. Calculate the potential energy of the dipole, if it has charge +8 nC.

[2014]

SOLUTION

First find the dipole moment. The dipole length is $\mathbf{2a = 4\ cm = 0.04\ m}$ and charge $\mathbf{q = +8\ nC = 8\times 10^{-9}\ C}$.

$$p = q\,(2a) = (8\times 10^{-9})(0.04)$$

$$p = 3.2\times 10^{-10}\ C\,m$$

The dipole is at $\theta = 60°$ to the field and the torque is $\tau = 4\sqrt{3}\ N\,m$. Using $\tau = pE\theta$, find E.

$$E = \frac{\tau}{p\theta} = \frac{4\sqrt{3}}{(3.2\times 10^{-10})\sin 60°}$$

$$E = \frac{4\sqrt{3}}{(3.2\times 10^{-10})(\frac{\sqrt{3}}{2})}$$

$$E = \frac{4\sqrt{3} \times 2}{3.2\sqrt{3} \times 10^{-10}}$$

$$E = \frac{8}{3.2 \times 10^{-10}}$$

$$E = 2.5 \times 10^{10}\ N\,C^{-1}$$

The potential energy of the dipole is $U = -pE\theta$.

$$U = -(3.2 \times 10^{-10})(2.5 \times 10^{10})\cos 60^\circ$$

$$U = -(3.2 \times 10^{-10})(2.5 \times 10^{10})(\frac{1}{2})$$

$$U = -(8)(\frac{1}{2})$$

$$U = -4\ J$$

Hence the potential energy of the dipole is $U = -4\ J$.

Final answer. Potential energy of the dipole $U = -4\ J$.

← *Back to Q87*

Q88

A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge 'Q'. A charge 'q' is placed at the centre of the shell.
(a) What is the surface charge density on the (i) inner surface, (ii) outer surface of the shell?
(b) Write the expression for the electric field at a point $x > r_2$ from the centre of the shell.

[2012]

SOLUTION

A spherical conducting shell has inner radius r_1, outer radius r_2, and carries charge Q. A point charge q is at the centre.

(a)(i) Inside the conducting material of the shell, the electrostatic field must be zero. Take a spherical Gaussian surface lying inside the conductor (radius between r_1 and r_2). Since $\vec{E} = 0$ everywhere on it, the flux is zero, so by Gauss's law the net charge enclosed is zero.

$$q + q_{\text{inner}} = 0 \Rightarrow q_{\text{inner}} = -q$$

So a charge $-q$ is induced on the inner surface (radius r_1). The surface charge density on the inner surface is

$$\sigma_{\text{inner}} = \frac{-q}{4\pi r_1^2}$$

(a)(ii) The shell carries total charge Q. Since $-q$ sits on the inner surface, by conservation of charge the outer surface (radius r_2) must carry

$$q_{\text{outer}} = Q - (-q) = Q + q$$

The surface charge density on the outer surface is

$$\sigma_{\text{outer}} = \frac{Q + q}{4\pi r_2^2}$$

(b) For a point at distance $x > r_2$, take a concentric spherical Gaussian surface of radius x. It encloses the central charge q, the inner-surface charge $-q$, and the outer-surface charge $(Q+q)$, giving net enclosed charge $Q + q$. By Gauss's law,

$$E\,(4\pi x^2) = \frac{Q + q}{\varepsilon_0}$$

$$E = \frac{1}{4\pi\varepsilon_0}\,\frac{Q + q}{x^2}$$

directed radially (outward if $Q + q > 0$).

Final answer. $\sigma_{\text{inner}} = \dfrac{-q}{4\pi r_1^2}$, $\sigma_{\text{outer}} = \dfrac{Q+q}{4\pi r_2^2}$, and for $x > r_2$, $E = \dfrac{1}{4\pi\varepsilon_0}\,\dfrac{Q+q}{x^2}$.

← *Back to Q88*

Q89

Show that the electric field at the surface of a charged conductor is given by $\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$, *where* σ *is the surface charge density and* $\hat{n}$ *is a unit vector normal to the surface in the outward direction.*

[2012]

SOLUTION

Consider a charged conductor with surface charge density σ at the point of interest. We find the field just outside its surface using Gauss's law.

Draw a small cylindrical Gaussian surface (a pillbox) straddling the conductor's surface: one flat face of area ΔS lies just outside the surface with its outward normal along $\hat{n}$, and the other flat face of equal area lies just inside the body of the conductor. The pillbox is very short, so its curved side has negligible area.

Now compute the flux through the three parts of the pillbox.

Inside the conductor the electrostatic field is zero, so the inner face contributes no flux.

In the thin-pillbox limit the curved side contributes negligible flux.

At the surface of a charged conductor the field is purely normal, so on the outer face $\vec{E}$ is parallel to $\hat{n}$ and the flux through it is

$$\Phi = E\,\Delta S$$

The charge enclosed by the pillbox is only the patch of surface charge,

$$q_{enc} = \sigma\,\Delta S$$

By Gauss's law $\Phi = \frac{q_{enc}}{\varepsilon_0}$,

$$E\,\Delta S = \frac{\sigma\,\Delta S}{\varepsilon_0}$$

$$E = \frac{\sigma}{\varepsilon_0}$$

Since the field is directed along the outward normal,

$$\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$$

which is the required result.

Final answer. By Gauss's law the field just outside a charged conductor is $\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$.

← *Back to Q89*

Q90

An electric dipole is held in a uniform electric field.
(i) Show that the net force acting on it is zero.
(ii) The dipole is aligned parallel to the field. Find the work done in rotating it through the angle of 180°.

[2012]

SOLUTION

(i) Consider a dipole of charges $+q$ and $-q$ in a uniform field $\vec{E}$.

The force on the positive charge is $\vec{F}_+ = +q\vec{E}$ and the force on the negative charge is $\vec{F}_- = -q\vec{E}$.

The net force is

$$\vec{F}_{net} = q\vec{E} + (-q\vec{E}) = \vec{0}$$

Since the field is uniform, the two forces are equal in magnitude and opposite in direction, so the net force on the dipole is zero. (They still form a couple, giving a torque, but zero translational force.)

(ii) The potential energy of a dipole at angle θ to the field is

$$U(\theta) = -pE\theta$$

The dipole is initially aligned parallel to the field, $\theta_i = 0°$, and is rotated through 180° to $\theta_f = 180°$. The work done by the external agent equals the change in potential energy.

$$W = U(\theta_f) - U(\theta_i)$$

$$W = (-pE\cos 180°) - (-pE\cos 0°)$$

$$W = (-pE)(-1) - (-pE)(1)$$

$$W = pE + pE$$

$$W = 2pE$$

Hence the net force is zero, and the work done to rotate the dipole through 180° from the aligned position is 2pE.

Final answer. Net force is zero; the work done in rotating the dipole through **180°** is $W = 2pE$.

← *Back to Q90*

Q91

Plot a graph showing the variation of coulomb force (F) versus $(\frac{1}{r^2})$*, where r is the distance between the two charges of each pair of charges:* (1 μC, 2 μC) *and* (2 μC, −3 μC). *Interpret the graphs obtained.*

[2011]

SOLUTION

By Coulomb's law, the force between two point charges q_a and q_b separated by a distance r is given by:

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_a q_b}{r^2}$$

Writing this in terms of $\frac{1}{r^2}$ as the variable,

$$F = (\frac{q_a q_b}{4\pi\varepsilon_0})\frac{1}{r^2}$$

This is of the form $F = m\left(\frac{1}{r^2}\right)$, which is a straight line passing through the origin with slope $m = \frac{q_a q_b}{4\pi\varepsilon_0}$. The sign of the slope is fixed by the product $q_a\, q_b$.

For the first pair (1 μC, 2 μC), both charges are positive, so the force is repulsive and the slope is positive:

$$m_1 = \frac{(1\ \mu C)(2\ \mu C)}{4\pi\varepsilon_0} = \frac{2\ \mu^2}{4\pi\varepsilon_0} > 0$$

For the second pair (2 μC, −3 μC), the charges are of opposite sign, so the force is attractive and the slope is negative:

$$m_2 = \frac{(2\ \mu C)(-3\ \mu C)}{4\pi\varepsilon_0} = -\frac{6\ \mu^2}{4\pi\varepsilon_0} < 0$$

Graph: Take $\frac{1}{r^2}$ on the x-axis and F on the y-axis. Draw two straight lines through the origin. The line for pair 1 rises into the first quadrant (positive slope, gentler). The line for pair 2 goes into the fourth quadrant (negative slope, steeper in magnitude). Label the upper line (1 μC, 2 μC) and the lower line (2 μC, −3 μC).

Interpretation: Both lines pass through the origin, meaning the Coulomb force vanishes as $\frac{1}{r^2} \to 0$ (i.e. $r \to \infty$). A positive slope denotes a repulsive force (like charges) and a negative slope denotes an attractive force (unlike charges). The second line has the steeper magnitude of slope because the product of its charge magnitudes (6 μ^2) is larger than that of the first (2 μ^2). **The slope of each line equals $\frac{q_a q_b}{4\pi\varepsilon_0}$.**

Final answer. Both are straight lines through the origin: pair 1 has a positive (smaller) slope, pair 2 a negative (larger magnitude) slope.

← *Back to Q91*

Q92

A thin straight infinitely long conducting wire having charge density λ is enclosed by a cylindrical surface of radius r and length l, its axis coinciding with the length of the

wire. Find the expression for the electric flux through the surface of the cylinder.

[2011]

SOLUTION

By Gauss's law, the net electric flux through any closed surface equals the total charge enclosed divided by ε_0:

$$\Phi_E = \frac{q_{enc}}{\varepsilon_0}$$

The cylindrical surface of length l is coaxial with the wire, so it encloses a length l of the wire. With linear charge density λ (charge per unit length), the charge enclosed is:

$$q_{enc} = \lambda\, l$$

Substituting this into Gauss's law,

$$\Phi_E = \frac{\lambda\, l}{\varepsilon_0}$$

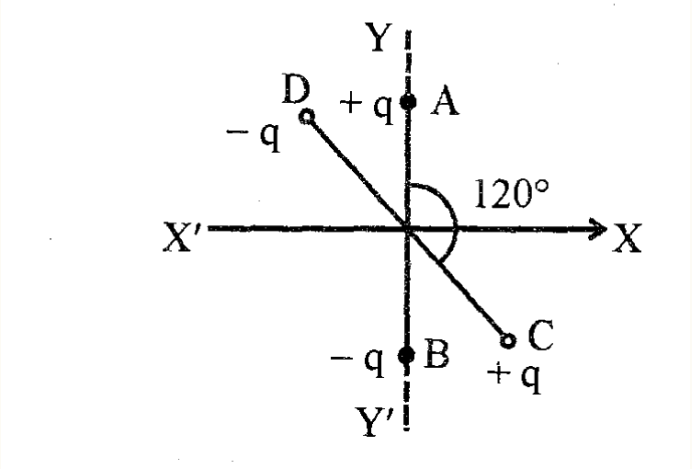
This is the required expression. **The flux through the cylinder is $\Phi_E = \dfrac{\lambda\, l}{\varepsilon_0}$,** which depends only on the enclosed length l and the linear charge density λ, and is independent of the radius r of the cylinder.

Final answer. The electric flux through the cylinder is $\Phi_E = \dfrac{\lambda\, l}{\varepsilon_0}$.

← *Back to Q92*

Q93

Two small identical electrical dipoles AB *and* CD*, each of dipole moment p, are kept at an angle of* 120° *as shown in the figure. What is the resultant dipole moment of this combination? If this system is subjected to electric field* $\vec{E}$ *directed along +X direction, what will be the magnitude and direction of the torque acting on this?*



[2011 • Set 55-1-1]

SOLUTION

From the figure, the two identical dipoles each of dipole moment p are inclined at 120° to one another, placed symmetrically about the +X-axis. So $\vec{p}_{AB}$ makes +60° with the +X-axis and $\vec{p}_{CD}$ makes −60° with it.

Resolving each dipole moment into components,

$$\vec{p}_{AB} = p\cos 60^\circ\,\hat{i} + p\sin 60^\circ\,\hat{j} = \frac{p}{2}\,\hat{i} + \frac{\sqrt{3}\,p}{2}\,\hat{j}$$

$$\vec{p}_{CD} = p\cos 60^\circ\,\hat{i} - p\sin 60^\circ\,\hat{j} = \frac{p}{2}\,\hat{i} - \frac{\sqrt{3}\,p}{2}\,\hat{j}$$

Adding, the $\hat{j}$ components cancel:

$$\vec{p}_{net} = \vec{p}_{AB} + \vec{p}_{CD} = p\,\hat{i}$$

Hence the resultant dipole moment has magnitude $|\vec{p}_{net}| = p$, directed along the +X-axis (along the internal bisector of the 120° angle).

When this system is placed in the field $\vec{E} = E\,\hat{i}$ (along +X), the torque is:

$$\vec{\tau} = \vec{p}_{net} \times \vec{E} = (p\,\hat{i}) \times (E\,\hat{i}) = \vec{0}$$

since the cross product of parallel vectors is zero. **The resultant dipole moment is p along +X, and the torque on the system is zero**, because $\vec{p}_{net}$ is parallel to $\vec{E}$ (the system is in stable equilibrium).

Final answer. Resultant dipole moment is p along +X; the torque is zero.

← Back to Q93

Q94

Calculate the amount of work done in rotating a dipole, of dipole moment 3×10^{-8} *C m, from its position of stable equilibrium to the position of unstable equilibrium, in a uniform electric field of intensity* 10^4 *N/C.*

[2011 • Set 55-2-1]

SOLUTION

The potential energy of a dipole of moment p making angle θ with a uniform field E is:

$$U(\theta) = -pE\theta$$

In stable equilibrium the dipole is aligned with the field, $\theta_i = 0°$:

$$U_i = -pE\cos 0° = -pE$$

In unstable equilibrium the dipole is anti-aligned, $\theta_f = 180°$:

$$U_f = -pE\cos 180° = +pE$$

The work done in rotating the dipole equals the change in potential energy:

$$W = U_f - U_i = (+pE) - (-pE) = 2pE$$

Substituting $p = 3\times 10^{-8}\ C\,m$ and $E = 10^4\ N\,C^{-1}$,

$$W = 2\times(3\times 10^{-8})\times(10^4)$$

$$W = 6\times 10^{-4}\ J$$

Hence the work done is $W = 6\times 10^{-4}\ J$ (= 0.6 mJ).

COMMON MISTAKE

Rotating from stable to unstable equilibrium needs $W = 2pE$*, not* pE*. The factor 2 comes from* $\cos 0° - \cos 180° = 2$*; dropping it is a common slip that loses a mark.*

Final answer. The work done is $6\times 10^{-4}\,J$.

← Back to Q94

Q95

Define electric flux. Write its S.I. unit.
A charge q is enclosed by a spherical surface of radius R. If the radius is reduced to half, how would the electric flux through the surface change?

[2009]

SOLUTION

Electric flux through a surface is defined as the surface integral of the electric field over that surface, i.e. the total number of electric field lines passing normally through the surface:

$$\Phi_E = \int_S \vec{E}\cdot \mathrm{d}\vec{S}$$

Its SI unit is $N\,\mathrm{m}^2\,\mathrm{C}^{-1}$ (equivalently $V\,m$).

By Gauss's law, the flux through any closed surface depends only on the net charge enclosed and not on the size or shape of the surface:

$$\Phi_E = \frac{q_{enc}}{\varepsilon_0}$$

When the radius of the spherical surface is reduced to half, the same charge q still lies inside the smaller sphere, so the enclosed charge is unchanged. Therefore:

$$\Phi_E^{\text{new}} = \frac{q}{\varepsilon_0} = \Phi_E^{\text{old}}$$

Hence the electric flux through the surface remains unchanged (equal to $\frac{q}{\varepsilon_0}$) when the radius is halved.

Final answer. The flux is unchanged; it stays equal to $\frac{q}{\varepsilon_0}$.

← Back to Q95

Q96

Derive an expression for the torque acting on an electric dipole, which is held in a uniform electric field, when the axis of the dipole makes an angle θ with the electric field.

[2008]

SOLUTION

Consider an electric dipole of charges $+q$ and $-q$ separated by a distance 2a, so its dipole moment $\vec{p}$ has magnitude $\boldsymbol{p = q(2a)}$ and points from $-q$ to $+q$. It is placed in a uniform external field $\vec{E}$ such that $\vec{p}$ makes an angle θ with $\vec{E}$.

Diagram: Draw $\vec{E}$ horizontal. Place the dipole inclined at angle θ to $\vec{E}$, with $+q$ at the top end and $-q$ at the bottom end of a line of length 2a. Show a force $+q\vec{E}$ on $+q$ along $\vec{E}$ and a force $-q\vec{E}$ on $-q$ opposite to $\vec{E}$. Mark the perpendicular distance between the two parallel forces as 2aθ.

The force on $+q$ is $q\vec{E}$ (along $\vec{E}$) and on $-q$ is $-q\vec{E}$ (opposite to $\vec{E}$). These two forces are equal in magnitude (qE each) and opposite in direction, so:

$$\vec{F}_{\text{net}} = q\vec{E} + (-q\vec{E}) = 0$$

The net force is zero, but the two forces have different lines of action and therefore form a couple, producing a torque. The magnitude of torque equals force times the perpendicular distance (arm of the couple):

$$\tau = (\text{force}) \times (\text{perpendicular distance})$$

$$\tau = (qE) \times (2a\theta)$$

$$\tau = (q \cdot 2a)E\theta$$

$$\tau = pE\theta$$

In vector form this is written as:

$$\vec{\tau} = \vec{p} \times \vec{E}$$

Hence the torque on the dipole is $\tau = pE\theta$, directed perpendicular to the plane of $\vec{p}$ and $\vec{E}$, tending to align $\vec{p}$ along $\vec{E}$.

Final answer. The torque on the dipole is $\vec{\tau} = \vec{p} \times \vec{E}$, of magnitude $\tau = pE\theta$.

← Back to Q96

Q97

The electric field E due to a point charge at any point near it is defined as $E = \lim_{q \to 0} \frac{F}{q}$, *where q is the test charge and F is the force acting on it. What is the physical significance of* $\lim_{q \to 0}$ *in this expression? Draw the electric field lines of a point charge Q when (i)* $Q > 0$ *and (ii)* $Q < 0$.

[2007]

SOLUTION

Physical significance of $_{q \to 0}$: The test charge q is itself a source of electric field. If it is large, it can disturb the original source charges, for example by inducing charges on nearby conductors or rearranging charges in the source distribution, thereby altering the very field we want to measure. By taking $q \to 0$, the test charge produces a negligible disturbance, so the measured value of $\frac{F}{q}$ gives the true field of the source as it exists in the absence of the test charge. It also guarantees that the field $\vec{E} = \frac{\vec{F}}{q}$ is a property of the source alone and is independent of the magnitude of the test charge.

Field lines of a point charge:

(i) For $Q > 0$: Draw the positive charge at the centre with straight radial lines pointing outward in all directions, arrowheads pointing away from the charge.

(ii) For $Q < 0$: Draw the negative charge at the centre with straight radial lines pointing inward from all directions, arrowheads pointing towards the charge.

Thus $_{q \to 0}$ ensures the test charge does not disturb the source, so $E = F/q$ measures the true field; the lines are radially outward for $Q>0$ and radially inward for $Q<0$.

Final answer. The limit ensures the test charge does not disturb the source field; lines are radially outward for $Q>0$ and radially inward for $Q<0$.

← *Back to Q97*

Q98

Define electric flux. Write its S.I. units. A spherical rubber balloon carries a charge that is uniformly distributed over its surface. As the balloon is blown up and increases in size, how does the total electric flux coming out of the surface change? Give reason.

[2007]

SOLUTION

Electric flux through a surface is the surface integral of the electric field over that surface, i.e. the total electric field lines crossing it normally:

$$\Phi_E = \int_S \vec{E} \cdot \mathrm{d}\vec{S}$$

Its SI unit is $N\,m^2\,C^{-1}$ (equivalently $V\,m$).

By Gauss's law, the total flux out of any closed surface depends only on the net charge enclosed, and not on the size or shape of the surface:

$$\Phi_E = \frac{q_{\text{enc}}}{\varepsilon_0}$$

As the balloon is blown up, no charge is added to or removed from it, so the total charge on its surface stays the same. Since the enclosed charge is unchanged:

$$\Phi_E = \frac{q}{\varepsilon_0} = \text{constant}$$

Hence the total electric flux coming out of the balloon does not change as it expands, because the flux depends only on the enclosed charge, which remains constant, not on the changing size of the surface.

Final answer. The total flux remains unchanged, since the enclosed charge is constant.

← Back to Q98

Q99

The electric field and electric potential at any point due to a point charge kept in air is $20\ NC^{-1}$ *and* $10\ JC^{-1}$ *respectively. Compute the magnitude of this charge.*

[2006]

SOLUTION

At a distance r from a point charge q, the electric field and electric potential are:

$$E = \frac{kq}{r^2} \quad V = \frac{kq}{r}$$

Dividing V by E gives the distance r directly:

$$\frac{V}{E} = \frac{kq/r}{kq/r^2} = r$$

Substituting the given values $\boldsymbol{V = 10\ J\ C^{-1}}$ and $\boldsymbol{E = 20\ N\ C^{-1}}$,

$$r = \frac{V}{E} = \frac{10}{20} = 0.5\ m$$

Now using $\boldsymbol{V = \frac{kq}{r}}$ to find q,

$$q = \frac{Vr}{k} = \frac{(10)(0.5)}{9\times 10^9}$$

$$q = \frac{5}{9\times 10^9}$$

$$q \approx 5.6\times 10^{-10}\ C$$

Hence the magnitude of the charge is $q \approx 5.6\times 10^{-10}\ C$ (≈ 0.56 nC).

TIP

The ratio $\boldsymbol{\frac{V}{E} = r}$ is a quick shortcut here: dividing potential by field gives the distance straight away, so you never need to know k until the last step.

Final answer. The magnitude of the charge is about $5.6\times 10^{-10}\ C$.

← Back to Q99

Q100

Define 'electric line of force' and give its two important properties.

[2005]

SOLUTION

An electric line of force is an imaginary smooth curve drawn in an electric field such that the tangent drawn at any point on the curve gives the direction of the electric field $\vec{E}$ at that point. The direction of the line (its arrow) shows the direction in which a small positive test charge would move when placed there.

Two important properties:

(i) Two electric lines of force can never intersect (cross) each other. If they did, two tangents could be drawn at the point of intersection, implying two directions of the field at the same point, which is impossible.

(ii) Electric lines of force start from positive charges and end on negative charges (or extend to infinity). They are continuous curves with no breaks and never form closed loops, because the electrostatic field is conservative ($\oint \vec{E}\cdot \mathrm{d}\vec{l} = 0$).

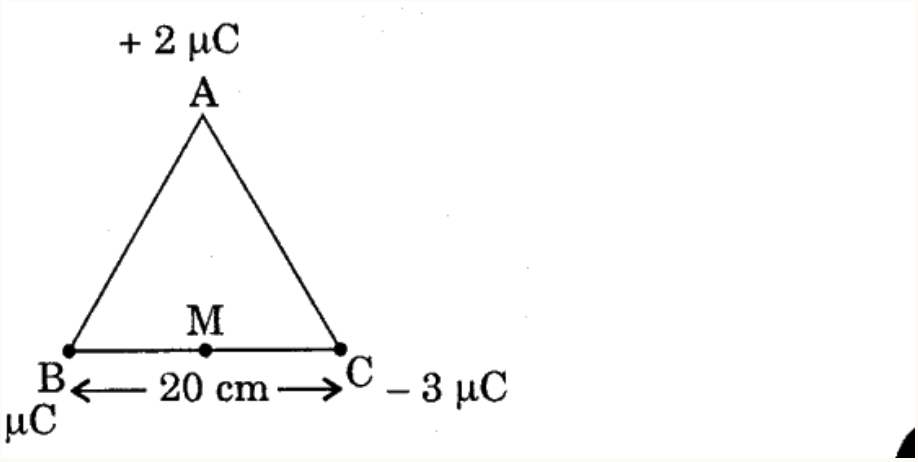
Thus an electric line of force is a curve whose tangent gives the field direction; it never crosses another line and always runs from positive to negative charge.

Final answer. An electric line of force is a curve whose tangent gives the field direction; it never intersects another line and runs from positive to negative charges.

← Back to Q100

Q101

Three point charges of +2 μC, −3 μC *and* −3 μC *are kept at the vertices A, B and C respectively of an equilateral triangle of side* 20 cm *as shown in the figure. What should be the sign and magnitude of the charge to be placed at the mid-point* (*M*) *of side* BC *so that the charge at A remains in equilibrium?*



[2005]

SOLUTION

Let the side of the equilateral triangle be $L = 20\ \text{cm} = 0.20\ m$, with charge +2 μC at *A*, and −3 μC each at *B* and *C*. *M* is the midpoint of BC.

The charge at *A* feels two attractive forces, one towards *B* and one towards *C*, each of magnitude

$$F_1 = \frac{k\,(2\ \mu C)(3\ \mu C)}{L^2} = \frac{6k\,\mu^2}{L^2}$$

These two forces are inclined at 60° to each other at *A*. Their resultant acts along AM, directed from *A* towards *M*, with magnitude

$$F_{BC} = 2F_1\cos 30° = \sqrt{3}\,F_1 = \frac{6\sqrt{3}\,k\,\mu^2}{L^2}$$

For the charge at *A* to be in equilibrium, the charge *Q* placed at *M* must exert on *A* an equal force directed along MA (away from *M*). Since the charge at *A* is positive, *Q* must also be **positive** so that the two like charges repel and push *A* away from *M*.

The distance from *A* to *M* is the height of the triangle: $\mathbf{AM} = \frac{\sqrt{3}}{2}\mathbf{L}$.

Equating the repulsion from Q to the resultant attraction:

$$\frac{k(2\,\mu C)\,Q}{(\frac{\sqrt{3}}{2}L)^2} = \frac{6\sqrt{3}\,k\,\mu^2}{L^2}$$

$$\frac{2\,\mu C\cdot Q}{\frac{3}{4}L^2} = \frac{6\sqrt{3}\,\mu^2}{L^2}$$

$$2\,\mu C\cdot Q = \frac{3}{4}\times 6\sqrt{3}\,\mu^2 = \frac{9\sqrt{3}}{2}\,\mu^2$$

$$Q = \frac{9\sqrt{3}}{4}\,\mu C \approx 3.9\,\mu C$$

Hence a **positive charge of magnitude $\frac{9\sqrt{3}}{4}$ μC ≈ 3.9 μC** placed at M keeps the charge at A in equilibrium.

Final answer. A positive charge of $\frac{9\sqrt{3}}{4}$ **μC ≈ 3.9 μC** must be placed at ***M***.

← *Back to Q101*

Q102

Mention any two properties of electric lines of force. Sketch them for an isolated positive point charge.

[2003]

SOLUTION

Two properties of electric lines of force:

(i) Two electric field lines can never intersect each other. If they did, the field at the point of intersection would have two different directions (along the two tangents), which is impossible for a single field.

(ii) Electric field lines always start from a positive charge and end on a negative charge. They are continuous open curves and never form closed loops, because the electrostatic field is conservative. (Any equivalent valid property, such as

lines being normal to the surface of a conductor, or their density being proportional to field strength, is also acceptable.)

Sketch for an isolated positive point charge:

Draw the charge $+q$ at the centre. From it, draw several straight arrows pointing radially outward in all directions, like the spokes of a wheel or a sunburst, with the arrowheads directed away from the charge and the lines extending to infinity.

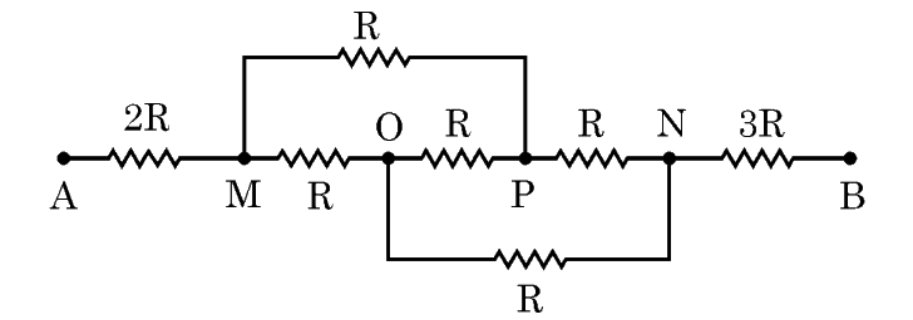
The lines are **straight, radial and directed outward**, and they spread apart as the distance increases, showing that the field weakens (as $1/r^2$) with distance from the charge.

Final answer. Lines never intersect and they begin on positive and end on negative charges; for $+q$ they are straight radial arrows pointing outward.

← *Back to Q102*

Q103

(a) *Using Gauss's law, deduce an expression for electric field at a point due to a uniformly charged infinite plane thin sheet.*
(b) *Two large thin plane sheets, each having surface charge density σ, are held close and parallel to each other in air. What is the net electric field at a point (i) inside and (ii) outside, the sheets?*
OR
(a) *Obtain the condition of balance of a Wheatstone bridge.*
(b) *Find net resistance of the network of resistors connected between A and B, as shown in figure.*



[2026 • Set 55-1-1]

SOLUTION

(a) Electric field due to an infinite uniformly charged thin plane sheet.

Let the sheet have uniform surface charge density σ. By symmetry, the field $\vec{E}$ is perpendicular to the sheet and equal in magnitude on both sides.

Choose a Gaussian surface in the form of a cylindrical pillbox of cross-sectional area A, passing perpendicularly through the sheet, with its two flat faces at equal distances on either side. The curved surface is parallel to $\vec{E}$, so the flux through it is zero. The two flat faces each contribute EA.

Total flux through the pillbox:

$$\Phi = EA + EA = 2EA$$

Charge enclosed by the pillbox is that lying on the area A of the sheet:

$$q_{enc} = \sigma A$$

By Gauss's law, $\Phi = \frac{q_{enc}}{\varepsilon_0}$:

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}$$

In vector form, $\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$, directed away from the sheet (for $\sigma > 0$). The field is uniform and independent of distance from the sheet.

(b) Two parallel sheets, each of charge density σ.

Each sheet produces a field of magnitude $\frac{\sigma}{2\varepsilon_0}$ directed away from it.

(i) Inside (in the region between the sheets): the field of the left sheet points to the right while the field of the right sheet points to the left. They are equal and opposite, so they cancel:

$$E_{\text{inside}} = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0$$

(ii) Outside the sheets: both fields point in the same direction (away from the pair), so they add:

$$E_{\text{outside}} = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$

Hence **the field between the two sheets is zero, and outside it is $\frac{\sigma}{\varepsilon_0}$**, directed away from the sheets.

COMMON MISTAKE

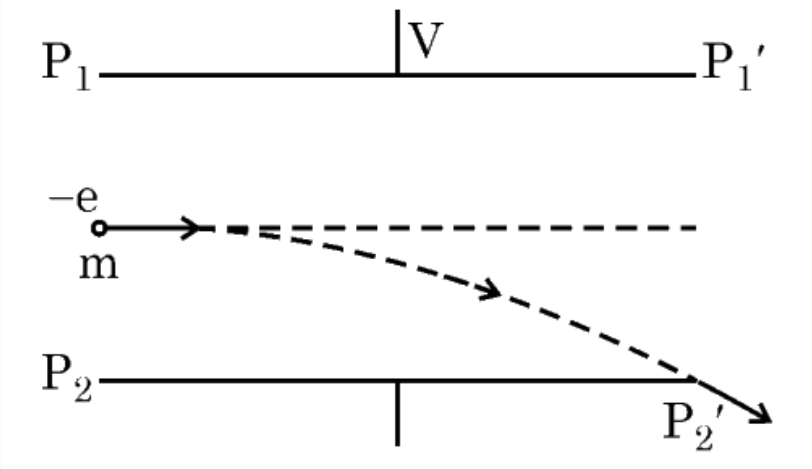
Both sheets here carry the same sign of σ, so the field cancels inside; if the two sheets carried equal and opposite charge, the answers would swap (field σ/ε_0 inside, zero outside).

Final answer. $E = \frac{\sigma}{2\varepsilon_0}$ for one sheet; between two like sheets the field is **0** and outside it is $\frac{\sigma}{\varepsilon_0}$.

← *Back to Q103*

Q104

Figure shows a narrow beam of electrons entering with a velocity of 3×10^7 *m/s, symmetrically through the space between two parallel horizontal plates* P_1 P_1' *and* P_2 P_2' *kept 2 cm apart.*



If each plate is 3 cm long, calculate the potential difference V applied between the plates so that the beam just strikes the end P_2'.

[2026 • Set 55-1-1]

SOLUTION

The electron beam enters horizontally along the central axis, midway between the plates. To 'just strike the end P_2", the electron must be deflected vertically by half the gap.

Given: horizontal speed $\mathbf{v_x = 3\times 10^7\ m/s}$, plate length $\boldsymbol{L = 3\ cm = 0.03\ m}$, plate separation $\boldsymbol{d = 2\ cm = 0.02\ m}$.

Required vertical deflection while crossing the plates:

$$y = \frac{d}{2} = \frac{0.02}{2} = 0.01\ m$$

The horizontal motion is uniform, so the time spent between the plates is

$$t = \frac{L}{v_x} = \frac{0.03}{3\times 10^7} = 1\times 10^{-9}\ s$$

The vertical motion starts from rest (no initial vertical velocity) under a uniform acceleration a_y:

$$y = \frac{1}{2} a_y t^2$$

$$a_y = \frac{2y}{t^2} = \frac{2(0.01)}{(1\times 10^{-9})^2} = \frac{0.02}{10^{-18}} = 2\times 10^{16}\ m/s^2$$

The vertical force on the electron is eE, so $\mathbf{a_y = \frac{eE}{m_e}}$, giving the field

$$E = \frac{m_e\, a_y}{e} = \frac{(9.1\times 10^{-31})(2\times 10^{16})}{1.6\times 10^{-19}}$$

$$E = \frac{1.82\times 10^{-14}}{1.6\times 10^{-19}} \approx 1.14\times 10^{5}\ V/m$$

The potential difference across the plates separated by d is

$$V = E\,d = (1.14\times 10^{5})(0.02)$$

$$V \approx 2.28\times 10^{3}\ V \approx 2.28\ \text{kV}$$

Hence the required potential difference is **about 2.28 kV**.

Final answer. The required potential difference is $\boldsymbol{V \approx 2.28\times 10^{3}\ V \approx 2.28\ \text{kV}}$.

← *Back to Q104*

Q105

Two point charges $\mathbf{q_1 = 2.5\times 10^{-7}\ C}$ *and* $\mathbf{q_2 = -2.5\times 10^{-7}\ C}$ *are located at points* (0, 0, −15 cm) *and* (0, 0, 15 cm) *respectively. Find:*
(a) the electric dipole moment of the system, and
(b) the magnitude and direction of electric field at the origin (0, 0, 0).

[2026 • Set 55-5-3]

SOLUTION

The charges are $\mathbf{q_1 = +2.5\times 10^{-7}\ C}$ at (0,0,−15 cm) and $\mathbf{q_2 = -2.5\times 10^{-7}\ C}$ at (0,0,+15 cm). The separation is $\mathbf{2a = 30\ cm = 0.30\ m}$, and the origin lies exactly midway between them.

(a) Electric dipole moment.

The dipole moment has magnitude $\boldsymbol{p = q\times 2a}$ and points from the negative charge to the positive charge.

$$p = q\times 2a = (2.5\times 10^{-7})(0.30)$$

$$p = 7.5\times 10^{-8}\ C\ m$$

Since $+q$ lies at $z = -15$ cm and $-q$ at $z = +15$ cm, the moment points from $+\hat{k}$ side towards $-\hat{k}$ side:

$$\vec{p} = -7.5\times 10^{-8}\ \hat{k}\ C\ m$$

So $|\vec{p}| = 7.5\times 10^{-8}\ C\ m$, directed along the $-z$ axis (from q_2 towards q_1).

(b) Electric field at the origin.

The origin is at distance $r = 0.15\ m$ from each charge. The field of q_1 ($+q$ at $z=-15$ cm) points away from it, i.e. along $+\hat{k}$. The field of q_2 ($-q$ at $z=+15$ cm) points towards it, i.e. also along $+\hat{k}$. The two fields are in the same direction and add up.

$$E = E_1 + E_2 = 2\times \frac{kq}{r^2}$$

$$E = 2\times \frac{(9\times 10^9)(2.5\times 10^{-7})}{(0.15)^2}$$

$$E = 2\times \frac{2250}{0.0225}$$

$$E = 2\times 1.0\times 10^5 = 2.0\times 10^5\ N/C$$

The direction is along $+\hat{k}$, i.e. **antiparallel to the dipole moment $\vec{p}$**, as expected for the field at a point on the equatorial line of a dipole.

Hence $|\vec{p}| = 7.5\times 10^{-8}\ C\ m$ along $-\hat{k}$, and $\vec{E} = 2.0\times 10^5\ N/C$ along $+\hat{k}$.

Final answer. $\vec{p} = 7.5\times 10^{-8}\ C\ m$ along $-\hat{k}$; $\vec{E} = 2.0\times 10^5\ N/C$ along $+\hat{k}$.

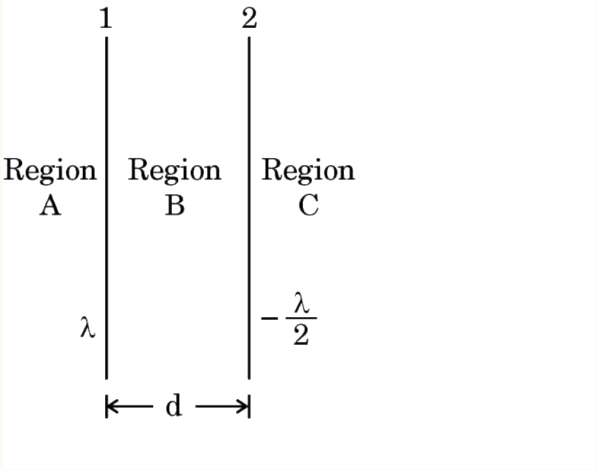
← *Back to Q105*

Q106

(a) *Two small solid metal balls A and B of radii R and* 2R *having charge densities* 2σ *and* 3σ *respectively are kept far apart. Find the charge densities on A and B after they are connected by a conducting wire.*

OR

(b) *Two infinitely long straight wires '1' and '2' are placed d distance apart, parallel to each other, as shown in the figure. They are uniformly charged having charge densities* λ *and* $-\frac{\lambda}{2}$ *respectively. Locate the position of the point from wire '1' at which the net electric field is zero and identify the region in which it lies.*



[2025 • Set 55-5-1]

SOLUTION

(a) Charge densities after the metal balls are connected.

Ball A has radius R and density 2σ; ball B has radius 2R and density 3σ. Initial charges:

$$Q_A = (2\sigma)\ 4\pi\ R^2 = 8\pi\sigma\ R^2$$

$$Q_B = (3\sigma)\ 4\pi(2R)^2 = 3\sigma\ 4\pi\ 4R^2 = 48\pi\sigma\ R^2$$

Total charge (conserved when connected):

$$Q = Q_A + Q_B = 8\pi\sigma\ R^2 + 48\pi\sigma\ R^2 = 56\pi\sigma\ R^2$$

When connected by a wire, both balls reach the same potential. For a sphere $V = \frac{kQ}{r}$, so

$$\frac{kQ'_A}{R} = \frac{kQ'_B}{2R} \Rightarrow Q'_B = 2Q'_A$$

Using charge conservation:

$$Q'_A + Q'_B = 56\pi\sigma\ R^2$$

$$Q'_A + 2Q'_A = 56\pi\sigma\,R^2 \Rightarrow Q'_A = \frac{56\pi\sigma\,R^2}{3},\ Q'_B = \frac{112\pi\sigma\,R^2}{3}$$

New surface charge densities:

$$\sigma'_A = \frac{Q'_A}{4\pi\,R^2} = \frac{56\pi\sigma\,R^2}{3\times 4\pi\,R^2} = \frac{14\sigma}{3}$$

$$\sigma'_B = \frac{Q'_B}{4\pi(2R)^2} = \frac{112\pi\sigma\,R^2}{3\times 16\pi\,R^2} = \frac{7\sigma}{3}$$

Thus $\boldsymbol{\sigma'_A = \frac{14\sigma}{3}}$ and $\boldsymbol{\sigma'_B = \frac{7\sigma}{3}}$ (ratio 2:1, since on connected conductors $\boldsymbol{\sigma \propto 1/r}$).

OR (b) Point of zero field for wires with densities λ and $-\frac{\lambda}{2}$.

Let wire 1 (density λ) be at $\boldsymbol{x = 0}$ and wire 2 (density $-\lambda/2$) at $\boldsymbol{x = d}$. The field of a line charge at perpendicular distance r has magnitude $\boldsymbol{E = \frac{|\lambda|}{2\pi\varepsilon_0 r}}$, pointing away from a positive wire and towards a negative wire.

The two fields can cancel only where they are equal in magnitude and opposite in direction. In the region between the wires both fields point the same way (to the right), so they cannot cancel there. The fields oppose each other in the two outer regions, and for unequal charges the null point must lie beyond the wire carrying the smaller magnitude of charge, i.e. beyond wire 2 ($\boldsymbol{x > d}$).

Take the point at distance x from wire 1 (so its distance from wire 2 is $x - d$). There the field of wire 1 points along $+x$ (away from the positive wire) and the field of wire 2 points along $-x$ (towards the negative wire). Equating magnitudes:

$$\frac{\lambda}{2\pi\varepsilon_0\,x} = \frac{\lambda/2}{2\pi\varepsilon_0\,(x-d)}$$

$$\frac{1}{x} = \frac{1}{2(x-d)}$$

$$2(x - d) = x$$

$$2x - 2d = x \Rightarrow x = 2d$$

Hence the net electric field is zero at a point at a distance 2d from wire 1, lying in the region beyond wire 2 (the wire with the smaller, negative charge), at a distance d from wire 2 on the side away from wire 1.

Final answer. $\sigma'_A = \frac{14\sigma}{3}$, $\sigma'_B = \frac{7\sigma}{3}$; OR the net field is zero at $x = 2d$ from wire 1, in the region beyond wire 2 (the smaller, negative charge), i.e. at distance d from wire 2 on the far side.

← *Back to Q106*

Q107

An electric field $\vec{E}$ *given by:*
$\vec{E} = 100\,\hat{i}\ N/C$ *for* $x > 0$
$\vec{E} = -100\,\hat{i}\ N/C$ *for* $x < 0$
exists in a region. A right circular cylinder of length 10 cm *and radius* 2 cm*, is placed in the region such that its axis coincides with x-axis and its two faces are at* $x = -5$ **cm** *and* $x = 5$ **cm**. *Calculate:*
(a) the net outward flux through the cylinder, and
(b) the net charge inside the cylinder.

[2025 • Set 55-7-3]

SOLUTION

The cylinder has length $L = 10$ **cm** $= 0.10\ m$ and radius $R = 2$ **cm** $= 0.02\ m$, with its axis along the x-axis and flat faces at $x = -5$ **cm** and $x = +5$ **cm**. Each flat face has area

$$A = \pi R^2 = \pi(0.02)^2 = 4\pi \times 10^{-4}\ \text{m}^2$$

(a) Net outward flux.

The field is everywhere along the x-axis, while the curved surface has its area vector perpendicular to the axis. So the flux through the curved surface is zero.

Right face at $x = +5$ **cm** (here $x>0$, so $\vec{E} = 100\,\hat{i}$), outward normal $+\hat{i}$:

$$\Phi_R = \vec{E}\cdot\vec{A} = (100)(4\pi \times 10^{-4}) = 4\pi \times 10^{-2}\ N\,\text{m}^2/C$$

Left face at $x = -5$ **cm** (here $x<0$, so $\vec{E} = -100\,\hat{i}$), outward normal $-\hat{i}$:

$$\Phi_L = (-100)(-1)(4\pi \times 10^{-4}) = 4\pi \times 10^{-2}\ N\,\text{m}^2/C$$

Both faces give positive (outward) flux, so the net flux is

$$\Phi_{\text{net}} = \Phi_R + \Phi_L = 4\pi\times 10^{-2} + 4\pi\times 10^{-2} = 8\pi\times 10^{-2}$$

$$\Phi_{\text{net}} \approx 0.251\ N\,\text{m}^2/C$$

(b) Net charge inside.

By Gauss's law, $\mathbf{q_{enc} = \varepsilon_0\,\Phi_{net}}$:

$$q_{\text{enc}} = (8.854\times 10^{-12})(0.251)$$

$$q_{\text{enc}} \approx 2.22\times 10^{-12}\ C = 2.22\ \text{pC}$$

Hence the net outward flux is $\approx \mathbf{0.251\ N\,m^2/C}$ and the charge enclosed is $\approx$ **2.22 pC**.

COMMON MISTAKE

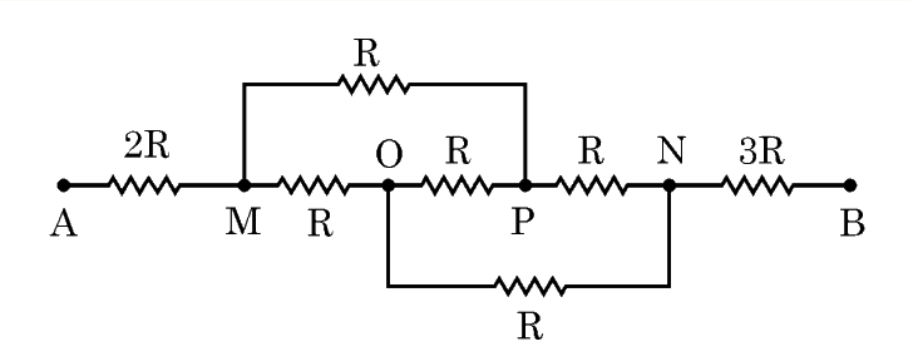
Both faces contribute positive (outward) flux because the field reverses direction across x**=0***; do not subtract them.*

Final answer. Net flux $\approx \mathbf{0.251\ N\,m^2/C}$ and enclosed charge $\approx$ **2.22 pC**.

← Back to Q107

Q108

A cube of side 0.1 *m is placed, as shown in the figure, in a region where electric field* $\vec{\boldsymbol{E}} = 500\,x\,\hat{\boldsymbol{i}}$ *exists. Here x is in metres and E in* NC^{-1}*. Calculate:*
(a) the flux passing through the cube, and
(b) the charge within the cube.



[2024 • Set 55-1-1]

SOLUTION

The cube of side 0.1 m is placed with its faces along the axes, one face at $\boldsymbol{x = 0}$ and the opposite face at $\boldsymbol{x = 0.1\ m}$. The field is $\vec{E} = \mathbf{500}\, \boldsymbol{x}\, \hat{\boldsymbol{i}}$, which varies with x and points along $\hat{i}$.

(a) Flux through the cube.

The four faces whose outward normals are $\pm\hat{j}$ or $\pm\hat{k}$ are perpendicular to $\vec{E}$, so each gives zero flux.

At the face $\boldsymbol{x = 0}$ (outward normal $-\hat{i}$): here $\boldsymbol{E = 500 \times 0 = 0}$, so

$$\Phi_1 = 0$$

At the face $\boldsymbol{x = 0.1\ m}$ (outward normal $+\hat{i}$): here $\boldsymbol{E = 500 \times 0.1 = 50\ N/C}$, and the face area is

$$A = (0.1)^2 = 1 \times 10^{-2}\ \mathrm{m}^2$$

$$\Phi_2 = E\,A = 50 \times 10^{-2} = 0.5\ N\,\mathrm{m}^2/C$$

Net flux through the cube:

$$\Phi_E = \Phi_1 + \Phi_2 = 0 + 0.5 = 0.5\ N\,\mathrm{m}^2/C$$

(b) Charge inside the cube.

By Gauss's law, $\boldsymbol{\Phi_E = \dfrac{q_{enc}}{\varepsilon_0}}$, so

$$q_{\mathrm{enc}} = \varepsilon_0\, \Phi_E = (8.854 \times 10^{-12})(0.5)$$

$$q_{\mathrm{enc}} \approx 4.43 \times 10^{-12}\ C = 4.43\ \mathrm{pC}$$

Hence the flux through the cube is $\boldsymbol{0.5\ N\,m^2/C}$ and the enclosed charge is $\boldsymbol{\approx 4.43\ pC}$.

Final answer. Flux through the cube = $\boldsymbol{0.5\ N\,m^2/C}$ and enclosed charge $\boldsymbol{\approx 4.43\ pC}$.

← *Back to Q108*

Q109

Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (1) maximum, and (2) minimum?

[2024 • Set 55-1-1]

SOLUTION

Consider a dipole of moment $\vec{p}$ placed in a uniform external field $\vec{E}$, making an angle θ with the field. The field exerts a torque on the dipole of magnitude

$$\tau = pE\theta$$

To rotate the dipole through a small angle dθ against this torque, an external agent does work

$$dW = \tau\, d\theta = pE\theta\, d\theta$$

This work is stored as the potential energy of the dipole. Taking the position $\theta_0 = 90°$ as the reference (zero potential energy), the energy at angle θ is

$$U(\theta) = \int_{90^\circ}^{\theta} pE\theta'\, d\theta'$$

$$U(\theta) = pE[-\theta']_{90^\circ}^{\theta}$$

$$U(\theta) = pE(-\theta + \cos 90^\circ)$$

$$U(\theta) = -pE\theta$$

In vector form, the potential energy of an electric dipole in a uniform field is

$$U = -\vec{p}\cdot\vec{E} = -pE\theta$$

(1) Maximum potential energy: U is maximum when $\theta = -1$, i.e. $\theta = 180°$ (dipole antiparallel to the field). Then $U_{max} = +pE$. This corresponds to **unstable equilibrium**.

(2) Minimum potential energy: U is minimum when $\theta = +1$, i.e. $\theta = 0°$ (dipole parallel to the field). Then $U_{min} = -pE$. This corresponds to **stable equilibrium**.

Final answer. $U = -\vec{p}\cdot\vec{E} = -pE\theta$; maximum ($+pE$) at $\theta = 180°$ and minimum ($-pE$) at $\theta = 0°$.

← *Back to Q109*

Q110

(a) Define the term 'electric flux' and write its dimensions.
(b) A plane surface, in shape of a square of side 1 cm *is placed in an electric field* $\vec{E} = (100\ N/C)\hat{i}$ *such that the unit vector normal to the surface is given by* $\hat{n} = 0.8\hat{i} + 0.6\hat{k}$. *Find the electric flux through the surface.*

[2024 • Set 55-5-1]

SOLUTION

(a) Definition and dimensions of electric flux.

Electric flux through a surface is the surface integral of the electric field over that surface:

$$\Phi_E = \int_S \vec{E}\cdot d\vec{S}$$

It is a scalar quantity that gives a measure of the number of electric field lines crossing the surface. Its SI unit is $N\,m^2C^{-1}$ (equivalently $V\,m$).

To find the dimensions, electric field has dimensions $[E] = \dfrac{\text{force}}{\text{charge}} = [M\,L\,T^{-3}\,A^{-1}]$ and area has dimensions $[L^2]$, so

$$[\Phi_E] = [M\,L\,T^{-3}\,A^{-1}][L^2] = [M\,L^3\,T^{-3}\,A^{-1}]$$

(b) Flux through the square surface.

Side of square $= 1\ \text{cm} = 10^{-2}\ m$, so its area is

$$A = (10^{-2})^2 = 10^{-4}\ m^2$$

The area vector is $\vec{A} = A\,\hat{n} = 10^{-4}(0.8\,\hat{i} + 0.6\,\hat{k})\ m^2$.

The flux is

$$\Phi_E = \vec{E}\cdot\vec{A} = (100\,\hat{i})\cdot 10^{-4}(0.8\,\hat{i} + 0.6\,\hat{k})$$

$$\Phi_E = 100\times 10^{-4}[(1)(0.8) + (0)(0.6)]$$

$$\Phi_E = 100\times 10^{-4}\times 0.8$$

$$\Phi_E = 8\times 10^{-3}\, N\, m^2C^{-1}$$

Hence the electric flux through the square surface is $\mathbf{8\times 10^{-3}\, N\, m^2C^{-1} = 0.008\, N\, m^2C^{-1}}$.

TIP

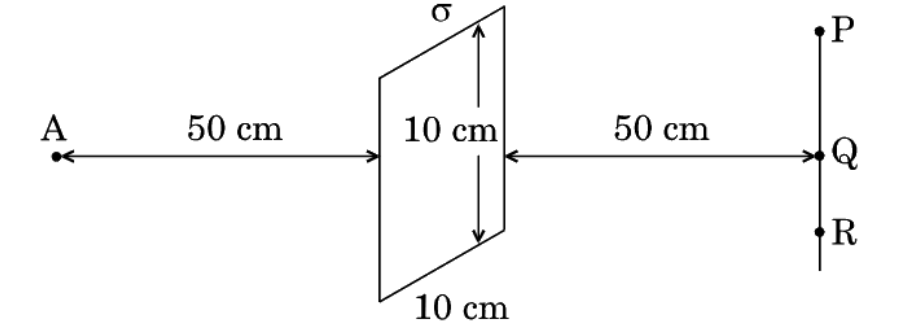
Only the $\hat{i}$ component of $\hat{n}$ contributes since $\vec{E}$ is purely along $\hat{i}$; the 0.6 $\hat{k}$ part dots to zero.

Final answer. $[\Phi_E] = [M\, L^3\, T^{-3}\, A^{-1}]$ and the flux through the square is $\mathbf{8\times 10^{-3}\, N\, m^2C^{-1}}$.

← *Back to Q110*

Q111

(a) A uniformly charged large plane sheet has charge density $\sigma = (\frac{1}{18\pi}) \times 10^{-9}\ C/m^2$. *Find the electric field at point A which is* 50 cm *from the sheet.*
(b) Consider a straight line with three points P, Q and R, placed 50 cm *from the charged sheet on the right side as shown in the figure. At which of these points, does the magnitude of the electric field due to the sheet remain the same as that at point A and why?*



[2021]

SOLUTION

A uniformly charged large (infinite) plane sheet produces a uniform electric field on either side whose magnitude depends only on the surface charge density, given by $E = \frac{\sigma}{2\varepsilon_0}$, directed perpendicular to the sheet.

(a) Here $\sigma = \frac{1}{18\pi} \times 10^{-9}\ C/m^2$ and $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ N\,m^2C^{-2}$, so $\frac{1}{2\varepsilon_0} = 2\pi \times \frac{1}{4\pi\varepsilon_0} = 18\pi \times 10^9\ N\,m^2C^{-2}$.

$$E_A = \frac{\sigma}{2\varepsilon_0}$$

$$E_A = \frac{1}{18\pi} \times 10^{-9} \times 18\pi \times 10^9$$

$$E_A = 1\ N\,C^{-1}$$

The field is directed perpendicular to the sheet, pointing away from it since the sheet is positively charged. Note that the distance of 50 cm does not enter the calculation, because the field of an infinite sheet is independent of distance.

(b) The electric field due to an infinite plane sheet, $E = \frac{\sigma}{2\varepsilon_0}$, is uniform: its magnitude depends only on σ and not on the perpendicular distance from the sheet nor on the position along the sheet. Since P, Q and R all lie in the same half-space on the right of the sheet, the magnitude of the field at each of them equals that at A.

Hence the field at all three points P, Q and R has the same magnitude as at A, namely $1\ N\,C^{-1}$, because the field of an infinite charged sheet is the same at every point regardless of distance.

COMMON MISTAKE

Many students plug the 50 cm distance into a $1/r^2$ formula. For an infinite sheet there is no distance term at all; the field is the same everywhere on one side.

Final answer. The field at A is 1 N C^{-1} (away from the sheet); P, Q and R all have the same magnitude 1 N C^{-1} because an infinite sheet's field is distance-independent.

← *Back to Q111*

Q112

Two small identical conducting spheres carrying charge 10 μC *and* −20 μC *when separated by a distance of r, experience a force F each. If they are brought in contact and then separated to a distance of* $\frac{r}{2}$, *what is the new force between them in terms of F?*

[2021]

SOLUTION

When two identical conducting spheres are brought into contact, they reach a common potential and share the total charge equally; Coulomb's law then gives the force between them.

Before contact the spheres carry $\mathbf{q_1 = +10\ \mu C}$ and $\mathbf{q_2 = -20\ \mu C}$ at separation r. By Coulomb's law the magnitude of the force on each is

$$F = \frac{1}{4\pi\varepsilon_0}\,\frac{|q_1\,q_2|}{r^2} = \frac{k\,(10\ \mu C)(20\ \mu C)}{r^2} = \frac{200\,k\,\mu C^2}{r^2}$$

This force is attractive because the charges are of opposite sign.

On contact, the total charge is shared equally between the two identical spheres. The total charge is

$$Q_{tot} = (+10) + (-20) = -10\ \mu C$$

$$q' = \frac{Q_{tot}}{2} = \frac{-10\ \mu C}{2} = -5\ \mu C \text{ on each sphere}$$

They are now separated to a distance $\frac{r}{2}$. Both carry −5 μC, so by Coulomb's law the new force is

$$F' = \frac{k\,(5\ \mu C)(5\ \mu C)}{(\frac{r}{2})^2} = \frac{25\,k\,\mu C^2}{\frac{r^2}{4}} = \frac{100\,k\,\mu C^2}{r^2}$$

Taking the ratio of the two forces,

$$\frac{F'}{F} = \frac{100\,k\,\mu C^2 / r^2}{200\,k\,\mu C^2 / r^2} = \frac{100}{200} = \frac{1}{2}$$

$$\Rightarrow F' = \frac{F}{2}$$

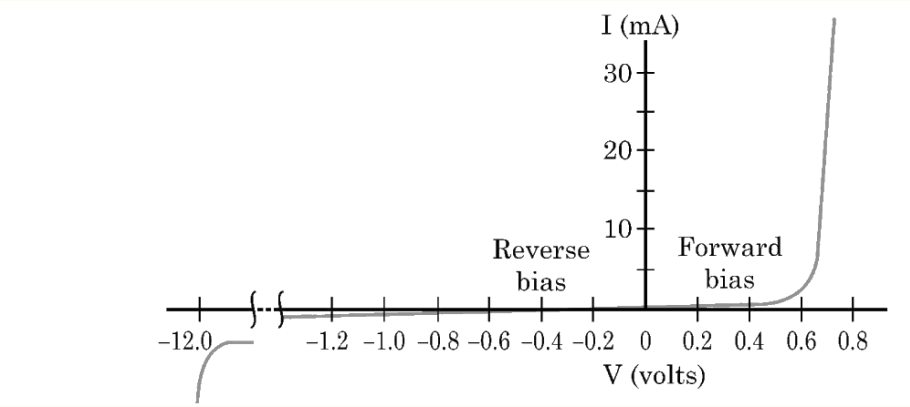
Hence the new force is $\boldsymbol{\frac{F}{2}}$**, and it is repulsive** since both spheres now carry like (negative) charges.

Final answer. The new force is F/2, and it is repulsive.

← *Back to Q112*

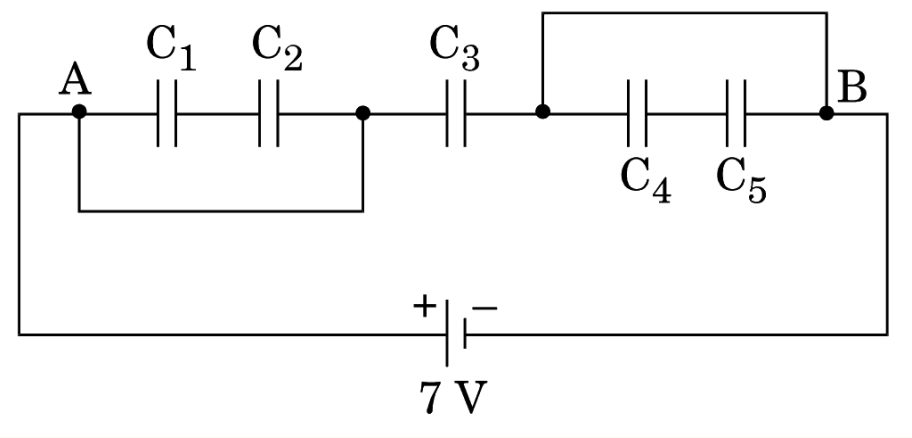
Q113

Two small identical electric dipoles AB and CD, each of dipole moment $\vec{p}$ *are kept at an angle of* 120° *to each other in an external electric field* $\vec{E}$ *pointing along the x-axis as shown in the figure. Find the*
(a) dipole moment of the arrangement, and
(b) magnitude and direction of the net torque acting on it.



OR

In the figure given below, find the
(a) equivalent capacitance of the network between points A and B. Given: $\mathbf{C_1 = C_5 = 8}$ μF, $\mathbf{C_2 = C_3 = C_4 = 4}$ μF.
(b) maximum charge supplied by the battery, and
(c) total energy stored in the network.



[2020 • Set 55-2-1]

SOLUTION

The two identical dipoles are placed symmetrically about the x-axis, each making 60° with it, so that the angle between them is 120°, in a field $\vec{E} = E\hat{i}$ along the +x-axis. Take $\vec{p}_{AB}$ at +60° above the axis and $\vec{p}_{CD}$ at −60° below it.

(a) Resolving each dipole moment into components,

$$\vec{p}_{AB} = p\cos 60^\circ\,\hat{i} + p\sin 60^\circ\,\hat{j} = \frac{p}{2}\hat{i} + \frac{\sqrt{3}\,p}{2}\hat{j}$$

$$\vec{p}_{CD} = p\cos 60^\circ\,\hat{i} - p\sin 60^\circ\,\hat{j} = \frac{p}{2}\hat{i} - \frac{\sqrt{3}\,p}{2}\hat{j}$$

Adding, the $\hat{j}$ (perpendicular) components cancel and the $\hat{i}$ components add,

$$\vec{p}_{net} = \vec{p}_{AB} + \vec{p}_{CD} = p\,\hat{i}$$

$$|\vec{p}_{net}| = p$$

So the resultant dipole moment has magnitude p and is directed along the +x-axis, i.e. along the bisector of the 120° angle.

(b) The torque on a dipole in a uniform field is $\vec{\tau} = \vec{p} \times \vec{E}$. Using the net dipole moment,

$$\vec{\tau} = \vec{p}_{net} \times \vec{E} = (p\,\hat{i}) \times (E\,\hat{i})$$

Since $\hat{i} \times \hat{i} = 0$,

$$\vec{\tau} = \vec{0}$$

The net torque is **zero**, because the resultant dipole moment $\vec{p}_{net}$ is parallel to the field $\vec{E}$. The arrangement is therefore in (stable) equilibrium.

Final answer. The net dipole moment is p along the +x-axis, and the net torque on the arrangement is zero.

← *Back to Q113*

Q114

(a) Two electric field lines cannot cross each other. Also, they cannot form closed loops. Give reasons.

(b) A particle of charge 2 μC and mass 1.6 g is moving with a velocity $4\hat{i}$ *m* s^{-1}. *At* $t = 0$ *the particle enters in a region having an electric field* $\vec{E}$ *(in* NC^{-1}*)* $= 80\hat{i} + 60\hat{j}$. *Find the velocity of the particle at* $t = 5$ *s.*

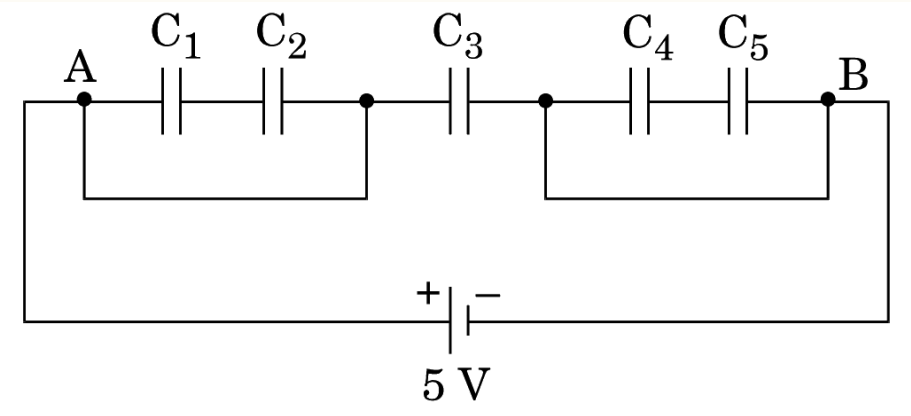
OR

In the figure given below, find the

(a) equivalent capacitance of the network between points A and B. Given: $C_1 = C_5 = 4$ *μF,* $C_2 = C_3 = C_4 = 2$ *μF.*

(b) maximum charge supplied by the battery, and

(c) total energy stored in the network.



[2020 • Set 55-2-2]

SOLUTION

(a) (i) Two electric field lines cannot cross each other. If they did, then at the point of intersection we could draw two tangents, giving two different

directions of the electric field at the same point. But the electric field at any point has a single, unique direction. Hence two field lines never cross.

(ii) Electric field lines cannot form closed loops. Electrostatic field lines always start from a positive charge and end on a negative charge (or extend to infinity). A closed loop would require the field to do non-zero work on a charge taken once round the loop, i.e. $\oint \vec{E}\cdot d\vec{l} \neq 0$. But the electrostatic field is conservative, so $\oint \vec{E}\cdot d\vec{l} = 0$ for every closed path. Hence the lines cannot close on themselves.

(b) The acceleration of the charged particle is given by $\vec{a} = \frac{q\vec{E}}{m}$. Here $q = 2\ \mu C = 2\times 10^{-6}\ C$, $m = 1.6\ g = 1.6\times 10^{-3}\ kg$, and $\vec{E} = (80\hat{i} + 60\hat{j})\ N\,C^{-1}$.

$$\vec{a} = \frac{(2\times 10^{-6})}{(1.6\times 10^{-3})}\ (80\hat{i} + 60\hat{j})$$

$$\vec{a} = (1.25\times 10^{-3})\ (80\hat{i} + 60\hat{j})$$

$$\vec{a} = (0.1\,\hat{i} + 0.075\,\hat{j})\ m\ s^{-2}$$

Using $\vec{v} = \vec{v}_0 + \vec{a}t$ with $\vec{v}_0 = 4\hat{i}\ m\ s^{-1}$ and $t = 5\ s$,

$$\vec{v} = 4\hat{i} + (0.1\hat{i} + 0.075\hat{j})(5)$$

$$\vec{v} = 4\hat{i} + 0.5\hat{i} + 0.375\hat{j}$$

$$\vec{v} = (4.5\,\hat{i} + 0.375\,\hat{j})\ m\ s^{-1}$$

Hence the velocity at $t = 5\ s$ is $(4.5\,\hat{i} + 0.375\,\hat{j})\ m\ s^{-1}$, with magnitude $|\vec{v}| = \sqrt{4.5^2 + 0.375^2} \approx 4.52\ m\ s^{-1}$.

COMMON MISTAKE

Convert mass to kg (1.6 g = 1.6×10^{-3} kg) and charge to coulomb before substituting; mixing grams and micro-coulombs is the usual slip here.

Final answer. The velocity at t = 5 s is (4.5î + 0.375ĵ) m s^{-1}, of magnitude about 4.52 m s^{-1}.

← Back to Q114

Q115

A hollow conducting sphere of inner radius r_1 and outer radius r_2 has a charge Q on its surface. A point charge −q is also placed at the centre of the sphere.
(a) What is the surface charge density on the (i) inner and (ii) outer surface of the sphere?
(b) Use Gauss' law of electrostatics to obtain the expression for the electric field at a point lying outside the sphere.
OR
(a) An infinitely long thin straight wire has a uniform linear charge density λ. Obtain the expression for the electric field E at a point lying at a distance x from the wire, using Gauss' law.
(b) Show graphically the variation of this electric field E as a function of distance x from the wire.

[2020 • Set 55-4-1]

SOLUTION

(a) A point charge $-q$ sits at the centre of a hollow conductor with inner radius r_1 and outer radius r_2, and the conductor carries a charge Q.

(i) Consider a Gaussian surface inside the metal wall (between r_1 and r_2). Inside a conductor in electrostatic equilibrium $\vec{E} = \mathbf{0}$, so the flux through this surface is zero, hence the charge it encloses is zero:

$$q_{inner} + (-q) = 0 \Rightarrow q_{inner} = +q$$

$$\sigma_{inner} = \frac{q_{inner}}{4\pi r_1^2} = \frac{+q}{4\pi r_1^2}$$

(ii) The total charge on the conductor is Q. Since $+q$ is induced on the inner surface, the outer surface must carry the remainder by charge conservation:

$$q_{outer} = Q - q_{inner} = Q - q$$

$$\sigma_{outer} = \frac{Q - q}{4\pi r_2^2}$$

(b) For a point outside the sphere, draw a concentric spherical Gaussian surface of radius $r > r_2$. By symmetry $\vec{E}$ is radial and uniform in magnitude over this surface. By Gauss's law,

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{enclosed}}{\varepsilon_0}$$

The total charge enclosed is the central charge plus all conductor charge:

$$q_{enclosed} = (-q) + (+q) + (Q - q) = Q - q$$

$$E\,(4\pi r^2) = \frac{Q - q}{\varepsilon_0}$$

$$E = \frac{Q - q}{4\pi\varepsilon_0 r^2}$$

Hence the field outside is $\boldsymbol{E = \frac{Q - q}{4\pi\varepsilon_0 r^2}}$, directed radially outward if $Q > q$ and radially inward if $Q < q$.

Final answer. σ_inner = +q/(4πr$_1^2$), σ_outer = (Q−q)/(4πr$_2^2$), and the field outside is E = (Q−q)/(4πε$_0$r^2).

← *Back to Q115*

Q116

Derive an expression for the electric field due to a dipole of dipole moment $\vec{p}$ at a point on its perpendicular bisector.

[2019 • Set 55-1-3]

SOLUTION

Consider a dipole made of charges $-q$ at A and $+q$ at B, separated by 2a, with dipole moment $\boldsymbol{p = q(2a)}$ directed from $-q$ to $+q$. Let P be a point on the perpendicular bisector (equatorial line) at distance r from the centre O.

Draw the dipole along a horizontal axis with O at its midpoint, and P directly above O on the perpendicular bisector. Each charge is at distance

$$AP = BP = \sqrt{r^2 + a^2}$$

The magnitude of the field at P due to each charge is

$$E_+ = E_- = \frac{1}{4\pi\varepsilon_0}\,\frac{q}{r^2 + a^2}$$

Resolve each field into a component perpendicular to the dipole axis and a component parallel to it. The perpendicular components are equal and opposite and cancel. The components parallel to the axis (both pointing from $+q$ towards $-q$, i.e. antiparallel to $\vec{p}$) add up. The parallel component of each field is $E_{\pm}\theta$, where $\theta = \frac{a}{\sqrt{r^2+a^2}}$. Hence the net field is

$$E = 2\,E_{+}\theta = 2\times\frac{1}{4\pi\varepsilon_0}\,\frac{q}{r^2+a^2}\times\frac{a}{\sqrt{r^2+a^2}}$$

$$E = \frac{1}{4\pi\varepsilon_0}\,\frac{2qa}{(r^2+a^2)^{3/2}}$$

Since $p = 2qa$,

$$E = \frac{1}{4\pi\varepsilon_0}\,\frac{p}{(r^2+a^2)^{3/2}}$$

For a short dipole, $r \gg a$, so a^2 is neglected against r^2,

$$E_{\text{equatorial}} = \frac{1}{4\pi\varepsilon_0}\,\frac{p}{r^3}$$

The field points **antiparallel to $\vec{p}$** and falls off as $\frac{1}{r^3}$. Its direction is opposite to the dipole moment, and its magnitude is exactly half the axial field at the same distance.

Final answer. On the perpendicular bisector, E = p/[4πε₀(r²+a²)^{3/2}], which for a short dipole gives E = p/(4πε₀r³), directed antiparallel to p.

← *Back to Q116*

Q117

Two large charged plane sheets of charge densities σ and -2σ C/m² are arranged vertically with a separation of d between them. Deduce expressions for the electric field at points (i) to the left of the first sheet, (ii) to the right of the second sheet, and (iii) between the two sheets.

OR

A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge Q.

(a) A charge q is placed at the centre of the shell. Find out the surface charge density on the inner and outer surfaces of the shell.
(b) Is the electric field inside a cavity (with no charge) zero; independent of the fact whether the shell is spherical or not? Explain.

[2019 • Set 55-2-1]

SOLUTION

Sheet 1 carries $\sigma > 0$ on the left; sheet 2 carries -2σ at separation d on the right. Each infinite sheet gives a uniform field $\frac{|\sigma|}{2\varepsilon_0}$, directed away from a positive sheet and towards a negative sheet. We superpose the two fields, taking rightward as positive.

Sheet 1 produces $\mathbf{E_1} = \frac{\sigma}{2\varepsilon_0}$, and sheet 2 produces $\mathbf{E_2} = \frac{2\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$.

(i) To the left of sheet 1: sheet 1 points left (away from +), sheet 2 points right (towards −).

$$E_{left} = \frac{\sigma}{\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{2\varepsilon_0}$$

directed to the right (towards the sheets).

(ii) To the right of sheet 2: sheet 1 points right (away from +), sheet 2 points left (towards −).

$$E_{right} = \frac{\sigma}{\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{2\varepsilon_0}$$

directed to the left (towards the sheets).

(iii) Between the sheets: sheet 1 points right (away from +), and sheet 2 also points right (towards the negative sheet, which lies on the right). The two add.

$$E_{between} = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{\varepsilon_0} = \frac{3\sigma}{2\varepsilon_0}$$

directed to the right, from sheet 1 towards sheet 2.

Hence $\mathbf{E_{left} = E_{right} = \frac{\sigma}{2\varepsilon_0}}$ (both pointing towards the gap) and $\mathbf{E_{between} = \frac{3\sigma}{2\varepsilon_0}}$ (pointing from the positive sheet to the negative sheet).

TIP

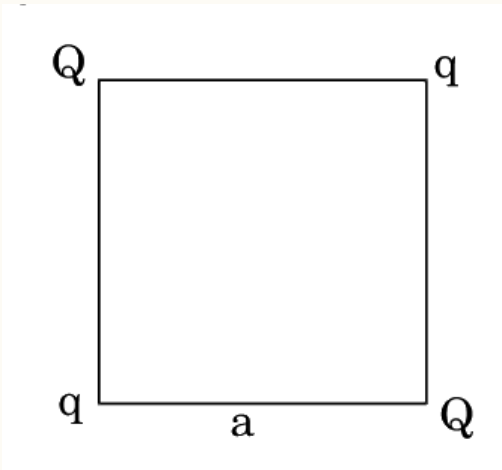
Always fix a positive direction (say rightward) first, write each sheet's contribution with its proper sign, then add. Guessing the net direction without the sign bookkeeping is where marks are lost.

Final answer. Left of sheet 1 and right of sheet 2 the field is $\sigma/2\varepsilon_0$ (pointing inward), and between the sheets it is $3\sigma/2\varepsilon_0$ (from the $+\sigma$ sheet to the -2σ sheet).

← *Back to Q117*

Q118

Four point charges Q, q, Q and q are placed at the corners of a square of side 'a' as shown in the figure.



Find the
(a) resultant electric force on a charge Q, and
(b) potential energy of this system.

[2018]

SOLUTION

In the square ABCD of side a, charge Q is at A and C (diagonally opposite) and charge q is at B and D (diagonally opposite). Find the force on the charge Q at A. Take Q and q as like (positive) charges, so all forces are repulsive.

(a) Three forces act on Q at A:

From q at B (distance a): $\mathbf{F_1 = \frac{kQq}{a^2}}$, directed along BA (i.e. along AB extended, away from B).

From q at D (distance a): $\mathbf{F_2 = \frac{kQq}{a^2}}$, directed along DA (away from D).

From Q at C (distance $a\sqrt{2}$): $\mathbf{F_3 = \frac{kQ^2}{(a\sqrt{2})^2} = \frac{kQ^2}{2a^2}}$, directed along CA (away from C, i.e. along the diagonal).

F_1 and F_2 are equal in magnitude and mutually perpendicular, so their resultant lies along the diagonal AC (the same line as F_3) and has magnitude

$$F_{12} = \sqrt{F_1^2 + F_2^2} = \sqrt{2}\ \frac{kQq}{a^2}$$

Since F_{12} and F_3 are collinear and both point outward along AC, they add:

$$F_{\text{net}} = F_{12} + F_3 = \frac{\sqrt{2}\ kQq}{a^2} + \frac{kQ^2}{2a^2}$$

$$F_{\text{net}} = \frac{kQ}{a^2}\left(\sqrt{2}\ q + \frac{Q}{2}\right) = \frac{Q}{4\pi\varepsilon_0\, a^2}\left(\sqrt{2}\ q + \frac{Q}{2}\right)$$

directed along the diagonal AC, away from the centre of the square.

(b) The potential energy is the sum over all six pairs: four sides (each a Qq pair at distance a), one diagonal QQ pair at distance $a\sqrt{2}$, and one diagonal qq pair at distance $a\sqrt{2}$.

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{4Qq}{a} + \frac{Q^2}{a\sqrt{2}} + \frac{q^2}{a\sqrt{2}}\right]$$

$$U = \frac{1}{4\pi\varepsilon_0\, a}\left[4Qq + \frac{Q^2+q^2}{\sqrt{2}}\right]$$

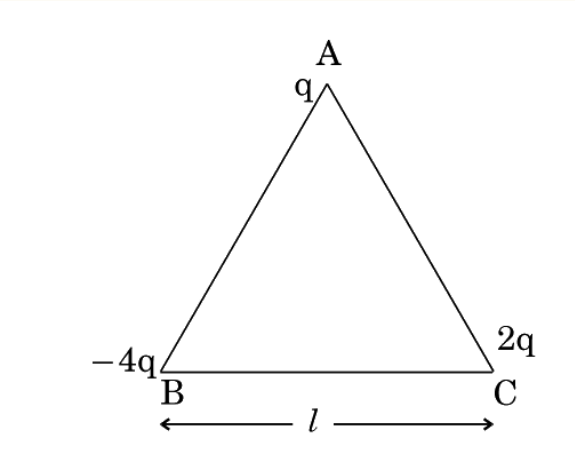
Hence the net force on Q is $\mathbf{\frac{Q}{4\pi\varepsilon_0\, a^2}\left(\sqrt{2}\ q + \frac{Q}{2}\right)}$ along AC, and the potential energy of the system is $\mathbf{\frac{1}{4\pi\varepsilon_0\, a}\left[4Qq + \frac{Q^2+q^2}{\sqrt{2}}\right]}$.

Final answer. Net force on Q is $(Q/4\pi\varepsilon_0 a^2)(\sqrt{2}\,q + Q/2)$ along the diagonal, and $U = (1/4\pi\varepsilon_0 a)[4Qq + (Q^2+q^2)/\sqrt{2}]$.

← *Back to Q118*

Q119

(a) Three point charges q, −4q *and* 2q *are placed at the vertices of an equilateral triangle* ABC *of side 'l' as shown in the figure. Obtain the expression for the magnitude of the resultant electric force acting on the charge q.*



(b) Find out the amount of the work done to separate the charges at infinite distance.

[2018]

SOLUTION

In the equilateral triangle ABC of side l, charge q is at A, −4q at B, and 2q at C. Find the force on q at A.

(a) Force from −4q at B: opposite charges attract, so this force on q is directed from A towards B, with magnitude

$$F_{AB} = \frac{k\,(q)(4q)}{l^2} = \frac{4kq^2}{l^2}$$

Force from 2q at C: like charges repel, so this force on q is directed from C towards A, i.e. away from C, with magnitude

$$F_{AC} = \frac{k\,(q)(2q)}{l^2} = \frac{2kq^2}{l^2}$$

The interior angle at A is 60°. The attractive force points along AB; the repulsive force points away from C (opposite to AC). The angle between these two force vectors is **180°− 60°= 120°**. Using the parallelogram law,

$$F^2 = F_{AB}{}^2 + F_{AC}{}^2 + 2F_{AB}F_{AC}\cos 120^\circ$$

$$F^2 = \left(\frac{4kq^2}{l^2}\right)^2 + \left(\frac{2kq^2}{l^2}\right)^2 + 2\left(\frac{4kq^2}{l^2}\right)\left(\frac{2kq^2}{l^2}\right)\left(-\frac{1}{2}\right)$$

$$F^2 = (16 + 4 - 8)\,\frac{k^2q^4}{l^4} = 12\,\frac{k^2q^4}{l^4}$$

$$F = 2\sqrt{3}\,\frac{kq^2}{l^2} = \frac{2\sqrt{3}\,q^2}{4\pi\varepsilon_0\,l^2}$$

(b) The work done to separate the charges to infinity equals the negative of the system's initial potential energy. The potential energy is the sum over the three pairs:

$$U = \frac{k}{l}[(q)(-4q) + (-4q)(2q) + (2q)(q)]$$

$$U = \frac{k}{l}(-4q^2 - 8q^2 + 2q^2) = -\frac{10kq^2}{l}$$

The work done by the external agent is

$$W = U_{\text{final}} - U_{\text{initial}} = 0 - \left(-\frac{10kq^2}{l}\right) = \frac{10kq^2}{l}$$

$$W = \frac{10\,q^2}{4\pi\varepsilon_0\,l}$$

Hence the force on q is $\dfrac{\mathbf{2\sqrt{3}\,kq^2}}{\mathbf{l^2}}$, and the work needed to separate the charges to infinity is $\dfrac{\mathbf{10kq^2}}{\boldsymbol{l}} = \dfrac{\mathbf{10\,q^2}}{\mathbf{4\pi\varepsilon_0}\,\boldsymbol{l}}$.

COMMON MISTAKE

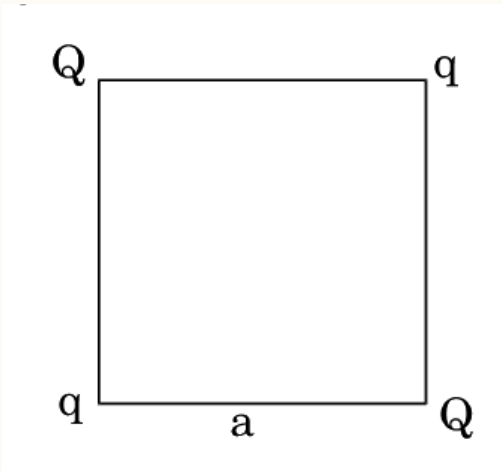
The work to dismantle equals −U_initial, and U here is negative, so the work comes out positive. A negative answer for the separation work signals a sign slip in the pairwise energy terms.

Final answer. The resultant force on q is $2\sqrt{3}\cdot kq^2/l^2$, and the work to separate the charges to infinity is $10kq^2/l$.

← Back to Q119

Q120

Four point charges Q, q, Q and q are placed at the corners of a square of side 'a' as shown in the figure.



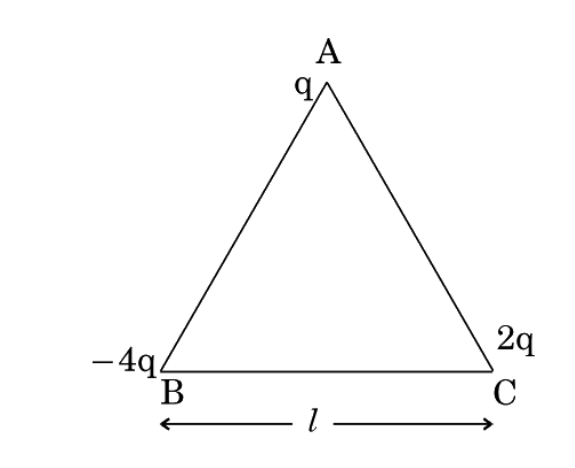
Find the

(a) resultant electric force on a charge Q, and

(b) potential energy of this system.

OR

(a) Three point charges q, −4q and 2q are placed at the vertices of an equilateral triangle ABC of side 'l' as shown in the figure. Obtain the expression for the magnitude of the resultant electric force acting on the charge q.



(b) Find out the amount of the work done to separate the charges at infinite distance.

[2018]

SOLUTION

In the square ABCD of side a, charge Q is at A and C (diagonally opposite) and charge q is at B and D (diagonally opposite). Find the force on the charge Q at A, treating Q and q as like charges so all forces are repulsive.

(a) Three forces act on Q at A:

From q at B (distance a): $\mathbf{F_1} = \frac{\mathbf{kQq}}{\mathbf{a^2}}$, directed along AB extended (away from B).

From q at D (distance a): $\mathbf{F_2} = \frac{\mathbf{kQq}}{\mathbf{a^2}}$, directed along AD extended (away from D).

From Q at C (distance $a\sqrt{2}$): $\mathbf{F_3} = \frac{\mathbf{kQ^2}}{\mathbf{(a\sqrt{2})^2}} = \frac{\mathbf{kQ^2}}{\mathbf{2a^2}}$, directed along CA (away from C, along the diagonal).

F_1 and F_2 are equal and perpendicular, so their resultant lies along the diagonal AC, collinear with F_3, with magnitude

$$F_{12} = \sqrt{F_1^2 + F_2^2} = \sqrt{2}\ \frac{kQq}{a^2}$$

Since F_{12} and F_3 point the same way (outward along AC), they add:

$$F_{net} = \frac{\sqrt{2}\ kQq}{a^2} + \frac{kQ^2}{2a^2} = \frac{Q}{4\pi\varepsilon_0\, a^2}\left(\sqrt{2}\ q + \frac{Q}{2}\right)$$

directed along the diagonal AC, away from the centre.

(b) The potential energy is the sum of the six pairwise terms: four side Qq pairs at distance a, one diagonal QQ pair at $a\sqrt{2}$, one diagonal qq pair at $a\sqrt{2}$.

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{4Qq}{a} + \frac{Q^2}{a\sqrt{2}} + \frac{q^2}{a\sqrt{2}}\right]$$

$$U = \frac{1}{4\pi\varepsilon_0\, a}\left[4Qq + \frac{Q^2+q^2}{\sqrt{2}}\right]$$

OR (Triangle alternative): For q, $-4q$, $2q$ at the vertices of an equilateral triangle of side l, the force on q has magnitude $\boldsymbol{F = 2\sqrt{3}\ \frac{kq^2}{l^2}}$ (vector sum of the attractive $\frac{4kq^2}{l^2}$ from $-4q$ and the repulsive $\frac{2kq^2}{l^2}$ from $2q$ at 120°), and the work to separate the charges to infinity is $\boldsymbol{W = \frac{10kq^2}{l}}$.

Hence the net force on Q is $\frac{Q}{4\pi\varepsilon_0 a^2}(\sqrt{2}\,q + \frac{Q}{2})$ along AC, and the potential energy is $\frac{1}{4\pi\varepsilon_0 a}[4Qq + \frac{Q^2+q^2}{\sqrt{2}}]$.

Final answer. Net force on Q is $(Q/4\pi\varepsilon_0 a^2)(\sqrt{2}\,q + Q/2)$ along the diagonal, and $U = (1/4\pi\varepsilon_0 a)[4Qq + (Q^2+q^2)/\sqrt{2}]$.

← *Back to Q120*

Q121

(i) Derive the expression for electric field at a point on the equatorial line of an electric dipole.
(ii) Depict the orientation of the dipole in (i) stable, (ii) unstable equilibrium in a uniform electric field.

[2017]

SOLUTION

(i) Consider an electric dipole made of charges $-q$ at A and $+q$ at B, separated by a distance 2a, so its dipole moment is $\vec{p}=q(2a)\hat{p}$, directed from $-q$ to $+q$. Let P be a point on the equatorial line (the perpendicular bisector of AB) at distance r from the centre O.

The distance of P from each charge is the same:

$$AP = BP = \sqrt{r^2+a^2}$$

The magnitude of the field at P due to each charge is therefore equal:

$$E_+ = E_- = \frac{1}{4\pi\varepsilon_0}\ \frac{q}{(r^2+a^2)}$$

Draw the two field vectors at P. The field E_+ (due to $+q$) points away from B, and E_- (due to $-q$) points towards A. Resolve each into a component along the dipole axis and a component perpendicular to it.

The components perpendicular to the dipole axis are equal and opposite, so they cancel. The components parallel to the axis (each of magnitude $E\theta$, where $\theta = \frac{a}{\sqrt{r^2+a^2}}$) point in the same direction (antiparallel to $\vec{p}$) and add up.

$$E_{eq} = 2E\theta = 2\times\frac{1}{4\pi\varepsilon_0}\ \frac{q}{(r^2+a^2)}\times\frac{a}{\sqrt{r^2+a^2}}$$

$$E_{eq} = \frac{1}{4\pi\varepsilon_0}\ \frac{q(2a)}{(r^2+a^2)^{3/2}} = \frac{1}{4\pi\varepsilon_0}\ \frac{p}{(r^2+a^2)^{3/2}}$$

For a short dipole, $r \gg a$, so a^2 is neglected:

$$\vec{E}_{eq} = -\frac{1}{4\pi\varepsilon_0}\ \frac{\vec{p}}{r^3}$$

The negative sign shows the equatorial field is **antiparallel to** $\vec{p}$.

(ii) For **stable equilibrium**, draw the dipole with $\vec{p}$ along $\vec{E}$, i.e. $\theta=0°$. Here the potential energy $U=-pE\theta=-pE$ is minimum, and any small displacement produces a restoring torque that brings $\vec{p}$ back to align with $\vec{E}$.

For **unstable equilibrium**, draw the dipole with $\vec{p}$ opposite to $\vec{E}$, i.e. $\theta=180°$. Here $U=+pE$ is maximum, and any small displacement produces a torque that turns the dipole away from this position.

COMMON MISTAKE

In stable equilibrium p is parallel to E (U minimum); a common slip is to write the unstable case as $\theta=90°$. *Unstable is antiparallel,* $\theta=180°$.

Final answer. $E_{eq}=\frac{1}{4\pi\varepsilon_0}\ \frac{p}{(r^2+a^2)^{3/2}}$, antiparallel to $\vec{p}$; stable at $\theta=0°$, unstable at $\theta=180°$.

← *Back to Q121*

Q122

(i) Obtain the expression for the torque $\vec{\tau}$ *experienced by an electric dipole of dipole moment* $\vec{p}$ *in a uniform electric field,* $\vec{E}$.
(ii) What will happen if the field were not uniform?

[2017]

SOLUTION

(i) Consider an electric dipole of charges $+q$ and $-q$ separated by 2a, placed in a uniform electric field $\vec{E}$ such that the dipole moment $\vec{p}=q(2a)$ makes an angle θ with $\vec{E}$.

The force on $+q$ is $+q\vec{E}$ and the force on $-q$ is $-q\vec{E}$. These two forces are equal in magnitude and opposite in direction:

$$\vec{F}_{\text{net}} = q\vec{E}+(-q\vec{E}) = 0$$

So there is no net translational force, but the two forces act along different lines, forming a couple that produces a torque.

The magnitude of the torque equals (force) × (perpendicular distance between the two forces):

$$\tau = qE\times(\text{perpendicular distance})$$

The perpendicular distance between the lines of action of the two forces is $2a\theta$. Therefore,

$$\tau = qE\,(2a\theta) = (q\,2a)E\theta = pE\theta$$

In vector form,

$$\vec{\tau} = \vec{p}\times\vec{E}$$

The direction of $\vec{\tau}$ is perpendicular to the plane of $\vec{p}$ and $\vec{E}$ (given by the right-hand rule), and it tends to align $\vec{p}$ along $\vec{E}$.

(ii) If the field were **not uniform**, the magnitude of $\vec{E}$ would be different at the two ends of the dipole. Then the forces $+q\vec{E}$ and $-q\vec{E}$ on the two charges would no longer be equal in magnitude, so they would not cancel.

As a result the dipole would experience a **net translational force** in addition to a torque. Hence in a non-uniform field the dipole both rotates and undergoes a net linear (translatory) motion, drifting towards the region of stronger field when $\vec{p}$ is parallel to $\vec{E}$.

Final answer. $\vec{\tau}=\vec{p}\times\vec{E}$ ($\tau=pE\theta$); in a non-uniform field the dipole also feels a net translational force.

← *Back to Q122*

Q123

A long charged cylinder of linear charge density $+\lambda_1$ is surrounded by a hollow coaxial conducting cylinder of linear charge density $-\lambda_2$. Use Gauss's law to obtain expressions for the electric field at a point (i) in the space between the cylinders, and (ii) outside the larger cylinder.

[2017]

SOLUTION

By symmetry, the field at any point is radial and its magnitude depends only on the perpendicular distance r from the common axis. To picture the setup, draw the inner cylinder ($+\lambda_1$) along the axis, the hollow conducting cylinder ($-\lambda_2$) around it, and a coaxial cylindrical Gaussian surface of radius r and length l.

(i) For a point **between the cylinders**, choose the Gaussian cylinder of radius r lying inside the outer cylinder. The charge enclosed is only that of the inner cylinder:

$$q_{enc} = \lambda_1 \, l$$

The flux passes only through the curved surface (the flat end caps give zero flux since $\vec{E} \perp d\vec{S}$ there):

$$\oint \vec{E} \cdot d\vec{S} = E(2\pi \, r \, l)$$

By Gauss's law,

$$E(2\pi \, r \, l) = \frac{\lambda_1 \, l}{\varepsilon_0}$$

$$E_{between} = \frac{\lambda_1}{2\pi\varepsilon_0 \, r}$$

directed radially outward (since $\lambda_1 > 0$).

(ii) For a point **outside the larger cylinder**, choose the Gaussian cylinder of radius r greater than the outer radius. Now both charges are enclosed:

$$q_{enc} = (\lambda_1 - \lambda_2) \, l$$

By Gauss's law,

$$E(2\pi\, r\, l) = \frac{(\lambda_1 - \lambda_2)\, l}{\varepsilon_0}$$

$$E_{\text{outside}} = \frac{\lambda_1 - \lambda_2}{2\pi\varepsilon_0\, r}$$

The field is radially outward if $\lambda_1 > \lambda_2$ and radially inward if $\lambda_2 > \lambda_1$. If $\lambda_1 = \lambda_2$, the field outside is zero.

Final answer. $E_{\text{between}} = \dfrac{\lambda_1}{2\pi\varepsilon_0\, r}$ and $E_{\text{outside}} = \dfrac{\lambda_1 - \lambda_2}{2\pi\varepsilon_0\, r}$.

← *Back to Q123*

Q124

Define electric flux. Write its SI unit.
Using Gauss's law, deduce an expression for electric field intensity due to an infinitely long straight uniformly charged wire.

[2017]

SOLUTION

Electric flux through a surface is defined as the total number of electric field lines passing normally through that surface. Mathematically, for a surface S,

$$\Phi_E = \int_S \vec{E} \cdot \mathrm{d}\vec{S}$$

It is a scalar quantity. Its **SI unit** is $N\,\mathrm{m}^2\,\mathrm{C}^{-1}$ (equivalently $V\,m$).

Now consider an infinitely long straight wire with uniform linear charge density λ. By symmetry, the field $\vec{E}$ is everywhere radial (perpendicular to the wire) and has the same magnitude at all points at the same distance r from the wire.

To picture the setup, draw the straight wire and a coaxial cylindrical Gaussian surface of radius r and length l wrapped around it, with two flat end caps.

The two flat end caps are parallel to $\vec{E}$, so $\vec{E} \perp \mathrm{d}\vec{S}$ on them and they contribute zero flux. The flux comes only from the curved surface:

$$\oint \vec{E} \cdot \mathrm{d}\vec{S} = E(2\pi\, r\, l)$$

The charge enclosed by this cylinder is

$$q_{enc} = \lambda\, l$$

By Gauss's law,

$$E(2\pi\, r\, l) = \frac{\lambda\, l}{\varepsilon_0}$$

$$E = \frac{\lambda}{2\pi\varepsilon_0\, r}$$

The field is directed radially outward for $\lambda>0$ and radially inward for $\lambda<0$, and it varies inversely as the distance r from the wire.

Final answer. $\Phi_E = \int \vec{E}\cdot d\vec{S}$ (unit $N\,m^2C^{-1}$); for the wire, $E = \frac{\lambda}{2\pi\varepsilon_0\, r}$.

← *Back to Q124*

Q125

Using Gauss's law in electrostatics, deduce an expression for electric field intensity due to a uniformly charged infinite plane sheet. If another identical sheet is placed parallel to it, show that there is no electric field in the region between the two sheets.

[2017]

SOLUTION

Consider an infinite plane sheet of charge with uniform surface charge density σ. By symmetry, $\vec{E}$ is perpendicular to the sheet, equal in magnitude on the two sides, and directed away from a positively charged sheet.

To picture the setup, draw the plane sheet and a cylindrical Gaussian pillbox of cross-sectional area A piercing the sheet symmetrically, so that one flat face lies on each side of the sheet.

The curved (side) surface of the pillbox is parallel to $\vec{E}$, so it gives zero flux. Each flat face contributes EA, so the total flux is

$$\oint \vec{E}\cdot d\vec{S} = EA + EA = 2EA$$

The charge enclosed within the pillbox is

$$q_{enc} = \sigma A$$

By Gauss's law,

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}$$

directed normally away from the sheet (for **σ>0**). Note that E is independent of the distance from the sheet.

Now place a second identical sheet (also +σ) parallel to the first. Each sheet sets up a field $\frac{\sigma}{2\varepsilon_0}$ directed away from itself.

In the region **between** the two sheets, the field of the left sheet points to the right, while the field of the right sheet points to the left. They are equal in magnitude and opposite in direction:

$$E_{between} = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0$$

Hence the resultant electric field in the region between two identical parallel charged sheets is **zero**.

Final answer. $E = \frac{\sigma}{2\varepsilon_0}$ for one sheet; between two identical sheets the fields cancel, so $E = 0$.

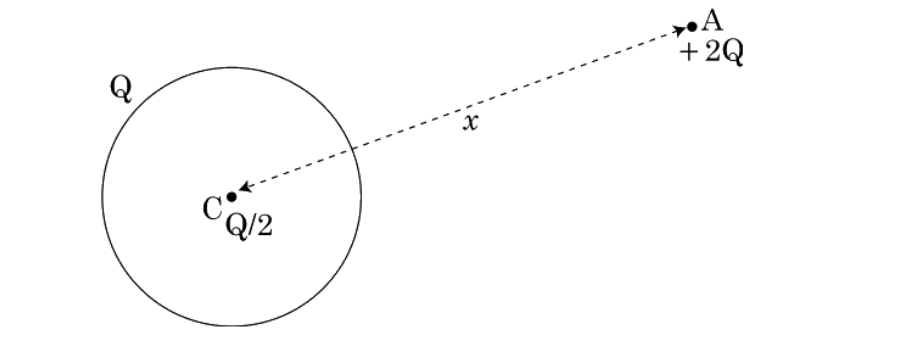
← *Back to Q125*

Q126

A thin metallic spherical shell of radius R carries a charge Q on its surface. A point charge Q/2 is placed at the centre C and another charge +2Q is placed outside the shell at A at a distance x from the centre as shown in the figure.
(i) Find the electric flux through the shell.

(ii) State the law used.

(iii) Find the force on the charges at the centre C of the shell and at the point A.



[2016]

SOLUTION

To picture the setup, draw the spherical shell of radius R carrying charge Q on its surface, the point charge $Q/2$ at the centre C, and the charge +2Q outside at point A, a distance x from C.

(i) Take the closed Gaussian surface to coincide with the shell. Only the central charge $Q/2$ lies inside this surface; the shell's own charge Q sits on the surface (not enclosed) and the external charge +2Q is outside. So the enclosed charge is $Q/2$.

$$\Phi_E = \frac{q_{enc}}{\varepsilon_0} = \frac{Q/2}{\varepsilon_0} = \frac{Q}{2\varepsilon_0}$$

(ii) The law used is **Gauss's law**: the net electric flux through any closed surface equals $\frac{1}{\varepsilon_0}$ times the net charge enclosed by it,

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}$$

(iii) **Force on the charge $Q/2$ at C.** The shell's own charge Q is uniformly spread on a sphere whose centre is C, so by the shell theorem it produces zero field at C. Hence only the external charge +2Q at A acts on $Q/2$:

$$F_C = \frac{1}{4\pi\varepsilon_0}\,\frac{(Q/2)(2Q)}{x^2} = \frac{1}{4\pi\varepsilon_0}\,\frac{Q^2}{x^2}$$

directed along CA (repulsive, away from the shell).

Force on the charge +2Q at A. For an external point, the shell charge Q together with the central charge $Q/2$ behave as a single charge $(Q+\frac{Q}{2})=\frac{3Q}{2}$ placed at the centre. So the force on +2Q is

$$F_A = \frac{1}{4\pi\varepsilon_0}\,\frac{(3Q/2)(2Q)}{x^2} = \frac{1}{4\pi\varepsilon_0}\,\frac{3Q^2}{x^2}$$

directed along CA produced (repulsive, away from C).

> **COMMON MISTAKE**
>
> *For the force on the outside charge, remember the shell charge AND the central charge both act as if concentrated at C, giving an effective $\frac{3Q}{2}$, not just Q.*

Final answer. $\Phi_E=\frac{Q}{2\varepsilon_0}$; $F_C=\frac{1}{4\pi\varepsilon_0}\,\frac{Q^2}{x^2}$ and $F_A=\frac{1}{4\pi\varepsilon_0}\,\frac{3Q^2}{x^2}$.

← *Back to Q126*

Q127

Find the electric field intensity due to a uniformly charged spherical shell at a point (i) outside the shell and (ii) inside the shell. Plot the graph of electric field with distance from the centre of the shell.

[2016]

SOLUTION

Consider a uniformly charged thin spherical shell of radius R carrying total charge Q. By spherical symmetry, the field is radial and its magnitude depends only on the distance r from the centre.

(i) **Outside the shell (r>R).** Draw a concentric spherical Gaussian surface of radius r. It encloses the entire charge Q, and the field is uniform and normal over it:

$$\oint \vec{E}\cdot \mathrm{d}\vec{S} = E(4\pi\, r^2)$$

By Gauss's law,

$$E(4\pi r^2) = \frac{Q}{\varepsilon_0}$$

$$E_{out} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$$

So outside, the shell behaves as if its whole charge were concentrated at the centre.

(ii) **Inside the shell (r).** Draw a concentric spherical Gaussian surface of radius **r. Since all the charge lies on the surface of the shell, the charge enclosed is zero:**

$$E(4\pi r^2) = \frac{0}{\varepsilon_0}$$

$$E_{in} = 0$$

Graph of *E* versus *r*. Draw the horizontal axis as r and the vertical axis as E. The curve is zero (along the r-axis) from $r=0$ up to $r=R$. At $r=R$ it rises to its maximum value $E_{max}=\frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}$. For $r>R$ it decreases as $\frac{1}{r^2}$, approaching zero as $r\to\infty$.

Final answer. $E_{out}=\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$ and $E_{in}=0$; E is zero up to R, peaks at R, then falls as $1/r^2$.

← *Back to Q127*

Q128

Derive an expression for the electric field intensity at a point on the equatorial line of an electric dipole of dipole moment $\vec{P}$ and length 2a. *What is the direction of this field?*

[2016]

SOLUTION

Consider an electric dipole with charge $-q$ at A and $+q$ at B, separated by distance 2a, so the dipole moment is $\vec{P}=q(2a)$, directed from $-q$ to $+q$. The equatorial line is the perpendicular bisector of AB. Take a point P on it at distance r from the centre O.

The distance of P from each charge is the same:

$$AP = BP = \sqrt{r^2+a^2}$$

The magnitude of the field at P due to each charge is therefore equal:

$$E_+ = E_- = \frac{1}{4\pi\varepsilon_0}\ \frac{q}{(r^2+a^2)}$$

Draw the two field vectors at P and resolve each along the dipole axis and perpendicular to it. The components perpendicular to the axis are equal and opposite, so they cancel. The components along the axis (each $E\theta$, with $\theta = \frac{a}{\sqrt{r^2+a^2}}$) add up.

$$E_{eq} = 2E\theta = 2\times\frac{1}{4\pi\varepsilon_0}\ \frac{q}{(r^2+a^2)}\times\frac{a}{\sqrt{r^2+a^2}}$$

$$E_{eq} = \frac{1}{4\pi\varepsilon_0}\ \frac{q(2a)}{(r^2+a^2)^{3/2}} = \frac{1}{4\pi\varepsilon_0}\ \frac{P}{(r^2+a^2)^{3/2}}$$

For a short dipole, $r \gg a$, so a^2 is neglected:

$$\vec{E}_{eq} = -\frac{1}{4\pi\varepsilon_0}\ \frac{\vec{P}}{r^3}$$

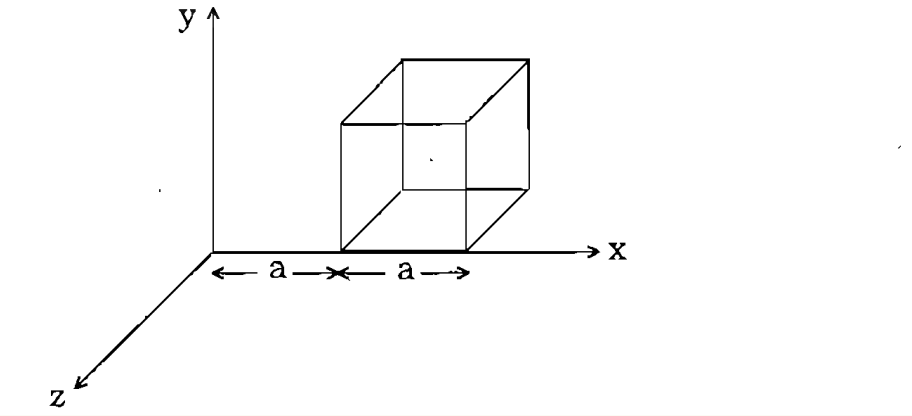
Direction: the field is along the dipole axis but **antiparallel to** $\vec{P}$, that is, it points from the $+q$ end towards the $-q$ end.

Final answer. $E_{eq} = \frac{1}{4\pi\varepsilon_0}\ \frac{P}{(r^2+a^2)^{3/2}}$, directed antiparallel to $\vec{P}$.

← *Back to Q128*

Q129

State Gauss's law in electrostatics. A cube with each side 'a' is kept in an electric field given by $\vec{E} = C\,x\,\hat{i}$, *(as is shown in the figure) where C is a positive dimensional constant. Find out*
(i) the electric flux through the cube, and
(ii) the net charge inside the cube.



[2012]

SOLUTION

Gauss's law states that the net electric flux through any closed surface equals $\frac{1}{\varepsilon_0}$ times the total charge enclosed by that surface:

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}$$

(i) The cube has side a with its left face at $x=0$ and right face at $x=a$, and the field is $\vec{E}=Cx\,\hat{i}$. To picture it, draw the cube with edges along the axes and the field pointing along $+x$, increasing as x increases.

The four side faces are perpendicular to $\hat{j}$ or $\hat{k}$, while $\vec{E}$ is along $\hat{i}$, so $\vec{E}\cdot d\vec{S}=0$ on each of them; they contribute zero flux.

At the **left face ($x=0$)**: $\vec{E}=C(0)\hat{i}=0$, with outward normal $-\hat{i}$, so

$$\Phi_L = 0$$

At the **right face ($x=a$)**: $\vec{E}=Ca\,\hat{i}$, with outward normal $+\hat{i}$, so

$$\Phi_R = (Ca)(a^2) = Ca^3$$

The net flux through the cube is

$$\Phi_{net} = \Phi_L + \Phi_R = 0 + Ca^3 = Ca^3$$

(ii) By Gauss's law, the net charge inside the cube is

$$q_{enc} = \varepsilon_0 \, \Phi_{net} = \varepsilon_0 \, C \, a^3$$

CONCEPT

The left face contributes zero flux because $E=0$ there ($x=0$), not because its normal is $-i$. Compute flux face by face and keep track of the outward normal direction.

Final answer. $\Phi_{net}=Ca^3$ and the net enclosed charge $q=\varepsilon_0\, Ca^3$.

← *Back to Q129*

Q130

Using Gauss's law obtain the expression for the electric field due to a uniformly charged thin spherical shell of radius R at a point outside the shell. Draw a graph showing the variation of electric field with r, for $r > R$ *and* $r < R$.

[2011 • Set 55-1-1]

SOLUTION

Consider a thin uniformly charged spherical shell of radius R carrying total charge Q. By spherical symmetry, the field is radial and its magnitude depends only on the distance r from the centre.

For a point **outside the shell ($r>R$)**, draw a concentric spherical Gaussian surface of radius r. The field is uniform and normal over this surface, and it encloses the full charge Q. The flux is

$$\oint \vec{E}\cdot \mathrm{d}\vec{S} = E(4\pi\, r^2)$$

By Gauss's law,

$$E(4\pi\, r^2) = \frac{Q}{\varepsilon_0}$$

$$E = \frac{1}{4\pi\varepsilon_0}\,\frac{Q}{r^2}$$

This is identical to the field of a point charge Q placed at the centre, directed radially outward for $Q>0$.

For a point **inside the shell (r)**, draw a concentric Gaussian surface of radius **r.** **Since all the charge resides on the surface, the enclosed charge is zero, so**

$$E = 0$$

Graph of *E* versus *r*. Draw the horizontal axis as r and the vertical axis as E. The field is zero from $r=0$ up to $r=R$. At $r=R$ it reaches its maximum value $\mathbf{E_{max}}= \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}$. For $r>R$ the curve falls off as $\frac{1}{r^2}$, tending to zero as $r\to\infty$.

Final answer. $E=\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$ for $r>R$ and $E=0$ for r**; the graph is flat at zero, jumps to a peak at** R**, then falls as** $1/r^2$**.**

← *Back to Q130*

Q131

A thin conducting spherical shell of radius R has charge Q spread uniformly over its surface. Using Gauss's law, derive an expression for an electric field at a point outside the shell.
Draw a graph of electric field $E(r)$ *with distance r from the centre of the shell for* $0 \le r \le \infty$.

[2009]

SOLUTION

Consider a thin conducting spherical shell of radius R carrying total charge Q spread uniformly over its surface. Draw the shell with its centre O, and a concentric spherical Gaussian surface of radius r passing through the point P where the field is required. By the spherical symmetry of the charge distribution, the electric field $\vec{E}$ at every point of this Gaussian surface is radial and has the same magnitude E.

For a point outside the shell, take $r > R$, so the Gaussian sphere encloses the whole charge Q.

Since $\vec{E}$ is parallel to $d\vec{S}$ at every point and E is constant over the surface, the flux is

$\oint \vec{E} \cdot d\vec{S} = E \oint dS = E \times 4\pi r^2$.

By Gauss's law, the total flux equals the enclosed charge divided by ε_0:

$$E \times 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$\Rightarrow E = \frac{Q}{4\pi\varepsilon_0 r^2}.$$

Thus, for a point outside the shell, $\boldsymbol{E = \frac{Q}{4\pi\varepsilon_0 r^2}}$, which is exactly the field of a point charge Q placed at the centre O.

For a point inside the shell ($r < R$), a concentric Gaussian sphere encloses no charge, so $E \times 4\pi r^2 = 0$, giving $E = 0$.

Graph of $E(r)$ versus r: Draw E on the vertical axis and r on the horizontal axis. From $r = 0$ to $r = R$ the curve lies flat along the r-axis ($E = 0$). At $r = R$ the field rises sharply to its maximum surface value $E_{max} = \frac{Q}{4\pi\varepsilon_0 R^2}$. For $r > R$ the curve falls off smoothly as $\frac{1}{r^2}$, approaching the r-axis as $r \to \infty$.

Final answer. For a point outside the shell, $E = \frac{Q}{4\pi\varepsilon_0 r^2}$; the field is zero inside and falls as $1/r^2$ outside.

← *Back to Q131*

Q132

State Gauss's theorem in electrostatics. Apply this theorem to derive an expression for electric field intensity at a point near an infinitely long straight charged wire.

[2007]

SOLUTION

Gauss's theorem states that the net electric flux through any closed surface in vacuum is equal to $\frac{1}{\varepsilon_0}$ times the total charge enclosed by that surface:

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}.$$

To find the field near an infinitely long straight wire of uniform linear charge density λ, draw the wire vertically and a coaxial cylindrical Gaussian surface of radius r (the perpendicular distance of the point from the wire) and length l around it, with two flat circular end caps. By the cylindrical symmetry of the wire, $\vec{E}$ is everywhere radial (perpendicular to the wire) and has the same magnitude at every point of the curved surface.

The flux has three contributions. At the two flat end caps $\vec{E}$ is parallel to the surface, so $\vec{E} \perp d\vec{S}$ and these contribute zero flux. Over the curved surface $\vec{E}$ is parallel to $d\vec{S}$, giving

$$\oint \vec{E} \cdot d\vec{S} = E \times 2\pi r l.$$

The charge enclosed by the cylinder is $q_{enc} = \lambda l$.

Applying Gauss's theorem,

$$E \times 2\pi r l = \frac{\lambda l}{\varepsilon_0}$$

$$\Rightarrow E = \frac{\lambda}{2\pi\varepsilon_0 r}.$$

Hence the electric field near an infinitely long straight charged wire is $E = \frac{\lambda}{2\pi\varepsilon_0 r}$, directed radially outward for $\lambda > 0$ and radially inward for $\lambda < 0$. The field falls off inversely with distance r from the wire.

Final answer. By Gauss's theorem, $E = \frac{\lambda}{2\pi\varepsilon_0 r}$, directed radially from the wire.

← *Back to Q132*

Q133

What is electric flux? Write its S.I. units.
Using Gauss's theorem, deduce an expression for the electric field at a point due to a uniformly charged infinite plane sheet.

[2006]

SOLUTION

Electric flux through a surface is the total number of electric field lines crossing that surface, measured as $\Phi_E = \int_S \vec{E}\cdot d\vec{S}$. It is a scalar quantity. Its SI unit is $N\,m^2\,C^{-1}$ (equivalently $V\,m$).

To find the field due to a uniformly charged infinite plane sheet of surface charge density σ, draw the sheet and a cylindrical Gaussian pillbox that pierces it symmetrically, with its two flat faces (each of area A) parallel to the sheet, one on each side. By symmetry, $\vec{E}$ is perpendicular to the sheet, has equal magnitude on both sides, and points away from the sheet (for $\sigma > 0$).

The curved side of the pillbox contributes no flux because there $\vec{E} \perp d\vec{S}$. Each flat face contributes EA (field parallel to $d\vec{S}$), so the total flux is

$$\oint \vec{E}\cdot d\vec{S} = EA + EA = 2EA.$$

The charge enclosed by the pillbox is $q_{enc} = \sigma A$.

By Gauss's theorem,

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$\Rightarrow E = \frac{\sigma}{2\varepsilon_0}.$$

In vector form, $\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$, where $\hat{n}$ is the unit vector normal to the sheet pointing away from it. The field is uniform and independent of the distance from the sheet.

COMMON MISTAKE

A common slip is writing $E = \frac{\sigma}{\varepsilon_0}$ for the sheet. That answer is for a charged conductor's surface; the standalone infinite sheet has the factor of 2 because field lines emerge from both faces.

Final answer. Electric flux is $\Phi_E = \int_S \vec{E}\cdot d\vec{S}$ in $N\,m^2\,C^{-1}$; the infinite sheet gives $E = \frac{\sigma}{2\varepsilon_0}$.

← Back to Q133

Q134

An electric dipole is held in a uniform electric field. ***(i)*** *Using a suitable diagram, show that it does not undergo any translatory motion, and* ***(ii)*** *derive an expression for the torque acting on it and specify its direction.*

[2005]

SOLUTION

Consider an electric dipole, charges $+q$ and $-q$ separated by distance 2a, placed in a uniform electric field $\vec{E}$ with its axis making an angle θ with $\vec{E}$. Draw the dipole as a short rod with $+q$ at one end and $-q$ at the other; show $\vec{E}$ as parallel horizontal field lines; draw the force $+q\vec{E}$ on the positive charge (along $\vec{E}$) and the force $q\vec{E}$ on the negative charge (opposite to $\vec{E}$).

(i) The force on the positive charge is $\vec{F}_+ = +q\vec{E}$ and the force on the negative charge is $\vec{F}_- = -q\vec{E}$. Their vector sum is

$\vec{F}_{net} = q\vec{E} + (-q\vec{E}) = \vec{0}$.

Since the field is uniform, these forces are equal in magnitude and opposite in direction, so the net force is zero. Hence the dipole experiences no translatory (linear) motion.

(ii) Although the net force is zero, the two equal and opposite forces act along different lines, forming a couple that produces a torque. The magnitude of the torque equals one force times the perpendicular distance between the two lines of action:

$$\tau = (\text{force}) \times (\text{perpendicular distance})$$

$$\tau = qE \times (2a\theta)$$

$\tau = (q \times 2a)\, E\theta$.

Since the dipole moment is $p = q \times 2a$,

$\tau = pE\theta$.

In vector form, $\vec{\tau} = \vec{p} \times \vec{E}$.

Direction: $\vec{\tau}$ is perpendicular to the plane containing $\vec{p}$ and $\vec{E}$, given by the right-hand rule. This torque tends to rotate the dipole so as to align $\vec{p}$ along $\vec{E}$.

Final answer. Net force is zero (no translation); torque is $\vec{\tau} = \vec{p} \times \vec{E}$ with magnitude $\tau = pE\theta$.

← *Back to Q134*

Q135

State Gauss' theorem in electrostatics. Using this theorem, derive the expression for the electric field intensity at any point outside a uniformly charged thin spherical shell.

[2004]

SOLUTION

Gauss's theorem states that the net electric flux through any closed surface in vacuum is equal to $\frac{1}{\varepsilon_0}$ times the total charge enclosed by it:

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}.$$

To find the field at a point outside a uniformly charged thin spherical shell of radius R carrying total charge Q, draw the shell with centre O and a concentric spherical Gaussian surface of radius $r > R$ passing through the point P where the field is wanted. By spherical symmetry, $\vec{E}$ is radial and has the same magnitude E at every point of the Gaussian surface.

Because $\vec{E}$ is parallel to $d\vec{S}$ everywhere and E is constant over the surface, the flux is

$$\oint \vec{E} \cdot d\vec{S} = E \oint dS = E \times 4\pi r^2.$$

The Gaussian sphere encloses the entire charge Q of the shell, so $q_{enc} = Q$.

Applying Gauss's theorem,

$$E \times 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$\Rightarrow E = \frac{Q}{4\pi\varepsilon_0 r^2}.$$

Hence the field at any point outside the uniformly charged thin spherical shell is $E = \frac{Q}{4\pi\varepsilon_0 r^2}$, directed radially. This is identical to the field of a point charge Q

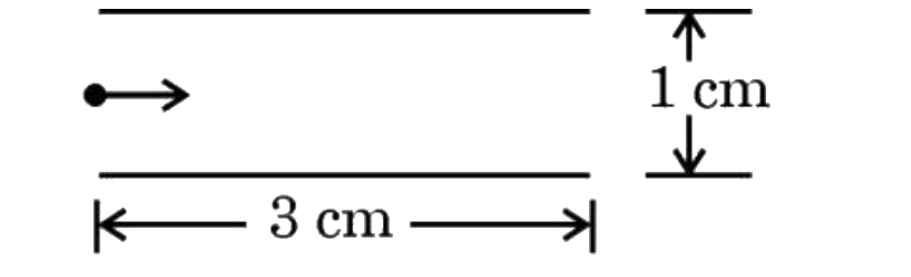
placed at the centre of the shell.

Final answer. By Gauss's theorem, the field outside the shell is $E = \frac{Q}{4\pi\varepsilon_0 r^2}$, as if all charge were at the centre.

← *Back to Q135*

Q136

A beam of electrons moving horizontally with a velocity of 3×10^7 *m/s enters a region between two plates as shown in the figure. A suitable potential difference is applied across the plates such that the electron beam just strikes the edge of the lower plate.*



Answer the following questions based on the above:

(a) How long does an electron take to strike the edge?

(b) What is the shape of the path followed by the electron and why?

(c) Find the potential difference applied.

OR

(c) Find the magnitude and direction of the magnetic field which should be created in the space between the plates so that the electron beam goes straight undeviated.

[2023 • Set 55-3-1]

SOLUTION

The electron enters horizontally with speed $v_x = 3\times 10^7\ m\ s^{-1}$ along the central axis between the plates. From the figure, the horizontal length of the plates is $L = 3\ cm = 0.03\ m$ and the plate separation is $d = 1\ cm = 0.01\ m$. Between the plates the only force is the vertical electric force, so the horizontal velocity stays v_x

while the electron is deflected vertically. To just strike the edge of the lower plate, an electron entering on the axis must deflect by $y = \frac{d}{2} = 0.005\ m$.

(a) The horizontal motion is uniform, so the time to traverse the plates is

$$t = \frac{L}{v_x} = \frac{0.03}{3\times 10^{7}}$$

$t = 1\times 10^{-9}\ s = 1\ \text{ns}$.

(b) Inside the plates the electron has constant horizontal velocity v_x (so $x = v_x t$) and constant vertical acceleration $a_y = \frac{eE}{m}$ due to the uniform field (so $y = \frac{1}{2} a_y t^2$). Eliminating t gives $y = \frac{a_y}{2v_x^2} x^2$, i.e. $y \propto x^2$. Hence the path is a **parabola**, exactly like projectile motion under gravity.

(c) The vertical deflection gives the acceleration:

$$y = \frac{1}{2} a_y t^2 \Rightarrow a_y = \frac{2y}{t^2} = \frac{2(0.005)}{(10^{-9})^2} = 1\times 10^{16}\ m\ s^{-2}.$$

The field needed is

$$E = \frac{m_e a_y}{e} = \frac{(9.1\times 10^{-31})(10^{16})}{1.6\times 10^{-19}} = 5.69\times 10^{4}\ V\ m^{-1}.$$

The potential difference across the plates is

$V = E\,d = (5.69\times 10^{4})(0.01) = 5.69\times 10^{2}\ V \approx 569\ V$.

OR (c) For the beam to go straight, the magnetic force must balance the electric force: $evB = eE$, so $B = \frac{E}{v} = \frac{5.69\times 10^{4}}{3\times 10^{7}} = 1.9\times 10^{-3}\ T$. The magnetic field must be perpendicular to both $\vec{v}$ and $\vec{E}$, directed so that the magnetic force on the electron exactly opposes the electric force (i.e. horizontal and perpendicular to the beam, pointing out of the plane for the given geometry).

COMMON MISTAKE

Always halve the plate gap. The electron enters on the central axis, so it only needs to fall d/2, not the full d, to graze the lower plate. Using the full d doubles the acceleration and the voltage.

Final answer. (a) $t = 1\times 10^{-9}\ s$; (b) parabolic path; (c) $V \approx 569\ V$ (OR $B \approx 1.9\times 10^{-3}\ T$ perpendicular to $\vec{v}$ and $\vec{E}$).

← Back to Q136

Q137

(a) *An electric dipole consists of two point charges q and −q separated by a distance* 2a. *Derive an expression for the electric field* $\vec{E}$ *due to this dipole at a point distant r from the centre of the dipole on the equatorial plane. Write the expression for the electric field at a far off point, i.e.* $\boldsymbol{r >> a}$.

(b) *A dipole is placed in x-y plane such that charges q and −q are located at* $\boldsymbol{x = a}$ *and* $\boldsymbol{x = b}$ *respectively. There exists an electric field* $\vec{\boldsymbol{E}} = \mathbf{2x}\,\hat{\boldsymbol{i}}$ *N/C in the region. Calculate the force* $\vec{F}$ *and torque* $\vec{\tau}$ *experienced by the dipole.*

OR

(a) *Two cells of emf* E_1 *and* E_2 *with internal resistances* r_1 *and* r_2 *respectively, are connected in parallel by connecting their positive terminals together and negative terminals together. Deduce an expression for equivalent emf and equivalent internal resistance of the combination.*

(b) *A parallel combination, as stated in (a) above, of two cells of emfs E and* 3E *and internal resistances R each is connected across a resistance* 2R. *Find the current that flows through resistance* 2R.

[2026 • Set 55-1-1]

SOLUTION

(a) Consider a dipole of charges $+q$ and $-q$ separated by 2a. Let P lie on the equatorial plane at distance r from the centre O. Each charge is at distance $\sqrt{r^2+a^2}$ from P, so each produces a field of magnitude

$$E_+ = E_- = \frac{1}{4\pi\varepsilon_0}\,\frac{q}{(r^2+a^2)}.$$

Resolve each field into a component along the dipole axis and one perpendicular to it. The perpendicular components are equal and opposite, so they cancel. The axial components add. Each axial component is $E_{\pm}\,\theta$ where $\theta = \frac{a}{\sqrt{r^2+a^2}}$.

$$E_{eq} = 2E_+\theta = 2\cdot\frac{1}{4\pi\varepsilon_0}\,\frac{q}{(r^2+a^2)}\cdot\frac{a}{\sqrt{r^2+a^2}}$$

$E_{eq} = \frac{1}{4\pi\varepsilon_0}\,\frac{q(2a)}{(r^2+a^2)^{3/2}} = \frac{1}{4\pi\varepsilon_0}\,\frac{p}{(r^2+a^2)^{3/2}}$, where $p = q(2a)$.

The direction of $\vec{E}_{eq}$ is antiparallel to $\vec{p}$, so in vector form $\vec{E}_{eq} = -\frac{1}{4\pi\varepsilon_0}\,\frac{\vec{p}}{(r^2+a^2)^{3/2}}$.

For a far-off point with $r \gg a$, $(r^2+a^2)^{3/2} \approx r^3$, so

$\vec{E}_{eq} = -\frac{1}{4\pi\varepsilon_0}\,\frac{\vec{p}}{r^3}$.

(b) The dipole has $+q$ at $x = a$ and $-q$ at $x = b$, in the field $\vec{E} = 2x\,\hat{i}$ N/C. The force on each charge is $q\vec{E}$ evaluated at its position.

$$\vec{F}_+ = (+q)(2a)\hat{i} = 2qa\,\hat{i}$$

$$\vec{F}_- = (-q)(2b)\hat{i} = -2qb\,\hat{i}$$

$\vec{F} = \vec{F}_+ + \vec{F}_- = 2q(a-b)\,\hat{i}\,N$.

Both forces and both position vectors lie along $\hat{i}$, so each torque term involves $\hat{i}\times\hat{i} = \vec{0}$. Hence

$\vec{\tau} = \vec{0}$.

So $\vec{F} = 2q(a-b)\,\hat{i}\,\mathrm{N}$ and $\vec{\tau} = \vec{0}$ (the dipole lies along the field, giving zero torque but a net force because the field is non-uniform).

Final answer. Equatorial field: $\vec{E}_{eq} = -\frac{\vec{p}}{4\pi\varepsilon_0(r^2+a^2)^{3/2}} \to -\frac{\vec{p}}{4\pi\varepsilon_0 r^3}$ for $r \gg a$; and $\vec{F} = 2q(a-b)\,\hat{i}$ N, $\vec{\tau} = \vec{0}$.

← *Back to Q137*

Q138

(i) Two infinitely long straight wires having linear charge densities $-\lambda$ and 3λ are held vertically parallel to each other, distance r apart in free space. Find the nature and magnitude of the force/length exerted by one wire on the other.

[2026 • Set 55-3-1]

SOLUTION

Two infinitely long parallel straight wires carry linear charge densities $\lambda_1 = -\lambda$ and $\lambda_2 = 3\lambda$ and are held a distance r apart in free space. We find the force per unit length one exerts on the other.

The field at perpendicular distance r from an infinitely long wire of linear charge density λ_1 is

$$E_1 = \frac{\lambda_1}{2\pi\varepsilon_0 r}.$$

The second wire (density λ_2) sits in this field, so the force per unit length on it is

$$f = \lambda_2 E_1 = \frac{\lambda_1 \lambda_2}{2\pi\varepsilon_0 r}.$$

Substituting the magnitudes $|\lambda_1| = \lambda$ and $|\lambda_2| = 3\lambda$,

$$|f| = \frac{(\lambda)(3\lambda)}{2\pi\varepsilon_0 r}$$

$$|f| = \frac{3\lambda^2}{2\pi\varepsilon_0 r}.$$

The two charge densities have opposite signs ($-\lambda$ and $+3\lambda$), so the wires attract each other.

Hence the force per unit length is $\dfrac{3\lambda^2}{2\pi\varepsilon_0 r}$ and its nature is **attractive**.

Final answer. The force per unit length is $\dfrac{3\lambda^2}{2\pi\varepsilon_0 r}$ and it is attractive (opposite signs).

← *Back to Q138*

Q139

(ii) A small hollow conducting sphere of radius r_1 *is given a charge Q. It is surrounded by a concentric conducting spherical shell of inner radius* r_2 *and outer radius* r_3*, having charge* $-3Q$.
If a point charge $2q$ *were kept at the centre, find:*
(I) the electric flux through a concentric spherical Gaussian surface of radius x for (1) $x < r_1$*, and (2)* $r_1 < x < r_2$.

(II) electric field at a point distant x from the centre for (1) $x > r_3$*, and (2)* $r_1 < x < r_2$*.*
(III) surface charge density on the inner surface of (1) sphere, and (2) shell.

[2026 • Set 55-3-1]

SOLUTION

There is an inner hollow conducting sphere of radius r_1 carrying charge Q, surrounded by a concentric conducting shell (inner radius r_2, outer radius r_3) carrying charge $-3Q$, with a point charge $2q$ placed at the common centre. Because the electric field inside the body of any conductor must be zero, charges are induced on each surface so that a Gaussian surface drawn inside the metal encloses zero net charge.

On the inner surface of the sphere (at r_1^-): the central $2q$ induces $-2q$ there. Since the total on the sphere is Q, its outer surface (at r_1^+) carries $Q + 2q$. On the inner surface of the shell (at r_2^-): it must cancel the enclosed $2q + (Q+2q)$ field within the shell metal, so it carries $-(Q+2q)$. The shell total is $-3Q$, so its outer surface (at r_3) carries $\mathbf{-3Q + (Q+2q) = -2Q + 2q}$.

(I) Flux through a concentric Gaussian sphere of radius x, by Gauss's law $\Phi = \frac{q_{enc}}{\varepsilon_0}$:

For $x < r_1$ (only the central charge enclosed): $\Phi = \frac{2q}{\varepsilon_0}$.

For $r_1 < x < r_2$ (central charge plus the sphere's charge): $\Phi = \frac{2q + Q}{\varepsilon_0}$.

(II) Electric field at distance x (radial, $E = \frac{q_{enc}}{4\pi\varepsilon_0 x^2}$):

For $x > r_3$, total enclosed $= 2q + Q + (-3Q) = 2q - 2Q$, so

$$E = \frac{2q - 2Q}{4\pi\varepsilon_0 x^2} = \frac{q - Q}{2\pi\varepsilon_0 x^2}.$$

For $r_1 < x < r_2$, enclosed $= 2q + Q$, so

$$E = \frac{Q + 2q}{4\pi\varepsilon_0 x^2}.$$

(III) Surface charge densities on the inner surfaces (charge divided by area):

Inner surface of the sphere (area $4\pi r_1^2$, charge $-2q$): $\sigma = \frac{-2q}{4\pi r_1^2}$.

Inner surface of the shell (area $4\pi r_2^2$, charge $-(Q+2q)$): $\sigma = \frac{-(Q+2q)}{4\pi r_2^2}$.

CONCEPT

Flux through a Gaussian surface depends only on the charge enclosed, not on charges outside it. So for **x2** the shell's charge never enters the flux, even though it sits close by.

Final answer. Flux: $\frac{2q}{\varepsilon_0}$ and $\frac{2q+Q}{\varepsilon_0}$; fields: $\frac{q-Q}{2\pi\varepsilon_0 x^2}$ and $\frac{Q+2q}{4\pi\varepsilon_0 x^2}$; inner densities: $\frac{-2q}{4\pi r_1^2}$ and $\frac{-(Q+2q)}{4\pi r_2^2}$.

← *Back to Q139*

Q140

(i) What is difference between an open surface and a closed surface? Draw elementary surface vector $\vec{dS}$ *for a spherical surface S.*
(ii) Define electric flux through a surface. Give the significance of a Gaussian surface. A charge outside a Gaussian surface does not contribute to total electric flux through the surface. Why?
(iii) A small spherical shell S_1 *has point charges* $q_1 = -3\ \mu C$, $q_2 = -2\ \mu C$ *and* $q_3 = 9\ \mu C$ *inside it. This shell is enclosed by another big spherical shell* S_2*. A point charge Q is placed in between the two surfaces* S_1 *and* S_2*. If the electric flux through the surface* S_2 *is four times the flux through surface* S_1*, find charge Q.*

[2025 • Set 55-2-1]

SOLUTION

(i) A closed surface completely encloses a volume of space and has no boundary edge (for example, the surface of a sphere). An open surface has a boundary curve and does not enclose any 3-D region (for example, a flat disc or a hemispherical cap). For a spherical surface S of radius R, the elementary surface vector is drawn pointing radially outward at each point: $\vec{dS} = dS\,\hat{n}$, where $\hat{n} = \hat{r}$ is the outward radial unit vector. Picture small patches all over the sphere, each with a tiny arrow $\vec{dS}$ sticking straight out from the centre.

(ii) Electric flux through a surface is the surface integral of the electric field over it: $\Phi_E = \int_S \vec{E} \cdot \vec{dS}$, i.e. the net number of field lines crossing the surface. A Gaussian surface is an imaginary closed surface chosen with the symmetry of

the charge distribution so that Gauss's law can be used to find $\vec{E}$ easily. A charge lying outside a Gaussian surface contributes zero net flux because every field line from it that enters the surface on one side must leave it on the other; the inward flux exactly cancels the outward flux, giving zero net flux.

(iii) The charge enclosed by S_1 is

$$\mathbf{q_{enc}(S_1) = q_1 + q_2 + q_3 = (-3) + (-2) + 9 = 4\ \mu C.}$$

The charge enclosed by S_2 includes these plus the charge Q placed between the surfaces:

$$\mathbf{q_{enc}(S_2) = 4\ \mu C + Q.}$$

By Gauss's law $\Phi = \frac{q_{enc}}{\varepsilon_0}$, the condition $\Phi_{S_2} = 4\,\Phi_{S_1}$ gives

$$\frac{4\ \mu C + Q}{\varepsilon_0} = 4 \times \frac{4\ \mu C}{\varepsilon_0}$$

$$4\ \mu C + Q = 16\ \mu C$$

$$Q = 16 - 4 = 12\ \mu C.$$

Hence the charge placed between the two surfaces is $\boldsymbol{Q = 12\ \mu C}$.

Final answer. $Q = 12\ \mu C$.

← *Back to Q140*

Q141

(a)

(i) Two point charges +q and −q are held at $(a, 0)$ *and* $(-a, 0)$ *in x-y plane. Obtain an expression for the net electric field due to the charges at a point* $(0, y)$*. Hence, find electric field at a far off point* $(y \gg a)$*.*

(ii) Three point charges of −2 nC, −1 nC, *and* +5 nC *are kept at the vertices A, B and C of an equilateral triangle of side* 0.2 *m. Find the total amount of work done in shifting the charges from A to* A_1, *B to* B_1 *and C to* C_1*. Here* A_1, B_1 *and* C_1 *are the midpoints of sides* AB, BC *and* CA, *respectively.*

OR

(b)

(i) Show that Gauss's theorem is consistent with Coulomb's law. Using it, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius r at a point at a distance y from the centre of the shell such that (I) $\mathbf{y > r}$*, and (II)* $\mathbf{y < r}$.

(ii) A point charge of +2 nC *is kept at the origin of a three-dimensional coordinate system. Find the type and magnitude of the charge which should be kept at* (0, 0, −6 *m*) *so that the potential due to the system becomes zero at* (0, 0, 2 *m*).

[2025 • Set 55-4-1]

SOLUTION

(a)(i) Net field at (0,*y*) due to +*q* at (*a*,0) and −*q* at (−*a*,0).

The distance of the point (0,*y*) from either charge is the same:

$$d = \sqrt{a^2+y^2}$$

So the magnitude of the field due to each charge is equal:

$$E_+ = E_- = \frac{1}{4\pi\varepsilon_0}\ \frac{q}{a^2+y^2}$$

The field $\vec{E}_+$ due to +*q* points away from +*q* (from (*a*,0) towards (0,*y*)), and $\vec{E}_-$ due to −*q* points towards −*q* (from (0,*y*) towards (−*a*,0)). When we resolve these two equal vectors, the *y*-components are equal and opposite and cancel, while the *x*-components are equal and both point along $-\hat{i}$, so they add.

Each *x*-component carries the geometric factor $\frac{a}{\sqrt{a^2+y^2}}$:

$$E_x = 2E_+\theta = 2\cdot\frac{1}{4\pi\varepsilon_0}\ \frac{q}{a^2+y^2}\cdot\frac{a}{\sqrt{a^2+y^2}}$$

$$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\ \frac{2qa}{(a^2+y^2)^{3/2}}\ \hat{i}$$

Writing the dipole moment $\boldsymbol{p = q(2a) = 2qa}$:

$$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\ \frac{p}{(a^2+y^2)^{3/2}}\ \hat{i}$$

Far-off point ($\boldsymbol{y \gg a}$): here $a^2+y^2 \approx y^2$, so

$$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\ \frac{p}{y^3}\ \hat{i} = -\frac{\vec{p}}{4\pi\varepsilon_0\, y^3}$$

This is the standard equatorial field of a dipole, antiparallel to $\vec{p}$ and falling off as $1/y^3$.

(a)(ii) Work to shift the three charges to the midpoints.

The original triangle ABC has side $L = 0.20\ m$. The midpoints A_1, B_1, C_1 of its sides form a smaller equilateral triangle of side $L/2 = 0.10\ m$. The work done equals the change in the total electrostatic potential energy of the system.

Charges: $q_A = -2\ nC$, $q_B = -1\ nC$, $q_C = +5\ nC$, with $k = 9\times 10^9\ N\,m^2C^{-2}$.

The sum of products of charge pairs is

$$\Sigma = q_A q_B + q_B q_C + q_C q_A = (-2)(-1)+(-1)(5)+(5)(-2)$$

$$\Sigma = 2 - 5 - 10 = -13 \text{ (in } 10^{-18}\ C^2)$$

Initial energy (separation $L = 0.20\ m$):

$$U_i = \frac{k}{L}\,\Sigma = \frac{9\times 10^9}{0.20}\times(-13\times 10^{-18})$$

$$U_i = (4.5\times 10^{10})(-13\times 10^{-18}) = -5.85\times 10^{-7}\ J$$

Final energy (separation $L/2 = 0.10\ m$, so the factor doubles):

$$U_f = \frac{k}{L/2}\,\Sigma = 2U_i = -1.17\times 10^{-6}\ J$$

Work done by the external agent:

$$W = U_f - U_i = -1.17\times 10^{-6} - (-5.85\times 10^{-7})$$

$$W = -1.17\times 10^{-6} + 0.585\times 10^{-6} = -5.85\times 10^{-7}\ J$$

The total work done in shifting the charges is $W = -5.85\times 10^{-7}\ J$. The negative sign means the configuration loses energy, so the field itself does positive work and no external work is needed to make the shift happen.

> **COMMON MISTAKE**
>
> *Work done* $= U_f - U_i$, *not just* U_f. *A common slip is to compute only the final energy and forget to subtract the initial energy already stored in triangle* ABC.

Final answer. Equatorial field $\vec{E} = -\frac{\vec{p}}{4\pi\varepsilon_0 y^3}$ for $y \gg a$; total work done $W = -5.85 \times 10^{-7}\ J$.

← Back to Q141

Q142

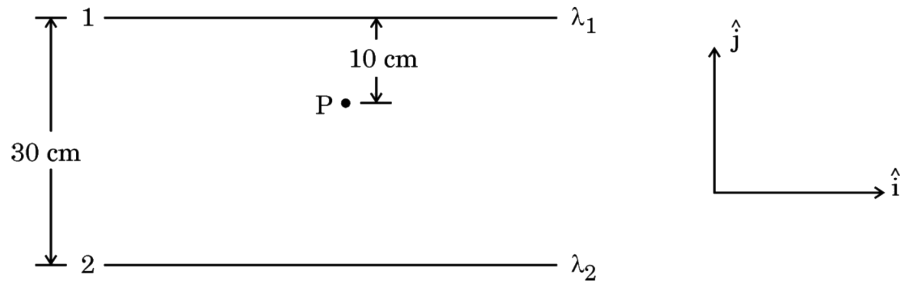
(a) (i) Obtain an expression for the electric potential due to a small dipole of dipole moment p, at a point r from its centre, for much larger distances compared to the size of the dipole.

(ii) Three point charges q, 2q and nq are placed at the vertices of an equilateral triangle. If the potential energy of the system is zero, find the value of n.

OR

(b) (i) State Gauss's Law in electrostatics. Apply this to obtain the electric field E at a point near a uniformly charged infinite plane sheet.

(ii) Two long straight wires 1 and 2 are kept as shown in the figure. The linear charge density of the two wires are $\lambda_1 = 10\ \mu C/m$ *and* $\lambda_2 = -20\ \mu C/m$*. Find the net force F experienced by an electron held at point P.*



[2024 • Set 55-2-1]

SOLUTION

(a)(i) Potential due to a short dipole at a point $\vec{r}$.

Take a dipole with $-q$ at A and $+q$ at B, separated by 2a, centre O. Let P be the point at distance r from O with $\angle$ **POB** = **θ**. Let $\mathbf{r_1}$ = **AP** and $\mathbf{r_2}$ = **BP**.

The potential at P is the algebraic sum of the two point-charge potentials:

$$V_P = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_2} - \frac{q}{r_1}\right)$$

$$V_P = \frac{q}{4\pi\varepsilon_0}\,\frac{r_1 - r_2}{r_1\, r_2}$$

For $r \gg a$, dropping perpendiculars from A and B onto OP gives the approximations

$$r_1 \approx r + a\theta,\ r_2 \approx r - a\theta$$

so that

$$r_1 - r_2 \approx 2a\theta,\ r_1\, r_2 \approx r^2$$

Substituting,

$$V_P = \frac{q}{4\pi\varepsilon_0}\,\frac{2a\theta}{r^2}$$

Using the dipole moment $\boldsymbol{p = q(2a)}$,

$$V_P = \frac{1}{4\pi\varepsilon_0}\,\frac{p\theta}{r^2} = \frac{1}{4\pi\varepsilon_0}\,\frac{\vec{p}\cdot\hat{r}}{r^2}$$

This is the required expression. Note V falls as $1/r^2$ (faster than the $1/r$ of a point charge), and $\boldsymbol{V = 0}$ on the equatorial plane where $\boldsymbol{\theta = 90°}$.

(a)(ii) Value of *n* for zero potential energy.

Charges q, 2q, nq sit at the vertices of an equilateral triangle of side a. The total energy is the sum of the three pair energies, all at the same separation a, with $\boldsymbol{k = \frac{1}{4\pi\varepsilon_0}}$:

$$U = \frac{k}{a}[(q)(2q) + (2q)(nq) + (nq)(q)]$$

$$U = \frac{kq^2}{a}\,(2 + 2n + n)$$

$$U = \frac{kq^2}{a}\,(2 + 3n)$$

Setting $\boldsymbol{U = 0}$ (and $q \neq 0$):

$$2 + 3n = 0$$

$$n = -\frac{2}{3}$$

Hence $n = -\frac{2}{3}$.

(b)(i) Gauss's law and field of an infinite plane sheet.

Gauss's law: the net electric flux through any closed surface equals $\frac{1}{\varepsilon_0}$ times the total charge enclosed:

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}$$

For an infinite plane sheet of uniform surface charge density σ, by symmetry $\vec{E}$ is perpendicular to the sheet, equal in magnitude on both sides, and points away from a positive sheet. Choose a Gaussian pillbox: a cylinder of cross-section A piercing the sheet, with its two flat faces parallel to the sheet, one on each side.

The curved side contributes no flux ($\vec{E} \perp d\vec{S}$). Each flat face contributes EA, so total flux = **2EA**. Enclosed charge = **σ *A***.

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$$

The field is uniform and independent of distance from the sheet.

(b)(ii) Net force on an electron at P from the two wires.

The field of a long charged wire at perpendicular distance r is $\boldsymbol{E = \frac{\lambda}{2\pi\varepsilon_0 r}} = \frac{\mathbf{2k\lambda}}{\boldsymbol{r}}$. From the figure, point P is at perpendicular distance r_1 from wire 1 and r_2 from wire 2 (commonly $\mathbf{r_1 = r_2 = 0.5\ m}$ in the standard version of this problem). Taking $\mathbf{r_1 = r_2 = 0.5\ m}$:

Field due to wire 1 ($\boldsymbol{\lambda_1 = +10\ \mu C/m}$), directed away from wire 1:

$$E_1 = \frac{2(9\times 10^9)(10\times 10^{-6})}{0.5} = 3.6\times 10^5\ N\,C^{-1}$$

Field due to wire 2 ($\boldsymbol{\lambda_2 = -20\ \mu C/m}$), directed towards wire 2:

$$E_2 = \frac{2(9\times 10^9)(20\times 10^{-6})}{0.5} = 7.2\times 10^5\ N\,C^{-1}$$

Since wire 1 is positive (field points away from it) and wire 2 is negative (field points towards it), at the midpoint P both fields point in the same direction (from wire 1 towards wire 2), so they add:

$$E_{net} = E_1 + E_2 = 3.6\times 10^5 + 7.2\times 10^5 = 1.08\times 10^6\ N\,C^{-1}$$

Force on the electron (charge magnitude $e = 1.6\times 10^{-19}\ C$):

$$F = eE_{net} = (1.6\times 10^{-19})(1.08\times 10^6)$$

$$F = 1.728\times 10^{-13}\ N$$

The electron, being negative, feels a force opposite to $\vec{E}_{net}$, i.e. directed from wire 2 towards wire 1.

The net force on the electron is $F \approx 1.73\times 10^{-13}\ N$, directed from the negative wire towards the positive wire.

Final answer. $V_P = \dfrac{p\theta}{4\pi\varepsilon_0 r^2}$ and $n = -\dfrac{2}{3}$; OR $\vec{E} = \dfrac{\sigma}{2\varepsilon_0}\hat{n}$ and net force on electron $\approx 1.73\times 10^{-13}\ N$.

← Back to Q142

Q143

(i) A charge +Q is placed on a thin conducting spherical shell of radius R. Use Gauss's theorem to derive an expression for the electric field at a point lying (i) inside and (ii) outside the shell.
(ii) Show that the electric field for same charge density (σ) is twice in case of a conducting plate or surface than in a nonconducting sheet.

[2024 • Set 55-3-1]

SOLUTION

(i) Field inside and outside a charged conducting spherical shell of charge +Q.

By the spherical symmetry of the shell, the field $\vec{E}$ at any point is radial and its magnitude depends only on the distance r from the centre. To find it at distance r, draw a concentric spherical Gaussian surface of radius r; over it $\vec{E}$ is uniform in magnitude and parallel to $d\vec{S}$, so

$$\oint \vec{E} \cdot d\vec{S} = E \cdot 4\pi r^2$$

(a) Inside the shell ($r < R$): all the charge $+Q$ resides on the outer surface of the conductor, so the Gaussian surface encloses no charge.

$$E \cdot 4\pi r^2 = \frac{0}{\varepsilon_0} = 0$$

$$E_{in} = 0$$

The interior of the shell is field-free.

(b) Outside the shell ($r > R$): the Gaussian surface now encloses the full charge Q.

$$E \cdot 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$E_{out} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}$$

The field outside is the same as if the whole charge were concentrated as a point charge at the centre.

(ii) Field of a conducting surface is twice that of a non-conducting sheet for the same σ.

Non-conducting (thin) sheet: charge σ lies on the sheet and the field comes out on both faces. Using a pillbox of cross-section A piercing the sheet, flux emerges through both flat faces, giving total flux 2EA, while the enclosed charge is σA:

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$E_{sheet} = \frac{\sigma}{2\varepsilon_0}$$

Conducting plate/surface: inside the conductor $\vec{E} = 0$, and the charge σ lies on the surface. Take a pillbox with one flat face just outside the conductor and the

other inside the metal. The inside face contributes zero flux (field is zero there), and the curved side contributes nothing, so the flux comes only through the single outer face:

$$EA = \frac{\sigma A}{\varepsilon_0}$$

$$E_{cond} = \frac{\sigma}{\varepsilon_0}$$

Comparing the two results,

$$\frac{E_{cond}}{E_{sheet}} = \frac{\sigma/\varepsilon_0}{\sigma/2\varepsilon_0} = 2$$

Hence the field just outside a charged conducting surface is twice that just outside a non-conducting sheet of the same surface charge density σ.
Physically, for the conductor the field exists only on one side (it is zero inside the metal), so all the flux from σ goes out through that one face, doubling the field.

CONCEPT

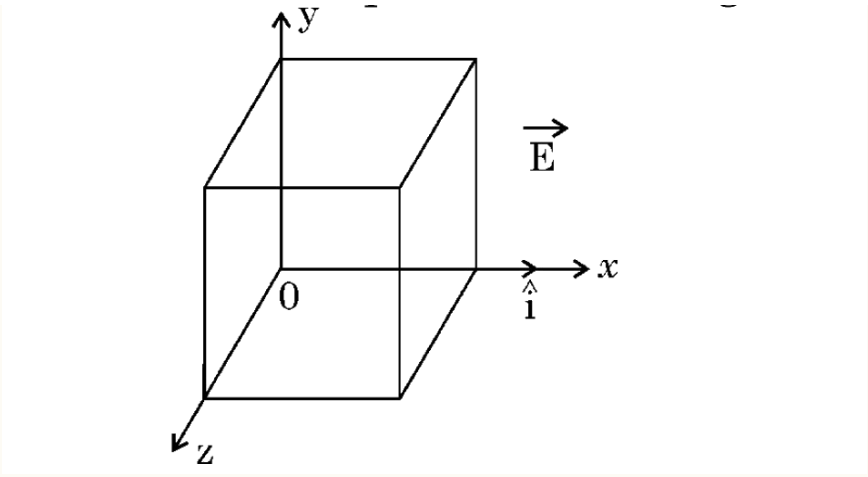
The factor of 2 comes purely from how many faces the flux escapes through: both faces for a thin sheet (2EA), one face for a conductor (EA), because the field is zero inside the metal.

Final answer. $E_{in}=0$, $E_{out}=\frac{Q}{4\pi\varepsilon_0 r^2}$; and $E_{cond}=\frac{\sigma}{\varepsilon_0}=2E_{sheet}$.

← *Back to Q143*

Q144

(i) Using Gauss's law, show that the electric field $\vec{E}$ *at a point due to a uniformly charged infinite plane sheet is given by* $\vec{E}=\frac{\sigma}{2\varepsilon_0}\hat{n}$ *where symbols have their usual meanings.*
(ii) Electric field $\vec{E}$ *in a region is given by* $\vec{E}=(5x^2+2)\hat{i}$*, where E is in N/C and x is in meters. A cube of side* 10 cm *is placed in the region as shown in figure. Calculate (1) the electric flux through the cube, and (2) the net charge enclosed by the cube.*



[2024 • Set 55-4-1]

SOLUTION

(i) Field due to a uniformly charged infinite plane sheet.

Let the sheet carry uniform surface charge density σ. By symmetry the field is everywhere perpendicular to the sheet, has equal magnitude on the two sides, and points away from a positively charged sheet.

Choose a Gaussian pillbox: a right cylinder of cross-section area A with its two flat faces parallel to the sheet, one on each side, the sheet passing through its middle.

The curved surface contributes no flux because $\vec{E}$ is parallel to it ($\vec{E} \perp d\vec{S}$ there). Each flat face contributes EA, so the total outward flux is

$$\Phi = EA + EA = 2EA$$

The charge enclosed by the pillbox is

$$q_{enc} = \sigma A$$

By Gauss's law $\Phi = \frac{q_{enc}}{\varepsilon_0}$:

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}$$

In vector form, with $\hat{n}$ the unit normal pointing away from the sheet,

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\,\hat{n}$$

The field is uniform and independent of distance from the sheet.

(ii) Cube in $\vec{E} = (5x^2+2)\hat{i}\ N\,C^{-1}$, side $L = 10$ cm $= 0.10\ m$.

Take the cube aligned with the axes, with its left face at $x = 0$ and right face at $x = L = 0.10\ m$. Each face has area

$$A = L^2 = (0.10)^2 = 10^{-2}\ m^2$$

Since $\vec{E}$ is along $\hat{i}$, the four faces whose normals are along $\pm\hat{j}$ and $\pm\hat{k}$ give zero flux. Only the left and right faces contribute.

At the left face ($x = 0$), outward normal $-\hat{i}$:

$$E_x = 5(0)^2 + 2 = 2\ N\,C^{-1}$$

$$\Phi_L = -E_x A = -(2)(10^{-2}) = -2\times 10^{-2}\ N\,m^2C^{-1}$$

At the right face ($x = L = 0.10\ m$), outward normal $+\hat{i}$:

$$E_x = 5(0.10)^2 + 2 = 5(0.01) + 2 = 2.05\ N\,C^{-1}$$

$$\Phi_R = +E_x A = (2.05)(10^{-2}) = 2.05\times 10^{-2}\ N\,m^2C^{-1}$$

(1) Net flux through the cube:

$$\Phi_{net} = \Phi_R + \Phi_L = 2.05\times 10^{-2} - 2\times 10^{-2}$$

$$\Phi_{net} = 0.05\times 10^{-2} = 5\times 10^{-4}\ N\,m^2C^{-1}$$

(2) Net charge enclosed, by Gauss's law $q_{enc} = \varepsilon_0\,\Phi_{net}$ with $\varepsilon_0 = 8.85\times 10^{-12}\ C^2N^{-1}m^{-2}$:

$$q_{enc} = (8.85\times 10^{-12})(5\times 10^{-4})$$

$$q_{enc} = 4.425\times 10^{-15}\ C$$

The net flux through the cube is $5\times 10^{-4}\ N\,m^2C^{-1}$ and the enclosed charge is $\approx 4.43\times 10^{-15}\ C$.

COMMON MISTAKE

Only the two faces perpendicular to $\hat{i}$ matter here, and you must evaluate E at each face's own x. Forgetting to convert L**=10 cm** *to* 0.10 *m before squaring is the usual error.*

Final answer. Net flux $=5\times 10^{-4}\ N\,m^2C^{-1}$ and enclosed charge $\approx 4.43\times 10^{-15}\ C$.

← Back to Q144

Q145

(i) A thin spherical shell of radius R has a uniform surface charge density σ. Using Gauss' law, deduce an expression for electric field (i) outside and (ii) inside the shell. (ii) Two long straight thin wires AB and CD have linear charge densities 10 μC/*m and* −20 μC/*m, respectively. They are kept parallel to each other at a distance* 1 *m. Find magnitude and direction of the net electric field at a point midway between them.*

[2024 • Set 55-5-1]

SOLUTION

(i) Field of a uniformly charged thin spherical shell (surface density σ, radius *R*).

Total charge on the shell:

$$Q = \sigma \cdot 4\pi R^2$$

By spherical symmetry the field $\vec{E}$ is radial and its magnitude depends only on the distance r from the centre. Draw a concentric spherical Gaussian surface of radius r; over it

$$\oint \vec{E} \cdot d\vec{S} = E \cdot 4\pi r^2$$

(i) Outside the shell ($r > R$): the Gaussian surface encloses the entire charge Q.

$$E \cdot 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$E_{out} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2} = \frac{\sigma R^2}{\varepsilon_0 r^2}$$

This is identical to the field of a point charge Q placed at the centre.

(ii) Inside the shell ($r < R$): the Gaussian surface encloses no charge (all charge is on the surface).

$$E \cdot 4\pi r^2 = \frac{0}{\varepsilon_0} = 0$$

$$E_{in} = 0$$

(ii) Net field at the midpoint of two parallel charged wires.

Wires AB and CD carry $\lambda_1 = +10\ \mu C/m$ and $\lambda_2 = -20\ \mu C/m$, separated by 1 m. The midpoint is at perpendicular distance

$$r = \frac{1}{2} = 0.5\ m$$

from each wire. The field of a long wire is $E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2k\lambda}{r}$, with $k = 9\times 10^9\ N\ m^2C^{-2}$.

Field due to wire AB (positive, points away from AB):

$$E_1 = \frac{2(9\times 10^9)(10\times 10^{-6})}{0.5}$$

$$E_1 = \frac{1.8\times 10^5}{0.5} = 3.6\times 10^5\ N\,C^{-1}$$

Field due to wire CD (negative, points towards CD):

$$E_2 = \frac{2(9\times 10^9)(20\times 10^{-6})}{0.5}$$

$$E_2 = \frac{3.6\times 10^5}{0.5} = 7.2\times 10^5\ N\,C^{-1}$$

At the midpoint, $\vec{E}_1$ points away from the positive wire AB and $\vec{E}_2$ points towards the negative wire CD; both therefore point in the same direction (from AB towards CD), so they add:

$$E_{net} = E_1 + E_2 = 3.6\times 10^5 + 7.2\times 10^5$$

$$E_{net} = 1.08\times 10^{6}\, N\, C^{-1}$$

The net field at the midpoint is $1.08\times 10^{6}\, N\, C^{-1}$, directed from the positively charged wire AB towards the negatively charged wire CD.

Final answer. $E_{out}=\dfrac{Q}{4\pi\varepsilon_0 r^2}$, $E_{in}=0$; net field at midpoint $=1.08\times 10^{6}\, N\, C^{-1}$ from AB towards CD.

← *Back to Q145*

Q146

(i) State Coulomb's law in electrostatics and write it in vector form, for two charges.
(ii) 'Gauss's law is based on the inverse-square dependence on distance contained in the Coulomb's law.' Explain.
(iii) Two charges A (charge q) and B (charge 2q) are located at points (0, 0) and (a, a) respectively. Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x-axis and y-axis respectively. Find the force exerted by A on B, in terms of $\hat{i}$ and $\hat{j}$.

[2023 • Set 55-1-1]

SOLUTION

(i) Coulomb's law (statement and vector form).

Two stationary point charges q_1 and q_2 separated by a distance r in vacuum attract or repel each other with a force whose magnitude is directly proportional to the product of the charges and inversely proportional to the square of the distance between them, directed along the line joining them.

Scalar form:

$$F = \frac{1}{4\pi\varepsilon_0}\,\frac{q_1 q_2}{r^2}$$

Vector form (force on q_2 due to q_1):

$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0}\,\frac{q_1 q_2}{r^2}\,\hat{r}_{12}$$

where $\hat{r}_{12}$ is the unit vector pointing from q_1 to q_2. By Newton's third law $\vec{F}_{12} = -\vec{F}_{21}$. The force is repulsive for like charges and attractive for unlike charges.

(ii) How Gauss's law rests on the $1/r^2$ dependence.

For a point charge q, draw a concentric sphere of radius r. The field on it is

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

and the surface area of the sphere is $4\pi r^2$. The flux is the product:

$$\Phi = E \times 4\pi r^2 = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \times 4\pi r^2 = \frac{q}{\varepsilon_0}$$

The two factors of r^2 cancel, so the flux is independent of r and equals q/ε_0. This cancellation works only because E falls off exactly as $1/r^2$ while the area grows exactly as r^2. If the force law were $1/r^3$ or any other power, the flux would depend on r and Gauss's law in its simple form would fail. Hence Gauss's law is a direct consequence of the inverse-square nature of Coulomb's law.

(iii) Force on charge *B* (2q) due to charge *A* (*q*).

A (*q*) is at (0,0) and *B* (2q) at (*a*,*a*). The separation vector from *A* to *B* is

$$\vec{r} = (a-0)\hat{i} + (a-0)\hat{j} = a\hat{i} + a\hat{j}$$

Its magnitude is

$$r = \sqrt{a^2+a^2} = a\sqrt{2}$$

The unit vector from *A* to *B* is

$$\hat{r} = \frac{a\hat{i} + a\hat{j}}{a\sqrt{2}} = \frac{\hat{i}+\hat{j}}{\sqrt{2}}$$

The force magnitude factor is

$$\frac{1}{4\pi\varepsilon_0}\frac{(q)(2q)}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{2q^2}{2a^2} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$$

Therefore the force on *B* due to *A* is

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}\cdot\frac{\hat{i}+\hat{j}}{\sqrt{2}}$$

$$\vec{F} = \frac{q^2}{4\sqrt{2}\,\pi\varepsilon_0 a^2}\,(\hat{i}+\hat{j})$$

This is the force exerted by A on B, directed along $(\hat{i}+\hat{j})$, i.e. away from A along the diagonal (both charges being positive, the force is repulsive).

TIP

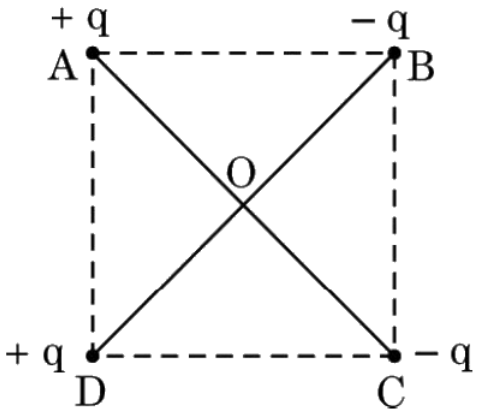
Always build the unit vector $\hat{r}=\vec{r}/|\vec{r}|$ first, then multiply by the scalar force. Plugging $\mathbf{r^2 = 2a^2}$ into the denominator is what cleanly halves the $2q^2$ to q^2.

Final answer. $\vec{F}_{AB} = \dfrac{q^2}{4\sqrt{2}\,\pi\varepsilon_0 a^2}\,(\hat{i}+\hat{j})$, directed away from A.

← *Back to Q146*

Q147

(i) Derive an expression for the electric field at a point on the equatorial plane of an electric dipole consisting of charges q and $-q$ separated by a distance 2a.
(ii) The distance of a far off point on the equatorial plane of an electric dipole is halved. How will the electric field be affected for the dipole?
(iii) Two identical electric dipoles are placed along the diagonals of a square ABCD *of side $\sqrt{2}$ m as shown in the figure. Obtain the magnitude and direction of the net electric field at the centre (O) of the square.*



[2023 • Set 55-1-1]

SOLUTION

(i) Field on the equatorial plane of a dipole (q, $-q$ separated by 2a).

Let $-q$ be at A, $+q$ at B, centre O, separation 2a. Take a point P on the equatorial plane at distance r from O. Then

$$AP = BP = \sqrt{r^2+a^2}$$

The field due to $+q$ at B is

$\mathbf{E_+} = \frac{1}{4\pi\varepsilon_0}\ \frac{q}{r^2+a^2}$, directed from B towards P

The field due to $-q$ at A is

$\mathbf{E_-} = \frac{1}{4\pi\varepsilon_0}\ \frac{q}{r^2+a^2}$, directed from P towards A

Resolve each into a component along the dipole axis and one perpendicular to it. The perpendicular (vertical) components are equal and opposite and cancel. The components parallel to the axis add; both point from $+q$ towards $-q$, i.e. antiparallel to $\vec{p}$. Each axial component carries the factor $\theta = \frac{a}{\sqrt{r^2+a^2}}$:

$$E_{eq} = 2E_+\theta = 2\cdot\frac{1}{4\pi\varepsilon_0}\ \frac{q}{r^2+a^2}\cdot\frac{a}{\sqrt{r^2+a^2}}$$

$$E_{eq} = \frac{1}{4\pi\varepsilon_0}\ \frac{2qa}{(r^2+a^2)^{3/2}} = \frac{1}{4\pi\varepsilon_0}\ \frac{p}{(r^2+a^2)^{3/2}}$$

For a short dipole ($r \gg a$), $(r^2+a^2)^{3/2} \approx r^3$:

$$\vec{E}_{eq} = -\frac{1}{4\pi\varepsilon_0}\ \frac{\vec{p}}{r^3}$$

The field is antiparallel to $\vec{p}$.

(ii) Effect of halving the distance of a far-off point.

On the equatorial plane $\mathbf{E_{eq}} \propto \frac{1}{r^3}$. Replacing r by $r/2$:

$$\frac{E'}{E} = \left(\frac{r}{r/2}\right)^3 = 2^3 = 8$$

So the field becomes 8 times its original value.

(iii) Net field at the centre O of the square (two identical dipoles along the diagonals).

The square ABCD has side $\sqrt{2}\ m$, so each diagonal has length

$$d = \sqrt{2} \times \sqrt{2} = 2\ m$$

and the distance from a vertex to the centre O along a diagonal is $r = \dfrac{d}{2} = 1\ m$. Each dipole lies along a diagonal, so O is an axial point of each dipole at distance $r = 1\ m$. The axial field of a short dipole is

$$E_{axial} = \frac{1}{4\pi\varepsilon_0}\ \frac{2p}{r^3}$$

Each dipole produces a field along its own diagonal, directed parallel to its dipole moment. The two diagonals of a square are perpendicular, so the two axial fields $\vec{E}_1$ and $\vec{E}_2$ at O are equal in magnitude and mutually perpendicular. Their resultant is

$$E_{net} = \sqrt{E_1^{\,2} + E_2^{\,2}} = \sqrt{2}\ E_{axial}$$

$$E_{net} = \sqrt{2} \cdot \frac{1}{4\pi\varepsilon_0}\ \frac{2p}{r^3} = \frac{1}{4\pi\varepsilon_0}\ \frac{2\sqrt{2}\ p}{r^3}$$

With $r = 1\ m$,

$$E_{net} = \frac{2\sqrt{2}\ p}{4\pi\varepsilon_0} = \frac{1}{4\pi\varepsilon_0}\ (2\sqrt{2}\ p)$$

The net field at O has magnitude $E_{net} = \dfrac{2\sqrt{2}\ p}{4\pi\varepsilon_0}\ N\,C^{-1}$, directed along the diagonal that bisects the two dipole moments (the resultant lies midway between the two perpendicular axial fields, pointing from the centre towards the side that both fields point to). Since the two component fields are equal and at 90°, the resultant makes 45° with each diagonal.

CONCEPT

Diagonal length of a square of side s is $s\sqrt{2}$, so for $s=2$ the vertex-to-centre distance is exactly 1 m, which makes the axial-field arithmetic clean.

Final answer. $\vec{E}_{eq} = -\dfrac{p}{4\pi\varepsilon_0\, r^3}$; halving r makes it 8 times; net field at O $= \dfrac{2\sqrt{2}\ p}{4\pi\varepsilon_0}$ at **45°** to each diagonal.

← Back to Q147

Q148

(i) Use Gauss' law to obtain an expression for the electric field due to an infinitely long thin straight wire with uniform linear charge density λ.
(ii) An infinitely long positively charged straight wire has a linear charge density λ. An electron is revolving in a circle with a constant speed v such that the wire passes through the centre, and is perpendicular to the plane, of the circle. Find the kinetic energy of the electron in terms of magnitudes of its charge and linear charge density λ on the wire.
(iii) Draw a graph of kinetic energy as a function of linear charge density λ.

[2023 • Set 55-2-1]

SOLUTION

(i) Field of an infinitely long uniformly charged straight wire (linear density λ).

By symmetry the field $\vec{E}$ is radial (perpendicular to the wire) and its magnitude depends only on the perpendicular distance r. Choose a coaxial cylindrical Gaussian surface of radius r and length l, with the wire along its axis.

The two flat end caps contribute no flux, since $\vec{E}$ is parallel to them ($\vec{E} \perp d\vec{S}$ there). The curved surface contributes

$$\Phi = E \cdot (2\pi r\, l)$$

The charge enclosed by the cylinder is

$$q_{enc} = \lambda\, l$$

By Gauss's law,

$$E \cdot 2\pi r\, l = \frac{\lambda\, l}{\varepsilon_0}$$

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

directed radially outward for $\lambda > 0$.

(ii) Kinetic energy of an electron revolving around the wire.

The electron of charge magnitude e moves in a circle of radius r about the positive wire. The attractive electrostatic force provides the centripetal force:

$$\frac{m_e v^2}{r} = eE = \frac{e\lambda}{2\pi\varepsilon_0 r}$$

Multiplying both sides by r:

$$m_e v^2 = \frac{e\lambda}{2\pi\varepsilon_0}$$

The kinetic energy is

$$K = \frac{1}{2} m_e v^2 = \frac{1}{2} \cdot \frac{e\lambda}{2\pi\varepsilon_0}$$

$$K = \frac{e\lambda}{4\pi\varepsilon_0}$$

Notice the kinetic energy is independent of the orbit radius r and is directly proportional to λ.

(iii) Graph of K versus λ.

Since

$$K = \left(\frac{e}{4\pi\varepsilon_0}\right)\lambda$$

K is a linear function of λ. Draw a graph with λ on the horizontal axis and K on the vertical axis: it is a straight line passing through the origin, with constant positive slope $\frac{e}{4\pi\varepsilon_0}$. As λ increases, K increases proportionally.

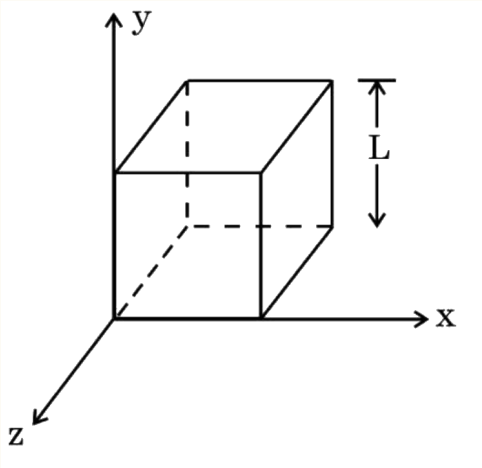
Thus $K = \frac{e\lambda}{4\pi\varepsilon_0}$, independent of orbit radius, and the K versus λ graph is a straight line through the origin.

Final answer. $E = \frac{\lambda}{2\pi\varepsilon_0 r}$, $K = \frac{e\lambda}{4\pi\varepsilon_0}$ (independent of r); K vs λ is a straight line through the origin.

← *Back to Q148*

Q149

(a) (i) Define electric flux and write its SI unit.
(ii) Use Gauss' law to obtain the expression for the electric field due to a uniformly charged infinite plane sheet.
(iii) A cube of side L is kept in space, as shown in the figure. An electric field $\vec{E} = (Ax + B)\,\hat{i}$ *exists in the region. Find the net charge enclosed by the cube.*



OR

(b) (i) Define electric potential at a point and write its SI unit.
(ii) Two capacitors are connected in series. Derive an expression of the equivalent capacitance of the combination.
(iii) Two point charges +q and −q are located at points (3a, 0) *and* (0, 4a) *respectively in x-y plane. A third charge Q is kept at the origin. Find the value of Q, in terms of q and a, so that the electrostatic potential energy of the system is zero.*

[2023 • Set 55-3-1]

SOLUTION

(a)(i) Electric flux and its SI unit.

Electric flux through a surface S is the surface integral of the field over it:

$$\Phi_E = \int_S \vec{E} \cdot d\vec{S}$$

It is a scalar quantity that measures the number of electric field lines crossing the surface. Its SI unit is $N\,m^2C^{-1}$ (equivalently $V\,m$).

(a)(ii) Field of a uniformly charged infinite plane sheet.

For a sheet of surface charge density σ, symmetry makes $\vec{E}$ perpendicular to the sheet and equal on both sides. Use a Gaussian pillbox of cross-section A piercing the sheet, with flat faces on either side. The curved side gives no flux; each flat face gives EA, so total flux = **2EA**, while the enclosed charge is σA:

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$$

directed away from the sheet (for **σ>0**), independent of distance.

(a)(iii) Net charge in a cube of side L in $\vec{E}=(Ax+B)\hat{i}$.

The field is along $\hat{i}$, so only the two faces perpendicular to the x-axis contribute. Let the left face be at $x=0$ and the right face at $x=L$; each has area L^2.

Left face ($x=0$), outward normal $-\hat{i}$:

$$\Phi_L = -(A\cdot 0 + B)L^2 = -BL^2$$

Right face ($x=L$), outward normal $+\hat{i}$:

$$\Phi_R = (AL + B)L^2$$

Net flux:

$$\Phi_{net} = \Phi_R + \Phi_L = (AL+B)L^2 - BL^2 = AL^3$$

By Gauss's law the enclosed charge is

$$q_{enc} = \varepsilon_0\,\Phi_{net} = \varepsilon_0\,A\,L^3$$

So the net charge enclosed by the cube is $q_{enc} = \varepsilon_0 A L^3$.

(b)(i) Electric potential at a point and its SI unit.

The electric potential at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point against the electric field, without acceleration:

$$V = \frac{W}{q_0}$$

Its SI unit is the volt (V), where $1\,V = 1\,J\,C^{-1}$.

(b)(ii) Equivalent capacitance of two capacitors in series.

In series the same charge Q sits on each capacitor, and the total voltage is the sum of the individual voltages:

$$V = V_1 + V_2$$

With $V_1 = \frac{Q}{C_1}$ and $V_2 = \frac{Q}{C_2}$,

$$V = \frac{Q}{C_1} + \frac{Q}{C_2}$$

If C_s is the equivalent capacitance, $V = \frac{Q}{C_s}$, so

$$\frac{Q}{C_s} = \frac{Q}{C_1} + \frac{Q}{C_2}$$

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C_s = \frac{C_1 C_2}{C_1 + C_2}$$

(b)(iii) Value of Q for zero potential energy.

Charges: $+q$ at (3a,0), $-q$ at (0,4a), and Q at the origin. The mutual distances are

$$r_{+q,\,Q} = 3a,\ r_{-q,\,Q} = 4a$$

$$r_{+q,\,-q} = \sqrt{(3a)^2+(4a)^2} = \sqrt{9a^2+16a^2} = 5a$$

Total potential energy (with $k = \frac{1}{4\pi\varepsilon_0}$):

$$U = k\left[\frac{(q)(Q)}{3a} + \frac{(-q)(Q)}{4a} + \frac{(q)(-q)}{5a}\right]$$

Setting $U = 0$ and cancelling $\frac{k}{a}$:

$$\frac{qQ}{3} - \frac{qQ}{4} - \frac{q^2}{5} = 0$$

$$qQ\left(\frac{1}{3} - \frac{1}{4}\right) = \frac{q^2}{5}$$

$$qQ\left(\frac{4-3}{12}\right) = \frac{q^2}{5}$$

$$\frac{qQ}{12} = \frac{q^2}{5}$$

$$Q = \frac{12q}{5} = 2.4\,q$$

Hence $Q = \frac{12q}{5} = 2.4\,q$ makes the total electrostatic potential energy of the system zero.

COMMON MISTAKE

Use the 3-4-5 right triangle: the +q to −q separation is 5a, *not* 7a. *Adding the distances instead of using Pythagoras is the most common slip in part (b)(iii).*

Final answer. $q_{enc}=\varepsilon_0 A L^3$; OR $C_s=\frac{C_1C_2}{C_1+C_2}$ and $Q=\frac{12q}{5}=2.4\,q$.

← *Back to Q149*

Q150

Using Gauss law, derive expression for electric field due to a spherical shell of uniform charge distribution σ and radius R at a point lying at a distance x from the centre of shell, such that
(i) **0 < x < R**, *and*
(ii) ***x > R***.

[2020 • Set 55-1-1]

SOLUTION

Setup. A spherical shell of radius R carries uniform surface charge density σ, so the total charge is

$$Q = \sigma \cdot 4\pi R^2$$

By spherical symmetry the field $\vec{E}$ is radial and its magnitude depends only on the distance x from the centre. To find the field at distance x, draw a concentric

spherical Gaussian surface of radius x. Over this surface $\vec{E}$ is uniform in magnitude and parallel to the area element $d\vec{S}$, so

$$\oint \vec{E} \cdot d\vec{S} = E(x) \cdot 4\pi x^2$$

Diagram to draw: a circle of radius R for the charged shell with + signs spread evenly on it; a smaller dashed concentric circle of radius x (for **x) and a larger dashed concentric circle of radius x (for x>R) as the two Gaussian surfaces; outward radial arrows for $\vec{E}$ on the outer surface.**

(i) Inside the shell ($0 \le x < R$): the Gaussian surface lies entirely inside the shell and encloses no charge, since all the charge sits on the surface at radius R.

$$q_{enc} = 0$$

By Gauss's law,

$$E(x) \cdot 4\pi x^2 = \frac{0}{\varepsilon_0} = 0$$

$$E_{in}(x) = 0$$

The interior of a uniformly charged spherical shell is field-free.

(ii) Outside the shell ($x > R$): the Gaussian surface now encloses the entire charge Q.

$$q_{enc} = Q = \sigma \cdot 4\pi R^2$$

By Gauss's law,

$$E(x) \cdot 4\pi x^2 = \frac{Q}{\varepsilon_0}$$

$$E_{out}(x) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{x^2}$$

Substituting $Q = \sigma \cdot 4\pi R^2$,

$$E_{out}(x) = \frac{1}{4\pi\varepsilon_0} \frac{\sigma \cdot 4\pi R^2}{x^2} = \frac{\sigma R^2}{\varepsilon_0 x^2}$$

directed radially outward (for $\sigma > 0$).

Hence $E_{in} = 0$ for $x < R$, and $E_{out} = \frac{Q}{4\pi\varepsilon_0 x^2} = \frac{\sigma R^2}{\varepsilon_0 x^2}$ for $x > R$. The field outside is exactly that of a point charge Q placed at the centre of the shell, which is the shell theorem.

Final answer. $E_{in}=0$ for x and $E_{out}=\frac{Q}{4\pi\varepsilon_0 x^2}=\frac{\sigma R^2}{\varepsilon_0 x^2}$ for $x>R$.

← Back to Q150

Q151

An electric field is uniform and acts along +x direction in the region of positive x. It is also uniform with the same magnitude but acts in −x direction in the region of negative x. The value of the field is $\boldsymbol{E = 200\ N/C}$ *for* $\boldsymbol{x > 0}$ *and* $\boldsymbol{E = -200\ N/C}$ *for* $\boldsymbol{x < 0}$. *A right circular cylinder of length* 20 cm *and radius* 5 cm *has its centre at the origin and its axis along the x-axis so that one flat face is at* $\boldsymbol{x = +10\ cm}$ *and the other is at* $\boldsymbol{x = -10\ cm}$.

Find:

(i) The net outward flux through the cylinder.

(ii) The net charge present inside the cylinder.

[2020 • Set 55-1-1]

SOLUTION

The cylinder has length $L = 20\ cm = 0.20\ m$ and radius $R = 5\ cm = 0.05\ m$, with its axis along the x-axis and centre at the origin. Its two flat circular faces lie at $x = +0.10\ m$ and $x = -0.10\ m$.

The area of each flat face is

$$A = \pi R^2 = \pi (0.05)^2$$

$$A = \pi \times 2.5 \times 10^{-3} \approx 7.85 \times 10^{-3}\ m^2$$

(i) On the curved (lateral) surface, the outward area vector $\vec{dS}$ points radially, perpendicular to the x-axis. Since $\vec{E}$ is along $\pm\hat{i}$ everywhere, $\vec{E} \cdot \vec{dS} = 0$, so the curved surface contributes zero flux.

On the right flat face ($x = +0.10\ m$): $\vec{E} = +200\ \hat{i}\ N/C$ and the outward normal is $+\hat{i}$, so

$$\Phi_R = E\,A = 200 \times 7.85\times 10^{-3} \approx +1.57\ N\,\mathrm{m^2C^{-1}}$$

On the left flat face ($x = -0.10\ m$): $\vec{E} = -200\ \hat{i}\ N/C$ and the outward normal is $-\hat{i}$, so $\vec{E}\cdot \mathbf{d}\vec{S} = (-200)(-1)\ \mathbf{dS}$ is positive,

$$\Phi_L = E\,A = 200 \times 7.85\times 10^{-3} \approx +1.57\ N\,\mathrm{m^2C^{-1}}$$

Hence the net outward flux is

$$\Phi_{\text{net}} = \Phi_R + \Phi_L + \Phi_{\text{curved}} = 2 \times 1.57 + 0$$

$$\Phi_{\text{net}} = 2EA = 2 \times 200 \times \pi(0.05)^2$$

$$\Phi_{\text{net}} = 400\pi \times 2.5\times 10^{-3} = \pi \approx 3.14\ N\,\mathrm{m^2C^{-1}}$$

(ii) By Gauss's law, $\Phi_{\text{net}} = \dfrac{q_{\text{in}}}{\varepsilon_0}$, so the net charge enclosed is

$$q_{\text{in}} = \varepsilon_0\,\Phi_{\text{net}} = 8.85\times 10^{-12} \times 3.14$$

$$q_{\text{in}} \approx 2.78\times 10^{-11}\ C = 27.8\ \mathrm{pC}$$

Therefore the net outward flux is about $\mathbf{3.14\ N\,m^2\,C^{-1}}$ and the net charge inside the cylinder is about $\mathbf{2.78\times 10^{-11}\ C}$.

COMMON MISTAKE

Both flat faces give POSITIVE flux here: on the left face the field and the outward normal both point in $-\hat{i}$, so their dot product is positive. Do not let them cancel.

Final answer. Net outward flux ≈ $\mathbf{3.14\ N\,m^2C^{-1}}$; net enclosed charge ≈ $\mathbf{2.78\times 10^{-11}\ C}$.

← Back to Q151

Q152

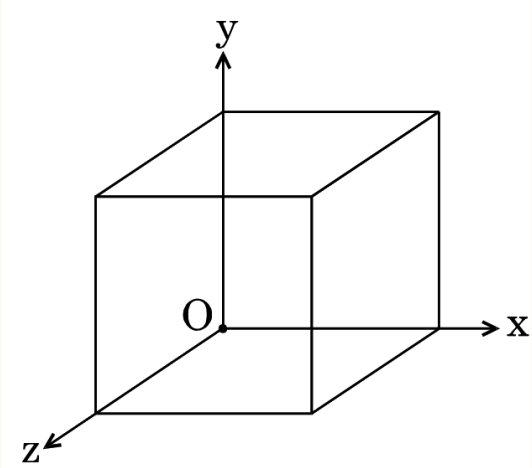
(a) Use Gauss's law to show that due to a uniformly charged spherical shell of radius R, the electric field at any point situated outside the shell at a distance r from its centre is equal to the electric field at the same point, when the entire charge on the shell were concentrated at its centre. Also plot the graph showing the variation of electric field with r, for $\boldsymbol{r < R}$ *and* $\boldsymbol{r \geq R}$.

(b) Two point charges of +1 μC *and* +4 μC *are kept* 30 *cm apart. How far from the* +1 μC *charge on the line joining the two charges, will the net electric field be zero?*

OR

(a) Two point charges q_1 *and* q_2 *are kept r distance apart in a uniform external electric field* $\vec{E}$. *Find the amount of work done in assembling this system of charges.*

(b) A cube of side 20 *cm is kept in a region as shown in the figure. An electric field* $\vec{E}$ *exists in the region such that the potential at a point is given by* $\boldsymbol{V = 10x + 5}$, *where V is in volt and x is in m.*



Find the (i) electric field $\vec{E}$, *and (ii) total electric flux through the cube.*

[2020 • Set 55-2-1]

SOLUTION

(a) By symmetry, the field of a uniformly charged spherical shell of radius R and total charge Q is radial, and its magnitude depends only on the distance r from the centre. To find the field at an outside point ($\boldsymbol{r > R}$), draw a concentric spherical Gaussian surface of radius r passing through that point. The field is uniform in magnitude and parallel to $d\vec{S}$ at every point of this surface.

By Gauss's law,

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q}{\varepsilon_0}$$

$$E \times 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$E = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}$$

This is exactly the field that a point charge Q placed at the centre would produce at the same distance r. Hence, for outside points, the shell behaves as if its entire charge were concentrated at its centre.

For an inside point ($r < R$), the Gaussian sphere encloses no charge, so $E \times 4\pi r^2 = 0$, giving $E = 0$ everywhere inside the shell.

Graph of E versus r: $E = 0$ for $0 \le r < R$ (a flat line on the r-axis), then E jumps to its maximum value $\frac{Q}{4\pi\varepsilon_0 R^2}$ at $r = R$, and for $r > R$ it falls off as a $1/r^2$ curve, decreasing smoothly towards zero.

(b) Both charges +1 μC and +4 μC are positive, so the null point lies between them, nearer the smaller charge. Let it be at distance x (in cm) from the +1 μC charge; then it is at $(30-x)$ cm from the +4 μC charge. Setting the field magnitudes equal,

$$\frac{1}{4\pi\varepsilon_0} \frac{(1\ \mu C)}{x^2} = \frac{1}{4\pi\varepsilon_0} \frac{(4\ \mu C)}{(30-x)^2}$$

$$\frac{(30-x)^2}{x^2} = 4$$

$$\frac{30-x}{x} = 2$$

$$30 - x = 2x \Rightarrow x = 10 \text{ cm}$$

So the net field is zero at a point **10 cm** from the +1 μC charge.

OR

(a) Work done in assembling two charges q_1 and q_2 kept distance r apart in an external field $\vec{E}$ (with external potential V): The work to bring in q_1 from infinity to its point $\vec{r}_1$ is $W_1 = q_1 V(\vec{r}_1)$. The work to bring in q_2 from infinity to $\vec{r}_2$ has two

parts: against the external field, $q_2\,V(\vec{r}_2)$, and against the field of q_1 already placed, $\frac{1}{4\pi\varepsilon_0}\frac{q_1\,q_2}{r}$. Hence the total work is

$$W = q_1\,V(\vec{r}_1) + q_2\,V(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\,\frac{q_1\,q_2}{r}$$

(b) The potential is $V = 10x + 5$. The electric field is $\vec{E} = -\frac{dV}{dx}\,\hat{i}$,

$$\vec{E} = -\frac{d}{dx}(10x + 5)\,\hat{i} = -10\,\hat{i}\ V/m$$

So the field is uniform, of magnitude 10 N/C, directed along $-x$. Since the field is uniform, the flux entering one face of the cube equals the flux leaving the opposite face, and there is no net charge inside. Therefore the total electric flux through the cube is

$$\Phi = 0$$

(This also follows from Gauss's law, as the cube encloses no charge.)

Final answer. (a) $E = \frac{Q}{4\pi\varepsilon_0\,r^2}$ outside; null point at 10 cm from the **+1 µC** charge. (OR) $\vec{E} = -10\,\hat{i}\ V/m$ and total flux through the cube = **0**.

← *Back to Q152*

Q153

(a) An electric dipole of dipole moment $\vec{P}$ is placed in a uniform electric field $\vec{E}$ at an angle θ with it. Derive the expression for torque $(\vec{\tau})$ acting on it. Find the orientation of the dipole relative to the electric field for which torque on it is (i) maximum, and (ii) half of maximum.

(b) Two point charges q_1 = +1 µC and q_2 = +4 µC are placed 2 m apart in air. At what distance from q_1 along the line joining the two charges, will the net electric field be zero?

OR

(a) Derive an expression for the energy stored in a parallel plate capacitor of capacitance C when charged up to voltage V. How is this energy stored in the capacitor?

(b) A capacitor of capacitance 1 μF *is charged by connecting a battery of negligible internal resistance and emf* 10 *V across it. Calculate the amount of charge supplied by the battery in charging the capacitor fully.*

[2020 • Set 55-3-1]

SOLUTION

(a) Consider a dipole made of charges $-q$ and $+q$ separated by 2a, with dipole moment $\vec{p} = q(2a)$ directed from $-q$ to $+q$. Place it in a uniform field $\vec{E}$ so that $\vec{p}$ makes an angle θ with $\vec{E}$.

Each charge feels a force: $+q\vec{E}$ on the positive charge and $-q\vec{E}$ on the negative charge. These forces are equal in magnitude (qE) and opposite in direction, so the net force on the dipole is zero. However, they act along different lines, forming a couple that produces a torque.

The magnitude of the torque equals (force) × (perpendicular distance between the two forces). The perpendicular distance between the lines of action is 2aθ,

$$\tau = (\text{force}) \times (\text{perpendicular distance})$$

$$\tau = qE \times 2a\theta$$

$$\tau = (q\ 2a)\ E\theta = pE\theta$$

In vector form, $\vec{\tau} = \vec{p} \times \vec{E}$.

(i) Torque is maximum when $\theta = 1$, that is $\theta = 90°$, giving $\tau_{max} = pE$.

(ii) For half the maximum torque, $pE\theta = \frac{1}{2}pE$, so $\theta = \frac{1}{2}$, giving $\theta = 30°$ (or 150°).

(b) Both charges are positive, so the null point lies between them, nearer the smaller charge. Let it be at distance x from $q_1 = +1\ \mu C$; then it is at $(2-x)$ m from $q_2 = +4\ \mu C$. Equating field magnitudes,

$$\frac{1}{4\pi\varepsilon_0}\frac{(1\ \mu C)}{x^2} = \frac{1}{4\pi\varepsilon_0}\frac{(4\ \mu C)}{(2-x)^2}$$

$$\frac{(2-x)^2}{x^2} = 4$$

$$\frac{2-x}{x} = 2 \Rightarrow 2 - x = 2x$$

$$x = \frac{2}{3}\, m \approx 0.67\, m$$

So the net field is zero at $\frac{2}{3}$ **m** from the +1 μC charge.

OR

(a) Let the capacitor of capacitance C be charged so that at some instant it holds charge q and its potential is $V' = \frac{q}{C}$. To transfer a further small charge dq from one plate to the other, the work done is

$$dW = V'\, dq = \frac{q}{C}\, dq$$

The total work to charge it from 0 to the final charge $Q = CV$ is

$$W = \int_0^Q \frac{q}{C}\, dq = \frac{1}{C} \cdot \frac{Q^2}{2}$$

$$W = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV$$

This work is stored as electrostatic potential energy in the capacitor,

$$U = \frac{1}{2} CV^2$$

It is stored in the form of energy of the electric field set up in the region between the plates.

(b) For $C = 1\ \mu F$ charged to $V = 10\ V$, the charge supplied by the battery is

$$Q = CV = (1 \times 10^{-6}\ F)(10\ V)$$

$$Q = 1 \times 10^{-5}\ C = 10\ \mu C$$

So the battery supplies a charge of **10 μ C**.

Final answer. (a) $\vec{\tau} = \vec{p} \times \vec{E}$, $\tau = pE\theta$; max at $\theta = 90°$, half-max at $\theta = 30°$; null point at $\frac{2}{3}$ m from q_1. (OR) $U = \frac{1}{2} CV^2$; charge = **10 μC**.

← *Back to Q153*

Q154

(a) Write two important characteristics of equipotential surfaces.
(b) A thin circular ring of radius r is charged uniformly so that its linear charge density becomes λ. Derive an expression for the electric field at a point P at a distance x from it along the axis of the ring. Hence, prove that at large distances ($x \gg r$), the ring behaves as a point charge.

OR

(a) State Gauss's law on electrostatics and derive an expression for the electric field due to a long straight thin uniformly charged wire (linear charge density λ) at a point lying at a distance r from the wire.
(b) The magnitude of electric field (in NC^{-1}) in a region varies with the distance r (in m) as

$\boldsymbol{E = 10r + 5}$

By how much does the electric potential increase in moving from point at $\boldsymbol{r = 1}$ *m to a point at* $\boldsymbol{r = 10}$ *m.*

[2020 • Set 55-5-1]

SOLUTION

(a) Two important characteristics of equipotential surfaces:

(i) The electric field $\vec{E}$ is always perpendicular (normal) to the equipotential surface at every point. If it had a component along the surface, that component would do work in moving a charge along the surface, which would change the potential and contradict the surface being equipotential.

(ii) No work is done in moving a test charge from one point to another on the same equipotential surface, since the potential difference between any two such points is zero ($W = q\,\Delta V = 0$).

(b) Consider a thin ring of radius r with uniform linear charge density λ, so its total charge is $Q = \lambda(2\pi r)$. Take an axial point P at distance x from the centre O, along the axis. A small element of length dl carries charge $\mathbf{dq = \lambda\, dl}$ and lies at distance

$$s = \sqrt{r^2 + x^2}$$

from P. The field due to this element at P has magnitude

$$dE = \frac{1}{4\pi\varepsilon_0}\ \frac{dq}{(r^2+x^2)}$$

Resolve dE into a component along the axis (dEα) and one perpendicular to the axis. For every element there is a diametrically opposite element whose perpendicular component is equal and opposite, so all perpendicular components cancel. Only the axial components survive and add up. With $\alpha = \frac{x}{\sqrt{r^2+x^2}}$,

$$E = \oint dE\alpha = \frac{1}{4\pi\varepsilon_0}\ \frac{x}{(r^2+x^2)^{3/2}} \oint dq$$

$$E = \frac{1}{4\pi\varepsilon_0}\ \frac{Q\,x}{(r^2+x^2)^{3/2}}$$

Putting $Q = 2\pi\, r\lambda$,

$$E = \frac{1}{4\pi\varepsilon_0}\ \frac{2\pi\, r\lambda\, x}{(r^2+x^2)^{3/2}} = \frac{\lambda\, r\, x}{2\varepsilon_0\,(r^2+x^2)^{3/2}}$$

directed along the axis, away from the ring (for $\lambda > 0$).

For $x \gg r$, we have $r^2 + x^2 \approx x^2$, so $(r^2+x^2)^{3/2} \approx x^3$,

$$E \approx \frac{1}{4\pi\varepsilon_0}\ \frac{Q\,x}{x^3} = \frac{1}{4\pi\varepsilon_0}\ \frac{Q}{x^2}$$

This is the field of a point charge Q at distance x. Hence at large distances the ring behaves as a point charge.

OR

(a) Gauss's law states that the total electric flux through any closed surface equals $\frac{1}{\varepsilon_0}$ times the net charge enclosed,

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{enclosed}}{\varepsilon_0}$$

For a long straight uniformly charged wire of linear density λ, by symmetry $\vec{E}$ is radial and its magnitude is the same at all points at distance r. Choose a coaxial cylindrical Gaussian surface of radius r and length l. The two flat end-caps have $\vec{E} \perp d\vec{S}$, so they contribute no flux; the curved surface gives $E(2\pi\, r\, l)$. The charge enclosed is $\lambda\, l$,

$$E\,(2\pi\, r\, l) = \frac{\lambda\, l}{\varepsilon_0}$$

$$E = \frac{\lambda}{2\pi\varepsilon_0\, r}$$

directed radially outward (for $\lambda > 0$).

(b) The increase in potential in moving from $r = 1$ m to $r = 10$ m, where $E = 10r + 5$, is

$$V_{10} - V_1 = -\int_1^{10} E\, dr = -\int_1^{10}(10r + 5)\, dr$$

$$= -[5r^2 + 5r]_1^{10}$$

$$= -[(500 + 50) - (5 + 5)]$$

$$= -(550 - 10) = -540\ V$$

So the potential decreases by 540 V; that is, it 'increases' by −540 V (the potential actually falls by **540 V** as one moves outward).

CONCEPT

Potential is found from $\Delta V = -\int E\, dr$. A positive, outward field always gives a fall in potential as you move away, so a negative answer is physically expected here.

Final answer. (b) $E = \frac{\lambda\, r\, x}{2\varepsilon_0(r^2+x^2)^{3/2}}$, reducing to $\frac{Q}{4\pi\varepsilon_0\, x^2}$ for $x \gg r$. (OR) $E = \frac{\lambda}{2\pi\varepsilon_0\, r}$; the potential changes by **−540 V**.

← *Back to Q154*

Q155

Derive an expression for the electric field at any point on the equatorial line of an electric dipole.

[2019 • Set 55-1-1]

SOLUTION

A dipole consists of charges $-q$ at A and $+q$ at B, separated by 2a, with dipole moment $\boldsymbol{p = q(2a)}$ pointing from $-q$ to $+q$. The equatorial line is the perpendicular bisector of AB. Take a point P on it at distance r from the centre O.

Draw the dipole horizontally with $-q$ at A (left) and $+q$ at B (right), centre O. The point P is directly above O at distance r. The distance from each charge to P is

$$AP = BP = \sqrt{r^2 + a^2}$$

The magnitude of the field at P due to each charge is equal,

$$E_+ = E_- = \frac{1}{4\pi\varepsilon_0} \frac{q}{(r^2 + a^2)}$$

E_+ points away from $+q$ (from B towards P and beyond), while E_- points towards $-q$ (from P towards A). Resolve each into a component along the axis (parallel to AB) and a component perpendicular to the axis (along OP).

The perpendicular (OP-direction) components are equal and opposite, so they cancel. The components parallel to the axis both point from $+q$ towards $-q$ (antiparallel to $\vec{p}$) and therefore add. With $\theta = \frac{a}{\sqrt{r^2+a^2}}$, the resultant field is

$$E = E_+\theta + E_-\theta = 2E_+\theta$$

$$E = 2\times\frac{1}{4\pi\varepsilon_0} \frac{q}{(r^2+a^2)}\times\frac{a}{\sqrt{r^2+a^2}}$$

$$E = \frac{1}{4\pi\varepsilon_0} \frac{2qa}{(r^2+a^2)^{3/2}}$$

Since $\boldsymbol{p = 2qa}$,

$$E = \frac{1}{4\pi\varepsilon_0} \frac{p}{(r^2+a^2)^{3/2}}$$

directed antiparallel to $\vec{p}$.

For a short dipole ($r \gg a$), $\mathbf{r^2 + a^2 \approx r^2}$, so $\mathbf{(r^2+a^2)^{3/2} \approx r^3}$,

$$\vec{E}_{eq} = -\frac{1}{4\pi\varepsilon_0} \frac{\vec{p}}{r^3}$$

The negative sign shows the equatorial field is opposite to the dipole moment. Note it is half the axial field at the same distance.

Final answer. $E_{eq} = \frac{1}{4\pi\varepsilon_0}\frac{p}{(r^2+a^2)^{3/2}}$, which for $r \gg a$ becomes $\vec{E}_{eq} = -\frac{1}{4\pi\varepsilon_0}\frac{\vec{p}}{r^3}$.

← Back to Q155

Q156

Two identical point charges, q each, are kept 2 m apart in air. A third point charge Q of unknown magnitude and sign is placed on the line joining the charges such that the system remains in equilibrium. Find the position and nature of Q.

[2019 • Set 55-1-1]

SOLUTION

Let the two identical charges $+q$ be at $A(0,0)$ and $B(2\ m,0)$, and place the third charge Q on the line joining them at a point $P(x,0)$.

First find the position of Q from its own equilibrium. The two equal charges $+q$ exert forces on Q along the line. For these forces to balance, Q must be equidistant from both, that is at the midpoint,

$$x = 1\ m$$

At this stage the sign and magnitude of Q are not yet fixed, because a charge at the midpoint feels zero net force from the two equal $+q$ for any value of Q.

Now apply the condition that the original charges must also be in equilibrium. Consider the charge $+q$ at A. The repulsive force on it due to the other $+q$ at B (distance 2 m) is

$$F_1 = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{(2)^2} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{4}$$

directed away from B (i.e. in the $-x$ direction). For equilibrium, the force on this $+q$ due to Q (distance 1 m) must be equal and opposite, i.e. attractive (towards Q, in $+x$). An attractive force requires Q to be negative. Its magnitude is

$$F_2 = \frac{1}{4\pi\varepsilon_0}\frac{q\,|Q|}{(1)^2}$$

Setting $F_2 = F_1$,

$$\frac{q\,|Q|}{1} = \frac{q^2}{4}$$

$$|Q| = \frac{q}{4}$$

By symmetry, the same condition makes the charge at B also in equilibrium. Hence the third charge must be placed at the midpoint and must be negative,

$$Q = -\frac{q}{4}$$

So Q is placed at **1 m** from each charge (the midpoint) and is **negative** with magnitude $\frac{q}{4}$.

TIP

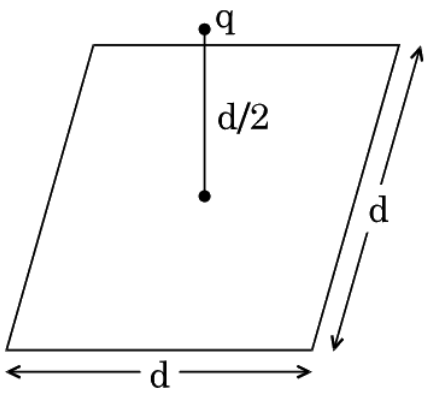
Midpoint placement comes from the equilibrium of Q; the value $\boldsymbol{Q = -q/4}$ comes from the equilibrium of one of the outer charges. You must use both conditions to score full marks.

Final answer. $\boldsymbol{Q}$ is at the midpoint (1 m from each charge) and is negative, $\boldsymbol{Q = -\frac{q}{4}}$.

← *Back to Q156*

Q157

(a) Define electric flux. Is it a scalar or a vector quantity ? A point charge q is at a distance of d/2 directly above the centre of a square of side d, as shown in the figure. Use Gauss' law to obtain the expression for the electric flux through the square.



(b) If the point charge is now moved to a distance 'd' from the centre of the square and the side of the square is doubled, explain how the electric flux will be affected.

[2018]

SOLUTION

(a) Electric flux through a surface is defined as the total number of electric field lines crossing that surface normally, measured as

$$\Phi_E = \int \vec{E} \cdot \mathrm{d}\vec{S}$$

It is a scalar quantity, since it is obtained from the dot product $\vec{E} \cdot \mathrm{d}\vec{S}$, which is a scalar at every point. Its SI unit is $N\,\mathrm{m}^2\mathrm{C}^{-1}$.

Now the point charge q is at a perpendicular distance $\frac{d}{2}$ directly above the centre of a square of side d. Imagine this square to be one face of a cube of side d, with the charge q exactly at the centre of that cube. (A charge at height $\frac{d}{2}$ above the centre of a face is precisely at the cube's centre, since $\frac{d}{2}$ is half the side.)

By Gauss's law, the total flux through the whole cube is

$$\Phi_{\text{cube}} = \frac{q}{\varepsilon_0}$$

By symmetry, the charge sits at the centre and the six faces of the cube are identical, so each face receives an equal share of the flux,

$$\Phi_{\text{square}} = \frac{1}{6}\,\Phi_{\text{cube}} = \frac{q}{6\varepsilon_0}$$

(b) When the charge is moved to a distance d from the centre and the side of the square is doubled to 2d, the new geometry is again the centre-of-cube situation: a charge at perpendicular distance d (which is half of 2d) above the centre of a 2d× 2d face places it exactly at the centre of a cube of side 2d.

By the identical Gauss's-law symmetry argument, each face still receives one-sixth of the total flux,

$$\Phi_{\text{square}}' = \frac{q}{6\varepsilon_0}$$

Hence the flux through the square is unchanged. This is expected, because the flux depends only on the enclosed charge and not on the size of the surface or the distance, as long as the symmetric centre-of-cube geometry is preserved.

CONCEPT

The trick is to recognise that a charge at half-the-side distance above a face centre sits at the cube's centre, so the flux is just $\frac{1}{6}$ of q/ε_0 by the six-face symmetry.

Final answer. $\boldsymbol{\Phi_{\text{square}} = \frac{q}{6\varepsilon_0}}$ in both cases; the flux is unaffected by moving the charge to $\boldsymbol{d}$ and doubling the side.

← *Back to Q157*

Q158

(a) Use Gauss' law to derive the expression for the electric field $(\vec{E})$ *due to a straight uniformly charged infinite line of charge density* λ *C/m.*
(b) Draw a graph to show the variation of E with perpendicular distance r from the line of charge.
(c) Find the work done in bringing a charge q from perpendicular distance r_1 *to* r_2 **(**$\mathbf{r_2 > r_1}$**).**

[2018]

SOLUTION

(a) Consider an infinite straight wire with uniform linear charge density $\lambda\ C/m$. By the cylindrical symmetry of the charge distribution, the field $\vec{E}$ must be radial (pointing straight out from the wire) and its magnitude depends only on the perpendicular distance r.

Choose a coaxial cylindrical Gaussian surface of radius r and length l, with the wire as its axis. It has a curved surface and two flat end-caps.

On the two end-caps, $\vec{E}$ is parallel to the surface (perpendicular to $d\vec{S}$), so they contribute zero flux. On the curved surface, $\vec{E}$ is everywhere parallel to $d\vec{S}$ and uniform in magnitude, so the flux is

$$\Phi = E \times (\text{curved area}) = E\,(2\pi\, r\, l)$$

The charge enclosed by this cylinder is $\mathbf{q_{enc} = \lambda\, l}$. By Gauss's law,

$$E\,(2\pi\, r\, l) = \frac{\lambda\, l}{\varepsilon_0}$$

$$E = \frac{\lambda}{2\pi\varepsilon_0\, r}$$

directed radially outward for $\boldsymbol{\lambda > 0}$. In vector form, $\vec{E} = \dfrac{\lambda}{2\pi\varepsilon_0\, r}\,\hat{r}$.

(b) Graph of E versus r: Since $E \propto \dfrac{1}{r}$, the curve is a rectangular hyperbola. E is very large near the wire (it tends to infinity as $r \to 0$) and falls off smoothly, approaching zero as r increases. Plot a decreasing curve in the first quadrant passing through points such as (r_0, E_0), $(2r_0, E_0/2)$, $(4r_0, E_0/4)$.

(c) The work done in bringing a charge q from perpendicular distance r_1 to r_2 (with $\mathbf{r_2 > r_1}$) is the integral of the force against the displacement,

$$W = \int_{r_1}^{r_2} q\, E\, dr = \int_{r_1}^{r_2} \frac{q\lambda}{2\pi\varepsilon_0\, r}\, dr$$

$$W = \frac{q\lambda}{2\pi\varepsilon_0}\,[\ln r]_{r_1}^{r_2}$$

$$W = \frac{q\lambda}{2\pi\varepsilon_0}\,\ln\left(\frac{r_2}{r_1}\right)$$

This is the work done by the electric force as the charge moves outward (for like signs of q and λ). The work done by an external agent in carrying it in from

r_1 to r_2 is the negative of this, $W_{ext} = \frac{q\lambda}{2\pi\varepsilon_0}\ln(\frac{r_1}{r_2})$.

Final answer. $E = \frac{\lambda}{2\pi\varepsilon_0 r}$ ($E \propto 1/r$); work done $= \frac{q\lambda}{2\pi\varepsilon_0}\ln(\frac{r_2}{r_1})$.

← *Back to Q158*

Q159

(a) Derive an expression for the electric field $\vec{E}$ *due to a dipole of length* '2a' *at a point distant r from the centre of the dipole on the axial line.*
(b) Draw a graph of E versus r for $r >> a$.
(c) If this dipole were kept in a uniform external electric field $\vec{E}_0$, *diagrammatically represent the position of the dipole in stable and unstable equilibrium and write the expressions for the torque acting on the dipole in both the cases.*

[2017]

SOLUTION

(a) Take a dipole of charges $-q$ at A and $+q$ at B, separated by 2a, lying along the x-axis with centre O at the origin. The point P is on the axial line at distance r from O, on the side of $+q$. Then

$$BP = r - a \quad AP = r + a$$

The field at P due to $+q$ (at B) points away from B (along $+x$),

$$E_+ = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r-a)^2}$$

The field at P due to $-q$ (at A) points towards A (along $-x$),

$$E_- = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r+a)^2}$$

Since E_+ and E_- are opposite, the net axial field has magnitude

$$E = E_+ - E_- = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2}\right]$$

$$E = \frac{q}{4\pi\varepsilon_0}\cdot\frac{(r+a)^2 - (r-a)^2}{(r^2-a^2)^2}$$

$$E = \frac{q}{4\pi\varepsilon_0} \cdot \frac{4ar}{(r^2 - a^2)^2}$$

With $p = 2qa$,

$$E = \frac{1}{4\pi\varepsilon_0} \frac{2pr}{(r^2 - a^2)^2}$$

directed along $\vec{p}$ (from $-q$ to $+q$). For a short dipole ($r \gg a$), $r^2 - a^2 \approx r^2$, so

$$\vec{E}_{ax} = \frac{1}{4\pi\varepsilon_0} \frac{2\vec{p}}{r^3}$$

(b) Graph of E versus r for $r \gg a$: a steeply decreasing curve since $E \propto \frac{1}{r^3}$. It falls off faster than the $1/r^2$ field of a point charge; doubling r reduces E by a factor of 8. Plot a decreasing curve approaching the r-axis quickly.

(c) Stable and unstable equilibrium in a uniform field $\vec{E}_0$:

Stable equilibrium: draw the dipole with $\vec{p}$ pointing along $\vec{E}_0$ ($\theta = 0°$); the $+q$ end faces the field direction. Potential energy is minimum, $U = -pE_0$. The torque is

$$\vec{\tau} = \vec{p} \times \vec{E}_0 = 0 \ (\theta = 0°)$$

Unstable equilibrium: draw the dipole with $\vec{p}$ pointing opposite to $\vec{E}_0$ ($\theta = 180°$). Potential energy is maximum, $U = +pE_0$. The torque is

$$\vec{\tau} = \vec{p} \times \vec{E}_0 = 0 \ (\theta = 180°)$$

In both cases $\tau = pE_0\theta = 0$, but the stable case has minimum PE (the dipole returns if disturbed) while the unstable case has maximum PE (the dipole flips away if disturbed).

Final answer. $E_{ax} = \frac{1}{4\pi\varepsilon_0} \frac{2pr}{(r^2 - a^2)^2}$, giving $\frac{2\vec{p}}{4\pi\varepsilon_0 r^3}$ for $r \gg a$; torque is zero in both equilibria.

← *Back to Q159*

Q160

(a) Use Gauss's theorem to find the electric field due to a uniformly charged infinitely large plane thin sheet with surface charge density σ.
(b) An infinitely large thin plane sheet has a uniform surface charge density +σ. Obtain the expression for the amount of work done in bringing a point charge q from infinity to a point, distant r, in front of the charged plane sheet.

[2017]

SOLUTION

(a) Consider an infinitely large thin plane sheet with uniform surface charge density σ. By symmetry, the field $\vec{E}$ must be directed perpendicular to the sheet and point outward on both sides, with the same magnitude E at equal distances.

Choose a Gaussian surface in the form of a cylinder (or pillbox) that pierces the sheet, with its two flat faces (each of area A) parallel to the sheet, one on each side, and its curved surface perpendicular to the sheet.

On the curved surface, $\vec{E}$ is parallel to the surface, so it contributes no flux. On the two flat faces, $\vec{E}$ is parallel to $d\vec{S}$ and uniform, so the total flux is

$$\Phi = E\,A + E\,A = 2EA$$

The charge enclosed by the pillbox is the charge on the area A of the sheet, $\mathbf{q_{enc} = \sigma A}$. By Gauss's law,

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}$$

directed away from the sheet (for $\sigma > 0$). In vector form, $\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$. Notice the field is uniform and independent of the distance from the sheet.

(b) The field of the infinite sheet is uniform, $E = \frac{\sigma}{2\varepsilon_0}$, and points away from the positively charged sheet. To bring a point charge q from a chosen reference towards the sheet up to a distance r, work must be done against (or by) this constant field.

Taking the sheet's surface as the reference of zero potential, the potential at perpendicular distance r is

$$V(r) = -\int_0^r E\, dr' = -\frac{\sigma}{2\varepsilon_0}\, r$$

The work done by an external agent in bringing the charge q to the point at distance r equals q times this potential (relative to the reference),

$$W = q\,|V(r)| = \frac{q\,\sigma\,r}{2\varepsilon_0}$$

So the magnitude of the work done is

$$W = \frac{q\,\sigma\,r}{2\varepsilon_0}$$

For a positive test charge moving towards the positively charged sheet, the external agent does positive work against the repulsive field. (For unlike signs the work done by the external agent would be negative.)

Final answer. $E = \dfrac{\sigma}{2\varepsilon_0}$ (uniform, normal to the sheet); work done $W = \dfrac{q\sigma\, r}{2\varepsilon_0}$.

← *Back to Q160*

Q161

(i) Use Gauss's law to find the electric field due to a uniformly charged infinite plane sheet. What is the direction of field for positive and negative charge densities?
(ii) Find the ratio of the potential differences that must be applied across the parallel and series combination of two capacitors C_1 *and* C_2 *with their capacitances in the ratio* 1 : 2 *so that the energy stored in the two cases becomes the same.*
OR
(i) If two similar large plates, each of area A having surface charge densities $+\sigma$ *and* $-\sigma$ *are separated by a distance d in air, find the expressions for*
(a) field at points between the two plates and on outer side of the plates. Specify the direction of the field in each case.
(b) the potential difference between the plates.
(c) the capacitance of the capacitor so formed.

(ii) Two metallic spheres of radii R and 2R are charged so that both of these have same surface charge density σ. If they are connected to each other with a conducting wire, in which direction will the charge flow and why?

[2016]

SOLUTION

(*i*) Consider an infinite plane sheet with uniform surface charge density σ. By symmetry, the field $\vec{E}$ is perpendicular to the sheet and has the same magnitude at equal distances on either side.

Draw a Gaussian pillbox (a small cylinder) of cross-sectional area A piercing the sheet, with its two flat faces parallel to the sheet at equal distances on each side. The field is perpendicular to both flat faces and parallel to the curved surface.

Flux through the curved surface is zero (since $\vec{E}$ is parallel to it). Flux through the two flat faces is $E\,A$ each.

$$\oint \vec{E} \cdot d\vec{S} = E\,A + E\,A = 2EA$$

The charge enclosed by the pillbox is $\mathbf{q_{enc} = \sigma A}$.

By Gauss's law, $\oint \vec{E} \cdot d\vec{S} = \dfrac{q_{enc}}{\varepsilon_0}$

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$\Rightarrow E = \frac{\sigma}{2\varepsilon_0}$$

This is independent of the distance from the sheet, so the field is uniform. For positive charge density ($\sigma > 0$) the field points **away from the sheet** on both sides; for negative charge density ($\sigma < 0$) it points **towards the sheet** on both sides.

(ii) Let $C_1 = C$ and $C_2 = 2C$ (ratio 1:2).

In parallel, $C_p = C_1 + C_2 = C + 2C = 3C$.

In series, $C_s = \dfrac{C_1 C_2}{C_1 + C_2} = \dfrac{(C)(2C)}{3C} = \dfrac{2C}{3}$.

The energy stored is $U = \dfrac{1}{2}CV^2$. For equal energies in the two cases,

$$\frac{1}{2} C_p V_p^2 = \frac{1}{2} C_s V_s^2$$

$$\frac{V_p^2}{V_s^2} = \frac{C_s}{C_p} = \frac{2C/3}{3C} = \frac{2}{9}$$

$$\Rightarrow \frac{V_p}{V_s} = \frac{\sqrt{2}}{3}$$

The required ratio of potential differences is $\frac{V_p}{V_s} = \frac{\sqrt{2}}{3}$.

OR

(*i*) (*a*) Each large plate with surface charge density σ produces a field of magnitude $\frac{\sigma}{2\varepsilon_0}$ on either side.

Between the plates the two fields add up (both point from the +σ plate to the −σ plate):

$\mathbf{E_{between}} = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$, directed from the positive plate towards the negative plate.

On the outer sides of the plates the two fields are equal and opposite, so they cancel:

$\mathbf{E_{outside}} = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0.$

(*b*) Potential difference between the plates: $V = E\,d = \frac{\sigma\, d}{\varepsilon_0}$.

(*c*) With charge $Q = \sigma A$ on the positive plate,

$C = \frac{Q}{V} = \frac{\sigma A}{\sigma\, d/\varepsilon_0} = \frac{\varepsilon_0 A}{d}$.

(ii) The two spheres have the same surface charge density σ, so their charges are $Q_R = \sigma\,(4\pi R^2)$ and $Q_{2R} = \sigma\,(4\pi\,(2R)^2) = 4\sigma\,(4\pi R^2)$.

The potential of an isolated sphere is $V = \frac{kQ}{r}$. For the small sphere,

$$V_R = \frac{k\,\sigma(4\pi R^2)}{R} = 4\pi k\sigma R$$

For the large sphere,

$$V_{2R} = \frac{k\,\sigma(4\pi \cdot 4R^2)}{2R} = 8\pi\, k\sigma\, R$$

Since $V_{2R} > V_R$, the larger sphere is at higher potential. When connected by a wire, **charge flows from the larger sphere (radius 2R) to the smaller sphere (radius R)**, because charge always flows from higher to lower potential until both reach the same potential.

COMMON MISTAKE

When two charged spheres are joined, it is the potential (not the charge or the charge density) that must equalise. Same surface charge density does NOT mean same potential, so charge still flows.

Final answer. Sheet field $E = \frac{\sigma}{2\varepsilon_0}$; voltage ratio $\frac{V_p}{V_s} = \frac{\sqrt{2}}{3}$; OR $C = \frac{\varepsilon_0 A}{d}$ and charge flows from the larger to the smaller sphere.

← *Back to Q161*

Q162

(a) "The outward electric flux due to charge +Q is independent of the shape and size of the surface which encloses it." Give two reasons to justify this statement.
(b) Two identical circular loops '1' and '2' of radius R each have linear charge densities −λ and +λ C/m respectively. The loops are placed coaxially with their centres $R\sqrt{3}$ distance apart. Find the magnitude and direction of the net electric field at the centre of loop '1'.

[2015]

SOLUTION

(*a*) The outward flux through a closed surface enclosing +Q is $\Phi = \frac{Q}{\varepsilon_0}$, independent of the shape and size of the surface. Two reasons:

(i) By Gauss's law, $\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}$. The right-hand side depends only on the enclosed charge and not on the geometry of the surface, so any closed surface enclosing the same charge +Q gives the same flux.

(ii) In terms of field lines, a charge $+Q$ sends out a fixed number of field lines that extend to infinity. Every closed surface surrounding the charge is crossed (net) by exactly the same number of these lines, so the flux is the same whatever the shape or size of the surface.

(*b*) Draw the two coaxial loops. Loop 1 carries $-\lambda$ and loop 2 carries $+\lambda$; their centres are separated by $\boldsymbol{d = R\sqrt{3}}$ along the common axis.

At its own centre, loop 1 produces zero field, because the contributions from diametrically opposite charge elements cancel by symmetry.

So the net field at the centre of loop 1 is due only to loop 2. The centre of loop 1 lies on the axis of loop 2 at axial distance $\boldsymbol{d = R\sqrt{3}}$.

Charge on loop 2: $\boldsymbol{Q = \lambda\,(2\pi\,R)}$.

Axial field of a charged ring at distance d from its centre:

$$E = \frac{1}{4\pi\varepsilon_0}\,\frac{Q\,d}{(R^2 + d^2)^{3/2}}$$

Substituting $\boldsymbol{Q = 2\pi\,R\lambda}$ and $\boldsymbol{d = R\sqrt{3}}$, so that $\boldsymbol{R^2 + d^2 = R^2 + 3R^2 = 4R^2}$:

$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{(2\pi\,R\lambda)(R\sqrt{3})}{(4R^2)^{3/2}}$$

$\boldsymbol{(4R^2)^{3/2} = 8R^3}$, so

$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2\pi\sqrt{3}\,\lambda\,R^2}{8R^3} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\sqrt{3}\,\pi\lambda}{4R}$$

$$E = \frac{\sqrt{3}\,\lambda}{16\,\varepsilon_0\,R}$$

Direction: loop 2 carries positive charge, so its axial field points away from loop 2, i.e. along the axis directed from loop 2 towards loop 1 (and beyond).

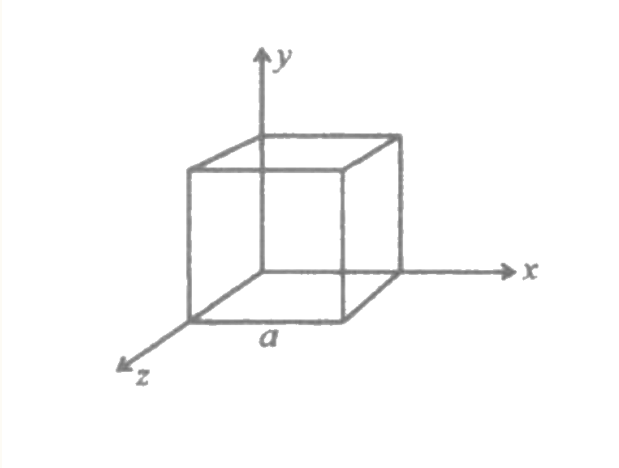
The net field at the centre of loop 1 is $\boldsymbol{E = \frac{\sqrt{3}\,\lambda}{16\,\varepsilon_0\,R}}$, directed along the axis pointing from loop 2 to loop 1.

Final answer. Net field at the centre of loop 1 is $\boldsymbol{\frac{\sqrt{3}\,\lambda}{16\,\varepsilon_0\,R}}$, directed along the axis from loop 2 towards loop 1.

← Back to Q162

Q163

(a) An electric dipole of dipole moment $\vec{p}$ consists of point charges +q and −q separated by a distance 2a *apart. Deduce the expression for the electric field $\vec{E}$ due to the dipole at a distance x from the centre of the dipole on its axial line in terms of the dipole moment $\vec{p}$. Hence show that in the limit $x \gg a$, $\vec{E} \to 2\vec{p}/(4\pi\varepsilon_0 x^3)$.*
(b) Given the electric field in the region $\vec{E} = 2x\hat{i}$, find the net electric flux through the cube and the charge enclosed by it.



[2015]

SOLUTION

(*a*) Draw the dipole along a horizontal axis: $-q$ on the left and $+q$ on the right, separated by 2a, with O the midpoint. Take the axial point P at distance x from O, on the side of $+q$.

Distance of P from $+q$ is $(x - a)$ and from $-q$ is $(x + a)$.

Field at P due to $+q$ (directed away from $+q$, i.e. along $\vec{p}$):

$$E_+ = \frac{1}{4\pi\varepsilon_0}\,\frac{q}{(x-a)^2}$$

Field at P due to $-q$ (directed towards $-q$, i.e. opposite to $\vec{p}$):

$$E_- = \frac{1}{4\pi\varepsilon_0}\,\frac{q}{(x+a)^2}$$

Net axial field (along $\vec{p}$):

$$E = E_+ - E_- = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(x-a)^2} - \frac{1}{(x+a)^2}\right]$$

$$E = \frac{q}{4\pi\varepsilon_0} \cdot \frac{(x+a)^2 - (x-a)^2}{(x^2 - a^2)^2}$$

$\mathbf{(x+a)^2 - (x-a)^2 = 4ax}$, so

$$E = \frac{q}{4\pi\varepsilon_0} \cdot \frac{4ax}{(x^2 - a^2)^2} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{2(2qa)\,x}{(x^2 - a^2)^2}$$

Using $\boldsymbol{p = 2qa}$,

$\vec{E} = \dfrac{1}{4\pi\varepsilon_0} \cdot \dfrac{2p\,x}{(x^2 - a^2)^2}$, directed along $\vec{p}$.

For a short dipole, $x \gg a$, so a^2 is negligible compared with x^2 and $\mathbf{(x^2 - a^2)^2 \approx x^4}$:

$$\vec{E} \to \frac{1}{4\pi\varepsilon_0} \cdot \frac{2p\,x}{x^4} = \frac{2\vec{p}}{4\pi\varepsilon_0\, x^3}$$

which is the required result.

(*b*) The field is $\vec{E} = 2x\,\hat{i}$. Place a cube of side a with one corner at the origin and edges along the axes; the left face is at $x = 0$ and the right face at $x = a$, each of area a^2.

Faces with outward normals along $\pm\hat{j}$ and $\pm\hat{k}$ are parallel to $\vec{E}$, so the flux through them is zero.

Left face ($x = 0$, outward normal $-\hat{i}$): here $\vec{E} = 0$, so $\Phi_L = 0$.

Right face ($x = a$, outward normal $+\hat{i}$): here $E = 2a$, so

$$\Phi_R = E\,a^2 = (2a)(a^2) = 2a^3$$

Net flux through the cube:

$$\Phi_{net} = \Phi_R + \Phi_L = 2a^3\ N\,m^2\,C^{-1}$$

By Gauss's law, charge enclosed:

$$q_{enc} = \varepsilon_0\,\Phi_{net} = 2\varepsilon_0\,a^3$$

Net flux = $2a^3\ N\,m^2C^{-1}$ and **enclosed charge** = $2\varepsilon_0\,a^3\ C$.

Final answer. Axial field $\vec{E}=\frac{2\vec{p}}{4\pi\varepsilon_0 x^3}$ for $x \gg a$; net flux $=2a^3\ N\,m^2C^{-1}$ and enclosed charge $=2\varepsilon_0 a^3\ C$.

← Back to Q163

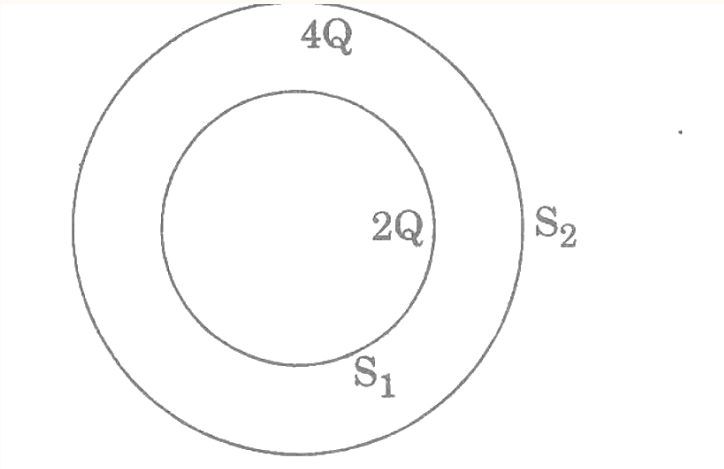
Q164

Draw a labelled diagram of Van de Graaff generator. State its working principle to show how by introducing a small charged sphere into a larger sphere, a large amount of charge can be transferred to the outer sphere. State the use of this machine and also point out its limitations.

OR

(a) Deduce the expression for the torque acting on a dipole of dipole moment $\vec{p}$ in the presence of a uniform electric field $\vec{E}$.

(b) Consider two hollow concentric spheres, S_1 and S_2, enclosing charges 2Q and 4Q respectively as shown in the figure. (i) Find out the ratio of the electric flux through them. (ii) How will the electric flux through the sphere S_1 change if a medium of dielectric constant 'ε_r' is introduced in the space inside S_1 in place of air ? Deduce the necessary expression.



[2014]

SOLUTION

Van de Graaff generator

Labelled diagram (describe): A large hollow metal dome (sphere) sits at the top of a tall insulating column. A long endless belt of insulating material runs over

two pulleys, a lower pulley near the base and an upper pulley inside the dome, driven by a motor. A sharp comb B_1 (spray comb) near the lower pulley is connected to a high-voltage source; a second sharp comb B_2 (collecting comb) near the upper pulley is connected to the inner surface of the dome. The whole assembly stands on an insulating support, often enclosed in a chamber of high-pressure gas.

Working principle: If a small charged conductor is placed inside a larger hollow conductor and the two are connected, the entire charge of the inner conductor flows to the outer surface of the larger conductor, however much charge the outer one already carries. This is because charge always resides on the outer surface of a conductor, and the inside of a hollow conductor is at the same potential as its surface, so charge keeps flowing outward to the dome.

Working: The comb B_1 sprays positive charge onto the moving belt by the action of sharp points (corona discharge). The belt carries this charge up into the dome, where the comb B_2 collects it (again by point/corona action) and passes it to the dome. The charge spreads over the outer surface of the dome. As the belt runs continuously, charge keeps accumulating and the potential of the dome rises to a few million volts.

Use: It is used to build up very high potentials (of the order of a few MV) needed to accelerate charged particles (protons, deuterons, ions) to high energies for use in nuclear physics experiments and particle accelerators.

Limitations: The maximum attainable voltage is limited by the dielectric breakdown (ionisation) of the surrounding air or gas. Once the surface field reaches the breakdown value (about 3 MV/m for air), the charge leaks away by sparking, fixing an upper limit on the potential. Enclosing the dome in high-pressure gas of high dielectric strength raises this limit.

OR

(a) Consider a dipole of charges $+q$ and $-q$ separated by 2a, with dipole moment $\vec{p}$, placed in a uniform field $\vec{E}$ such that $\vec{p}$ makes angle θ with $\vec{E}$.

Force on $+q$ is $+q\vec{E}$ and on $-q$ is $-q\vec{E}$, each of magnitude qE. They are equal, opposite and parallel, so the net force is zero but they form a couple.

Torque (couple) = force × perpendicular distance between the lines of action.

$$\tau = qE \times (2a\theta)$$

$$\tau = (q \cdot 2a)\, E\theta = pE\theta$$

In vector form, $\vec{\tau} = \vec{p} \times \vec{E}$.

The torque acts perpendicular to the plane of $\vec{p}$ and $\vec{E}$ and tends to align the dipole moment along the field.

(*b*) (*i*) By Gauss's law, the flux through a closed surface depends only on the charge enclosed.

$\Phi_{S_1} = \dfrac{2Q}{\varepsilon_0}$ (sphere S_1 encloses 2Q).

$\Phi_{S_2} = \dfrac{2Q + 4Q}{\varepsilon_0} = \dfrac{6Q}{\varepsilon_0}$ (sphere S_2 encloses both charges).

$$\frac{\Phi_{S_1}}{\Phi_{S_2}} = \frac{2Q/\varepsilon_0}{6Q/\varepsilon_0} = \frac{1}{3}$$

Ratio of fluxes $\Phi_{S_1} : \Phi_{S_2} = 1 : 3$.

(ii) If a medium of dielectric constant ε_r fills the space inside S_1, the permittivity becomes $\varepsilon = \varepsilon_r \varepsilon_0$. The flux through S_1 becomes

$$\Phi_{S_1}' = \frac{2Q}{\varepsilon_r \varepsilon_0} = \frac{1}{\varepsilon_r}\,\Phi_{S_1}$$

So the flux through S_1 **decreases by a factor ε_r**, becoming $\dfrac{2Q}{\varepsilon_r \varepsilon_0}$.

Final answer. Van de Graaff builds up a few MV (limited by air breakdown ≈ **3 MV/*m***); OR $\vec{\tau} = \vec{p} \times \vec{E}$, $\Phi_{S_1} : \Phi_{S_2} = 1:3$, and with dielectric $\Phi_{S_1}' = \dfrac{2Q}{\varepsilon_r \varepsilon_0}$.

← *Back to Q164*

Q165

Explain the principle of a device that can build up high voltages of the order of a few million volts.
Draw a schematic diagram and explain the working of this device.
Is there any restriction on the upper limit of the high voltages set up in this machine? Explain.
OR

(a) Define electric flux. Write its S.I. units.
(b) Using Gauss's law, prove that the electric field at a point due to a uniformly charged infinite plane sheet is independent of the distance from it.
(c) How is the field directed if (i) the sheet is positively charged, (ii) negatively charged?

[2012]

SOLUTION

The device that builds up voltages of the order of a few million volts is the **Van de Graaff generator**.

Principle: If a small charged conductor is placed inside a larger hollow conductor and connected to it, the whole of its charge passes to the outer surface of the larger conductor, no matter how much charge that conductor already holds. This is because in a conductor charge resides only on the outer surface, and the inside of a hollow conductor is shielded and at the same potential as its surface, so charge keeps flowing outward. Repeating this transfer builds up a very large charge, and hence a very high potential, on the outer dome.

Schematic diagram (describe): A large hollow metallic dome mounted on an insulating column. An endless insulating belt runs over a lower pulley (at the base) and an upper pulley (inside the dome). A sharp spray comb B_1 near the base is connected to a high-voltage source; a sharp collecting comb B_2 near the upper pulley is connected to the dome.

Working: The comb B_1 sprays positive charge onto the moving belt by corona discharge from its sharp points. The belt carries this charge upward into the dome. There the comb B_2 collects the charge (again by point/corona action) and transfers it to the outer surface of the dome. As the belt runs continuously, charge keeps accumulating on the dome and its potential rises to a few million volts. These high potentials are used to accelerate charged particles.

Upper limit: Yes, there is a restriction. The maximum potential is limited by the dielectric breakdown (ionisation) of the surrounding gas. When the electric field at the dome reaches the breakdown strength of air (about 3 MV/m), the charge leaks away through sparking and the potential cannot rise further. Enclosing the generator in a vessel of high-pressure gas of high dielectric strength raises this limit.

OR

(*a*) Electric flux through a surface is the total number of electric field lines passing normally through it. It is defined as $\Phi_E = \int \vec{E} \cdot \vec{dS}$. Its SI unit is $N\ m^2\ C^{-1}$ (equivalently $V\ m$).

(*b*) Consider an infinite plane sheet with uniform surface charge density σ. By symmetry, the field is perpendicular to the sheet and equal on both sides.

Choose a Gaussian pillbox (cylinder) of cross-sectional area A piercing the sheet, with its flat faces parallel to the sheet at equal distances on either side.

Flux through the curved surface is zero (field is parallel to it). Flux through each flat face is EA, so total flux = **2EA**.

Charge enclosed = **σ *A***.

By Gauss's law,

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$\Rightarrow E = \frac{\sigma}{2\varepsilon_0}$$

This expression contains no distance term, so the field is independent of the distance from the sheet, i.e. it is uniform.

(*c*) Direction: (i) If the sheet is positively charged, the field points **away from the sheet** on both sides. (ii) If the sheet is negatively charged, the field points **towards the sheet** on both sides.

Final answer. The Van de Graaff generator builds up a few MV, limited by dielectric breakdown of air (≈ **3 MV/*m***); OR sheet field $E = \frac{\sigma}{2\varepsilon_0}$ is distance-independent.

← *Back to Q165*

Q166

State Gauss' theorem in electrostatics. Apply it to find the electric field strength at a

point near a long straight conductor carrying electric charge.

[2010]

SOLUTION

Gauss's theorem: The net electric flux through any closed surface is equal to $\frac{1}{\varepsilon_0}$ times the total charge enclosed by that surface.

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}$$

Field near a long straight charged conductor: Let the conductor carry a uniform linear charge density λ (charge per unit length). We find the field at a point at perpendicular distance r from it.

Draw the wire along the axis. By cylindrical symmetry, the field $\vec{E}$ is radial (pointing straight out from the wire) and has the same magnitude at every point on a cylinder coaxial with the wire.

Choose a coaxial Gaussian cylinder of radius r and length l around the conductor, with two flat end caps.

On the two end caps, $\vec{E}$ is parallel to the surface, so the flux through them is zero. On the curved surface, $\vec{E}$ is perpendicular (radial) and constant, so the flux is

$$\Phi = E \times (\text{curved area}) = E\,(2\pi\, r\, l)$$

Charge enclosed by the cylinder: $\mathbf{q_{enc} = \lambda\, l}$.

By Gauss's law,

$$E\,(2\pi\, r\, l) = \frac{\lambda\, l}{\varepsilon_0}$$

$$\Rightarrow E = \frac{\lambda}{2\pi\varepsilon_0\, r}$$

The field is radial, directed outward for $\lambda > 0$ (inward for $\lambda < 0$), and falls off as $\frac{1}{r}$.

The electric field near a long straight charged conductor is $E = \frac{\lambda}{2\pi\varepsilon_0\, r}$.

Final answer. Field near a long straight charged conductor is $E = \frac{\lambda}{2\pi\varepsilon_0 r}$, directed radially.

← *Back to Q166*

Q167

What is the magnitude of a point electric charge chosen so that the electric field due to it at a point 50 cm away has magnitude $2.0\ N\ C^{-1}$*?*

[2010]

SOLUTION

The electric field due to a point charge at distance r is

$E = \frac{1}{4\pi\varepsilon_0}\frac{|q|}{r^2} = \frac{k|q|}{r^2}$, with $k = 9\times 10^9\ N\,m^2C^{-2}$.

Rearranging for the charge,

$$|q| = \frac{E\,r^2}{k}$$

Given $E = 2.0\ N\ C^{-1}$ and $r = 50\ cm = 0.50\ m$.

Substituting the values,

$$|q| = \frac{(2.0)(0.50)^2}{9\times 10^9}$$

$$|q| = \frac{(2.0)(0.25)}{9\times 10^9}$$

$$|q| = \frac{0.50}{9\times 10^9}$$

$$|q| = 5.56\times 10^{-11}\ C$$

The required charge is $|q| \approx 5.6\times 10^{-11}\ C = 56\ pC$.

COMMON MISTAKE

Convert the distance to metres before squaring. Using $r=50$ *instead of* 0.50 *m gives an answer wrong by a factor of* 10^4.

Final answer. The point charge is $|q| \approx 5.6 \times 10^{-11}$ C (about **56 pC**).

← Back to Q167

Q168

(a) Using Gauss' law, derive an expression for the electric field intensity at any point outside a uniformly charged thin spherical shell of radius R and charge density σ C/m^2. *Draw the field lines when the charge density of the sphere is (i) positive, (ii) negative.*
(b) A uniformly charged conducting sphere of 2.5 *m in diameter has a surface charge density of* $100\ \mu C/m^2$. *Calculate the*
(i) Charge on the sphere
(ii) Total electric flux passing through the sphere

[2008]

SOLUTION

(*a*) Consider a thin spherical shell of radius R carrying charge uniformly with surface charge density σ, so the total charge is $Q = \sigma\,(4\pi R^2)$. We find the field at a point outside, at distance $r > R$ from the centre.

By spherical symmetry, the field $\vec{E}$ is radial and has the same magnitude at every point of a concentric sphere of radius r.

Choose a Gaussian sphere of radius r ($r > R$) concentric with the shell.

Flux through it: $\oint \vec{E} \cdot d\vec{S} = E\,(4\pi r^2)$.

Charge enclosed = Q.

By Gauss's law,

$$E\,(4\pi r^2) = \frac{Q}{\varepsilon_0}$$

$$\Rightarrow E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\sigma (4\pi R^2)}{r^2} = \frac{\sigma R^2}{\varepsilon_0 r^2}$$

So outside the shell the field is the same as if the whole charge were concentrated at the centre.

Field lines: (i) For a positively charged shell ($\sigma > 0$), draw straight radial lines emerging outward from the surface in all directions, ending at infinity. (ii) For a negatively charged shell ($\sigma < 0$), draw straight radial lines pointing inward, coming from infinity and ending on the surface.

(*b*) Diameter = **2.5 *m***, so radius ***R* = 1.25 *m***. Surface charge density $\sigma = 100\ \mu C\ m^{-2} = 1.0\times 10^{-4}\ C\ m^{-2}$.

(*i*) Charge on the sphere:

$$Q = \sigma (4\pi R^2)$$

$$Q = (1.0\times 10^{-4})(4\pi)(1.25)^2$$

$$Q = (1.0\times 10^{-4})(4\pi)(1.5625)$$

$$Q = 1.96\times 10^{-3}\ C \approx 1.96\ mC$$

(ii) Total electric flux through the sphere, by Gauss's law:

$$\Phi_E = \frac{Q}{\varepsilon_0}$$

$$\Phi_E = \frac{1.96\times 10^{-3}}{8.854\times 10^{-12}}$$

$$\Phi_E = 2.22\times 10^{8}\ N\ m^2\ C^{-1}$$

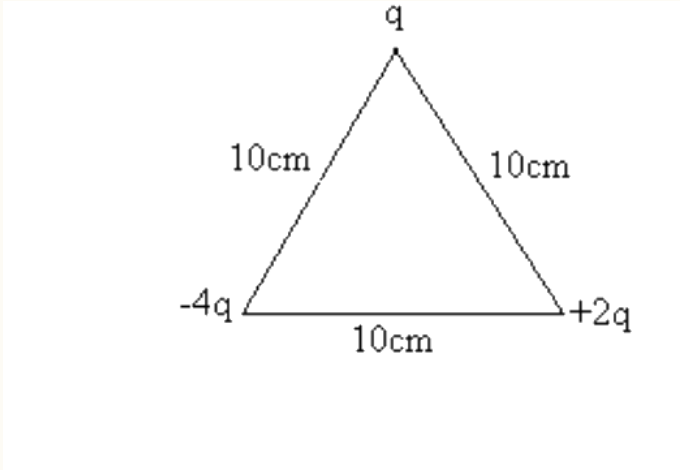
Charge $Q \approx 1.96$ mC and **total flux $\Phi_E \approx 2.22\times 10^8\ N\ m^2 C^{-1}$**.

Final answer. Outside field $E = \frac{\sigma R^2}{\varepsilon_0 r^2}$; charge $Q \approx 1.96$ mC and total flux $\Phi_E \approx 2.22\times 10^8\ N\ m^2 C^{-1}$.

← *Back to Q168*

Q169

(a) Derive an expression for the torque experienced by an electric dipole kept in a uniform electric field.
(b) Calculate the work done to dissociate the system of three charges placed on the vertices of a triangle as shown.



[2008]

SOLUTION

(*a*) Consider a dipole of charges +*q* and −*q* separated by 2a, with dipole moment $\vec{p}$ (magnitude $\boldsymbol{p = q \cdot 2a}$), placed in a uniform field $\vec{E}$ so that $\vec{p}$ makes angle θ with $\vec{E}$.

Force on +*q* is $+q\vec{E}$ and on −*q* is $-q\vec{E}$, each of magnitude qE. These are equal, opposite and act along different lines, so the net force is zero but together they form a couple.

Torque of the couple = (either force) × (perpendicular distance between the two forces).

The perpendicular distance between the lines of action is 2aθ, so

$$\tau = qE \times (2a\theta)$$

$$\tau = (q \cdot 2a)\, E\theta = pE\theta$$

In vector form, $\vec{\tau} = \vec{p} \times \vec{E}$.

The torque is perpendicular to the plane of $\vec{p}$ and $\vec{E}$ and tends to align the dipole with the field.

(*b*) The three charges q, q, $-4q$ sit at the vertices of an equilateral triangle of side $l = 10\ \text{cm} = 0.1\ m$. Let $k = \dfrac{1}{4\pi\varepsilon_0} = 9\times 10^9\ N\,m^2C^{-2}$.

The potential energy of the system is the sum over the three pairs:

$$U = \frac{k}{l}[(q)(q) + (q)(-4q) + (-4q)(q)]$$

$$U = \frac{k}{l}[q^2 - 4q^2 - 4q^2]$$

$$U = -\frac{7kq^2}{l}$$

To dissociate the system, every charge is taken to infinity, where the final potential energy is zero. The work done by the external agent equals the increase in potential energy:

$$W_{ext} = U_\infty - U = 0 - \left(-\frac{7kq^2}{l}\right) = \frac{7kq^2}{l}$$

$$W_{ext} = \frac{7\,(9\times 10^9)\,q^2}{0.1}$$

$W_{ext} = 6.3\times 10^{11}\ q^2\ J$ (with q in coulomb).

The work required to dissociate the system is $W_{ext} = \dfrac{7kq^2}{l} = 6.3\times 10^{11}\ q^2\ J$.

Final answer. Torque $\vec{\tau} = \vec{p}\times\vec{E}$; work to dissociate $W_{ext} = \dfrac{7kq^2}{l} = 6.3\times 10^{11}\ q^2\ J$.

← *Back to Q169*